

Global Anomaly-Free Spectra Catalogue

Dimension notation version: combined $n = 1$ through $n = 5$ scan results

This PDF combines the spectra retained by the complete scans with one through five simple non- R gauge factors. Each term is displayed as $\mathbf{m}(\mathbf{R}_1, \dots, \mathbf{R}_n)$, where the entries are representation dimensions under the ordered gauge factors. Neutral singlets are included as 1. For repeated dimensions in the same group, primes distinguish inequivalent irreducible representations; the table below defines those symbols. For half-hypermultiplets, the multiplicity is multiplied by 1/2. For D_4 , triality labels are written as subscripts, such as 8_v , 8_s , and 8_c .

n	Spectra retained
1	646
2	1968
3	1092
4	388
≥ 5	0
Total	4094

$n = 1$ lattice filter. The local scan contains 652 spectra; six denominator-3 G_2 branches fail the common odd unimodular lattice embedding, leaving 646 spectra in the combined catalogue.

No spectra for $n \geq 5$. The complete $n = 5$ scan is empty. Since every model with $n > 5$ would have an $n = 5$ shadow, the catalogue is empty for all $n \geq 5$.

Anomaly-vector convention: $\eta = ((0,1), (1,0))$, $a = (-2, -2)$, $b_R = (3/2, -5)$.

Notation table for repeated dimensions and D_4 triality labels

Group	Symbol	Dynkin label	Dimension
A3	20	(0,2,0)	20
A3	20'	(1,1,0)	20
A3	20''	(3,0,0)	20
C2	35	(2,1)	35
C2	35'	(4,0)	35
C3	14	(0,0,1)	14
C3	14'	(0,1,0)	14
D4	8_v	(1,0,0,0)	8
D4	8_s	(0,0,1,0)	8
D4	8_c	(0,0,0,1)	8
D4	35_v	(2,0,0,0)	35
D4	35_s	(0,0,2,0)	35
D4	35_c	(0,0,0,2)	35
D4	56_c	(1,0,1,0)	56
D4	56_s	(1,0,0,1)	56
D4	56_v	(0,0,1,1)	56

n = 1 detailed spectra

A3

Spectrum 1; $b = (2, 4)$; singlets = 64

$$64(\mathbf{1}) + 64(\mathbf{4}) + 1(\mathbf{20})$$

Spectrum 2; $b = (2, 4)$; singlets = 84

$$84(\mathbf{1}) + 43(\mathbf{4}) + 9(\mathbf{6}) + 1(\mathbf{10}) + 1(\mathbf{20}')$$

Spectrum 3; $b = (2, 4)$; singlets = 85

$$85(\mathbf{1}) + 43(\mathbf{4}) + 1(\mathbf{15}) + 8(\mathbf{6}) + 1(\mathbf{20}')$$

Spectrum 4; $b = (2, 4)$; singlets = 92

$$92(\mathbf{1}) + 32(\mathbf{4}) + 15(\mathbf{6}) + 3(\mathbf{10})$$

Spectrum 5; $b = (2, 4)$; singlets = 93

$$93(\mathbf{1}) + 32(\mathbf{4}) + 1(\mathbf{15}) + 14(\mathbf{6}) + 2(\mathbf{10})$$

Spectrum 6; $b = (2, 4)$; singlets = 94

$$94(\mathbf{1}) + 32(\mathbf{4}) + 2(\mathbf{15}) + 13(\mathbf{6}) + 1(\mathbf{10})$$

Spectrum 7; $b = (2, 4)$; singlets = 95

$$95(\mathbf{1}) + 32(\mathbf{4}) + 3(\mathbf{15}) + 12(\mathbf{6})$$

Spectrum 1; $b = (7/2, 9)$; singlets = 0

$$18(\mathbf{4}) + 6(\mathbf{15}) + 18(\mathbf{6}) + 1(\mathbf{10}) + 1(\mathbf{20}'') + 1(\mathbf{20}') + 1(\mathbf{20})$$

Spectrum 2; $b = (7/2, 9)$; singlets = 0

$$19(\mathbf{4}) + 2(\mathbf{15}) + 19(\mathbf{6}) + 2(\mathbf{10}) + 1(\mathbf{20}'') + 4(\mathbf{20}')$$

Spectrum 3; $b = (7/2, 9)$; singlets = 0

$$19(\mathbf{4}) + 6(\mathbf{15}) + 18(\mathbf{6}) + 1(\mathbf{10}) + 1(\mathbf{20}) + 1(\mathbf{36})$$

Spectrum 4; $b = (7/2, 9)$; singlets = 0

$$20(\mathbf{4}) + 2(\mathbf{15}) + 19(\mathbf{6}) + 2(\mathbf{10}) + 3(\mathbf{20}') + 1(\mathbf{36})$$

Spectrum 5; $b = (7/2, 9)$; singlets = 1

$$1(\mathbf{1}) + 18(\mathbf{4}) + 7(\mathbf{15}) + 17(\mathbf{6}) + 1(\mathbf{20}'') + 1(\mathbf{20}') + 1(\mathbf{20})$$

Spectrum 6; $b = (7/2, 9)$; singlets = 1

$$1(\mathbf{1}) + 19(\mathbf{4}) + 3(\mathbf{15}) + 18(\mathbf{6}) + 1(\mathbf{10}) + 1(\mathbf{20}'') + 4(\mathbf{20}')$$

Spectrum 7; $b = (7/2, 9)$; singlets = 1

$$1(\mathbf{1}) + 19(\mathbf{4}) + 7(\mathbf{15}) + 17(\mathbf{6}) + 1(\mathbf{20}) + 1(\mathbf{36})$$

Spectrum 8; $b = (7/2, 9)$; singlets = 1

$$1(\mathbf{1}) + 20(\mathbf{4}) + 3(\mathbf{15}) + 18(\mathbf{6}) + 1(\mathbf{10}) + 3(\mathbf{20}') + 1(\mathbf{36})$$

Spectrum 9; $b = (7/2, 9)$; singlets = 2

$$2(\mathbf{1}) + 7(\mathbf{4}) + 30(\mathbf{6}) + 9(\mathbf{10}) + 1(\mathbf{20}'') + 1(\mathbf{20})$$

Spectrum 10; $b = (7/2, 9)$; singlets = 2

$$2(\mathbf{1}) + 19(\mathbf{4}) + 4(\mathbf{15}) + 17(\mathbf{6}) + 1(\mathbf{20}'') + 4(\mathbf{20}')$$

Spectrum 11; $b = (7/2, 9)$; singlets = 2

$$2(\mathbf{1}) + 20(\mathbf{4}) + 4(\mathbf{15}) + 17(\mathbf{6}) + 3(\mathbf{20}') + 1(\mathbf{36})$$

Spectrum 12; $b = (7/2, 9)$; singlets = 3

$$3(\mathbf{1}) + 7(\mathbf{4}) + 1(\mathbf{15}) + 29(\mathbf{6}) + 8(\mathbf{10}) + 1(\mathbf{20}'') + 1(\mathbf{20})$$

Spectrum 13; $b = (7/2, 9)$; singlets = 4

$$4(\mathbf{1}) + 7(\mathbf{4}) + 2(\mathbf{15}) + 28(\mathbf{6}) + 7(\mathbf{10}) + 1(\mathbf{20}'') + 1(\mathbf{20})$$

Spectrum 14; $b = (7/2, 9)$; singlets = 5

$$5(\mathbf{1}) + 7(\mathbf{4}) + 3(\mathbf{15}) + 27(\mathbf{6}) + 6(\mathbf{10}) + 1(\mathbf{20}'') + 1(\mathbf{20})$$

Spectrum 15; $b = (7/2, 9)$; singlets = 6

$$6(\mathbf{1}) + 7(\mathbf{4}) + 4(\mathbf{15}) + 26(\mathbf{6}) + 5(\mathbf{10}) + 1(\mathbf{20}'') + 1(\mathbf{20})$$

Spectrum 16; $b = (7/2, 9)$; singlets = 6

$$6(\mathbf{1}) + 8(\mathbf{4}) + 27(\mathbf{6}) + 6(\mathbf{10}) + 1(\mathbf{20}'') + 3(\mathbf{20}')$$

Spectrum 17; $b = (7/2, 9)$; singlets = 6

$$6(\mathbf{1}) + 9(\mathbf{4}) + 27(\mathbf{6}) + 6(\mathbf{10}) + 2(\mathbf{20}') + 1(\mathbf{36})$$

Spectrum 18; $b = (7/2, 9)$; singlets = 6

$$6(\mathbf{1}) + 34(\mathbf{4}) + 3(\mathbf{6}) + 10(\mathbf{10}) + 2(\mathbf{20}') + 2(\mathbf{20})$$

Spectrum 19; $b = (7/2, 9)$; singlets = 7

$$7(\mathbf{1}) + 7(\mathbf{4}) + 5(\mathbf{15}) + 25(\mathbf{6}) + 4(\mathbf{10}) + 1(\mathbf{20}'') + 1(\mathbf{20})$$

Spectrum 20; $b = (7/2, 9)$; singlets = 7

$$7(\mathbf{1}) + 8(\mathbf{4}) + 1(\mathbf{15}) + 26(\mathbf{6}) + 5(\mathbf{10}) + 1(\mathbf{20}'') + 3(\mathbf{20}')$$

Spectrum 21; $b = (7/2, 9)$; singlets = 7

$$7(\mathbf{1}) + 9(\mathbf{4}) + 1(\mathbf{15}) + 26(\mathbf{6}) + 5(\mathbf{10}) + 2(\mathbf{20}') + 1(\mathbf{36})$$

Spectrum 22; $b = (7/2, 9)$; singlets = 7

$$7(\mathbf{1}) + 34(\mathbf{4}) + 1(\mathbf{15}) + 2(\mathbf{6}) + 9(\mathbf{10}) + 2(\mathbf{20}') + 2(\mathbf{20})$$

Spectrum 23; $b = (7/2, 9)$; singlets = 8

$$8(\mathbf{1}) + 7(\mathbf{4}) + 6(\mathbf{15}) + 24(\mathbf{6}) + 3(\mathbf{10}) + 1(\mathbf{20}'') + 1(\mathbf{20})$$

Spectrum 24; $b = (7/2, 9)$; singlets = 8

$$8(\mathbf{1}) + 8(\mathbf{4}) + 2(\mathbf{15}) + 25(\mathbf{6}) + 4(\mathbf{10}) + 1(\mathbf{20}'') + 3(\mathbf{20}')$$

Spectrum 25; $b = (7/2, 9)$; singlets = 8

$$8(\mathbf{1}) + 9(\mathbf{4}) + 2(\mathbf{15}) + 25(\mathbf{6}) + 4(\mathbf{10}) + 2(\mathbf{20}') + 1(\mathbf{36})$$

Spectrum 26; b = (7/2, 9); singlets = 8

$$8(\mathbf{1}) + 34(\mathbf{4}) + 2(\mathbf{15}) + 1(\mathbf{6}) + 8(\mathbf{10}) + 2(\mathbf{20}') + 2(\mathbf{20})$$

Spectrum 27; b = (7/2, 9); singlets = 9

$$9(\mathbf{1}) + 7(\mathbf{4}) + 7(\mathbf{15}) + 23(\mathbf{6}) + 2(\mathbf{10}) + 1(\mathbf{20}'') + 1(\mathbf{20})$$

Spectrum 28; b = (7/2, 9); singlets = 9

$$9(\mathbf{1}) + 8(\mathbf{4}) + 3(\mathbf{15}) + 24(\mathbf{6}) + 3(\mathbf{10}) + 1(\mathbf{20}'') + 3(\mathbf{20}')$$

Spectrum 29; b = (7/2, 9); singlets = 9

$$9(\mathbf{1}) + 9(\mathbf{4}) + 3(\mathbf{15}) + 24(\mathbf{6}) + 3(\mathbf{10}) + 2(\mathbf{20}') + 1(\mathbf{36})$$

Spectrum 30; b = (7/2, 9); singlets = 9

$$9(\mathbf{1}) + 34(\mathbf{4}) + 3(\mathbf{15}) + 7(\mathbf{10}) + 2(\mathbf{20}') + 2(\mathbf{20})$$

Spectrum 31; b = (7/2, 9); singlets = 10

$$10(\mathbf{1}) + 7(\mathbf{4}) + 8(\mathbf{15}) + 22(\mathbf{6}) + 1(\mathbf{10}) + 1(\mathbf{20}'') + 1(\mathbf{20})$$

Spectrum 32; b = (7/2, 9); singlets = 10

$$10(\mathbf{1}) + 8(\mathbf{4}) + 4(\mathbf{15}) + 23(\mathbf{6}) + 2(\mathbf{10}) + 1(\mathbf{20}'') + 3(\mathbf{20}')$$

Spectrum 33; b = (7/2, 9); singlets = 10

$$10(\mathbf{1}) + 9(\mathbf{4}) + 4(\mathbf{15}) + 23(\mathbf{6}) + 2(\mathbf{10}) + 2(\mathbf{20}') + 1(\mathbf{36})$$

Spectrum 34; b = (7/2, 9); singlets = 10

$$10(\mathbf{1}) + 35(\mathbf{4}) + 7(\mathbf{10}) + 5(\mathbf{20}') + 1(\mathbf{20})$$

Spectrum 35; b = (7/2, 9); singlets = 11

$$11(\mathbf{1}) + 7(\mathbf{4}) + 9(\mathbf{15}) + 21(\mathbf{6}) + 1(\mathbf{20}'') + 1(\mathbf{20})$$

Spectrum 36; b = (7/2, 9); singlets = 11

$$11(\mathbf{1}) + 8(\mathbf{4}) + 5(\mathbf{15}) + 22(\mathbf{6}) + 1(\mathbf{10}) + 1(\mathbf{20}'') + 3(\mathbf{20}')$$

Spectrum 37; b = (7/2, 9); singlets = 11

$$11(\mathbf{1}) + 9(\mathbf{4}) + 5(\mathbf{15}) + 22(\mathbf{6}) + 1(\mathbf{10}) + 2(\mathbf{20}') + 1(\mathbf{36})$$

Spectrum 38; b = (7/2, 9); singlets = 12

$$12(\mathbf{1}) + 8(\mathbf{4}) + 6(\mathbf{15}) + 21(\mathbf{6}) + 1(\mathbf{20}'') + 3(\mathbf{20}')$$

Spectrum 39; b = (7/2, 9); singlets = 12

$$12(\mathbf{1}) + 9(\mathbf{4}) + 6(\mathbf{15}) + 21(\mathbf{6}) + 2(\mathbf{20}') + 1(\mathbf{36})$$

Spectrum 40; b = (7/2, 9); singlets = 14

$$14(\mathbf{1}) + 23(\mathbf{4}) + 9(\mathbf{6}) + 12(\mathbf{10}) + 1(\mathbf{20}') + 2(\mathbf{20})$$

Spectrum 41; b = (7/2, 9); singlets = 15

$$15(\mathbf{1}) + 23(\mathbf{4}) + 1(\mathbf{15}) + 8(\mathbf{6}) + 11(\mathbf{10}) + 1(\mathbf{20}') + 2(\mathbf{20})$$

Spectrum 42; b = (7/2, 9); singlets = 16

$$16(\mathbf{1}) + 23(\mathbf{4}) + 2(\mathbf{15}) + 7(\mathbf{6}) + 10(\mathbf{10}) + 1(\mathbf{20}') + 2(\mathbf{20})$$

Spectrum 43; b = (7/2, 9); singlets = 17

$$17(\mathbf{1}) + 23(\mathbf{4}) + 3(\mathbf{15}) + 6(\mathbf{6}) + 9(\mathbf{10}) + 1(\mathbf{20}') + 2(\mathbf{20})$$

Spectrum 44; b = (7/2, 9); singlets = 18

$$18(\mathbf{1}) + 23(\mathbf{4}) + 4(\mathbf{15}) + 5(\mathbf{6}) + 8(\mathbf{10}) + 1(\mathbf{20}') + 2(\mathbf{20})$$

Spectrum 45; b = (7/2, 9); singlets = 18

$$18(\mathbf{1}) + 24(\mathbf{4}) + 6(\mathbf{6}) + 9(\mathbf{10}) + 4(\mathbf{20}') + 1(\mathbf{20})$$

Spectrum 46; b = (7/2, 9); singlets = 19

$$19(\mathbf{1}) + 23(\mathbf{4}) + 5(\mathbf{15}) + 4(\mathbf{6}) + 7(\mathbf{10}) + 1(\mathbf{20}') + 2(\mathbf{20})$$

Spectrum 47; b = (7/2, 9); singlets = 19

$$19(\mathbf{1}) + 24(\mathbf{4}) + 1(\mathbf{15}) + 5(\mathbf{6}) + 8(\mathbf{10}) + 4(\mathbf{20}') + 1(\mathbf{20})$$

Spectrum 48; b = (7/2, 9); singlets = 20

$$20(\mathbf{1}) + 23(\mathbf{4}) + 6(\mathbf{15}) + 3(\mathbf{6}) + 6(\mathbf{10}) + 1(\mathbf{20}') + 2(\mathbf{20})$$

Spectrum 49; b = (7/2, 9); singlets = 20

$$20(\mathbf{1}) + 24(\mathbf{4}) + 2(\mathbf{15}) + 4(\mathbf{6}) + 7(\mathbf{10}) + 4(\mathbf{20}') + 1(\mathbf{20})$$

Spectrum 50; b = (7/2, 9); singlets = 21

$$21(\mathbf{1}) + 23(\mathbf{4}) + 7(\mathbf{15}) + 2(\mathbf{6}) + 5(\mathbf{10}) + 1(\mathbf{20}') + 2(\mathbf{20})$$

Spectrum 51; b = (7/2, 9); singlets = 21

$$21(\mathbf{1}) + 24(\mathbf{4}) + 3(\mathbf{15}) + 3(\mathbf{6}) + 6(\mathbf{10}) + 4(\mathbf{20}') + 1(\mathbf{20})$$

Spectrum 52; b = (7/2, 9); singlets = 22

$$22(\mathbf{1}) + 12(\mathbf{4}) + 15(\mathbf{6}) + 14(\mathbf{10}) + 2(\mathbf{20})$$

Spectrum 53; b = (7/2, 9); singlets = 22

$$22(\mathbf{1}) + 23(\mathbf{4}) + 8(\mathbf{15}) + 1(\mathbf{6}) + 4(\mathbf{10}) + 1(\mathbf{20}') + 2(\mathbf{20})$$

Spectrum 54; b = (7/2, 9); singlets = 22

$$22(\mathbf{1}) + 24(\mathbf{4}) + 4(\mathbf{15}) + 2(\mathbf{6}) + 5(\mathbf{10}) + 4(\mathbf{20}') + 1(\mathbf{20})$$

Spectrum 55; b = (7/2, 9); singlets = 22

$$22(\mathbf{1}) + 25(\mathbf{4}) + 3(\mathbf{6}) + 6(\mathbf{10}) + 7(\mathbf{20}')$$

Spectrum 56; b = (7/2, 9); singlets = 23

$$23(\mathbf{1}) + 12(\mathbf{4}) + 1(\mathbf{15}) + 14(\mathbf{6}) + 13(\mathbf{10}) + 2(\mathbf{20})$$

Spectrum 57; b = (7/2, 9); singlets = 23

$$23(\mathbf{1}) + 23(\mathbf{4}) + 9(\mathbf{15}) + 3(\mathbf{10}) + 1(\mathbf{20}') + 2(\mathbf{20})$$

Spectrum 58; b = (7/2, 9); singlets = 23

$$23(\mathbf{1}) + 24(\mathbf{4}) + 5(\mathbf{15}) + 1(\mathbf{6}) + 4(\mathbf{10}) + 4(\mathbf{20}') + 1(\mathbf{20})$$

Spectrum 59; b = (7/2, 9); singlets = 23

$$23(\mathbf{1}) + 25(\mathbf{4}) + 1(\mathbf{15}) + 2(\mathbf{6}) + 5(\mathbf{10}) + 7(\mathbf{20}')$$

Spectrum 60; b = (7/2, 9); singlets = 24
24(1) + 12(4) + 2(15) + 13(6) + 12(10) + 2(20)
Spectrum 61; b = (7/2, 9); singlets = 24
24(1) + 24(4) + 6(15) + 3(10) + 4(20') + 1(20)
Spectrum 62; b = (7/2, 9); singlets = 24
24(1) + 25(4) + 2(15) + 1(6) + 4(10) + 7(20')
Spectrum 63; b = (7/2, 9); singlets = 25
25(1) + 12(4) + 3(15) + 12(6) + 11(10) + 2(20)
Spectrum 64; b = (7/2, 9); singlets = 25
25(1) + 25(4) + 3(15) + 3(10) + 7(20')
Spectrum 65; b = (7/2, 9); singlets = 26
26(1) + 12(4) + 4(15) + 11(6) + 10(10) + 2(20)
Spectrum 66; b = (7/2, 9); singlets = 26
26(1) + 13(4) + 12(6) + 11(10) + 3(20') + 1(20)
Spectrum 67; b = (7/2, 9); singlets = 27
27(1) + 12(4) + 5(15) + 10(6) + 9(10) + 2(20)
Spectrum 68; b = (7/2, 9); singlets = 27
27(1) + 13(4) + 1(15) + 11(6) + 10(10) + 3(20') + 1(20)
Spectrum 69; b = (7/2, 9); singlets = 28
28(1) + 12(4) + 6(15) + 9(6) + 8(10) + 2(20)
Spectrum 70; b = (7/2, 9); singlets = 28
28(1) + 13(4) + 2(15) + 10(6) + 9(10) + 3(20') + 1(20)
Spectrum 71; b = (7/2, 9); singlets = 29
29(1) + 12(4) + 7(15) + 8(6) + 7(10) + 2(20)
Spectrum 72; b = (7/2, 9); singlets = 29
29(1) + 13(4) + 3(15) + 9(6) + 8(10) + 3(20') + 1(20)
Spectrum 73; b = (7/2, 9); singlets = 30
30(1) + 12(4) + 8(15) + 7(6) + 6(10) + 2(20)
Spectrum 74; b = (7/2, 9); singlets = 30
30(1) + 13(4) + 4(15) + 8(6) + 7(10) + 3(20') + 1(20)
Spectrum 75; b = (7/2, 9); singlets = 30
30(1) + 14(4) + 9(6) + 8(10) + 6(20')
Spectrum 76; b = (7/2, 9); singlets = 31
31(1) + 12(4) + 9(15) + 6(6) + 5(10) + 2(20)
Spectrum 77; b = (7/2, 9); singlets = 31
31(1) + 13(4) + 5(15) + 7(6) + 6(10) + 3(20') + 1(20)
Spectrum 78; b = (7/2, 9); singlets = 31
31(1) + 14(4) + 1(15) + 8(6) + 7(10) + 6(20')
Spectrum 79; b = (7/2, 9); singlets = 32
32(1) + 12(4) + 10(15) + 5(6) + 4(10) + 2(20)
Spectrum 80; b = (7/2, 9); singlets = 32
32(1) + 13(4) + 6(15) + 6(6) + 5(10) + 3(20') + 1(20)
Spectrum 81; b = (7/2, 9); singlets = 32
32(1) + 14(4) + 2(15) + 7(6) + 6(10) + 6(20')
Spectrum 82; b = (7/2, 9); singlets = 33
33(1) + 12(4) + 11(15) + 4(6) + 3(10) + 2(20)
Spectrum 83; b = (7/2, 9); singlets = 33
33(1) + 13(4) + 7(15) + 5(6) + 4(10) + 3(20') + 1(20)
Spectrum 84; b = (7/2, 9); singlets = 33
33(1) + 14(4) + 3(15) + 6(6) + 5(10) + 6(20')
Spectrum 85; b = (7/2, 9); singlets = 34
34(1) + 2(4) + 18(6) + 13(10) + 2(20') + 1(20)
Spectrum 86; b = (7/2, 9); singlets = 34
34(1) + 12(4) + 12(15) + 3(6) + 2(10) + 2(20)
Spectrum 87; b = (7/2, 9); singlets = 34
34(1) + 13(4) + 8(15) + 4(6) + 3(10) + 3(20') + 1(20)
Spectrum 88; b = (7/2, 9); singlets = 34
34(1) + 14(4) + 4(15) + 5(6) + 4(10) + 6(20')
Spectrum 89; b = (7/2, 9); singlets = 35
35(1) + 2(4) + 1(15) + 17(6) + 12(10) + 2(20') + 1(20)
Spectrum 90; b = (7/2, 9); singlets = 35
35(1) + 12(4) + 13(15) + 2(6) + 1(10) + 2(20)
Spectrum 91; b = (7/2, 9); singlets = 35
35(1) + 13(4) + 9(15) + 3(6) + 2(10) + 3(20') + 1(20)
Spectrum 92; b = (7/2, 9); singlets = 35
35(1) + 14(4) + 5(15) + 4(6) + 3(10) + 6(20')
Spectrum 93; b = (7/2, 9); singlets = 36
36(1) + 2(4) + 2(15) + 16(6) + 11(10) + 2(20') + 1(20)

Spectrum 94; $b = (7/2, 9)$; *singlets* = 36
36(1) + 12(4) + 14(15) + 1(6) + 2(20)
Spectrum 95; $b = (7/2, 9)$; *singlets* = 36
36(1) + 13(4) + 10(15) + 2(6) + 1(10) + 3(20') + 1(20)
Spectrum 96; $b = (7/2, 9)$; *singlets* = 36
36(1) + 14(4) + 6(15) + 3(6) + 2(10) + 6(20')
Spectrum 97; $b = (7/2, 9)$; *singlets* = 37
37(1) + 2(4) + 3(15) + 15(6) + 10(10) + 2(20') + 1(20)
Spectrum 98; $b = (7/2, 9)$; *singlets* = 37
37(1) + 13(4) + 11(15) + 1(6) + 3(20') + 1(20)
Spectrum 99; $b = (7/2, 9)$; *singlets* = 37
37(1) + 14(4) + 7(15) + 2(6) + 1(10) + 6(20')
Spectrum 100; $b = (7/2, 9)$; *singlets* = 38
38(1) + 2(4) + 4(15) + 14(6) + 9(10) + 2(20') + 1(20)
Spectrum 101; $b = (7/2, 9)$; *singlets* = 38
38(1) + 3(4) + 15(6) + 10(10) + 5(20')
Spectrum 102; $b = (7/2, 9)$; *singlets* = 38
38(1) + 14(4) + 8(15) + 1(6) + 6(20')
Spectrum 103; $b = (7/2, 9)$; *singlets* = 39
39(1) + 2(4) + 5(15) + 13(6) + 8(10) + 2(20') + 1(20)
Spectrum 104; $b = (7/2, 9)$; *singlets* = 39
39(1) + 3(4) + 1(15) + 14(6) + 9(10) + 5(20')
Spectrum 105; $b = (7/2, 9)$; *singlets* = 40
40(1) + 2(4) + 6(15) + 12(6) + 7(10) + 2(20') + 1(20)
Spectrum 106; $b = (7/2, 9)$; *singlets* = 40
40(1) + 3(4) + 2(15) + 13(6) + 8(10) + 5(20')
Spectrum 107; $b = (7/2, 9)$; *singlets* = 41
41(1) + 2(4) + 7(15) + 11(6) + 6(10) + 2(20') + 1(20)
Spectrum 108; $b = (7/2, 9)$; *singlets* = 41
41(1) + 3(4) + 3(15) + 12(6) + 7(10) + 5(20')
Spectrum 109; $b = (7/2, 9)$; *singlets* = 42
42(1) + 2(4) + 8(15) + 10(6) + 5(10) + 2(20') + 1(20)
Spectrum 110; $b = (7/2, 9)$; *singlets* = 42
42(1) + 3(4) + 4(15) + 11(6) + 6(10) + 5(20')
Spectrum 111; $b = (7/2, 9)$; *singlets* = 43
43(1) + 2(4) + 9(15) + 9(6) + 4(10) + 2(20') + 1(20)
Spectrum 112; $b = (7/2, 9)$; *singlets* = 43
43(1) + 3(4) + 5(15) + 10(6) + 5(10) + 5(20')
Spectrum 113; $b = (7/2, 9)$; *singlets* = 44
44(1) + 2(4) + 10(15) + 8(6) + 3(10) + 2(20') + 1(20)
Spectrum 114; $b = (7/2, 9)$; *singlets* = 44
44(1) + 3(4) + 6(15) + 9(6) + 4(10) + 5(20')
Spectrum 115; $b = (7/2, 9)$; *singlets* = 45
45(1) + 2(4) + 11(15) + 7(6) + 2(10) + 2(20') + 1(20)
Spectrum 116; $b = (7/2, 9)$; *singlets* = 45
45(1) + 3(4) + 7(15) + 8(6) + 3(10) + 5(20')
Spectrum 117; $b = (7/2, 9)$; *singlets* = 46
46(1) + 2(4) + 12(15) + 6(6) + 1(10) + 2(20') + 1(20)
Spectrum 118; $b = (7/2, 9)$; *singlets* = 46
46(1) + 3(4) + 8(15) + 7(6) + 2(10) + 5(20')
Spectrum 119; $b = (7/2, 9)$; *singlets* = 47
47(1) + 2(4) + 13(15) + 5(6) + 2(20') + 1(20)
Spectrum 120; $b = (7/2, 9)$; *singlets* = 47
47(1) + 3(4) + 9(15) + 6(6) + 1(10) + 5(20')
Spectrum 121; $b = (7/2, 9)$; *singlets* = 48
48(1) + 3(4) + 10(15) + 5(6) + 5(20')
Spectrum 1; $b = (5, 14)$; *singlets* = 0
1(4) + 4(15) + 1(6) + 19(10) + 1(20) + 1(60)
Spectrum 2; $b = (5, 14)$; *singlets* = 0
2(4) + 2(6) + 20(10) + 3(20') + 1(60)
Spectrum 3; $b = (5, 14)$; *singlets* = 1
1(1) + 1(4) + 5(15) + 18(10) + 1(20) + 1(60)
Spectrum 4; $b = (5, 14)$; *singlets* = 1
1(1) + 2(4) + 1(15) + 1(6) + 19(10) + 3(20') + 1(60)
Spectrum 5; $b = (5, 14)$; *singlets* = 2
2(1) + 2(4) + 2(15) + 18(10) + 3(20') + 1(60)
Spectrum 6; $b = (5, 14)$; *singlets* = 3
3(1) + 6(4) + 3(6) + 13(10) + 1(20'') + 1(20') + 4(20) + 1(45)

$$8(1) + 10(4) + 6(10) + 1(20'') + 3(20') + 4(20) + 2(36)$$

Spectrum 41; $b = \{5, 14\}$; singlets = 8

$$8(\mathbf{1}) + 11(\mathbf{4}) + 6(\mathbf{10}) + 2(\mathbf{20}') + 4(\mathbf{20}) + 3(\mathbf{36})$$

Spectrum 42; $b = \{5, 14\}$; singlets = 16

$$16(\mathbf{1}) + 6(\mathbf{6}) + 8(\mathbf{10}) + 1(\mathbf{20}') + 4(\mathbf{20}) + 3(\mathbf{36})$$

Spectrum 43; $b = \{5, 14\}$; singlets = 17

$$17(\mathbf{1}) + 1(\mathbf{15}) + 5(\mathbf{6}) + 7(\mathbf{10}) + 1(\mathbf{20}') + 4(\mathbf{20}) + 3(\mathbf{36})$$

Spectrum 44; $b = \{5, 14\}$; singlets = 18

$$18(\mathbf{1}) + 2(\mathbf{15}) + 4(\mathbf{6}) + 6(\mathbf{10}) + 1(\mathbf{20}') + 4(\mathbf{20}) + 3(\mathbf{36})$$

Spectrum 45; $b = \{5, 14\}$; singlets = 19

$$19(\mathbf{1}) + 3(\mathbf{15}) + 3(\mathbf{6}) + 5(\mathbf{10}) + 1(\mathbf{20}') + 4(\mathbf{20}) + 3(\mathbf{36})$$

Spectrum 46; $b = \{5, 14\}$; singlets = 20

$$20(\mathbf{1}) + 3(\mathbf{6}) + 5(\mathbf{10}) + 1(\mathbf{20}'') + 5(\mathbf{20}') + 3(\mathbf{20}) + 2(\mathbf{36})$$

Spectrum 47; $b = \{5, 14\}$; singlets = 20

$$20(\mathbf{1}) + 4(\mathbf{15}) + 2(\mathbf{6}) + 4(\mathbf{10}) + 1(\mathbf{20}') + 4(\mathbf{20}) + 3(\mathbf{36})$$

Spectrum 48; $b = \{5, 14\}$; singlets = 20

$$20(\mathbf{1}) + 1(\mathbf{4}) + 3(\mathbf{6}) + 5(\mathbf{10}) + 4(\mathbf{20}') + 3(\mathbf{20}) + 3(\mathbf{36})$$

Spectrum 49; $b = \{5, 14\}$; singlets = 21

$$21(\mathbf{1}) + 1(\mathbf{15}) + 2(\mathbf{6}) + 4(\mathbf{10}) + 1(\mathbf{20}'') + 5(\mathbf{20}') + 3(\mathbf{20}) + 2(\mathbf{36})$$

Spectrum 50; $b = \{5, 14\}$; singlets = 21

$$21(\mathbf{1}) + 5(\mathbf{15}) + 1(\mathbf{6}) + 3(\mathbf{10}) + 1(\mathbf{20}') + 4(\mathbf{20}) + 3(\mathbf{36})$$

Spectrum 51; $b = \{5, 14\}$; singlets = 21

$$21(\mathbf{1}) + 1(\mathbf{4}) + 1(\mathbf{15}) + 2(\mathbf{6}) + 4(\mathbf{10}) + 4(\mathbf{20}') + 3(\mathbf{20}) + 3(\mathbf{36})$$

Spectrum 52; $b = \{5, 14\}$; singlets = 22

$$22(\mathbf{1}) + 2(\mathbf{15}) + 1(\mathbf{6}) + 3(\mathbf{10}) + 1(\mathbf{20}'') + 5(\mathbf{20}') + 3(\mathbf{20}) + 2(\mathbf{36})$$

Spectrum 53; $b = \{5, 14\}$; singlets = 22

$$22(\mathbf{1}) + 6(\mathbf{15}) + 2(\mathbf{10}) + 1(\mathbf{20}') + 4(\mathbf{20}) + 3(\mathbf{36})$$

Spectrum 54; $b = \{5, 14\}$; singlets = 22

$$22(\mathbf{1}) + 1(\mathbf{4}) + 2(\mathbf{15}) + 1(\mathbf{6}) + 3(\mathbf{10}) + 4(\mathbf{20}') + 3(\mathbf{20}) + 3(\mathbf{36})$$

Spectrum 55; $b = \{5, 14\}$; singlets = 23

$$23(\mathbf{1}) + 3(\mathbf{15}) + 2(\mathbf{10}) + 1(\mathbf{20}'') + 5(\mathbf{20}') + 3(\mathbf{20}) + 2(\mathbf{36})$$

Spectrum 56; $b = \{5, 14\}$; singlets = 23

$$23(\mathbf{1}) + 1(\mathbf{4}) + 3(\mathbf{15}) + 2(\mathbf{10}) + 4(\mathbf{20}') + 3(\mathbf{20}) + 3(\mathbf{36})$$

Spectrum 57; $b = \{5, 14\}$; singlets = 24

$$24(\mathbf{1}) + 2(\mathbf{10}) + 2(\mathbf{20}'') + 9(\mathbf{20}') + 2(\mathbf{20}) + 1(\mathbf{36})$$

Spectrum 58; $b = \{5, 14\}$; singlets = 24

$$24(\mathbf{1}) + 1(\mathbf{4}) + 2(\mathbf{10}) + 1(\mathbf{20}'') + 8(\mathbf{20}') + 2(\mathbf{20}) + 2(\mathbf{36})$$

Spectrum 59; $b = \{5, 14\}$; singlets = 24

$$24(\mathbf{1}) + 2(\mathbf{4}) + 2(\mathbf{10}) + 7(\mathbf{20}') + 2(\mathbf{20}) + 3(\mathbf{36})$$

A4

Spectrum 1; $b = \{1, 0\}$; singlets = 240

$$240(\mathbf{1}) + 16(\mathbf{5}) + 2(\mathbf{10})$$

Spectrum 1; $b = \{5/2, 5\}$; singlets = 15

$$15(\mathbf{1}) + 35(\mathbf{5}) + 7(\mathbf{10}) + 2(\mathbf{40})$$

Spectrum 2; $b = \{5/2, 5\}$; singlets = 15

$$15(\mathbf{1}) + 36(\mathbf{5}) + 6(\mathbf{10}) + 1(\mathbf{40}) + 1(\mathbf{45})$$

Spectrum 3; $b = \{5/2, 5\}$; singlets = 15

$$15(\mathbf{1}) + 37(\mathbf{5}) + 5(\mathbf{10}) + 2(\mathbf{45})$$

Spectrum 4; $b = \{5/2, 5\}$; singlets = 40

$$40(\mathbf{1}) + 15(\mathbf{5}) + 14(\mathbf{10}) + 3(\mathbf{15}) + 1(\mathbf{40})$$

Spectrum 5; $b = \{5/2, 5\}$; singlets = 40

$$40(\mathbf{1}) + 16(\mathbf{5}) + 13(\mathbf{10}) + 3(\mathbf{15}) + 1(\mathbf{45})$$

Spectrum 6; $b = \{5/2, 5\}$; singlets = 41

$$41(\mathbf{1}) + 15(\mathbf{5}) + 1(\mathbf{24}) + 13(\mathbf{10}) + 2(\mathbf{15}) + 1(\mathbf{40})$$

Spectrum 7; $b = \{5/2, 5\}$; singlets = 41

$$41(\mathbf{1}) + 16(\mathbf{5}) + 1(\mathbf{24}) + 12(\mathbf{10}) + 2(\mathbf{15}) + 1(\mathbf{45})$$

Spectrum 8; $b = \{5/2, 5\}$; singlets = 42

$$42(\mathbf{1}) + 15(\mathbf{5}) + 2(\mathbf{24}) + 12(\mathbf{10}) + 1(\mathbf{15}) + 1(\mathbf{40})$$

Spectrum 9; $b = \{5/2, 5\}$; singlets = 42

$$42(\mathbf{1}) + 16(\mathbf{5}) + 2(\mathbf{24}) + 11(\mathbf{10}) + 1(\mathbf{15}) + 1(\mathbf{45})$$

Spectrum 10; $b = \{5/2, 5\}$; singlets = 43

$$43(\mathbf{1}) + 15(\mathbf{5}) + 3(\mathbf{24}) + 11(\mathbf{10}) + 1(\mathbf{40})$$

Spectrum 11; $b = \{5/2, 5\}$; singlets = 43

$$43(\mathbf{1}) + 16(\mathbf{5}) + 3(\mathbf{24}) + 10(\mathbf{10}) + 1(\mathbf{45})$$

A5

Spectrum 1; $b = \{3/2, 1\}$; singlets = 128

$$128(\mathbf{1}) + 27(\mathbf{6}) + 5/2(\mathbf{20})$$

Spectrum 2; $b = (3/2, 1)$; singlets = 129

$$129(\mathbf{1}) + 26(\mathbf{6}) + 1(\mathbf{15}) + 2(\mathbf{20})$$

Spectrum 3; $b = (3/2, 1)$; singlets = 130

$$130(\mathbf{1}) + 25(\mathbf{6}) + 2(\mathbf{15}) + 3/2(\mathbf{20})$$

Spectrum 4; $b = (3/2, 1)$; singlets = 131

$$131(\mathbf{1}) + 24(\mathbf{6}) + 3(\mathbf{15}) + 1(\mathbf{20})$$

Spectrum 5; $b = (3/2, 1)$; singlets = 132

$$132(\mathbf{1}) + 23(\mathbf{6}) + 4(\mathbf{15}) + 1/2(\mathbf{20})$$

Spectrum 6; $b = (3/2, 1)$; singlets = 133

$$133(\mathbf{1}) + 22(\mathbf{6}) + 5(\mathbf{15})$$

Spectrum 1; $b = (3, 6)$; singlets = 4

$$4(\mathbf{1}) + 14(\mathbf{20}) + 1(\mathbf{56})$$

Spectrum 2; $b = (3, 6)$; singlets = 23

$$23(\mathbf{1}) + 3(\mathbf{6}) + 11/2(\mathbf{20}) + 4(\mathbf{21}) + 1(\mathbf{105})$$

Spectrum 3; $b = (3, 6)$; singlets = 24

$$24(\mathbf{1}) + 2(\mathbf{6}) + 1(\mathbf{15}) + 5(\mathbf{20}) + 4(\mathbf{21}) + 1(\mathbf{105})$$

Spectrum 4; $b = (3, 6)$; singlets = 25

$$25(\mathbf{1}) + 1(\mathbf{6}) + 2(\mathbf{15}) + 9/2(\mathbf{20}) + 4(\mathbf{21}) + 1(\mathbf{105})$$

Spectrum 5; $b = (3, 6)$; singlets = 25

$$25(\mathbf{1}) + 2(\mathbf{6}) + 1(\mathbf{35}) + 5(\mathbf{20}) + 3(\mathbf{21}) + 1(\mathbf{105})$$

Spectrum 6; $b = (3, 6)$; singlets = 26

$$26(\mathbf{1}) + 3(\mathbf{15}) + 4(\mathbf{20}) + 4(\mathbf{21}) + 1(\mathbf{105})$$

Spectrum 7; $b = (3, 6)$; singlets = 26

$$26(\mathbf{1}) + 1(\mathbf{6}) + 1(\mathbf{35}) + 1(\mathbf{15}) + 9/2(\mathbf{20}) + 3(\mathbf{21}) + 1(\mathbf{105})$$

Spectrum 8; $b = (3, 6)$; singlets = 26

$$26(\mathbf{1}) + 2(\mathbf{6}) + 6(\mathbf{20}) + 2(\mathbf{21}) + 2(\mathbf{70})$$

Spectrum 9; $b = (3, 6)$; singlets = 26

$$26(\mathbf{1}) + 3(\mathbf{6}) + 5(\mathbf{20}) + 2(\mathbf{21}) + 1(\mathbf{70}) + 1(\mathbf{84})$$

Spectrum 10; $b = (3, 6)$; singlets = 26

$$26(\mathbf{1}) + 4(\mathbf{6}) + 4(\mathbf{20}) + 2(\mathbf{21}) + 2(\mathbf{84})$$

Spectrum 11; $b = (3, 6)$; singlets = 27

$$27(\mathbf{1}) + 1(\mathbf{35}) + 2(\mathbf{15}) + 4(\mathbf{20}) + 3(\mathbf{21}) + 1(\mathbf{105})$$

Spectrum 12; $b = (3, 6)$; singlets = 27

$$27(\mathbf{1}) + 1(\mathbf{6}) + 1(\mathbf{15}) + 11/2(\mathbf{20}) + 2(\mathbf{21}) + 2(\mathbf{70})$$

Spectrum 13; $b = (3, 6)$; singlets = 27

$$27(\mathbf{1}) + 1(\mathbf{6}) + 2(\mathbf{35}) + 9/2(\mathbf{20}) + 2(\mathbf{21}) + 1(\mathbf{105})$$

Spectrum 14; $b = (3, 6)$; singlets = 27

$$27(\mathbf{1}) + 2(\mathbf{6}) + 1(\mathbf{15}) + 9/2(\mathbf{20}) + 2(\mathbf{21}) + 1(\mathbf{70}) + 1(\mathbf{84})$$

Spectrum 15; $b = (3, 6)$; singlets = 27

$$27(\mathbf{1}) + 3(\mathbf{6}) + 1(\mathbf{15}) + 7/2(\mathbf{20}) + 2(\mathbf{21}) + 2(\mathbf{84})$$

Spectrum 16; $b = (3, 6)$; singlets = 28

$$28(\mathbf{1}) + 2(\mathbf{15}) + 5(\mathbf{20}) + 2(\mathbf{21}) + 2(\mathbf{70})$$

Spectrum 17; $b = (3, 6)$; singlets = 28

$$28(\mathbf{1}) + 2(\mathbf{35}) + 1(\mathbf{15}) + 4(\mathbf{20}) + 2(\mathbf{21}) + 1(\mathbf{105})$$

Spectrum 18; $b = (3, 6)$; singlets = 28

$$28(\mathbf{1}) + 1(\mathbf{6}) + 2(\mathbf{15}) + 4(\mathbf{20}) + 2(\mathbf{21}) + 1(\mathbf{70}) + 1(\mathbf{84})$$

Spectrum 19; $b = (3, 6)$; singlets = 28

$$28(\mathbf{1}) + 1(\mathbf{6}) + 1(\mathbf{35}) + 11/2(\mathbf{20}) + 1(\mathbf{21}) + 2(\mathbf{70})$$

Spectrum 20; $b = (3, 6)$; singlets = 28

$$28(\mathbf{1}) + 2(\mathbf{6}) + 2(\mathbf{15}) + 3(\mathbf{20}) + 2(\mathbf{21}) + 2(\mathbf{84})$$

Spectrum 21; $b = (3, 6)$; singlets = 28

$$28(\mathbf{1}) + 2(\mathbf{6}) + 1(\mathbf{35}) + 9/2(\mathbf{20}) + 1(\mathbf{21}) + 1(\mathbf{70}) + 1(\mathbf{84})$$

Spectrum 22; $b = (3, 6)$; singlets = 28

$$28(\mathbf{1}) + 3(\mathbf{6}) + 1(\mathbf{35}) + 7/2(\mathbf{20}) + 1(\mathbf{21}) + 2(\mathbf{84})$$

Spectrum 23; $b = (3, 6)$; singlets = 29

$$29(\mathbf{1}) + 3(\mathbf{15}) + 7/2(\mathbf{20}) + 2(\mathbf{21}) + 1(\mathbf{70}) + 1(\mathbf{84})$$

Spectrum 24; $b = (3, 6)$; singlets = 29

$$29(\mathbf{1}) + 1(\mathbf{35}) + 1(\mathbf{15}) + 5(\mathbf{20}) + 1(\mathbf{21}) + 2(\mathbf{70})$$

Spectrum 25; $b = (3, 6)$; singlets = 29

$$29(\mathbf{1}) + 3(\mathbf{35}) + 4(\mathbf{20}) + 1(\mathbf{21}) + 1(\mathbf{105})$$

Spectrum 26; $b = (3, 6)$; singlets = 29

$$29(\mathbf{1}) + 1(\mathbf{6}) + 3(\mathbf{15}) + 5/2(\mathbf{20}) + 2(\mathbf{21}) + 2(\mathbf{84})$$

Spectrum 27; $b = (3, 6)$; singlets = 29

$$29(\mathbf{1}) + 1(\mathbf{6}) + 1(\mathbf{35}) + 1(\mathbf{15}) + 4(\mathbf{20}) + 1(\mathbf{21}) + 1(\mathbf{70}) + 1(\mathbf{84})$$

Spectrum 28; $b = (3, 6)$; singlets = 29

$$29(\mathbf{1}) + 2(\mathbf{6}) + 1(\mathbf{35}) + 1(\mathbf{15}) + 3(\mathbf{20}) + 1(\mathbf{21}) + 2(\mathbf{84})$$

Spectrum 29; $b = (3, 6)$; singlets = 30

$$30(\mathbf{1}) + 4(\mathbf{15}) + 2(\mathbf{20}) + 2(\mathbf{21}) + 2(\mathbf{84})$$

Spectrum 30; $b = (3, 6)$; singlets = 30

$$30(\mathbf{1}) + 1(\mathbf{35}) + 2(\mathbf{15}) + 7/2(\mathbf{20}) + 1(\mathbf{21}) + 1(\mathbf{70}) + 1(\mathbf{84})$$

Spectrum 31; $b = (3, 6)$; singlets = 30

$$30(\mathbf{1}) + 2(\mathbf{35}) + 5(\mathbf{20}) + 2(\mathbf{70})$$

Spectrum 32; $b = (3, 6)$; singlets = 30

$$30(\mathbf{1}) + 1(\mathbf{6}) + 1(\mathbf{35}) + 2(\mathbf{15}) + 5/2(\mathbf{20}) + 1(\mathbf{21}) + 2(\mathbf{84})$$

Spectrum 33; $b = (3, 6)$; singlets = 30

$$30(\mathbf{1}) + 1(\mathbf{6}) + 2(\mathbf{35}) + 4(\mathbf{20}) + 1(\mathbf{70}) + 1(\mathbf{84})$$

Spectrum 34; $b = (3, 6)$; singlets = 30

$$30(\mathbf{1}) + 2(\mathbf{6}) + 2(\mathbf{35}) + 3(\mathbf{20}) + 2(\mathbf{84})$$

Spectrum 35; $b = (3, 6)$; singlets = 31

$$31(\mathbf{1}) + 1(\mathbf{35}) + 3(\mathbf{15}) + 2(\mathbf{20}) + 1(\mathbf{21}) + 2(\mathbf{84})$$

Spectrum 36; $b = (3, 6)$; singlets = 31

$$31(\mathbf{1}) + 2(\mathbf{35}) + 1(\mathbf{15}) + 7/2(\mathbf{20}) + 1(\mathbf{70}) + 1(\mathbf{84})$$

Spectrum 37; $b = (3, 6)$; singlets = 31

$$31(\mathbf{1}) + 1(\mathbf{6}) + 2(\mathbf{35}) + 1(\mathbf{15}) + 5/2(\mathbf{20}) + 2(\mathbf{84})$$

Spectrum 38; $b = (3, 6)$; singlets = 32

$$32(\mathbf{1}) + 2(\mathbf{35}) + 2(\mathbf{15}) + 2(\mathbf{20}) + 2(\mathbf{84})$$

A6

Spectrum 1; $b = (2, 2)$; singlets = 46

$$46(\mathbf{1}) + 23(\mathbf{7}) + 1(\mathbf{28}) + 3(\mathbf{35})$$

Spectrum 2; $b = (2, 2)$; singlets = 53

$$53(\mathbf{1}) + 18(\mathbf{7}) + 3(\mathbf{21}) + 1(\mathbf{28}) + 2(\mathbf{35})$$

Spectrum 3; $b = (2, 2)$; singlets = 54

$$54(\mathbf{1}) + 18(\mathbf{7}) + 1(\mathbf{48}) + 2(\mathbf{21}) + 2(\mathbf{35})$$

Spectrum 4; $b = (2, 2)$; singlets = 60

$$60(\mathbf{1}) + 13(\mathbf{7}) + 6(\mathbf{21}) + 1(\mathbf{28}) + 1(\mathbf{35})$$

Spectrum 5; $b = (2, 2)$; singlets = 61

$$61(\mathbf{1}) + 13(\mathbf{7}) + 1(\mathbf{48}) + 5(\mathbf{21}) + 1(\mathbf{35})$$

Spectrum 6; $b = (2, 2)$; singlets = 67

$$67(\mathbf{1}) + 8(\mathbf{7}) + 9(\mathbf{21}) + 1(\mathbf{28})$$

Spectrum 7; $b = (2, 2)$; singlets = 68

$$68(\mathbf{1}) + 8(\mathbf{7}) + 1(\mathbf{48}) + 8(\mathbf{21})$$

A7

Spectrum 1; $b = (5/2, 3)$; singlets = 34

$$34(\mathbf{1}) + 2(\mathbf{8}) + 3(\mathbf{36}) + 2(\mathbf{56}) + 1(\mathbf{70})$$

Spectrum 2; $b = (5/2, 3)$; singlets = 36

$$36(\mathbf{1}) + 2(\mathbf{28}) + 3(\mathbf{36}) + 2(\mathbf{70})$$

Spectrum 3; $b = (5/2, 3)$; singlets = 37

$$37(\mathbf{1}) + 1(\mathbf{63}) + 1(\mathbf{28}) + 2(\mathbf{36}) + 2(\mathbf{70})$$

Spectrum 4; $b = (5/2, 3)$; singlets = 38

$$38(\mathbf{1}) + 2(\mathbf{63}) + 1(\mathbf{36}) + 2(\mathbf{70})$$

Spectrum 1; $b = (4, 8)$; singlets = 4

$$4(\mathbf{1}) + 1(\mathbf{336})$$

B3

Spectrum 1; $b = (1, 0)$; singlets = 255

$$255(\mathbf{1}) + 3(\mathbf{7}) + 8(\mathbf{8})$$

Spectrum 1; $b = (5/2, 5)$; singlets = 35

$$35(\mathbf{1}) + 5(\mathbf{7}) + 26(\mathbf{8}) + 1(\mathbf{27}) + 1(\mathbf{35})$$

Spectrum 2; $b = (5/2, 5)$; singlets = 42

$$42(\mathbf{1}) + 3(\mathbf{21}) + 26(\mathbf{8}) + 1(\mathbf{27})$$

Spectrum 3; $b = (5/2, 5)$; singlets = 48

$$48(\mathbf{1}) + 6(\mathbf{7}) + 2(\mathbf{21}) + 20(\mathbf{8}) + 1(\mathbf{48})$$

Spectrum 4; $b = (5/2, 5)$; singlets = 50

$$50(\mathbf{1}) + 20(\mathbf{7}) + 10(\mathbf{8}) + 2(\mathbf{35})$$

Spectrum 5; $b = (5/2, 5)$; singlets = 57

$$57(\mathbf{1}) + 15(\mathbf{7}) + 3(\mathbf{21}) + 10(\mathbf{8}) + 1(\mathbf{35})$$

Spectrum 6; $b = (5/2, 5)$; singlets = 64

$$64(\mathbf{1}) + 10(\mathbf{7}) + 6(\mathbf{21}) + 10(\mathbf{8})$$

Spectrum 1; $b = (4, 10)$; singlets = 35

$$35(\mathbf{1}) + 2(\mathbf{7}) + 3(\mathbf{27}) + 6(\mathbf{35})$$

B4

Spectrum 1; $b = (2, 2)$; singlets = 104

$$104(\mathbf{1}) + 8(\mathbf{9}) + 1(\mathbf{36}) + 8(\mathbf{16})$$

Spectrum 1; $b = (7/2, 7)$; singlets = 10

$$10(\mathbf{1}) + 8(\mathbf{9}) + 2(\mathbf{36}) + 1(\mathbf{16}) + 1(\mathbf{44}) + 1(\mathbf{126})$$

Spectrum 2; $b = (7/2, 7)$; singlets = 31

$$31(\mathbf{1}) + 1(\mathbf{9}) + 2(\mathbf{36}) + 1(\mathbf{16}) + 1(\mathbf{84}) + 1(\mathbf{128})$$

Spectrum 3; $b = (7/2, 7)$; singlets = 31

$$31(\mathbf{1}) + 1(\mathbf{9}) + 2(\mathbf{36}) + 1(\mathbf{16}) + 1(\mathbf{44}) + 2(\mathbf{84})$$

C2

Spectrum 1; $b = (3/2, 3)$; singlets = 134

$$134(\mathbf{1}) + 42(\mathbf{4}) + 6(\mathbf{5}) + 1/2(\mathbf{16})$$

Spectrum 2; $b = (3/2, 3)$; singlets = 141

$$141(\mathbf{1}) + 36(\mathbf{4}) + 1(\mathbf{10}) + 9(\mathbf{5})$$

Spectrum 1; $b = (3, 8)$; singlets = 2

$$2(\mathbf{1}) + 95/2(\mathbf{4}) + 4(\mathbf{10}) + 12(\mathbf{5}) + 1(\mathbf{14}) + 3/2(\mathbf{16}) + 1/2(\mathbf{20})$$

Spectrum 2; $b = (3, 8)$; singlets = 2

$$2(\mathbf{1}) + 103/2(\mathbf{4}) + 1(\mathbf{10}) + 8(\mathbf{5}) + 9/2(\mathbf{16}) + 1/2(\mathbf{20})$$

Spectrum 3; $b = (3, 8)$; singlets = 3

$$3(\mathbf{1}) + 73/2(\mathbf{4}) + 29(\mathbf{5}) + 1(\mathbf{16}) + 3/2(\mathbf{20})$$

Spectrum 4; $b = (3, 8)$; singlets = 5

$$5(\mathbf{1}) + 48(\mathbf{4}) + 8(\mathbf{10}) + 7(\mathbf{5}) + 2(\mathbf{14})$$

Spectrum 5; $b = (3, 8)$; singlets = 5

$$5(\mathbf{1}) + 52(\mathbf{4}) + 5(\mathbf{10}) + 3(\mathbf{5}) + 1(\mathbf{14}) + 3(\mathbf{16})$$

Spectrum 6; $b = (3, 8)$; singlets = 6

$$6(\mathbf{1}) + 41(\mathbf{4}) + 1(\mathbf{10}) + 20(\mathbf{5}) + 5/2(\mathbf{16}) + 1(\mathbf{20})$$

Spectrum 7; $b = (3, 8)$; singlets = 9

$$9(\mathbf{1}) + 83/2(\mathbf{4}) + 5(\mathbf{10}) + 15(\mathbf{5}) + 1(\mathbf{14}) + 1(\mathbf{16}) + 1/2(\mathbf{20})$$

Spectrum 8; $b = (3, 8)$; singlets = 9

$$9(\mathbf{1}) + 91/2(\mathbf{4}) + 2(\mathbf{10}) + 11(\mathbf{5}) + 4(\mathbf{16}) + 1/2(\mathbf{20})$$

Spectrum 9; $b = (3, 8)$; singlets = 10

$$10(\mathbf{1}) + 61/2(\mathbf{4}) + 1(\mathbf{10}) + 32(\mathbf{5}) + 1/2(\mathbf{16}) + 3/2(\mathbf{20})$$

Spectrum 10; $b = (3, 8)$; singlets = 12

$$12(\mathbf{1}) + 46(\mathbf{4}) + 6(\mathbf{10}) + 6(\mathbf{5}) + 1(\mathbf{14}) + 5/2(\mathbf{16})$$

Spectrum 11; $b = (3, 8)$; singlets = 12

$$12(\mathbf{1}) + 50(\mathbf{4}) + 3(\mathbf{10}) + 2(\mathbf{5}) + 11/2(\mathbf{16})$$

Spectrum 12; $b = (3, 8)$; singlets = 13

$$13(\mathbf{1}) + 35(\mathbf{4}) + 2(\mathbf{10}) + 23(\mathbf{5}) + 2(\mathbf{16}) + 1(\mathbf{20})$$

Spectrum 13; $b = (3, 8)$; singlets = 16

$$16(\mathbf{1}) + 71/2(\mathbf{4}) + 6(\mathbf{10}) + 18(\mathbf{5}) + 1(\mathbf{14}) + 1/2(\mathbf{16}) + 1/2(\mathbf{20})$$

Spectrum 14; $b = (3, 8)$; singlets = 16

$$16(\mathbf{1}) + 79/2(\mathbf{4}) + 3(\mathbf{10}) + 14(\mathbf{5}) + 7/2(\mathbf{16}) + 1/2(\mathbf{20})$$

Spectrum 15; $b = (3, 8)$; singlets = 17

$$17(\mathbf{1}) + 49/2(\mathbf{4}) + 2(\mathbf{10}) + 35(\mathbf{5}) + 3/2(\mathbf{20})$$

Spectrum 16; $b = (3, 8)$; singlets = 19

$$19(\mathbf{1}) + 40(\mathbf{4}) + 7(\mathbf{10}) + 9(\mathbf{5}) + 1(\mathbf{14}) + 2(\mathbf{16})$$

Spectrum 17; $b = (3, 8)$; singlets = 19

$$19(\mathbf{1}) + 44(\mathbf{4}) + 4(\mathbf{10}) + 5(\mathbf{5}) + 5(\mathbf{16})$$

Spectrum 18; $b = (3, 8)$; singlets = 20

$$20(\mathbf{1}) + 29(\mathbf{4}) + 3(\mathbf{10}) + 26(\mathbf{5}) + 3/2(\mathbf{16}) + 1(\mathbf{20})$$

Spectrum 19; $b = (3, 8)$; singlets = 23

$$23(\mathbf{1}) + 59/2(\mathbf{4}) + 7(\mathbf{10}) + 21(\mathbf{5}) + 1(\mathbf{14}) + 1/2(\mathbf{20})$$

Spectrum 20; $b = (3, 8)$; singlets = 23

$$23(\mathbf{1}) + 67/2(\mathbf{4}) + 4(\mathbf{10}) + 17(\mathbf{5}) + 3(\mathbf{16}) + 1/2(\mathbf{20})$$

Spectrum 21; $b = (3, 8)$; singlets = 26

$$26(\mathbf{1}) + 34(\mathbf{4}) + 8(\mathbf{10}) + 12(\mathbf{5}) + 1(\mathbf{14}) + 3/2(\mathbf{16})$$

Spectrum 22; $b = (3, 8)$; singlets = 26

$$26(\mathbf{1}) + 38(\mathbf{4}) + 5(\mathbf{10}) + 8(\mathbf{5}) + 9/2(\mathbf{16})$$

Spectrum 23; $b = (3, 8)$; singlets = 27

$$27(\mathbf{1}) + 23(\mathbf{4}) + 4(\mathbf{10}) + 29(\mathbf{5}) + 1(\mathbf{16}) + 1(\mathbf{20})$$

Spectrum 24; $b = (3, 8)$; singlets = 30

$$30(\mathbf{1}) + 55/2(\mathbf{4}) + 5(\mathbf{10}) + 20(\mathbf{5}) + 5/2(\mathbf{16}) + 1/2(\mathbf{20})$$

Spectrum 25; $b = (3, 8)$; singlets = 33

$$33(\mathbf{1}) + 28(\mathbf{4}) + 9(\mathbf{10}) + 15(\mathbf{5}) + 1(\mathbf{14}) + 1(\mathbf{16})$$

Spectrum 26; $b = (3, 8)$; singlets = 33

$$33(\mathbf{1}) + 32(\mathbf{4}) + 6(\mathbf{10}) + 11(\mathbf{5}) + 4(\mathbf{16})$$

Spectrum 27; $b = (3, 8)$; singlets = 34

$$34(\mathbf{1}) + 17(\mathbf{4}) + 5(\mathbf{10}) + 32(\mathbf{5}) + 1/2(\mathbf{16}) + 1(\mathbf{20})$$

Spectrum 28; $b = (3, 8)$; singlets = 37

$37(1) + 43/2(4) + 6(10) + 23(5) + 2(16) + 1/2(20)$
Spectrum 29; b = (3, 8); singlets = 40
 $40(1) + 22(4) + 10(10) + 18(5) + 1(14) + 1/2(16)$
Spectrum 30; b = (3, 8); singlets = 40
 $40(1) + 26(4) + 7(10) + 14(5) + 7/2(16)$
Spectrum 31; b = (3, 8); singlets = 41
 $41(1) + 11(4) + 6(10) + 35(5) + 1(20)$
Spectrum 32; b = (3, 8); singlets = 44
 $44(1) + 31/2(4) + 7(10) + 26(5) + 3/2(16) + 1/2(20)$
Spectrum 33; b = (3, 8); singlets = 47
 $47(1) + 16(4) + 11(10) + 21(5) + 1(14)$
Spectrum 34; b = (3, 8); singlets = 47
 $47(1) + 20(4) + 8(10) + 17(5) + 3(16)$
Spectrum 35; b = (3, 8); singlets = 51
 $51(1) + 19/2(4) + 8(10) + 29(5) + 1(16) + 1/2(20)$
Spectrum 36; b = (3, 8); singlets = 54
 $54(1) + 14(4) + 9(10) + 20(5) + 5/2(16)$
Spectrum 37; b = (3, 8); singlets = 58
 $58(1) + 7/2(4) + 9(10) + 32(5) + 1/2(16) + 1/2(20)$
Spectrum 38; b = (3, 8); singlets = 61
 $61(1) + 8(4) + 10(10) + 23(5) + 2(16)$
Spectrum 39; b = (3, 8); singlets = 68
 $68(1) + 2(4) + 11(10) + 26(5) + 3/2(16)$
Spectrum 1; b = (9/2, 13); singlets = 0
 $18(4) + 10(10) + 8(5) + 4(14) + 2(16) + 2(20)$
Spectrum 2; b = (9/2, 13); singlets = 0
 $22(4) + 7(10) + 4(5) + 3(14) + 5(16) + 2(20)$
Spectrum 3; b = (9/2, 13); singlets = 0
 $26(4) + 4(10) + 2(14) + 8(16) + 2(20)$
Spectrum 4; b = (9/2, 13); singlets = 1
 $1(1) + 7(4) + 6(10) + 25(5) + 3(14) + 3/2(16) + 3(20)$
Spectrum 5; b = (9/2, 13); singlets = 1
 $1(1) + 11(4) + 3(10) + 21(5) + 2(14) + 9/2(16) + 3(20)$
Spectrum 6; b = (9/2, 13); singlets = 1
 $1(1) + 15(4) + 17(5) + 1(14) + 15/2(16) + 3(20)$
Spectrum 7; b = (9/2, 13); singlets = 2
 $2(1) + 17/2(4) + 18(10) + 10(5) + 2(14) + 1(16) + 1/2(20) + 1/2(40)$
Spectrum 8; b = (9/2, 13); singlets = 2
 $2(1) + 19/2(4) + 12(10) + 15(5) + 2(14) + 2(16) + 1/2(20) + 1(35)$
Spectrum 9; b = (9/2, 13); singlets = 2
 $2(1) + 25/2(4) + 15(10) + 6(5) + 1(14) + 4(16) + 1/2(20) + 1/2(40)$
Spectrum 10; b = (9/2, 13); singlets = 2
 $2(1) + 27/2(4) + 9(10) + 11(5) + 1(14) + 5(16) + 1/2(20) + 1(35)$
Spectrum 11; b = (9/2, 13); singlets = 2
 $2(1) + 33/2(4) + 12(10) + 2(5) + 7(16) + 1/2(20) + 1/2(40)$
Spectrum 12; b = (9/2, 13); singlets = 2
 $2(1) + 35/2(4) + 6(10) + 7(5) + 8(16) + 1/2(20) + 1(35)$
Spectrum 13; b = (9/2, 13); singlets = 3
 $3(1) + 3/2(4) + 11(10) + 23(5) + 7/2(16) + 3/2(20) + 1/2(40)$
Spectrum 14; b = (9/2, 13); singlets = 3
 $3(1) + 5/2(4) + 5(10) + 28(5) + 9/2(16) + 3/2(20) + 1(35)$
Spectrum 15; b = (9/2, 13); singlets = 3
 $3(1) + 37/2(4) + 14(10) + 3(5) + 5(14) + 1/2(16) + 3/2(20)$
Spectrum 16; b = (9/2, 13); singlets = 4
 $4(1) + 15/2(4) + 10(10) + 20(5) + 4(14) + 5/2(20)$
Spectrum 17; b = (9/2, 13); singlets = 4
 $4(1) + 23/2(4) + 7(10) + 16(5) + 3(16) + 5/2(20)$
Spectrum 18; b = (9/2, 13); singlets = 4
 $4(1) + 31/2(4) + 4(10) + 12(5) + 2(14) + 6(16) + 5/2(20)$
Spectrum 19; b = (9/2, 13); singlets = 4
 $4(1) + 39/2(4) + 1(10) + 8(5) + 1(14) + 9(16) + 5/2(20)$
Spectrum 20; b = (9/2, 13); singlets = 5
 $5(1) + 1/2(4) + 3(10) + 33(5) + 2(14) + 5/2(16) + 7/2(20)$
Spectrum 21; b = (9/2, 13); singlets = 5
 $5(1) + 9/2(4) + 29(5) + 1(14) + 11/2(16) + 7/2(20)$
Spectrum 22; b = (9/2, 13); singlets = 5
 $5(1) + 10(4) + 16(10) + 10(5) + 3(14) + 1/2(16) + 1(35)$
Spectrum 23; b = (9/2, 13); singlets = 5

$5(\mathbf{1}) + 13(\mathbf{4}) + 19(\mathbf{10}) + 1(\mathbf{5}) + 2(\mathbf{14}) + 5/2(\mathbf{16}) + 1/2(\mathbf{40})$
Spectrum 24; $b = (9/2, 13)$; singlets = 5
 $5(\mathbf{1}) + 14(\mathbf{4}) + 13(\mathbf{10}) + 6(\mathbf{5}) + 2(\mathbf{14}) + 7/2(\mathbf{16}) + 1(\mathbf{35})$
Spectrum 25; $b = (9/2, 13)$; singlets = 5
 $5(\mathbf{1}) + 18(\mathbf{4}) + 10(\mathbf{10}) + 2(\mathbf{5}) + 1(\mathbf{14}) + 13/2(\mathbf{16}) + 1(\mathbf{35})$
Spectrum 26; $b = (9/2, 13)$; singlets = 6
 $6(\mathbf{1}) + 2(\mathbf{4}) + 15(\mathbf{10}) + 18(\mathbf{5}) + 1(\mathbf{14}) + 2(\mathbf{16}) + 1(\mathbf{20}) + 1/2(\mathbf{40})$
Spectrum 27; $b = (9/2, 13)$; singlets = 6
 $6(\mathbf{1}) + 3(\mathbf{4}) + 9(\mathbf{10}) + 23(\mathbf{5}) + 1(\mathbf{14}) + 3(\mathbf{16}) + 1(\mathbf{20}) + 1(\mathbf{35})$
Spectrum 28; $b = (9/2, 13)$; singlets = 6
 $6(\mathbf{1}) + 6(\mathbf{4}) + 12(\mathbf{10}) + 14(\mathbf{5}) + 5(\mathbf{16}) + 1(\mathbf{20}) + 1/2(\mathbf{40})$
Spectrum 29; $b = (9/2, 13)$; singlets = 6
 $6(\mathbf{1}) + 7(\mathbf{4}) + 6(\mathbf{10}) + 19(\mathbf{5}) + 6(\mathbf{16}) + 1(\mathbf{20}) + 1(\mathbf{35})$
Spectrum 30; $b = (9/2, 13)$; singlets = 7
 $7(\mathbf{1}) + 12(\mathbf{4}) + 11(\mathbf{10}) + 11(\mathbf{5}) + 4(\mathbf{14}) + 3/2(\mathbf{16}) + 2(\mathbf{20})$
Spectrum 31; $b = (9/2, 13)$; singlets = 7
 $7(\mathbf{1}) + 16(\mathbf{4}) + 8(\mathbf{10}) + 7(\mathbf{5}) + 3(\mathbf{14}) + 9/2(\mathbf{16}) + 2(\mathbf{20})$
Spectrum 32; $b = (9/2, 13)$; singlets = 7
 $7(\mathbf{1}) + 20(\mathbf{4}) + 5(\mathbf{10}) + 3(\mathbf{5}) + 2(\mathbf{14}) + 15/2(\mathbf{16}) + 2(\mathbf{20})$
Spectrum 33; $b = (9/2, 13)$; singlets = 8
 $8(\mathbf{1}) + 1(\mathbf{4}) + 7(\mathbf{10}) + 28(\mathbf{5}) + 3(\mathbf{14}) + 1(\mathbf{16}) + 3(\mathbf{20})$
Spectrum 34; $b = (9/2, 13)$; singlets = 8
 $8(\mathbf{1}) + 5(\mathbf{4}) + 4(\mathbf{10}) + 24(\mathbf{5}) + 2(\mathbf{14}) + 4(\mathbf{16}) + 3(\mathbf{20})$
Spectrum 35; $b = (9/2, 13)$; singlets = 8
 $8(\mathbf{1}) + 9(\mathbf{4}) + 1(\mathbf{10}) + 20(\mathbf{5}) + 1(\mathbf{14}) + 7(\mathbf{16}) + 3(\mathbf{20})$
Spectrum 36; $b = (9/2, 13)$; singlets = 9
 $9(\mathbf{1}) + 5/2(\mathbf{4}) + 19(\mathbf{10}) + 13(\mathbf{5}) + 2(\mathbf{14}) + 1/2(\mathbf{16}) + 1/2(\mathbf{20}) + 1/2(\mathbf{40})$
Spectrum 37; $b = (9/2, 13)$; singlets = 9
 $9(\mathbf{1}) + 7/2(\mathbf{4}) + 13(\mathbf{10}) + 18(\mathbf{5}) + 2(\mathbf{14}) + 3/2(\mathbf{16}) + 1/2(\mathbf{20}) + 1(\mathbf{35})$
Spectrum 38; $b = (9/2, 13)$; singlets = 9
 $9(\mathbf{1}) + 13/2(\mathbf{4}) + 16(\mathbf{10}) + 9(\mathbf{5}) + 1(\mathbf{14}) + 7/2(\mathbf{16}) + 1/2(\mathbf{20}) + 1/2(\mathbf{40})$
Spectrum 39; $b = (9/2, 13)$; singlets = 9
 $9(\mathbf{1}) + 15/2(\mathbf{4}) + 10(\mathbf{10}) + 14(\mathbf{5}) + 1(\mathbf{14}) + 9/2(\mathbf{16}) + 1/2(\mathbf{20}) + 1(\mathbf{35})$
Spectrum 40; $b = (9/2, 13)$; singlets = 9
 $9(\mathbf{1}) + 21/2(\mathbf{4}) + 13(\mathbf{10}) + 5(\mathbf{5}) + 13/2(\mathbf{16}) + 1/2(\mathbf{20}) + 1/2(\mathbf{40})$
Spectrum 41; $b = (9/2, 13)$; singlets = 9
 $9(\mathbf{1}) + 23/2(\mathbf{4}) + 7(\mathbf{10}) + 10(\mathbf{5}) + 15/2(\mathbf{16}) + 1/2(\mathbf{20}) + 1(\mathbf{35})$
Spectrum 42; $b = (9/2, 13)$; singlets = 10
 $10(\mathbf{1}) + 25/2(\mathbf{4}) + 15(\mathbf{10}) + 6(\mathbf{5}) + 5(\mathbf{14}) + 3/2(\mathbf{20})$
Spectrum 43; $b = (9/2, 13)$; singlets = 10
 $10(\mathbf{1}) + 33/2(\mathbf{4}) + 12(\mathbf{10}) + 2(\mathbf{5}) + 4(\mathbf{14}) + 3(\mathbf{16}) + 3/2(\mathbf{20})$
Spectrum 44; $b = (9/2, 13)$; singlets = 11
 $11(\mathbf{1}) + 11/2(\mathbf{4}) + 8(\mathbf{10}) + 19(\mathbf{5}) + 3(\mathbf{14}) + 5/2(\mathbf{16}) + 5/2(\mathbf{20})$
Spectrum 45; $b = (9/2, 13)$; singlets = 11
 $11(\mathbf{1}) + 19/2(\mathbf{4}) + 5(\mathbf{10}) + 15(\mathbf{5}) + 2(\mathbf{14}) + 11/2(\mathbf{16}) + 5/2(\mathbf{20})$
Spectrum 46; $b = (9/2, 13)$; singlets = 11
 $11(\mathbf{1}) + 27/2(\mathbf{4}) + 2(\mathbf{10}) + 11(\mathbf{5}) + 1(\mathbf{14}) + 17/2(\mathbf{16}) + 5/2(\mathbf{20})$
Spectrum 47; $b = (9/2, 13)$; singlets = 12
 $12(\mathbf{1}) + 4(\mathbf{4}) + 17(\mathbf{10}) + 13(\mathbf{5}) + 3(\mathbf{14}) + 1(\mathbf{35})$
Spectrum 48; $b = (9/2, 13)$; singlets = 12
 $12(\mathbf{1}) + 7(\mathbf{4}) + 20(\mathbf{10}) + 4(\mathbf{5}) + 2(\mathbf{14}) + 2(\mathbf{16}) + 1/2(\mathbf{40})$
Spectrum 49; $b = (9/2, 13)$; singlets = 12
 $12(\mathbf{1}) + 8(\mathbf{4}) + 14(\mathbf{10}) + 9(\mathbf{5}) + 2(\mathbf{14}) + 3(\mathbf{16}) + 1(\mathbf{35})$
Spectrum 50; $b = (9/2, 13)$; singlets = 12
 $12(\mathbf{1}) + 11(\mathbf{4}) + 17(\mathbf{10}) + 1(\mathbf{14}) + 5(\mathbf{16}) + 1/2(\mathbf{40})$
Spectrum 51; $b = (9/2, 13)$; singlets = 12
 $12(\mathbf{1}) + 12(\mathbf{4}) + 11(\mathbf{10}) + 5(\mathbf{5}) + 1(\mathbf{14}) + 6(\mathbf{16}) + 1(\mathbf{35})$
Spectrum 52; $b = (9/2, 13)$; singlets = 12
 $12(\mathbf{1}) + 16(\mathbf{4}) + 8(\mathbf{10}) + 1(\mathbf{5}) + 9(\mathbf{16}) + 1(\mathbf{35})$
Spectrum 53; $b = (9/2, 13)$; singlets = 13
 $13(\mathbf{1}) + 13(\mathbf{10}) + 17(\mathbf{5}) + 9/2(\mathbf{16}) + 1(\mathbf{20}) + 1/2(\mathbf{40})$
Spectrum 54; $b = (9/2, 13)$; singlets = 13
 $13(\mathbf{1}) + 1(\mathbf{4}) + 7(\mathbf{10}) + 22(\mathbf{5}) + 11/2(\mathbf{16}) + 1(\mathbf{20}) + 1(\mathbf{35})$
Spectrum 55; $b = (9/2, 13)$; singlets = 14
 $14(\mathbf{1}) + 6(\mathbf{4}) + 12(\mathbf{10}) + 14(\mathbf{5}) + 4(\mathbf{14}) + 1(\mathbf{16}) + 2(\mathbf{20})$
Spectrum 56; $b = (9/2, 13)$; singlets = 14
 $14(\mathbf{1}) + 10(\mathbf{4}) + 9(\mathbf{10}) + 10(\mathbf{5}) + 3(\mathbf{14}) + 4(\mathbf{16}) + 2(\mathbf{20})$
Spectrum 57; $b = (9/2, 13)$; singlets = 14

14(1) + 14(4) + 6(10) + 6(5) + 2(14) + 7(16) + 2(20)
Spectrum 58; b = (9/2, 13); singlets = 14
14(1) + 18(4) + 3(10) + 2(5) + 1(14) + 10(16) + 2(20)
Spectrum 59; b = (9/2, 13); singlets = 15
15(1) + 3(4) + 2(10) + 23(5) + 1(14) + 13/2(16) + 3(20)
Spectrum 60; b = (9/2, 13); singlets = 16
16(1) + 1/2(4) + 17(10) + 12(5) + 1(14) + 3(16) + 1/2(20) + 1/2(40)
Spectrum 61; b = (9/2, 13); singlets = 16
16(1) + 3/2(4) + 11(10) + 17(5) + 1(14) + 4(16) + 1/2(20) + 1(35)
Spectrum 62; b = (9/2, 13); singlets = 16
16(1) + 9/2(4) + 14(10) + 8(5) + 6(16) + 1/2(20) + 1/2(40)
Spectrum 63; b = (9/2, 13); singlets = 16
16(1) + 11/2(4) + 8(10) + 13(5) + 7(16) + 1/2(20) + 1(35)
Spectrum 64; b = (9/2, 13); singlets = 17
17(1) + 21/2(4) + 13(10) + 5(5) + 4(14) + 5/2(16) + 3/2(20)
Spectrum 65; b = (9/2, 13); singlets = 17
17(1) + 29/2(4) + 10(10) + 1(5) + 3(14) + 11/2(16) + 3/2(20)
Spectrum 66; b = (9/2, 13); singlets = 18
18(1) + 7/2(4) + 6(10) + 18(5) + 2(14) + 5(16) + 5/2(20)
Spectrum 67; b = (9/2, 13); singlets = 18
18(1) + 15/2(4) + 3(10) + 14(5) + 1(14) + 8(16) + 5/2(20)
Spectrum 68; b = (9/2, 13); singlets = 18
18(1) + 23/2(4) + 10(5) + 11(16) + 5/2(20)
Spectrum 69; b = (9/2, 13); singlets = 19
19(1) + 1(4) + 21(10) + 7(5) + 2(14) + 3/2(16) + 1/2(40)
Spectrum 70; b = (9/2, 13); singlets = 19
19(1) + 2(4) + 15(10) + 12(5) + 2(14) + 5/2(16) + 1(35)
Spectrum 71; b = (9/2, 13); singlets = 19
19(1) + 5(4) + 18(10) + 3(5) + 1(14) + 9/2(16) + 1/2(40)
Spectrum 72; b = (9/2, 13); singlets = 19
19(1) + 6(4) + 12(10) + 8(5) + 1(14) + 11/2(16) + 1(35)
Spectrum 73; b = (9/2, 13); singlets = 19
19(1) + 10(4) + 9(10) + 4(5) + 17/2(16) + 1(35)
Spectrum 74; b = (9/2, 13); singlets = 20
20(1) + 11(4) + 17(10) + 5(14) + 1(16) + 1(20)
Spectrum 75; b = (9/2, 13); singlets = 21
21(1) + 13(10) + 17(5) + 4(14) + 1/2(16) + 2(20)
Spectrum 76; b = (9/2, 13); singlets = 21
21(1) + 4(4) + 10(10) + 13(5) + 3(14) + 7/2(16) + 2(20)
Spectrum 77; b = (9/2, 13); singlets = 21
21(1) + 8(4) + 7(10) + 9(5) + 2(14) + 13/2(16) + 2(20)
Spectrum 78; b = (9/2, 13); singlets = 21
21(1) + 12(4) + 4(10) + 5(5) + 1(14) + 19/2(16) + 2(20)
Spectrum 79; b = (9/2, 13); singlets = 21
21(1) + 16(4) + 1(10) + 1(5) + 25/2(16) + 2(20)
Spectrum 80; b = (9/2, 13); singlets = 22
22(1) + 1(4) + 22(5) + 9(16) + 3(20)
Spectrum 81; b = (9/2, 13); singlets = 24
24(1) + 9/2(4) + 14(10) + 8(5) + 4(14) + 2(16) + 3/2(20)
Spectrum 82; b = (9/2, 13); singlets = 24
24(1) + 17/2(4) + 11(10) + 4(5) + 3(14) + 5(16) + 3/2(20)
Spectrum 83; b = (9/2, 13); singlets = 24
24(1) + 25/2(4) + 8(10) + 2(14) + 8(16) + 3/2(20)
Spectrum 84; b = (9/2, 13); singlets = 25
25(1) + 3/2(4) + 4(10) + 17(5) + 1(14) + 15/2(16) + 5/2(20)
Spectrum 85; b = (9/2, 13); singlets = 25
25(1) + 11/2(4) + 1(10) + 13(5) + 21/2(16) + 5/2(20)
Spectrum 86; b = (9/2, 13); singlets = 26
26(1) + 13(10) + 11(5) + 1(14) + 5(16) + 1(35)
Spectrum 87; b = (9/2, 13); singlets = 26
26(1) + 3(4) + 16(10) + 2(5) + 7(16) + 1/2(40)
Spectrum 88; b = (9/2, 13); singlets = 26
26(1) + 4(4) + 10(10) + 7(5) + 8(16) + 1(35)
Spectrum 89; b = (9/2, 13); singlets = 27
27(1) + 5(4) + 18(10) + 3(5) + 5(14) + 1/2(16) + 1(20)
Spectrum 90; b = (9/2, 13); singlets = 28
28(1) + 2(4) + 8(10) + 12(5) + 2(14) + 6(16) + 2(20)
Spectrum 91; b = (9/2, 13); singlets = 28

28(1) + 6(4) + 5(10) + 8(5) + 1(14) + 9(16) + 2(20)
Spectrum 92; b = (9/2, 13); singlets = 28
28(1) + 10(4) + 2(10) + 4(5) + 12(16) + 2(20)
Spectrum 93; b = (9/2, 13); singlets = 31
31(1) + 5/2(4) + 12(10) + 7(5) + 3(14) + 9/2(16) + 3/2(20)
Spectrum 94; b = (9/2, 13); singlets = 31
31(1) + 13/2(4) + 9(10) + 3(5) + 2(14) + 15/2(16) + 3/2(20)
Spectrum 95; b = (9/2, 13); singlets = 34
34(1) + 3(4) + 16(10) + 2(5) + 4(14) + 3(16) + 1(20)
Spectrum 96; b = (9/2, 13); singlets = 35
35(1) + 6(10) + 11(5) + 1(14) + 17/2(16) + 2(20)
Spectrum 97; b = (9/2, 13); singlets = 35
35(1) + 4(4) + 3(10) + 7(5) + 23/2(16) + 2(20)
Spectrum 98; b = (9/2, 13); singlets = 38
38(1) + 1/2(4) + 10(10) + 6(5) + 2(14) + 7(16) + 3/2(20)
Spectrum 99; b = (9/2, 13); singlets = 38
38(1) + 9/2(4) + 7(10) + 2(5) + 1(14) + 10(16) + 3/2(20)
Spectrum 100; b = (9/2, 13); singlets = 41
41(1) + 1(4) + 14(10) + 1(5) + 3(14) + 11/2(16) + 1(20)
Spectrum 101; b = (9/2, 13); singlets = 45
45(1) + 5/2(4) + 5(10) + 1(5) + 25/2(16) + 3/2(20)
Spectrum 1; b = (6, 18); singlets = 0
7/2(4) + 5(10) + 13(5) + 4(14) + 5/2(20) + 3(35)
Spectrum 2; b = (6, 18); singlets = 0
13/2(4) + 8(10) + 4(5) + 3(14) + 2(16) + 5/2(20) + 2(35) + 1/2(40)
Spectrum 3; b = (6, 18); singlets = 0
15/2(4) + 2(10) + 9(5) + 3(14) + 3(16) + 5/2(20) + 3(35)
Spectrum 4; b = (6, 18); singlets = 0
21/2(4) + 5(10) + 2(14) + 5(16) + 5/2(20) + 2(35) + 1/2(40)
Spectrum 5; b = (6, 18); singlets = 0
13(4) + 4(10) + 2(5) + 7(14) + 6(20) + 1/2(40)
Spectrum 6; b = (6, 18); singlets = 1
1(1) + 5(10) + 7(5) + 4(14) + 8(16) + 1(35') + 1(35)
Spectrum 7; b = (6, 18); singlets = 1
1(1) + 3/2(4) + 10(10) + 3(5) + 11/2(16) + 7/2(20) + 3/2(40)
Spectrum 8; b = (6, 18); singlets = 1
1(1) + 5/2(4) + 4(10) + 8(5) + 13/2(16) + 7/2(20) + 1(35) + 1(40)
Spectrum 9; b = (6, 18); singlets = 1
1(1) + 4(4) + 2(10) + 3(5) + 3(14) + 11(16) + 1(35') + 1(35)
Spectrum 10; b = (6, 18); singlets = 1
1(1) + 6(4) + 11(10) + 3(5) + 3(14) + 1/2(16) + 4(35)
Spectrum 11; b = (6, 18); singlets = 2
2(1) + 1/2(4) + 9(10) + 4(14) + 8(16) + 3/2(20) + 1/2(64)
Spectrum 12; b = (6, 18); singlets = 2
2(1) + 1(4) + 13(10) + 2(5) + 4(16) + 1(20) + 2(35) + 1(40)
Spectrum 13; b = (6, 18); singlets = 2
2(1) + 1(4) + 13(10) + 3(5) + 9(14) + 1/2(16) + 1(20) + 1(35')
Spectrum 14; b = (6, 18); singlets = 2
2(1) + 2(4) + 7(10) + 7(5) + 5(16) + 1(20) + 3(35) + 1/2(40)
Spectrum 15; b = (6, 18); singlets = 2
2(1) + 3(4) + 1(10) + 12(5) + 6(16) + 1(20) + 4(35)
Spectrum 16; b = (6, 18); singlets = 2
2(1) + 9/2(4) + 6(10) + 9(5) + 5(14) + 9/2(20) + 1(35) + 1/2(40)
Spectrum 17; b = (6, 18); singlets = 2
2(1) + 11/2(4) + 14(5) + 5(14) + 1(16) + 9/2(20) + 2(35)
Spectrum 18; b = (6, 18); singlets = 2
2(1) + 15/2(4) + 9(10) + 4(14) + 2(16) + 9/2(20) + 1(40)
Spectrum 19; b = (6, 18); singlets = 2
2(1) + 17/2(4) + 3(10) + 5(5) + 4(14) + 3(16) + 9/2(20) + 1(35) + 1/2(40)
Spectrum 20; b = (6, 18); singlets = 2
2(1) + 25/2(4) + 1(5) + 3(14) + 6(16) + 9/2(20) + 1(35) + 1/2(40)
Spectrum 21; b = (6, 18); singlets = 3
3(1) + 1/2(4) + 2(10) + 13(5) + 2(14) + 9/2(16) + 11/2(20) + 1(40)
Spectrum 22; b = (6, 18); singlets = 3
3(1) + 2(4) + 8(5) + 5(14) + 9(16) + 2(20) + 1(35')
Spectrum 23; b = (6, 18); singlets = 3
3(1) + 8(4) + 6(10) + 4(5) + 4(14) + 3/2(16) + 2(20) + 3(35)
Spectrum 24; b = (6, 18); singlets = 3

$3(1) + 12(4) + 3(10) + 3(14) + 9/2(16) + 2(20) + 3(35)$
Spectrum 25; b = (6, 18); singlets = 4
 $4(1) + 5(10) + 12(5) + 2(14) + 3(16) + 3(20) + 2(35) + 1/2(40)$
Spectrum 26; b = (6, 18); singlets = 4
 $4(1) + 3(4) + 8(10) + 3(5) + 1(14) + 5(16) + 3(20) + 1(35) + 1(40)$
Spectrum 27; b = (6, 18); singlets = 4
 $4(1) + 4(4) + 2(10) + 8(5) + 1(14) + 6(16) + 3(20) + 2(35) + 1/2(40)$
Spectrum 28; b = (6, 18); singlets = 4
 $4(1) + 13/2(4) + 1(10) + 10(5) + 6(14) + 1(16) + 13/2(20) + 1/2(40)$
Spectrum 29; b = (6, 18); singlets = 5
 $5(1) + 1/2(4) + 2(10) + 2(5) + 1(14) + 14(16) + 1/2(20) + 1(35') + 1/2(40)$
Spectrum 30; b = (6, 18); singlets = 5
 $5(1) + 5/2(4) + 11(10) + 2(5) + 1(14) + 7/2(16) + 1/2(20) + 3(35) + 1/2(40)$
Spectrum 31; b = (6, 18); singlets = 5
 $5(1) + 7/2(4) + 5(10) + 7(5) + 1(14) + 9/2(16) + 1/2(20) + 4(35)$
Spectrum 32; b = (6, 18); singlets = 5
 $5(1) + 15/2(4) + 2(10) + 3(5) + 15/2(16) + 1/2(20) + 4(35)$
Spectrum 33; b = (6, 18); singlets = 5
 $5(1) + 9(4) + 7(10) + 5(14) + 3/2(16) + 4(20) + 1(35) + 1/2(40)$
Spectrum 34; b = (6, 18); singlets = 5
 $5(1) + 10(4) + 1(10) + 5(5) + 5(14) + 5/2(16) + 4(20) + 2(35)$
Spectrum 35; b = (6, 18); singlets = 6
 $6(1) + 1(4) + 6(10) + 8(5) + 3(14) + 3(16) + 5(20) + 1(40)$
Spectrum 36; b = (6, 18); singlets = 6
 $6(1) + 2(4) + 1(14) + 15(16) + 2(20) + 1/2(64)$
Spectrum 37; b = (6, 18); singlets = 6
 $6(1) + 2(4) + 13(5) + 3(14) + 4(16) + 5(20) + 1(35) + 1/2(40)$
Spectrum 38; b = (6, 18); singlets = 6
 $6(1) + 5/2(4) + 4(10) + 3(5) + 6(14) + 15/2(16) + 3/2(20) + 1(35')$
Spectrum 39; b = (6, 18); singlets = 6
 $6(1) + 5(4) + 3(10) + 4(5) + 2(14) + 6(16) + 5(20) + 1(40)$
Spectrum 40; b = (6, 18); singlets = 6
 $6(1) + 9(4) + 1(14) + 9(16) + 5(20) + 1(40)$
Spectrum 41; b = (6, 18); singlets = 7
 $7(1) + 1/2(4) + 9(10) + 7(5) + 3(14) + 3/2(16) + 5/2(20) + 2(35) + 1/2(40)$
Spectrum 42; b = (6, 18); singlets = 7
 $7(1) + 3/2(4) + 3(10) + 12(5) + 3(14) + 5/2(16) + 5/2(20) + 3(35)$
Spectrum 43; b = (6, 18); singlets = 7
 $7(1) + 9/2(4) + 6(10) + 3(5) + 2(14) + 9/2(16) + 5/2(20) + 2(35) + 1/2(40)$
Spectrum 44; b = (6, 18); singlets = 7
 $7(1) + 11/2(4) + 8(5) + 2(14) + 11/2(16) + 5/2(20) + 3(35)$
Spectrum 45; b = (6, 18); singlets = 7
 $7(1) + 11(4) + 2(10) + 1(5) + 6(14) + 5/2(16) + 6(20) + 1/2(40)$
Spectrum 46; b = (6, 18); singlets = 8
 $8(1) + 12(10) + 6(5) + 3(14) + 4(35)$
Spectrum 47; b = (6, 18); singlets = 8
 $8(1) + 2(4) + 2(5) + 2(14) + 27/2(16) + 1(35') + 1(35)$
Spectrum 48; b = (6, 18); singlets = 8
 $8(1) + 4(4) + 9(10) + 2(5) + 2(14) + 3(16) + 4(35)$
Spectrum 49; b = (6, 18); singlets = 8
 $8(1) + 21/2(4) + 5(10) + 6(14) + 1(16) + 7/2(20) + 2(35)$
Spectrum 50; b = (6, 18); singlets = 9
 $9(1) + 3/2(4) + 10(10) + 3(5) + 4(14) + 3/2(16) + 9/2(20) + 1(40)$
Spectrum 51; b = (6, 18); singlets = 9
 $9(1) + 5/2(4) + 4(10) + 8(5) + 4(14) + 5/2(16) + 9/2(20) + 1(35) + 1/2(40)$
Spectrum 52; b = (6, 18); singlets = 9
 $9(1) + 13/2(4) + 1(10) + 4(5) + 3(14) + 11/2(16) + 9/2(20) + 1(35) + 1/2(40)$
Spectrum 53; b = (6, 18); singlets = 10
 $10(1) + 1(4) + 13(10) + 2(5) + 4(14) + 2(20) + 2(35) + 1/2(40)$
Spectrum 54; b = (6, 18); singlets = 10
 $10(1) + 2(4) + 7(10) + 7(5) + 4(14) + 1(16) + 2(20) + 3(35)$
Spectrum 55; b = (6, 18); singlets = 10
 $10(1) + 6(4) + 4(10) + 3(5) + 3(14) + 4(16) + 2(20) + 3(35)$
Spectrum 56; b = (6, 18); singlets = 10
 $10(1) + 25/2(4) + 1(5) + 7(14) + 2(16) + 11/2(20) + 1(35)$
Spectrum 57; b = (6, 18); singlets = 11
 $11(1) + 1/2(4) + 2(10) + 13(5) + 6(14) + 1/2(16) + 13/2(20) + 1/2(40)$
Spectrum 58; b = (6, 18); singlets = 11

$11(1) + 1(4) + 6(10) + 2(5) + 15/2(16) + 3(20) + 1(35) + 1(40)$
Spectrum 59; b = (6, 18); singlets = 11
 $11(1) + 2(4) + 7(5) + 17/2(16) + 3(20) + 2(35) + 1/2(40)$
Spectrum 60; b = (6, 18); singlets = 12
 $12(1) + 1/2(4) + 9(10) + 1(5) + 6(16) + 1/2(20) + 3(35) + 1/2(40)$
Spectrum 61; b = (6, 18); singlets = 12
 $12(1) + 3/2(4) + 3(10) + 6(5) + 7(16) + 1/2(20) + 4(35)$
Spectrum 62; b = (6, 18); singlets = 12
 $12(1) + 3(4) + 8(10) + 3(5) + 5(14) + 1(16) + 4(20) + 1(35) + 1/2(40)$
Spectrum 63; b = (6, 18); singlets = 12
 $12(1) + 4(4) + 2(10) + 8(5) + 5(14) + 2(16) + 4(20) + 2(35)$
Spectrum 64; b = (6, 18); singlets = 13
 $13(1) + 1/2(4) + 2(10) + 2(5) + 5(14) + 10(16) + 3/2(20) + 1(35')$
Spectrum 65; b = (6, 18); singlets = 13
 $13(1) + 3(4) + 1(10) + 3(5) + 1(14) + 17/2(16) + 5(20) + 1(40)$
Spectrum 66; b = (6, 18); singlets = 14
 $14(1) + 2(4) + 13(5) + 7(14) + 6(20) + 1(35)$
Spectrum 67; b = (6, 18); singlets = 14
 $14(1) + 5/2(4) + 4(10) + 2(5) + 1(14) + 7(16) + 5/2(20) + 2(35) + 1/2(40)$
Spectrum 68; b = (6, 18); singlets = 14
 $14(1) + 5(4) + 3(10) + 4(5) + 6(14) + 2(16) + 6(20) + 1/2(40)$
Spectrum 69; b = (6, 18); singlets = 14
 $14(1) + 9(4) + 5(14) + 5(16) + 6(20) + 1/2(40)$
Spectrum 70; b = (6, 18); singlets = 15
 $15(1) + 2(4) + 7(10) + 1(5) + 1(14) + 11/2(16) + 4(35)$
Spectrum 71; b = (6, 18); singlets = 15
 $15(1) + 9/2(4) + 6(10) + 3(5) + 6(14) + 1/2(16) + 7/2(20) + 2(35)$
Spectrum 72; b = (6, 18); singlets = 16
 $16(1) + 1/2(4) + 2(10) + 7(5) + 3(14) + 5(16) + 9/2(20) + 1(35) + 1/2(40)$
Spectrum 73; b = (6, 18); singlets = 17
 $17(1) + 5(10) + 6(5) + 3(14) + 7/2(16) + 2(20) + 3(35)$
Spectrum 74; b = (6, 18); singlets = 17
 $17(1) + 4(4) + 2(10) + 2(5) + 2(14) + 13/2(16) + 2(20) + 3(35)$
Spectrum 75; b = (6, 18); singlets = 17
 $17(1) + 13/2(4) + 1(10) + 4(5) + 7(14) + 3/2(16) + 11/2(20) + 1(35)$
Spectrum 76; b = (6, 18); singlets = 19
 $19(1) + 1(4) + 6(10) + 2(5) + 4(14) + 7/2(16) + 4(20) + 1(35) + 1/2(40)$
Spectrum 77; b = (6, 18); singlets = 19
 $19(1) + 2(4) + 7(5) + 4(14) + 9/2(16) + 4(20) + 2(35)$
Spectrum 78; b = (6, 18); singlets = 20
 $20(1) + 1/2(4) + 9(10) + 1(5) + 4(14) + 2(16) + 3/2(20) + 3(35)$
Spectrum 79; b = (6, 18); singlets = 21
 $21(1) + 1/2(4) + 2(10) + 1(5) + 19/2(16) + 5/2(20) + 2(35) + 1/2(40)$
Spectrum 80; b = (6, 18); singlets = 21
 $21(1) + 3(4) + 1(10) + 3(5) + 5(14) + 9/2(16) + 6(20) + 1/2(40)$
Spectrum 81; b = (6, 18); singlets = 22
 $22(1) + 5(10) + 8(16) + 4(35)$
Spectrum 82; b = (6, 18); singlets = 22
 $22(1) + 5/2(4) + 4(10) + 2(5) + 5(14) + 3(16) + 7/2(20) + 2(35)$
Spectrum 83; b = (6, 18); singlets = 22
 $22(1) + 9(4) + 9(14) + 1(16) + 7(20)$
Spectrum 84; b = (6, 18); singlets = 24
 $24(1) + 1/2(4) + 2(10) + 7(5) + 7(14) + 1(16) + 11/2(20) + 1(35)$
Spectrum 85; b = (6, 18); singlets = 24
 $24(1) + 2(4) + 1(5) + 1(14) + 9(16) + 2(20) + 3(35)$
Spectrum 86; b = (6, 18); singlets = 29
 $29(1) + 1/2(4) + 2(10) + 1(5) + 4(14) + 11/2(16) + 7/2(20) + 2(35)$
Spectrum 87; b = (6, 18); singlets = 29
 $29(1) + 3(4) + 1(10) + 3(5) + 9(14) + 1/2(16) + 7(20)$
Spectrum 1; b = (15/2, 23); singlets = 0
 $6(10) + 3(5) + 8(14) + 1/2(16) + 2(35') + 1(35) + 1(40)$
Spectrum 2; b = (15/2, 23); singlets = 0
 $1(4) + 8(5) + 8(14) + 3/2(16) + 2(35') + 2(35) + 1/2(40)$
Spectrum 3; b = (15/2, 23); singlets = 0
 $5/2(4) + 5(10) + 1(5) + 5(14) + 7/2(20) + 1(30) + 1(35') + 2(35)$
Spectrum 4; b = (15/2, 23); singlets = 0
 $9/2(4) + 5(14) + 5/2(16) + 15/2(20) + 1(30) + 1/2(64)$
Spectrum 5; b = (15/2, 23); singlets = 0

$$13/2(4) + 2(10) + 3(14) + 2(16) + 7/2(20) + 1(35') + 1(35) + 2(40)$$

Spectrum 6; b = (15/2, 23); singlets = 0

$$13/2(4) + 2(10) + 1(5) + 2(14) + 3/2(16) + 1/2(20) + 5(35) + 1/2(40) + 1/2(64)$$

Spectrum 7; b = (15/2, 23); singlets = 1

$$1(1) + 3/2(4) + 4(10) + 1(5) + 8(14) + 3/2(16) + 3/2(20) + 1(35') + 1(35) + 1/2(40) + 1/2(64)$$

Spectrum 8; b = (15/2, 23); singlets = 1

$$1(1) + 4(4) + 3(10) + 2(5) + 4(14) + 5(20) + 1(35) + 3/2(40) + 1/2(64)$$

Spectrum 9; b = (15/2, 23); singlets = 2

$$2(1) + 2(4) + 1(10) + 1(14) + 6(16) + 2(20) + 1(30) + 1(35') + 3(35)$$

Spectrum 10; b = (15/2, 23); singlets = 2

$$2(1) + 2(4) + 1(10) + 3(5) + 5(16) + 2(20) + 1(35') + 2(35) + 2(40)$$

Spectrum 11; b = (15/2, 23); singlets = 2

$$2(1) + 2(4) + 1(10) + 4(5) + 9(14) + 3/2(16) + 2(20) + 2(35') + 1(40)$$

Spectrum 12; b = (15/2, 23); singlets = 2

$$2(1) + 9/2(4) + 2(5) + 6(14) + 1(16) + 11/2(20) + 1(30) + 1(35') + 1(35)$$

Spectrum 13; b = (15/2, 23); singlets = 2

$$2(1) + 9/2(4) + 5(5) + 5(14) + 11/2(20) + 1(35') + 2(40)$$

Spectrum 14; b = (15/2, 23); singlets = 3

$$3(1) + 1(4) + 4(14) + 15/2(16) + 1(35') + 2(35) + 1/2(40) + 1/2(64)$$

Spectrum 15; b = (15/2, 23); singlets = 3

$$3(1) + 8(4) + 4(14) + 3/2(16) + 3(20) + 1(35') + 2(35) + 3/2(40)$$

Spectrum 16; b = (15/2, 23); singlets = 3

$$3(1) + 8(4) + 1(5) + 3(14) + 1(16) + 6(35) + 1/2(64)$$

Spectrum 17; b = (15/2, 23); singlets = 4

$$4(1) + 1(4) + 4(5) + 10(14) + 1/2(16) + 5(20) + 1(64)$$

Spectrum 18; b = (15/2, 23); singlets = 4

$$4(1) + 3(4) + 2(10) + 9/2(16) + 1(20) + 4(35) + 1(40) + 1/2(64)$$

Spectrum 19; b = (15/2, 23); singlets = 4

$$4(1) + 3(4) + 2(10) + 1(5) + 9(14) + 1(16) + 1(20) + 1(35') + 2(35) + 1/2(64)$$

Spectrum 20; b = (15/2, 23); singlets = 5

$$5(1) + 1/2(4) + 3(10) + 1(5) + 2(14) + 3(16) + 11/2(20) + 2(40) + 1/2(64)$$

Spectrum 21; b = (15/2, 23); singlets = 6

$$6(1) + 6(10) + 2(14) + 3/2(16) + 3(20) + 2(35) + 3/2(40) + 1/2(64)$$

Spectrum 22; b = (15/2, 23); singlets = 6

$$6(1) + 1(4) + 1(5) + 4(14) + 4(16) + 6(20) + 1(30) + 1(35') + 1/2(40)$$

Spectrum 23; b = (15/2, 23); singlets = 6

$$6(1) + 1(4) + 5(5) + 2(14) + 5/2(16) + 3(20) + 3(35) + 1(40) + 1/2(64)$$

Spectrum 24; b = (15/2, 23); singlets = 7

$$7(1) + 1/2(4) + 3(10) + 4(14) + 5/2(16) + 7/2(20) + 1(30) + 1(35') + 2(35)$$

Spectrum 25; b = (15/2, 23); singlets = 7

$$7(1) + 1/2(4) + 3(10) + 3(5) + 3(14) + 3/2(16) + 7/2(20) + 1(35') + 1(35) + 2(40)$$

Spectrum 26; b = (15/2, 23); singlets = 7

$$7(1) + 1/2(4) + 3(10) + 4(5) + 2(14) + 1(16) + 1/2(20) + 5(35) + 1/2(40) + 1/2(64)$$

Spectrum 27; b = (15/2, 23); singlets = 7

$$7(1) + 3(4) + 2(10) + 1(5) + 1(16) + 7(20) + 1(30) + 2(35) + 1(40)$$

Spectrum 28; b = (15/2, 23); singlets = 7

$$7(1) + 9/2(4) + 1(14) + 4(16) + 1/2(20) + 5(35) + 1/2(40) + 1/2(64)$$

Spectrum 29; b = (15/2, 23); singlets = 8

$$8(1) + 6(10) + 2(5) + 3(14) + 1(20) + 1(35') + 3(35) + 3/2(40)$$

Spectrum 30; b = (15/2, 23); singlets = 8

$$8(1) + 1(4) + 7(5) + 3(14) + 1(16) + 1(20) + 1(35') + 4(35) + 1(40)$$

Spectrum 31; b = (15/2, 23); singlets = 8

$$8(1) + 2(4) + 1(10) + 1(5) + 3(14) + 5/2(16) + 5(20) + 1(35) + 3/2(40) + 1/2(64)$$

Spectrum 32; b = (15/2, 23); singlets = 9

$$9(1) + 3/2(4) + 4(10) + 3(14) + 1(16) + 5/2(20) + 3(35) + 1(40) + 1/2(64)$$

Spectrum 33; b = (15/2, 23); singlets = 10

$$10(1) + 2(4) + 1(10) + 3(5) + 4(14) + 1(16) + 3(20) + 1(35') + 2(35) + 3/2(40)$$

Spectrum 34; b = (15/2, 23); singlets = 10

$$10(1) + 2(4) + 1(10) + 4(5) + 3(14) + 1/2(16) + 6(35) + 1/2(64)$$

Spectrum 35; b = (15/2, 23); singlets = 10

$$10(1) + 9/2(4) + 1(5) + 1(14) + 1/2(16) + 13/2(20) + 1(30) + 3(35) + 1/2(40)$$

Spectrum 36; b = (15/2, 23); singlets = 11

$$11(1) + 1(4) + 8(14) + 7/2(16) + 1(20) + 1(35') + 2(35) + 1/2(64)$$

Spectrum 37; b = (15/2, 23); singlets = 12

$$12(1) + 3(4) + 2(10) + 4(14) + 1/2(16) + 2(20) + 4(35) + 1/2(40) + 1/2(64)$$

Spectrum 38; b = (15/2, 23); singlets = 14

$$14(1) + 1(4) + 1(5) + 8(14) + 7(20) + 1(30) + 1(35')$$

Spectrum 39; b = (15/2, 23); singlets = 15

$$15(\mathbf{1}) + 9/2(\mathbf{4}) + 5(\mathbf{14}) + 3/2(\mathbf{20}) + 5(\mathbf{35}) + 1/2(\mathbf{64})$$

Spectrum 40; b = (15/2, 23); singlets = 21

$$21(\mathbf{1}) + 1(\mathbf{4}) + 1(\mathbf{5}) + 4(\mathbf{14}) + 3/2(\mathbf{16}) + 1(\mathbf{35}') + 5(\mathbf{35}) + 1/2(\mathbf{40})$$

Spectrum 1; b = (9, 28); singlets = 0

$$7/2(\mathbf{4}) + 2(\mathbf{10}) + 4(\mathbf{14}) + 3/2(\mathbf{20}) + 2(\mathbf{30}) + 3(\mathbf{35}') + 1(\mathbf{35}) + 1/2(\mathbf{40})$$

Spectrum 2; b = (9, 28); singlets = 1

$$1(\mathbf{1}) + 5(\mathbf{4}) + 1(\mathbf{5}) + 3(\mathbf{14}) + 3(\mathbf{20}) + 1(\mathbf{30}) + 2(\mathbf{35}') + 2(\mathbf{40}) + 1/2(\mathbf{64})$$

Spectrum 3; b = (9, 28); singlets = 3

$$3(\mathbf{1}) + 3/2(\mathbf{4}) + 8(\mathbf{14}) + 1/2(\mathbf{16}) + 5/2(\mathbf{20}) + 1(\mathbf{30}) + 1(\mathbf{35}') + 3/2(\mathbf{64})$$

Spectrum 4; b = (9, 28); singlets = 5

$$5(\mathbf{1}) + 3/2(\mathbf{4}) + 1(\mathbf{5}) + 5/2(\mathbf{16}) + 1/2(\mathbf{20}) + 1(\mathbf{30}) + 1(\mathbf{35}') + 3(\mathbf{35}) + 1(\mathbf{40}) + 1(\mathbf{64})$$

Spectrum 5; b = (9, 28); singlets = 6

$$6(\mathbf{1}) + 1(\mathbf{4}) + 3(\mathbf{10}) + 2(\mathbf{5}) + 1/2(\mathbf{16}) + 1(\mathbf{20}) + 2(\mathbf{35}') + 4(\mathbf{40}) + 1/2(\mathbf{64})$$

Spectrum 6; b = (9, 28); singlets = 7

$$7(\mathbf{1}) + 3/2(\mathbf{4}) + 2(\mathbf{5}) + 2(\mathbf{14}) + 3/2(\mathbf{16}) + 3/2(\mathbf{20}) + 1(\mathbf{30}) + 3(\mathbf{35}') + 5/2(\mathbf{40})$$

Spectrum 7; b = (9, 28); singlets = 7

$$7(\mathbf{1}) + 5/2(\mathbf{4}) + 1(\mathbf{10}) + 3/2(\mathbf{16}) + 5/2(\mathbf{20}) + 1(\mathbf{35}') + 7/2(\mathbf{40}) + 1(\mathbf{64})$$

Spectrum 8; b = (9, 28); singlets = 9

$$9(\mathbf{1}) + 5/2(\mathbf{4}) + 1(\mathbf{10}) + 2(\mathbf{5}) + 1(\mathbf{14}) + 1/2(\mathbf{20}) + 2(\mathbf{35}') + 1(\mathbf{35}) + 7/2(\mathbf{40}) + 1/2(\mathbf{64})$$

Spectrum 1; b = (21/2, 33); singlets = 6

$$6(\mathbf{1}) + 1(\mathbf{10}) + 2(\mathbf{14}) + 1(\mathbf{16}) + 1(\mathbf{30}) + 2(\mathbf{35}') + 1(\mathbf{35}) + 1(\mathbf{64}) + 1(\mathbf{81})$$

Spectrum 2; b = (21/2, 33); singlets = 6

$$6(\mathbf{1}) + 1(\mathbf{10}) + 3(\mathbf{5}) + 1(\mathbf{14}) + 2(\mathbf{35}') + 2(\mathbf{40}) + 1(\mathbf{64}) + 1(\mathbf{81})$$

C3

Spectrum 1; b = (2, 4); singlets = 31

$$31(\mathbf{1}) + 31(\mathbf{6}) + 1(\mathbf{21}) + 5(\mathbf{14}) + 1/2(\mathbf{64})$$

Spectrum 2; b = (2, 4); singlets = 33

$$33(\mathbf{1}) + 59/2(\mathbf{6}) + 1(\mathbf{21}) + 1(\mathbf{14}') + 9/2(\mathbf{14}) + 1/2(\mathbf{64})$$

Spectrum 3; b = (2, 4); singlets = 35

$$35(\mathbf{1}) + 28(\mathbf{6}) + 1(\mathbf{21}) + 2(\mathbf{14}') + 4(\mathbf{14}) + 1/2(\mathbf{64})$$

Spectrum 4; b = (2, 4); singlets = 37

$$37(\mathbf{1}) + 53/2(\mathbf{6}) + 1(\mathbf{21}) + 3(\mathbf{14}') + 7/2(\mathbf{14}) + 1/2(\mathbf{64})$$

Spectrum 5; b = (2, 4); singlets = 39

$$39(\mathbf{1}) + 25(\mathbf{6}) + 1(\mathbf{21}) + 4(\mathbf{14}') + 3(\mathbf{14}) + 1/2(\mathbf{64})$$

Spectrum 6; b = (2, 4); singlets = 41

$$41(\mathbf{1}) + 47/2(\mathbf{6}) + 1(\mathbf{21}) + 5(\mathbf{14}') + 5/2(\mathbf{14}) + 1/2(\mathbf{64})$$

Spectrum 7; b = (2, 4); singlets = 43

$$43(\mathbf{1}) + 22(\mathbf{6}) + 1(\mathbf{21}) + 6(\mathbf{14}') + 2(\mathbf{14}) + 1/2(\mathbf{64})$$

Spectrum 8; b = (2, 4); singlets = 45

$$45(\mathbf{1}) + 41/2(\mathbf{6}) + 1(\mathbf{21}) + 7(\mathbf{14}') + 3/2(\mathbf{14}) + 1/2(\mathbf{64})$$

Spectrum 9; b = (2, 4); singlets = 47

$$47(\mathbf{1}) + 19(\mathbf{6}) + 1(\mathbf{21}) + 8(\mathbf{14}') + 1(\mathbf{14}) + 1/2(\mathbf{64})$$

Spectrum 10; b = (2, 4); singlets = 49

$$49(\mathbf{1}) + 35/2(\mathbf{6}) + 1(\mathbf{21}) + 9(\mathbf{14}') + 1/2(\mathbf{14}) + 1/2(\mathbf{64})$$

Spectrum 11; b = (2, 4); singlets = 51

$$51(\mathbf{1}) + 16(\mathbf{6}) + 1(\mathbf{21}) + 10(\mathbf{14}') + 1/2(\mathbf{64})$$

Spectrum 12; b = (2, 4); singlets = 53

$$53(\mathbf{1}) + 21(\mathbf{6}) + 3(\mathbf{21}) + 7(\mathbf{14})$$

Spectrum 13; b = (2, 4); singlets = 55

$$55(\mathbf{1}) + 39/2(\mathbf{6}) + 3(\mathbf{21}) + 1(\mathbf{14}') + 13/2(\mathbf{14})$$

Spectrum 14; b = (2, 4); singlets = 57

$$57(\mathbf{1}) + 18(\mathbf{6}) + 3(\mathbf{21}) + 2(\mathbf{14}') + 6(\mathbf{14})$$

Spectrum 15; b = (2, 4); singlets = 59

$$59(\mathbf{1}) + 33/2(\mathbf{6}) + 3(\mathbf{21}) + 3(\mathbf{14}') + 11/2(\mathbf{14})$$

Spectrum 16; b = (2, 4); singlets = 61

$$61(\mathbf{1}) + 15(\mathbf{6}) + 3(\mathbf{21}) + 4(\mathbf{14}') + 5(\mathbf{14})$$

Spectrum 17; b = (2, 4); singlets = 63

$$63(\mathbf{1}) + 27/2(\mathbf{6}) + 3(\mathbf{21}) + 5(\mathbf{14}') + 9/2(\mathbf{14})$$

Spectrum 18; b = (2, 4); singlets = 65

$$65(\mathbf{1}) + 12(\mathbf{6}) + 3(\mathbf{21}) + 6(\mathbf{14}') + 4(\mathbf{14})$$

Spectrum 19; b = (2, 4); singlets = 67

$$67(\mathbf{1}) + 21/2(\mathbf{6}) + 3(\mathbf{21}) + 7(\mathbf{14}') + 7/2(\mathbf{14})$$

Spectrum 20; b = (2, 4); singlets = 69

$$69(\mathbf{1}) + 9(\mathbf{6}) + 3(\mathbf{21}) + 8(\mathbf{14}') + 3(\mathbf{14})$$

Spectrum 21; b = (2, 4); singlets = 71

$$71(\mathbf{1}) + 15/2(\mathbf{6}) + 3(\mathbf{21}) + 9(\mathbf{14}') + 5/2(\mathbf{14})$$

Spectrum 22; b = (2, 4); singlets = 73

$73(1) + 6(6) + 3(21) + 10(14') + 2(14)$
Spectrum 23; $b = (2, 4)$; singlets = 75
 $75(1) + 9/2(6) + 3(21) + 11(14') + 3/2(14)$
Spectrum 24; $b = (2, 4)$; singlets = 77
 $77(1) + 3(6) + 3(21) + 12(14') + 1(14)$
Spectrum 25; $b = (2, 4)$; singlets = 79
 $79(1) + 3/2(6) + 3(21) + 13(14') + 1/2(14)$
Spectrum 26; $b = (2, 4)$; singlets = 81
 $81(1) + 3(21) + 14(14')$
Spectrum 1; $b = (7/2, 9)$; singlets = 0
 $3(6) + 6(14') + 3(14) + 1(56) + 2(70)$
Spectrum 2; $b = (7/2, 9)$; singlets = 1
 $1(1) + 3(21) + 5(14') + 4(14) + 1/2(56) + 1/2(64) + 1(90)$
Spectrum 3; $b = (7/2, 9)$; singlets = 1
 $1(1) + 17/2(6) + 1(21) + 1(14') + 3/2(14) + 1/2(56) + 1(64) + 2(70)$
Spectrum 4; $b = (7/2, 9)$; singlets = 2
 $2(1) + 3/2(6) + 7(14') + 5/2(14) + 1(56) + 2(70)$
Spectrum 5; $b = (7/2, 9)$; singlets = 2
 $2(1) + 3(6) + 6(21) + 5/2(14) + 3/2(64) + 1/2(126)$
Spectrum 6; $b = (7/2, 9)$; singlets = 2
 $2(1) + 11/2(6) + 4(21) + 5/2(14) + 3/2(64) + 1(90)$
Spectrum 7; $b = (7/2, 9)$; singlets = 3
 $3(1) + 7(6) + 1(21) + 2(14') + 1(14) + 1/2(56) + 1(64) + 2(70)$
Spectrum 8; $b = (7/2, 9)$; singlets = 4
 $4(1) + 8(14') + 2(14) + 1(56) + 2(70)$
Spectrum 9; $b = (7/2, 9)$; singlets = 4
 $4(1) + 3/2(6) + 6(21) + 1(14') + 2(14) + 3/2(64) + 1/2(126)$
Spectrum 10; $b = (7/2, 9)$; singlets = 4
 $4(1) + 4(6) + 4(21) + 1(14') + 2(14) + 3/2(64) + 1(90)$
Spectrum 11; $b = (7/2, 9)$; singlets = 5
 $5(1) + 11/2(6) + 1(21) + 3(14') + 1/2(14) + 1/2(56) + 1(64) + 2(70)$
Spectrum 12; $b = (7/2, 9)$; singlets = 6
 $6(1) + 6(21) + 2(14') + 3/2(14) + 3/2(64) + 1/2(126)$
Spectrum 13; $b = (7/2, 9)$; singlets = 6
 $6(1) + 5/2(6) + 4(21) + 2(14') + 3/2(14) + 3/2(64) + 1(90)$
Spectrum 14; $b = (7/2, 9)$; singlets = 7
 $7(1) + 4(6) + 1(21) + 4(14') + 1/2(56) + 1(64) + 2(70)$
Spectrum 15; $b = (7/2, 9)$; singlets = 8
 $8(1) + 1(6) + 4(21) + 3(14') + 1(14) + 3/2(64) + 1(90)$
Spectrum 16; $b = (7/2, 9)$; singlets = 21
 $21(1) + 3(21) + 4(14) + 1/2(56) + 1/2(64) + 2(70)$
Spectrum 17; $b = (7/2, 9)$; singlets = 26
 $26(1) + 4(14) + 1/2(56) + 5/2(64) + 1(70)$

C4

Spectrum 1; $b = (1, 0)$; singlets = 185
 $185(1) + 16(8) + 1(27)$
Spectrum 1; $b = (5/2, 5)$; singlets = 0
 $19/2(8) + 2(42) + 5/2(48) + 1/2(120)$
Spectrum 2; $b = (5/2, 5)$; singlets = 0
 $12(8) + 2(42) + 1(160)$
Spectrum 3; $b = (5/2, 5)$; singlets = 5
 $5(1) + 11/2(8) + 3(27) + 3(42) + 1/2(48) + 1/2(120)$
Spectrum 4; $b = (5/2, 5)$; singlets = 10
 $10(1) + 9/2(8) + 2(27) + 2(42) + 2(48) + 1/2(120)$
Spectrum 5; $b = (5/2, 5)$; singlets = 15
 $15(1) + 1/2(8) + 5(27) + 3(42) + 1/2(120)$
Spectrum 6; $b = (5/2, 5)$; singlets = 15
 $15(1) + 7/2(8) + 1(27) + 1(42) + 7/2(48) + 1/2(120)$
Spectrum 7; $b = (5/2, 5)$; singlets = 15
 $15(1) + 6(8) + 1(27) + 1(42) + 1(48) + 1(160)$
Spectrum 8; $b = (5/2, 5)$; singlets = 20
 $20(1) + 5/2(8) + 5(48) + 1/2(120)$
Spectrum 9; $b = (5/2, 5)$; singlets = 20
 $20(1) + 5(8) + 5/2(48) + 1(160)$
Spectrum 10; $b = (5/2, 5)$; singlets = 25
 $25(1) + 1(8) + 3(27) + 1(42) + 1/2(48) + 1(160)$
Spectrum 11; $b = (5/2, 5)$; singlets = 30

$$30(\mathbf{1}) + 2(2\mathbf{7}) + 2(4\mathbf{8}) + 1(16\mathbf{0})$$

C5

Spectrum 1; $b = (3/2, 1)$; singlets = 36

$$36(\mathbf{1}) + 41/2(1\mathbf{0}) + 1(4\mathbf{4}) + 1/2(11\mathbf{0})$$

Spectrum 2; $b = (3/2, 1)$; singlets = 64

$$64(\mathbf{1}) + 10(1\mathbf{0}) + 4(4\mathbf{4})$$

C6

Spectrum 1; $b = (2, 2)$; singlets = 6

$$6(\mathbf{1}) + 4(1\mathbf{2}) + 1(7\mathbf{8}) + 1(20\mathbf{8})$$

D4

Spectrum 1; $b = (3/2, 1)$; singlets = 172

$$172(\mathbf{1}) + 7(\mathbf{8}^-) + 7(\mathbf{8}^-) + 7(\mathbf{8}^-)$$

Spectrum 1; $b = (3, 6)$; singlets = 14

$$14(\mathbf{1}) + 25(\mathbf{8}^-) + 1(3\mathbf{5}^-) + 1(3\mathbf{5}^-) + 1(5\mathbf{6}_c)$$

Spectrum 2; $b = (3, 6)$; singlets = 14

$$14(\mathbf{1}) + 9(\mathbf{8}^-) + 16(\mathbf{8}^-) + 1(3\mathbf{5}^-) + 1(3\mathbf{5}_s) + 1(5\mathbf{6}_s)$$

Spectrum 3; $b = (3, 6)$; singlets = 14

$$14(\mathbf{1}) + 16(\mathbf{8}^-) + 9(\mathbf{8}^-) + 1(3\mathbf{5}^-) + 1(3\mathbf{5}_c) + 1(5\mathbf{6}_c)$$

Spectrum 4; $b = (3, 6)$; singlets = 14

$$14(\mathbf{1}) + 25(\mathbf{8}^-) + 1(3\mathbf{5}^-) + 1(3\mathbf{5}^-) + 1(5\mathbf{6}_s)$$

Spectrum 5; $b = (3, 6)$; singlets = 14

$$14(\mathbf{1}) + 9(\mathbf{8}^-) + 16(\mathbf{8}^-) + 1(3\mathbf{5}^-) + 1(3\mathbf{5}_s) + 1(5\mathbf{6}_v)$$

Spectrum 6; $b = (3, 6)$; singlets = 14

$$14(\mathbf{1}) + 9(\mathbf{8}^-) + 16(\mathbf{8}^-) + 1(3\mathbf{5}^-) + 1(3\mathbf{5}_c) + 1(5\mathbf{6}_v)$$

Spectrum 7; $b = (3, 6)$; singlets = 14

$$14(\mathbf{1}) + 16(\mathbf{8}^-) + 9(\mathbf{8}^-) + 1(3\mathbf{5}^-) + 1(3\mathbf{5}_c) + 1(5\mathbf{6}_c)$$

Spectrum 8; $b = (3, 6)$; singlets = 14

$$14(\mathbf{1}) + 16(\mathbf{8}^-) + 9(\mathbf{8}^-) + 1(3\mathbf{5}^-) + 1(3\mathbf{5}_c) + 1(5\mathbf{6}_s)$$

Spectrum 9; $b = (3, 6)$; singlets = 14

$$14(\mathbf{1}) + 25(\mathbf{8}^-) + 1(3\mathbf{5}^-) + 1(3\mathbf{5}^-) + 1(5\mathbf{6}_v)$$

Spectrum 10; $b = (3, 6)$; singlets = 15

$$15(\mathbf{1}) + 8(\mathbf{8}^-) + 1(2\mathbf{8}) + 8(\mathbf{8}^-) + 8(\mathbf{8}^-) + 1(3\mathbf{5}_v) + 1(3\mathbf{5}_s) + 1(3\mathbf{5}_c)$$

Spectrum 11; $b = (3, 6)$; singlets = 28

$$28(\mathbf{1}) + 2(2\mathbf{8}) + 18(\mathbf{8}^-) + 2(5\mathbf{6}^-)$$

Spectrum 12; $b = (3, 6)$; singlets = 28

$$28(\mathbf{1}) + 2(2\mathbf{8}) + 9(\mathbf{8}^-) + 9(\mathbf{8}^-) + 1(5\mathbf{6}_c) + 1(5\mathbf{6}_s)$$

Spectrum 13; $b = (3, 6)$; singlets = 28

$$28(\mathbf{1}) + 2(2\mathbf{8}) + 18(\mathbf{8}^-) + 2(5\mathbf{6}^-)$$

Spectrum 14; $b = (3, 6)$; singlets = 28

$$28(\mathbf{1}) + 9(\mathbf{8}^-) + 2(2\mathbf{8}) + 9(\mathbf{8}^-) + 1(5\mathbf{6}_c) + 1(5\mathbf{6}_v)$$

Spectrum 15; $b = (3, 6)$; singlets = 28

$$28(\mathbf{1}) + 9(\mathbf{8}^-) + 2(2\mathbf{8}) + 9(\mathbf{8}^-) + 1(5\mathbf{6}_s) + 1(5\mathbf{6}_v)$$

Spectrum 16; $b = (3, 6)$; singlets = 28

$$28(\mathbf{1}) + 18(\mathbf{8}^-) + 2(2\mathbf{8}) + 2(5\mathbf{6}^-)$$

Spectrum 17; $b = (3, 6)$; singlets = 29

$$29(\mathbf{1}) + 1(\mathbf{8}^-) + 3(2\mathbf{8}) + 8(\mathbf{8}^-) + 8(\mathbf{8}^-) + 1(3\mathbf{5}_v) + 1(5\mathbf{6}_v)$$

Spectrum 18; $b = (3, 6)$; singlets = 29

$$29(\mathbf{1}) + 8(\mathbf{8}^-) + 3(2\mathbf{8}) + 1(\mathbf{8}^-) + 8(\mathbf{8}^-) + 1(3\mathbf{5}_s) + 1(5\mathbf{6}_s)$$

Spectrum 19; $b = (3, 6)$; singlets = 29

$$29(\mathbf{1}) + 8(\mathbf{8}^-) + 3(2\mathbf{8}) + 8(\mathbf{8}^-) + 1(\mathbf{8}^-) + 1(3\mathbf{5}_c) + 1(5\mathbf{6}_c)$$

Spectrum 20; $b = (3, 6)$; singlets = 30

$$30(\mathbf{1}) + 4(2\mathbf{8}) + 16(\mathbf{8}^-) + 1(3\mathbf{5}^-) + 1(3\mathbf{5}_s)$$

Spectrum 21; $b = (3, 6)$; singlets = 30

$$30(\mathbf{1}) + 4(2\mathbf{8}) + 16(\mathbf{8}^-) + 1(3\mathbf{5}^-) + 1(3\mathbf{5}_c)$$

Spectrum 22; $b = (3, 6)$; singlets = 30

$$30(\mathbf{1}) + 16(\mathbf{8}^-) + 4(2\mathbf{8}) + 1(3\mathbf{5}^-) + 1(3\mathbf{5}_c)$$

Spectrum 23; $b = (3, 6)$; singlets = 44

$$44(\mathbf{1}) + 6(2\mathbf{8}) + 9(\mathbf{8}^-) + 1(5\mathbf{6}^-)$$

Spectrum 24; $b = (3, 6)$; singlets = 44

$$44(\mathbf{1}) + 6(2\mathbf{8}) + 9(\mathbf{8}^-) + 1(5\mathbf{6}^-)$$

Spectrum 25; $b = (3, 6)$; singlets = 44

$$44(\mathbf{1}) + 9(\mathbf{8}^-) + 6(2\mathbf{8}) + 1(5\mathbf{6}^-)$$

Spectrum 26; $b = (3, 6)$; singlets = 60

$$60(\mathbf{1}) + 10(2\mathbf{8})$$

D5

Spectrum 1; $b = (1, -2)$; singlets = 320

320(1) + 2(10)

Spectrum 1; $b = (5/2, 3)$; singlets = 63

63(1) + 3(10) + 3(45) + 7(16)

Spectrum 1; $b = (4, 8)$; singlets = 22

22(1) + 2(54) + 1(210)

D7

Spectrum 1; $b = (3, 2)$; singlets = 2

2(1) + 2(14) + 2(91) + 2(64)

G2

Spectrum 1; $b = (2, 4)$; singlets = 96

96(1) + 31(7) + 1(27)

Spectrum 2; $b = (2, 4)$; singlets = 102

102(1) + 28(7) + 3(14)

Spectrum 1; $b = (7/2, 9)$; singlets = 31

31(1) + 17(7) + 2(14) + 6(27)

Spectrum 2; $b = (7/2, 9)$; singlets = 37

37(1) + 14(7) + 5(14) + 5(27)

Spectrum 3; $b = (7/2, 9)$; singlets = 43

43(1) + 11(7) + 8(14) + 4(27)

Spectrum 4; $b = (7/2, 9)$; singlets = 49

49(1) + 8(7) + 11(14) + 3(27)

Spectrum 5; $b = (7/2, 9)$; singlets = 55

55(1) + 5(7) + 14(14) + 2(27)

Spectrum 6; $b = (7/2, 9)$; singlets = 61

61(1) + 2(7) + 17(14) + 1(27)

Spectrum 1; $b = (5, 14)$; singlets = 2

2(1) + 6(7) + 12(14) + 2(64)

Spectrum 2; $b = (5, 14)$; singlets = 5

5(1) + 1(7) + 16(14) + 1(27) + 1(77)

E6

Spectrum 1; $b = (3, 2)$; singlets = 22

22(1) + 6(27) + 2(78)

n = 2 detailed spectra

A3 x A3

Spectrum 1; b1=(7/2,9), b2=(2,4); singlets = 10

10(1,1) + 3(4,4) + 2(4,1) + 1(1,20) + 2(6,4) + 1(6,1) + 2(10,4) + 1(20',4) + 2(20',1)

Spectrum 2; b1=(7/2,9), b2=(2,4); singlets = 10

10(1,1) + 1(15,4) + 3(4,4) + 2(4,1) + 1(1,4) + 1(1,20) + 1(6,4) + 1(6,1) + 1(10,4) + 1(20',4) + 2(20',1)

Spectrum 3; b1=(7/2,9), b2=(2,4); singlets = 6

6(1,1) + 1(15,6) + 3(4,4) + 2(4,1) + 3(1,10) + 3(6,1) + 2(10,1) + 1(20',4) + 2(20',1)

Spectrum 4; b1=(7/2,9), b2=(2,4); singlets = 7

7(1,1) + 1(15,1) + 1(15,6) + 3(4,4) + 2(4,1) + 3(1,10) + 2(6,1) + 1(10,1) + 1(20',4) + 2(20',1)

Spectrum 5; b1=(7/2,9), b2=(2,4); singlets = 10

10(1,1) + 2(15,4) + 3(4,4) + 2(4,1) + 2(1,4) + 1(1,20) + 1(6,1) + 1(20',4) + 2(20',1)

Spectrum 6; b1=(7/2,9), b2=(2,4); singlets = 8

8(1,1) + 2(15,1) + 1(15,6) + 3(4,4) + 2(4,1) + 3(1,10) + 1(6,1) + 1(20',4) + 2(20',1)

Spectrum 7; b1=(2,4), b2=(2,4); singlets = 44

44(1,1) + 16(4,4) + 1(1,20) + 1(20,1)

Spectrum 8; b1=(2,4), b2=(2,4); singlets = 40

40(1,1) + 4(4,4) + 4(4,1) + 2(4,6) + 4(1,4) + 1(1,6) + 3(1,10) + 2(6,4) + 1(6,1) + 1(6,6) + 3(10,1)

Spectrum 9; b1=(2,4), b2=(2,4); singlets = 44

44(1,1) + 8(4,4) + 3(1,6) + 3(1,10) + 3(6,1) + 2(6,6) + 3(10,1)

Spectrum 10; b1=(2,4), b2=(2,4); singlets = 41

41(1,1) + 4(4,4) + 4(4,1) + 2(4,6) + 1(1,15) + 4(1,4) + 2(1,10) + 2(6,4) + 1(6,1) + 1(6,6) + 3(10,1)

Spectrum 11; b1=(2,4), b2=(2,4); singlets = 45

45(1,1) + 8(4,4) + 1(1,15) + 2(1,6) + 2(1,10) + 3(6,1) + 2(6,6) + 3(10,1)

Spectrum 12; b1=(2,4), b2=(2,4); singlets = 46

46(1,1) + 8(4,4) + 2(1,15) + 1(1,6) + 1(1,10) + 3(6,1) + 2(6,6) + 3(10,1)

Spectrum 13; b1=(2,4), b2=(2,4); singlets = 47

47(1,1) + 8(4,4) + 3(1,15) + 3(6,1) + 2(6,6) + 3(10,1)

Spectrum 14; b1=(2,4), b2=(2,4); singlets = 42

42(1,1) + 1(15,1) + 4(4,4) + 4(4,1) + 2(4,6) + 1(1,15) + 4(1,4) + 2(1,10) + 2(6,4) + 1(6,6) + 2(10,1)

Spectrum 15; b1=(2,4), b2=(2,4); singlets = 46

46(1,1) + 1(15,1) + 8(4,4) + 1(1,15) + 2(1,6) + 2(1,10) + 2(6,1) + 2(6,6) + 2(10,1)

Spectrum 16; b1=(2,4), b2=(2,4); singlets = 47

47(1,1) + 1(15,1) + 8(4,4) + 2(1,15) + 1(1,6) + 1(1,10) + 2(6,1) + 2(6,6) + 2(10,1)

Spectrum 17; b1=(2,4), b2=(2,4); singlets = 48

48(1,1) + 1(15,1) + 8(4,4) + 3(1,15) + 2(6,1) + 2(6,6) + 2(10,1)

Spectrum 18; b1=(2,4), b2=(2,4); singlets = 48

48(1,1) + 2(15,1) + 8(4,4) + 2(1,15) + 1(1,6) + 1(1,10) + 1(6,1) + 2(6,6) + 1(10,1)

Spectrum 19; b1=(2,4), b2=(2,4); singlets = 49

49(1,1) + 2(15,1) + 8(4,4) + 3(1,15) + 1(6,1) + 2(6,6) + 1(10,1)

Spectrum 20; b1=(2,4), b2=(2,4); singlets = 50

50(1,1) + 3(15,1) + 8(4,4) + 3(1,15) + 2(6,6)

A3 x A4

Spectrum 21; b1=(2,4), b2=(1,0); singlets = 44

44(1,1) + 4(4,5) + 44(4,1) + 2(1,10) + 1(20,1)

Spectrum 22; b1=(2,4), b2=(1,0); singlets = 64

64(1,1) + 4(4,5) + 23(4,1) + 2(1,10) + 9(6,1) + 1(10,1) + 1(20',1)

Spectrum 23; b1=(2,4), b2=(1,0); singlets = 54

54(1,1) + 2(4,5) + 33(4,1) + 2(1,5) + 2(1,10) + 1(6,5) + 4(6,1) + 1(10,1) + 1(20',1)

Spectrum 24; b1=(2,4), b2=(1,0); singlets = 65

65(1,1) + 1(15,1) + 4(4,5) + 23(4,1) + 2(1,10) + 8(6,1) + 1(20',1)

Spectrum 25; b1=(2,4), b2=(1,0); singlets = 55

55(1,1) + 1(15,1) + 2(4,5) + 33(4,1) + 2(1,5) + 2(1,10) + 1(6,5) + 3(6,1) + 1(20',1)

Spectrum 26; b1=(2,4), b2=(1,0); singlets = 72

72(1,1) + 4(4,5) + 12(4,1) + 2(1,10) + 15(6,1) + 3(10,1)

Spectrum 27; b1=(2,4), b2=(1,0); singlets = 62

62(1,1) + 2(4,5) + 22(4,1) + 2(1,5) + 2(1,10) + 1(6,5) + 10(6,1) + 3(10,1)

Spectrum 28; b1=(2,4), b2=(1,0); singlets = 52

52(1,1) + 32(4,1) + 4(1,5) + 2(1,10) + 2(6,5) + 5(6,1) + 3(10,1)

Spectrum 29; b1=(2,4), b2=(1,0); singlets = 73

73(1,1) + 1(15,1) + 4(4,5) + 12(4,1) + 2(1,10) + 14(6,1) + 2(10,1)

Spectrum 30; b1=(2,4), b2=(1,0); singlets = 63

63(1,1) + 1(15,1) + 2(4,5) + 22(4,1) + 2(1,5) + 2(1,10) + 1(6,5) + 9(6,1) + 2(10,1)

Spectrum 31; b1=(2,4), b2=(1,0); singlets = 53

53(1,1) + 1(15,1) + 32(4,1) + 4(1,5) + 2(1,10) + 2(6,5) + 4(6,1) + 2(10,1)

Spectrum 32; b1=(2,4), b2=(1,0); singlets = 74

74(1,1) + 2(15,1) + 4(4,5) + 12(4,1) + 2(1,10) + 13(6,1) + 1(10,1)

Spectrum 33; b1=(2,4), b2=(1,0); singlets = 64

64(1,1) + 2(15,1) + 2(4,5) + 22(4,1) + 2(1,5) + 2(1,10) + 1(6,5) + 8(6,1) + 1(10,1)

Spectrum 34; b1=(2,4), b2=(1,0); singlets = 54

54(1,1) + 2(15,1) + 32(4,1) + 4(1,5) + 2(1,10) + 2(6,5) + 3(6,1) + 1(10,1)

Spectrum 35; b1=(2,4), b2=(1,0); singlets = 75

75(1,1) + 3(15,1) + 4(4,5) + 12(4,1) + 2(1,10) + 12(6,1)

Spectrum 36; b1=(2,4), b2=(1,0); singlets = 65

65(1,1) + 3(15,1) + 2(4,5) + 22(4,1) + 2(1,5) + 2(1,10) + 1(6,5) + 7(6,1)

Spectrum 37; b1=(2,4), b2=(1,0); singlets = 55

55(1,1) + 3(15,1) + 32(4,1) + 4(1,5) + 2(1,10) + 2(6,5) + 2(6,1)

A3 x A5

Spectrum 38; b1=(2,4), b2=(3/2,1); singlets = 12

12(1,1) + 4(4,6) + 25(4,1) + 1(4,15) + 7(1,6) + 1/2(1,20) + 1(20,1)

Spectrum 39; b1=(2,4), b2=(3/2,1); singlets = 13

13(1,1) + 4(4,6) + 25(4,1) + 1(4,15) + 6(1,6) + 1(1,15) + 1(20,1)

Spectrum 40; b1=(2,4), b2=(3/2,1); singlets = 32

32(1,1) + 4(4,6) + 4(4,1) + 1(4,15) + 7(1,6) + 1/2(1,20) + 9(6,1) + 1(10,1) + 1(20',1)

Spectrum 41; b1=(2,4), b2=(3/2,1); singlets = 20

20(1,1) + 2(4,6) + 16(4,1) + 1(4,15) + 9(1,6) + 1/2(1,20) + 1(6,6) + 3(6,1) + 1(10,1) + 1(20',1)

Spectrum 42; b1=(2,4), b2=(3/2,1); singlets = 33

33(1,1) + 4(4,6) + 4(4,1) + 1(4,15) + 6(1,6) + 1(1,15) + 9(6,1) + 1(10,1) + 1(20',1)

Spectrum 43; b1=(2,4), b2=(3/2,1); singlets = 21

21(1,1) + 2(4,6) + 16(4,1) + 1(4,15) + 8(1,6) + 1(1,15) + 1(6,6) + 3(6,1) + 1(10,1) + 1(20',1)

Spectrum 44; b1=(2,4), b2=(3/2,1); singlets = 33

33(1,1) + 1(15,1) + 4(4,6) + 4(4,1) + 1(4,15) + 7(1,6) + 1/2(1,20) + 8(6,1) + 1(20',1)

Spectrum 45; b1=(2,4), b2=(3/2,1); singlets = 21

21(1,1) + 1(15,1) + 2(4,6) + 16(4,1) + 1(4,15) + 9(1,6) + 1/2(1,20) + 1(6,6) + 2(6,1) + 1(20',1)

Spectrum 46; b1=(2,4), b2=(3/2,1); singlets = 34

34(1,1) + 1(15,1) + 4(4,6) + 4(4,1) + 1(4,15) + 6(1,6) + 1(1,15) + 8(6,1) + 1(20',1)

Spectrum 47; b1=(2,4), b2=(3/2,1); singlets = 22

22(1,1) + 1(15,1) + 2(4,6) + 16(4,1) + 1(4,15) + 8(1,6) + 1(1,15) + 1(6,6) + 2(6,1) + 1(20',1)

Spectrum 48; b1=(2,4), b2=(3/2,1); singlets = 28

28(1,1) + 2(4,6) + 5(4,1) + 1(4,15) + 9(1,6) + 1/2(1,20) + 1(6,6) + 9(6,1) + 3(10,1)

Spectrum 49; b1=(2,4), b2=(3/2,1); singlets = 16

16(1,1) + 17(4,1) + 1(4,15) + 11(1,6) + 1/2(1,20) + 2(6,6) + 3(6,1) + 3(10,1)

Spectrum 50; b1=(2,4), b2=(3/2,1); singlets = 29

29(1,1) + 2(4,6) + 5(4,1) + 1(4,15) + 8(1,6) + 1(1,15) + 1(6,6) + 9(6,1) + 3(10,1)

Spectrum 51; b1=(2,4), b2=(3/2,1); singlets = 17

17(1,1) + 17(4,1) + 1(4,15) + 10(1,6) + 1(1,15) + 2(6,6) + 3(6,1) + 3(10,1)

Spectrum 52; b1=(2,4), b2=(3/2,1); singlets = 29

29(1,1) + 1(15,1) + 2(4,6) + 5(4,1) + 1(4,15) + 9(1,6) + 1/2(1,20) + 1(6,6) + 8(6,1) + 2(10,1)

Spectrum 53; b1=(2,4), b2=(3/2,1); singlets = 17

17(1,1) + 1(15,1) + 17(4,1) + 1(4,15) + 11(1,6) + 1/2(1,20) + 2(6,6) + 2(6,1) + 2(10,1)

Spectrum 54; b1=(2,4), b2=(3/2,1); singlets = 30

30(1,1) + 1(15,1) + 2(4,6) + 5(4,1) + 1(4,15) + 8(1,6) + 1(1,15) + 1(6,6) + 8(6,1) + 2(10,1)

Spectrum 55; b1=(2,4), b2=(3/2,1); singlets = 18

18(1,1) + 1(15,1) + 17(4,1) + 1(4,15) + 10(1,6) + 1(1,15) + 2(6,6) + 2(6,1) + 2(10,1)

Spectrum 56; b1=(2,4), b2=(3/2,1); singlets = 30

30(1,1) + 2(15,1) + 2(4,6) + 5(4,1) + 1(4,15) + 9(1,6) + 1/2(1,20) + 1(6,6) + 7(6,1) + 1(10,1)

Spectrum 57; b1=(2,4), b2=(3/2,1); singlets = 18

18(1,1) + 2(15,1) + 17(4,1) + 1(4,15) + 11(1,6) + 1/2(1,20) + 2(6,6) + 1(6,1) + 1(10,1)

Spectrum 58; b1=(2,4), b2=(3/2,1); singlets = 31

31(1,1) + 2(15,1) + 2(4,6) + 5(4,1) + 1(4,15) + 8(1,6) + 1(1,15) + 1(6,6) + 7(6,1) + 1(10,1)

Spectrum 59; b1=(2,4), b2=(3/2,1); singlets = 19

19(1,1) + 2(15,1) + 17(4,1) + 1(4,15) + 10(1,6) + 1(1,15) + 2(6,6) + 1(6,1) + 1(10,1)

Spectrum 60; b1=(2,4), b2=(3/2,1); singlets = 31

31(1,1) + 3(15,1) + 2(4,6) + 5(4,1) + 1(4,15) + 9(1,6) + 1/2(1,20) + 1(6,6) + 6(6,1)

Spectrum 61; b1=(2,4), b2=(3/2,1); singlets = 19

19(1,1) + 3(15,1) + 17(4,1) + 1(4,15) + 11(1,6) + 1/2(1,20) + 2(6,6)

Spectrum 62; b1=(2,4), b2=(3/2,1); singlets = 32

32(1,1) + 3(15,1) + 2(4,6) + 5(4,1) + 1(4,15) + 8(1,6) + 1(1,15) + 1(6,6) + 6(6,1)

Spectrum 63; b1=(2,4), b2=(3/2,1); singlets = 20

20(1,1) + 3(15,1) + 17(4,1) + 1(4,15) + 10(1,6) + 1(1,15) + 2(6,6)

A3 x A6

Spectrum 64; b1=(2,4), b2=(2,2); singlets = 15

15(1,1) + 2(4,7) + 8(4,1) + 2(4,21) + 1(1,21) + 1(1,28) + 1(20,1)

Spectrum 65; b1=(2,4), b2=(2,2); singlets = 16

16(1,1) + 2(4,7) + 8(4,1) + 2(4,21) + 1(1,48) + 1(20,1)

Spectrum 66; b1=(2,4), b2=(2,2); singlets = 21

21(1,1) + 1(4,1) + 2(4,21) + 2(1,7) + 1(1,21) + 1(1,28) + 1(6,7) + 2(6,1) + 1(10,1) + 1(20',1)

Spectrum 67; b1=(2,4), b2=(2,2); singlets = 22

22(1,1) + 1(4,1) + 2(4,21) + 1(1,48) + 2(1,7) + 1(6,7) + 2(6,1) + 1(10,1) + 1(20',1)

Spectrum 68; b1=(2,4), b2=(2,2); singlets = 22

22(1,1) + 1(15,1) + 1(4,1) + 2(4,21) + 2(1,7) + 1(1,21) + 1(1,28) + 1(6,7) + 1(6,1) + 1(20',1)

Spectrum 69; b1=(2,4), b2=(2,2); singlets = 23

23(1,1) + 1(15,1) + 1(4,1) + 2(4,21) + 1(1,48) + 2(1,7) + 1(6,7) + 1(6,1) + 1(20',1)

A3 x B3

Spectrum 70; b1=(2,4), b2=(5/2,5); singlets = 28

28(1,1) + 1(4,21) + 2(4,7) + 1(4,8) + 2(1,21) + 2(1,7) + 1(6,1) + 1(6,8) + 1(10,1) + 1(20',1)

Spectrum 71; b1=(2,4), b2=(5/2,5); singlets = 29

29(1,1) + 1(15,1) + 1(4,21) + 2(4,7) + 1(4,8) + 2(1,21) + 2(1,7) + 1(6,8) + 1(20',1)

Spectrum 72; b1=(2,4), b2=(5/2,5); singlets = 22

22(1,1) + 1(4,21) + 3(4,1) + 1(4,8) + 2(1,21) + 4(1,7) + 1(6,7) + 1(6,8) + 3(10,1)

Spectrum 73; b1=(2,4), b2=(5/2,5); singlets = 18

18(1,1) + 1(4,21) + 1(4,7) + 4(4,1) + 2(1,21) + 4(1,8) + 1(6,7) + 1(6,8) + 3(10,1)

Spectrum 74; b1=(2,4), b2=(1,0); singlets = 43

43(1,1) + 48(4,1) + 2(4,8) + 3(1,7) + 1(20,1)

Spectrum 75; b1=(2,4), b2=(1,0); singlets = 63

63(1,1) + 27(4,1) + 2(4,8) + 3(1,7) + 9(6,1) + 1(10,1) + 1(20',1)

Spectrum 76; b1=(2,4), b2=(1,0); singlets = 47

47(1,1) + 43(4,1) + 3(1,7) + 2(1,8) + 1(6,1) + 1(6,8) + 1(10,1) + 1(20',1)

Spectrum 77; b1=(2,4), b2=(1,0); singlets = 64

64(1,1) + 1(15,1) + 27(4,1) + 2(4,8) + 3(1,7) + 8(6,1) + 1(20',1)

Spectrum 78; b1=(2,4), b2=(1,0); singlets = 48

48(1,1) + 1(15,1) + 43(4,1) + 3(1,7) + 2(1,8) + 1(6,8) + 1(20',1)

Spectrum 79; b1=(2,4), b2=(1,0); singlets = 71

71(1,1) + 16(4,1) + 2(4,8) + 3(1,7) + 15(6,1) + 3(10,1)

Spectrum 80; b1=(2,4), b2=(1,0); singlets = 55

55(1,1) + 32(4,1) + 3(1,7) + 2(1,8) + 7(6,1) + 1(6,8) + 3(10,1)

Spectrum 81; b1=(2,4), b2=(1,0); singlets = 72

72(1,1) + 1(15,1) + 16(4,1) + 2(4,8) + 3(1,7) + 14(6,1) + 2(10,1)

Spectrum 82; b1=(2,4), b2=(1,0); singlets = 56

56(1,1) + 1(15,1) + 32(4,1) + 3(1,7) + 2(1,8) + 6(6,1) + 1(6,8) + 2(10,1)

Spectrum 83; b1=(2,4), b2=(1,0); singlets = 73

73(1,1) + 2(15,1) + 16(4,1) + 2(4,8) + 3(1,7) + 13(6,1) + 1(10,1)

Spectrum 84; b1=(2,4), b2=(1,0); singlets = 57

57(1,1) + 2(15,1) + 32(4,1) + 3(1,7) + 2(1,8) + 5(6,1) + 1(6,8) + 1(10,1)

Spectrum 85; b1=(2,4), b2=(1,0); singlets = 74

74(1,1) + 3(15,1) + 16(4,1) + 2(4,8) + 3(1,7) + 12(6,1)

Spectrum 86; b1=(2,4), b2=(1,0); singlets = 58

58(1,1) + 3(15,1) + 32(4,1) + 3(1,7) + 2(1,8) + 4(6,1) + 1(6,8)

A3 x B4

Spectrum 87; b1=(2,4), b2=(2,2); singlets = 28

28(1,1) + 2(4,9) + 14(4,1) + 2(4,16) + 1(1,36) + 1(20,1)

Spectrum 88; b1=(2,4), b2=(2,2); singlets = 30

30(1,1) + 11(4,1) + 2(4,16) + 1(1,36) + 2(1,9) + 1(6,9) + 1(10,1) + 1(20',1)

Spectrum 89; b1=(2,4), b2=(2,2); singlets = 38

38(1,1) + 2(4,16) + 1(1,36) + 2(1,9) + 1(6,9) + 6(6,1) + 3(10,1)

Spectrum 90; b1=(2,4), b2=(2,2); singlets = 39

39(1,1) + 1(15,1) + 2(4,16) + 1(1,36) + 2(1,9) + 1(6,9) + 5(6,1) + 2(10,1)

Spectrum 91; b1=(2,4), b2=(2,2); singlets = 40

40(1,1) + 2(15,1) + 2(4,16) + 1(1,36) + 2(1,9) + 1(6,9) + 4(6,1) + 1(10,1)

Spectrum 92; b1=(2,4), b2=(2,2); singlets = 41

41(1,1) + 3(15,1) + 2(4,16) + 1(1,36) + 2(1,9) + 1(6,9) + 3(6,1)

A3 x C2

Spectrum 93; b1=(7/2,9), b2=(3,8); singlets = 5

5(1,1) + 1(4,10) + 2(4,1) + 1/2(1,4) + 1(1,5) + 1/2(1,20) + 1/2(6,4) + 1(6,5) + 1/2(6,16) + 2(10,4) + 1(10,1) + 1(10,5) + 1/2(20,4)

Spectrum 94; b1=(7/2,9), b2=(3,8); singlets = 5

5(1,1) + 1(15,5) + 1(4,10) + 2(4,1) + 1/2(1,4) + 2(1,5) + 1/2(1,20) + 1/2(6,4) + 1/2(6,16) + 2(10,4) + 1(10,1) + 1/2(20,4)

$$1(15.1) + 1(4.4) + 11(4.1) + 2(4.5) + 1(1.10) + 2(1.4) + 1(1.5) + 2(6.1) + 1(10.4) + 1(10.1) + 1(20'.4) + 3(20'.1)$$

$$S(1,1) = 1(15,4) = 1(15,1) = 2(4,1) = 2(4,5) = 1(1,15) = 1(1,4) = 1(1,5) = 15(5,1) = 2(15,4) = 1(15,1) = 2(20,1)$$

Spectrum 197; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 3
 $3(1,1) + 1/2(15,4) + 6(15,1) + 2(4,4) + 4(4,1) + 1(1,10) + 1/2(1,4) + 3(1,5) + 2(6,1) + 1(6,5) + 1(10,4) + 2(10,1) + 1/2(20,4)$

Spectrum 198; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 1
 $1(1,1) + 3/2(15,4) + 2(15,1) + 2(4,1) + 2(4,5) + 1(1,10) + 7/2(1,4) + 1(1,5) + 7(6,1) + 6(10,1) + 1/2(20,4)$

Spectrum 199; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 8
 $8(1,1) + 2(4,4) + 1(4,1) + 1(4,5) + 1(1,4) + 2(1,5) + 1/2(1,16) + 1/2(6,4) + 7(6,1) + 1(10,4) + 4(10,1) + 1(20',4) + 2(20',1)$

Spectrum 200; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 4
 $4(1,1) + 1(4,4) + 5(4,1) + 1(4,5) + 2(1,4) + 2(1,5) + 1/2(1,16) + 1(6,4) + 5(6,1) + 1(10,4) + 4(10,1) + 1(20',4) + 2(20',1)$

Spectrum 201; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 0
 $9(4,1) + 1(4,5) + 3(1,4) + 2(1,5) + 1/2(1,16) + 3/2(6,4) + 3(6,1) + 1(10,4) + 4(10,1) + 1(20',4) + 2(20',1)$

Spectrum 202; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 7
 $7(1,1) + 1(4,4) + 2(4,5) + 1(1,10) + 2(1,4) + 1(1,5) + 9(6,1) + 1(10,4) + 4(10,1) + 1(20',4) + 2(20',1)$

Spectrum 203; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 3
 $3(1,1) + 4(4,1) + 2(4,5) + 1(1,10) + 3(1,4) + 1(1,5) + 1/2(6,4) + 7(6,1) + 1(10,4) + 4(10,1) + 1(20',4) + 2(20',1)$

Spectrum 204; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 17
 $17(1,1) + 3/2(15,4) + 3(15,1) + 2(4,4) + 4(4,1) + 3/2(1,4) + 1/2(1,16) + 1(6,1) + 1(6,5) + 1(10,4) + 1(10,1) + 2(20,1)$

Spectrum 205; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 5
 $5(1,1) + 2(15,4) + 1(15,1) + 7(4,1) + 1(4,5) + 2(1,4) + 2(1,5) + 1/2(1,16) + 6(6,1) + 1(10,4) + 1(10,1) + 2(20,1)$

Spectrum 206; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 7
 $7(1,1) + 1/2(15,4) + 7(15,1) + 1(4,4) + 3(4,1) + 1(4,5) + 3/2(1,4) + 2(1,5) + 1/2(1,16) + 3/2(6,4) + 1(10,4) + 1(10,1) + 1/2(20,4)$

Spectrum 207; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 3
 $3(1,1) + 3/2(15,4) + 3(15,1) + 1(4,4) + 3(4,1) + 1(4,5) + 5/2(1,4) + 2(1,5) + 1/2(1,16) + 1/2(6,4) + 4(6,1) + 5(10,1) + 1/2(20,4)$

Spectrum 208; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 6
 $6(1,1) + 1/2(15,4) + 7(15,1) + 2(4,1) + 2(4,5) + 1(1,10) + 5/2(1,4) + 1(1,5) + 1(6,4) + 2(6,1) + 1(10,4) + 1(10,1) + 1/2(20,4)$

Spectrum 209; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 4
 $4(1,1) + 1/2(15,4) + 7(15,1) + 2(4,4) + 4(4,1) + 1(1,10) + 1/2(1,4) + 3(1,5) + 1(6,1) + 1(6,5) + 1(10,4) + 1(10,1) + 1/2(20,4)$

Spectrum 210; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 2
 $2(1,1) + 3/2(15,4) + 3(15,1) + 2(4,1) + 2(4,5) + 1(1,10) + 7/2(1,4) + 1(1,5) + 6(6,1) + 5(10,1) + 1/2(20,4)$

Spectrum 211; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 9
 $9(1,1) + 1(15,1) + 2(4,4) + 1(4,1) + 1(4,5) + 1(1,4) + 2(1,5) + 1/2(1,16) + 1/2(6,4) + 6(6,1) + 1(10,4) + 3(10,1) + 1(20',4) + 2(20',1)$

Spectrum 212; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 5
 $5(1,1) + 1(15,1) + 1(4,4) + 5(4,1) + 1(4,5) + 2(1,4) + 2(1,5) + 1/2(1,16) + 1(6,4) + 4(6,1) + 1(10,4) + 3(10,1) + 1(20',4) + 2(20',1)$

Spectrum 213; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 1
 $1(1,1) + 1(15,1) + 9(4,1) + 1(4,5) + 3(1,4) + 2(1,5) + 1/2(1,16) + 3/2(6,4) + 2(6,1) + 1(10,4) + 3(10,1) + 1(20',4) + 2(20',1)$

Spectrum 214; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 8
 $8(1,1) + 1(15,1) + 1(4,4) + 2(4,5) + 1(1,10) + 2(1,4) + 1(1,5) + 8(6,1) + 1(10,4) + 3(10,1) + 1(20',4) + 2(20',1)$

Spectrum 215; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 4
 $4(1,1) + 1(15,1) + 4(4,1) + 2(4,5) + 1(1,10) + 3(1,4) + 1(1,5) + 1/2(6,4) + 6(6,1) + 1(10,4) + 3(10,1) + 1(20',4) + 2(20',1)$

Spectrum 216; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 18
 $18(1,1) + 3/2(15,4) + 4(15,1) + 2(4,4) + 4(4,1) + 3/2(1,4) + 1/2(1,16) + 1(6,5) + 1(10,4) + 2(20,1)$

Spectrum 217; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 6
 $6(1,1) + 2(15,4) + 2(15,1) + 7(4,1) + 1(4,5) + 2(1,4) + 2(1,5) + 1/2(1,16) + 5(6,1) + 1(10,4) + 2(20,1)$

Spectrum 218; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 6
 $6(1,1) + 5/2(15,4) + 12(4,1) + 9/2(1,4) + 1/2(1,16) + 1(6,5) + 4(10,1) + 2(20,1)$

Spectrum 219; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 4
 $4(1,1) + 3/2(15,4) + 4(15,1) + 1(4,4) + 3(4,1) + 1(4,5) + 5/2(1,4) + 2(1,5) + 1/2(1,16) + 1/2(6,4) + 3(6,1) + 4(10,1) + 1/2(20,4)$

Spectrum 220; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 0
 $3/2(15,4) + 4(15,1) + 7(4,1) + 1(4,5) + 7/2(1,4) + 2(1,5) + 1/2(1,16) + 1(6,4) + 1(6,1) + 4(10,1) + 1/2(20,4)$

Spectrum 221; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 7
 $7(1,1) + 1/2(15,4) + 8(15,1) + 2(4,1) + 2(4,5) + 1(1,10) + 5/2(1,4) + 1(1,5) + 1(6,4) + 1(6,1) + 1(10,4) + 1/2(20,4)$

Spectrum 222; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 5
 $5(1,1) + 1/2(15,4) + 8(15,1) + 2(4,4) + 4(4,1) + 1(1,10) + 1/2(1,4) + 3(1,5) + 1(6,5) + 1(10,4) + 1/2(20,4)$

Spectrum 223; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 3
 $3(1,1) + 3/2(15,4) + 4(15,1) + 2(4,1) + 2(4,5) + 1(1,10) + 7/2(1,4) + 1(1,5) + 5(6,1) + 4(10,1) + 1/2(20,4)$

Spectrum 224; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 18
 $18(1,1) + 3/2(15,4) + 2(4,4) + 5(4,1) + 3/2(1,4) + 1/2(1,16) + 1(6,1) + 1(6,5) + 1(10,4) + 1(10,1) + 3(20',1) + 1(20,1)$

Spectrum 225; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 10
 $10(1,1) + 2(15,1) + 2(4,4) + 1(4,1) + 1(4,5) + 2(1,4) + 2(1,5) + 1/2(1,16) + 1/2(6,4) + 5(6,1) + 1(10,4) + 2(10,1) + 1(20',4) + 2(20',1)$

Spectrum 226; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 6
 $6(1,1) + 2(15,1) + 1(4,4) + 5(4,1) + 1(4,5) + 2(1,4) + 2(1,5) + 1/2(1,16) + 1(6,4) + 3(6,1) + 1(10,4) + 2(10,1) + 1(20',4) + 2(20',1)$

Spectrum 227; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 2
 $2(1,1) + 2(15,1) + 9(4,1) + 1(4,5) + 3(1,4) + 2(1,5) + 1/2(1,16) + 3/2(6,4) + 1(6,1) + 1(10,4) + 2(10,1) + 1(20',4) + 2(20',1)$

Spectrum 228; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 14
 $14(1,1) + 1/2(15,4) + 3(4,4) + 2(4,1) + 5/2(1,4) + 1/2(1,16) + 2(6,1) + 1(6,5) + 6(10,1) + 1(20',4) + 2(20',1)$

Spectrum 229; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 10
 $10(1,1) + 1/2(15,4) + 2(4,4) + 6(4,1) + 7/2(1,4) + 1/2(1,16) + 1/2(6,4) + 1(6,5) + 6(10,1) + 1(20',4) + 2(20',1)$

Spectrum 230; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 9
 $9(1,1) + 2(15,1) + 1(4,4) + 2(4,5) + 1(1,10) + 2(1,4) + 1(1,5) + 7(6,1) + 1(10,4) + 2(10,1) + 1(20',4) + 2(20',1)$

Spectrum 231; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 5

$5(1,1) + 2(15,1) + 4(4,1) + 2(4,5) + 1(1,10) + 3(1,4) + 1(1,5) + 1/2(6,4) + 5(6,1) + 1(10,4) + 2(10,1) + 1(20',4) + 2(20',1)$

Spectrum 232; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 5

$5(1,1) + 3/2(15,4) + 5(15,1) + 1(4,4) + 3(4,1) + 1(4,5) + 5/2(1,4) + 2(1,5) + 1/2(1,16) + 1/2(6,4) + 2(6,1) + 3(10,1) + 1/2(20,4)$

Spectrum 233; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 1

$1(1,1) + 3/2(15,4) + 5(15,1) + 7(4,1) + 1(4,5) + 7/2(1,4) + 2(1,5) + 1/2(1,16) + 1(6,4) + 3(10,1) + 1/2(20,4)$

Spectrum 234; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 4

$4(1,1) + 3/2(15,4) + 5(15,1) + 2(4,1) + 2(4,5) + 1(1,10) + 7/2(1,4) + 1(1,5) + 4(6,1) + 3(10,1) + 1/2(20,4)$

Spectrum 235; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 19

$19(1,1) + 3/2(15,4) + 1(15,1) + 2(4,4) + 5(4,1) + 3/2(1,4) + 1/2(1,16) + 1(6,5) + 1(10,4) + 3(20',1) + 1(20,1)$

Spectrum 236; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 11

$11(1,1) + 3(15,1) + 2(4,4) + 1(4,1) + 1(4,5) + 1(1,4) + 2(1,5) + 1/2(1,16) + 1/2(6,4) + 4(6,1) + 1(10,4) + 1(10,1) + 1(20',4) + 2(20',1)$

Spectrum 237; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 7

$7(1,1) + 3(15,1) + 1(4,4) + 5(4,1) + 1(4,5) + 2(1,4) + 2(1,5) + 1/2(1,16) + 1(6,4) + 2(6,1) + 1(10,4) + 1(10,1) + 1(20',4) + 2(20',1)$

Spectrum 238; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 3

$3(1,1) + 3(15,1) + 9(4,1) + 1(4,5) + 3(1,4) + 2(1,5) + 1/2(1,16) + 3/2(6,4) + 1(10,4) + 1(10,1) + 1(20',4) + 2(20',1)$

Spectrum 239; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 15

$15(1,1) + 1/2(15,4) + 1(15,1) + 3(4,4) + 2(4,1) + 5/2(1,4) + 1/2(1,16) + 1(6,1) + 1(6,5) + 5(10,1) + 1(20',4) + 2(20',1)$

Spectrum 240; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 10

$10(1,1) + 3(15,1) + 1(4,4) + 2(4,5) + 1(1,10) + 2(1,4) + 1(1,5) + 6(6,1) + 1(10,4) + 1(10,1) + 1(20',4) + 2(20',1)$

Spectrum 241; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 6

$6(1,1) + 3(15,1) + 4(4,1) + 2(4,5) + 1(1,10) + 3(1,4) + 1(1,5) + 1/2(6,4) + 4(6,1) + 1(10,4) + 1(10,1) + 1(20',4) + 2(20',1)$

Spectrum 242; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 0

$3(15,1) + 1(4,4) + 10(4,1) + 1(1,10) + 2(1,4) + 3(1,5) + 1(6,1) + 1(6,5) + 1(10,4) + 1(10,1) + 1(20',4) + 2(20',1)$

Spectrum 243; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 2

$2(1,1) + 2(4,1) + 6(1,4) + 1/2(1,16) + 1(6,4) + 9(6,1) + 1(6,5) + 3(10,4) + 1(10,1) + 2(20',1) + 1(20,1)$

Spectrum 244; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 9

$9(1,1) + 2(4,1) + 1(1,10) + 3(1,5) + 1(6,4) + 9(6,1) + 1(6,5) + 3(10,4) + 1(10,1) + 2(20',1) + 1(20,1)$

Spectrum 245; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 6

$6(1,1) + 3/2(15,4) + 6(15,1) + 1(4,4) + 3(4,1) + 1(4,5) + 5/2(1,4) + 2(1,5) + 1/2(1,16) + 1/2(6,4) + 1(6,1) + 2(10,1) + 1/2(20,4)$

Spectrum 246; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 5

$5(1,1) + 3/2(15,4) + 6(15,1) + 2(4,1) + 2(4,5) + 1(1,10) + 7/2(1,4) + 1(1,5) + 3(6,1) + 2(10,1) + 1/2(20,4)$

Spectrum 247; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 12

$12(1,1) + 4(15,1) + 2(4,4) + 1(4,1) + 1(4,5) + 1(1,4) + 2(1,5) + 1/2(1,16) + 1/2(6,4) + 3(6,1) + 1(10,4) + 1(20',4) + 2(20',1)$

Spectrum 248; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 8

$8(1,1) + 4(15,1) + 1(4,4) + 5(4,1) + 1(4,5) + 2(1,4) + 2(1,5) + 1/2(1,16) + 1(6,4) + 1(6,1) + 1(10,4) + 1(20',4) + 2(20',1)$

Spectrum 249; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 16

$16(1,1) + 1/2(15,4) + 2(15,1) + 3(4,4) + 2(4,1) + 5/2(1,4) + 1/2(1,16) + 1(6,5) + 4(10,1) + 1(20',4) + 2(20',1)$

Spectrum 250; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 4

$4(1,1) + 1(15,4) + 1(4,4) + 5(4,1) + 1(4,5) + 3(1,4) + 2(1,5) + 1/2(1,16) + 5(6,1) + 4(10,1) + 1(20',4) + 2(20',1)$

Spectrum 251; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 0

$1(15,4) + 9(4,1) + 1(4,5) + 4(1,4) + 2(1,5) + 1/2(1,16) + 1/2(6,4) + 3(6,1) + 4(10,1) + 1(20',4) + 2(20',1)$

Spectrum 252; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 11

$11(1,1) + 4(15,1) + 1(4,4) + 2(4,5) + 1(1,10) + 2(1,4) + 1(1,5) + 5(6,1) + 1(10,4) + 1(20',4) + 2(20',1)$

Spectrum 253; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 7

$7(1,1) + 4(15,1) + 4(4,1) + 2(4,5) + 1(1,10) + 3(1,4) + 1(1,5) + 1/2(6,4) + 3(6,1) + 1(10,4) + 1(20',4) + 2(20',1)$

Spectrum 254; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 1

$1(1,1) + 4(15,1) + 1(4,4) + 10(4,1) + 1(1,10) + 2(1,4) + 3(1,5) + 1(6,5) + 1(10,4) + 1(20',4) + 2(20',1)$

Spectrum 255; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 3

$3(1,1) + 1(15,1) + 2(4,1) + 6(1,4) + 1/2(1,16) + 1(6,4) + 8(6,1) + 1(6,5) + 3(10,4) + 2(20',1) + 1(20,1)$

Spectrum 256; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 10

$10(1,1) + 1(15,1) + 2(4,1) + 1(1,10) + 3(1,5) + 1(6,4) + 8(6,1) + 1(6,5) + 3(10,4) + 2(20',1) + 1(20,1)$

Spectrum 257; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 7

$7(1,1) + 3/2(15,4) + 7(15,1) + 1(4,4) + 3(4,1) + 1(4,5) + 5/2(1,4) + 2(1,5) + 1/2(1,16) + 1/2(6,4) + 1(10,1) + 1/2(20,4)$

Spectrum 258; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 6

$6(1,1) + 3/2(15,4) + 7(15,1) + 2(4,1) + 2(4,5) + 1(1,10) + 7/2(1,4) + 1(1,5) + 2(6,1) + 1(10,1) + 1/2(20,4)$

Spectrum 259; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 5

$5(1,1) + 1(15,4) + 1(15,1) + 1(4,4) + 5(4,1) + 1(4,5) + 3(1,4) + 2(1,5) + 1/2(1,16) + 4(6,1) + 3(10,1) + 1(20',4) + 2(20',1)$

Spectrum 260; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 1

$1(1,1) + 1(15,4) + 1(15,1) + 9(4,1) + 1(4,5) + 4(1,4) + 2(1,5) + 1/2(1,16) + 1/2(6,4) + 2(6,1) + 3(10,1) + 1(20',4) + 2(20',1)$

Spectrum 261; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 18

$18(1,1) + 1/2(15,4) + 2(4,1) + 5/2(1,4) + 1/2(1,16) + 2(6,4) + 3(6,1) + 1(6,5) + 2(10,4) + 3(10,1) + 2(20',1) + 1(20,1)$

Spectrum 262; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 7

$7(1,1) + 3/2(15,4) + 8(15,1) + 2(4,1) + 2(4,5) + 1(1,10) + 7/2(1,4) + 1(1,5) + 1(6,1) + 1/2(20,4)$

Spectrum 263; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 6

$6(1,1) + 1(15,4) + 2(15,1) + 1(4,4) + 1(4,5) + 3(1,4) + 2(1,5) + 1/2(1,16) + 3(6,1) + 2(10,1) + 1(20',4) + 2(20',1)$

Spectrum 264; $b1=(7/2,9)$, $b2=(3/2,3)$; singlets = 2

$2(1,1) + 1(15,4) + 2(15,1) + 9(4,1) + 1(4,5) + 4(1,4) + 2(1,5) + 1/2(1,16) + 1/2(6,4) + 1(6,1) + 2(10,1) + 1(20',4) + 2(20',1)$

Spectrum 299; b1=(2,4), b2=(3,8); singlets = 2

2(1,1) + 1(4,10) + 8(4,4) + 1(4,1) + 1(4,5) + 1(4,16) + 1(1,10) + 6(1,4) + 4(1,5) + 1/2(1,16) + 1(20,1)

Spectrum 300; b1=(2,4), b2=(3,8); singlets = 9

9(1,1) + 1(4,10) + 8(4,4) + 1(4,1) + 1(4,5) + 1(4,16) + 2(1,10) + 7(1,5) + 1(20,1)

Spectrum 301; b1=(2,4), b2=(3,8); singlets = 27

27(1,1) + 2(4,10) + 2(4,4) + 1(4,1) + 7(4,5) + 3/2(1,4) + 1(1,5) + 1(1,16) + 1/2(1,20) + 1(20,1)

Spectrum 302; b1=(2,4), b2=(3,8); singlets = 22

22(1,1) + 2(4,10) + 4(4,1) + 8(4,5) + 1(1,10) + 7/2(1,4) + 1/2(1,16) + 1/2(1,20) + 1(20,1)

Spectrum 303; b1=(2,4), b2=(3,8); singlets = 3

3(1,1) + 1(4,10) + 4(4,4) + 2(4,1) + 3(4,5) + 4(1,10) + 4(1,4) + 5(1,5) + 1(6,1) + 1/2(6,16) + 1(10,1) + 1(20,1)

Spectrum 304; b1=(2,4), b2=(3,8); singlets = 4

4(1,1) + 1(15,1) + 1(4,10) + 4(4,4) + 2(4,1) + 3(4,5) + 4(1,10) + 4(1,4) + 5(1,5) + 1/2(6,16) + 1(20,1)

Spectrum 305; b1=(2,4), b2=(3,8); singlets = 6

6(1,1) + 1(4,10) + 4(4,4) + 1(4,1) + 1(4,5) + 6(1,4) + 8(1,5) + 1(1,14) + 1/2(1,16) + 1(6,10) + 1(6,5) + 3(10,1)

Spectrum 306; b1=(2,4), b2=(3,8); singlets = 4

4(1,1) + 1(4,10) + 3(4,4) + 2(4,5) + 3(1,10) + 5(1,4) + 6(1,5) + 1/2(1,16) + 3/2(6,4) + 1(6,1) + 1/2(6,16) + 3(10,1)

Spectrum 307; b1=(2,4), b2=(3,8); singlets = 2

2(1,1) + 1(4,10) + 5(4,4) + 2(4,1) + 3(1,10) + 3(1,4) + 8(1,5) + 1/2(1,16) + 1/2(6,4) + 1(6,5) + 1/2(6,16) + 3(10,1)

Spectrum 308; b1=(2,4), b2=(3,8); singlets = 3

3(1,1) + 2(4,1) + 6(4,5) + 11(1,4) + 5(1,5) + 1(1,20) + 1(6,10) + 1(6,5) + 3(10,1)

Spectrum 309; b1=(2,4), b2=(3,8); singlets = 3

3(1,1) + 1(4,10) + 2(4,1) + 4(4,5) + 1(1,10) + 10(1,4) + 5(1,5) + 1(1,14) + 1(6,10) + 1(6,4) + 1(6,1) + 3(10,1)

Spectrum 310; b1=(2,4), b2=(3,8); singlets = 13

13(1,1) + 1(4,10) + 4(4,4) + 1(4,1) + 1(4,5) + 1(1,10) + 11(1,5) + 1(1,14) + 1(6,10) + 1(6,5) + 3(10,1)

Spectrum 311; b1=(2,4), b2=(3,8); singlets = 1

1(1,1) + 1(4,10) + 2(4,4) + 4(4,1) + 2(4,5) + 1(1,10) + 8(1,4) + 7(1,5) + 1(1,14) + 1(6,10) + 1(6,5) + 3(10,1)

Spectrum 312; b1=(2,4), b2=(3,8); singlets = 1

1(1,1) + 1(4,10) + 4(4,4) + 1(4,1) + 1(4,5) + 4(1,10) + 4(1,4) + 7(1,5) + 2(6,1) + 1(6,5) + 1/2(6,16) + 3(10,1)

Spectrum 313; b1=(2,4), b2=(3,8); singlets = 13

13(1,1) + 2(4,10) + 2(4,1) + 2(4,5) + 19/2(1,4) + 3(1,5) + 1(1,16) + 1/2(1,20) + 3(6,5) + 3(10,1)

Spectrum 314; b1=(2,4), b2=(3,8); singlets = 20

20(1,1) + 2(4,10) + 2(4,1) + 2(4,5) + 1(1,10) + 7/2(1,4) + 6(1,5) + 1/2(1,16) + 1/2(1,20) + 3(6,5) + 3(10,1)

Spectrum 315; b1=(2,4), b2=(3,8); singlets = 20

20(1,1) + 2(4,1) + 6(4,5) + 3(1,10) + 7/2(1,4) + 2(1,5) + 1/2(1,16) + 1/2(1,20) + 1(6,10) + 1(6,5) + 3(10,1)

Spectrum 316; b1=(2,4), b2=(3,8); singlets = 17

17(1,1) + 1(4,10) + 2(4,1) + 4(4,5) + 2(1,4) + 7(1,5) + 2(1,16) + 1(6,10) + 1(6,4) + 1(6,1) + 3(10,1)

Spectrum 317; b1=(2,4), b2=(3,8); singlets = 15

15(1,1) + 1(4,10) + 2(4,4) + 4(4,1) + 2(4,5) + 9(1,5) + 2(1,16) + 1(6,10) + 1(6,5) + 3(10,1)

Spectrum 318; b1=(2,4), b2=(3,8); singlets = 3

3(1,1) + 1(4,10) + 7(4,1) + 3(4,5) + 8(1,4) + 5(1,5) + 2(1,16) + 1(6,10) + 1(6,5) + 3(10,1)

Spectrum 319; b1=(2,4), b2=(3,8); singlets = 30

30(1,1) + 2(4,10) + 2(4,1) + 2(4,5) + 3(1,10) + 2(1,4) + 3/2(1,16) + 3(6,5) + 3(10,1)

Spectrum 320; b1=(2,4), b2=(3,8); singlets = 10

10(1,1) + 1(4,10) + 7(4,1) + 3(4,5) + 1(1,10) + 2(1,4) + 8(1,5) + 3/2(1,16) + 1(6,10) + 1(6,5) + 3(10,1)

Spectrum 321; b1=(2,4), b2=(3,8); singlets = 5

5(1,1) + 1(15,1) + 1(4,10) + 3(4,4) + 2(4,5) + 3(1,10) + 5(1,4) + 6(1,5) + 1/2(1,16) + 3/2(6,4) + 1/2(6,16) + 2(10,1)

Spectrum 322; b1=(2,4), b2=(3,8); singlets = 4

4(1,1) + 1(15,1) + 1(4,10) + 2(4,1) + 4(4,5) + 1(1,10) + 10(1,4) + 5(1,5) + 1(1,14) + 1(6,10) + 1(6,4) + 2(10,1)

Spectrum 323; b1=(2,4), b2=(3,8); singlets = 0

1(15,1) + 1(4,10) + 1(4,4) + 3(4,1) + 3(4,5) + 4(1,10) + 7(1,4) + 5(1,5) + 3/2(6,4) + 1/2(6,16) + 2(10,1)

Spectrum 324; b1=(2,4), b2=(3,8); singlets = 2

2(1,1) + 1(15,1) + 1(4,10) + 4(4,4) + 1(4,1) + 1(4,5) + 4(1,10) + 4(1,4) + 7(1,5) + 1(6,1) + 1(6,5) + 1/2(6,16) + 2(10,1)

Spectrum 325; b1=(2,4), b2=(3,8); singlets = 18

18(1,1) + 1(15,1) + 1(4,10) + 2(4,1) + 4(4,5) + 2(1,4) + 7(1,5) + 2(1,16) + 1(6,10) + 1(6,4) + 2(10,1)

Spectrum 326; b1=(2,4), b2=(3,8); singlets = 0

1/2(15,4) + 3(4,4) + 4(4,5) + 13(1,4) + 2(1,5) + 1(1,14) + 1/2(1,16) + 1/2(1,20) + 1(6,10) + 1/2(6,4) + 1(6,1) + 1(10,1)

Spectrum 327; b1=(2,4), b2=(3,8); singlets = 6

6(1,1) + 1/2(15,4) + 8(4,4) + 3(1,10) + 8(1,5) + 1/2(1,16) + 1/2(1,20) + 1(6,5) + 1/2(6,16) + 1(10,1)

Spectrum 328; b1=(2,4), b2=(3,8); singlets = 7

7(1,1) + 1/2(15,4) + 3(4,4) + 4(4,5) + 1(1,10) + 7(1,4) + 5(1,5) + 1(1,14) + 1/2(1,20) + 1(6,10) + 1/2(6,4) + 1(6,1) + 1(10,1)

Spectrum 329; b1=(2,4), b2=(3,8); singlets = 3

3(1,1) + 1/2(15,4) + 4(4,4) + 1(4,1) + 3(4,5) + 4(1,10) + 4(1,4) + 5(1,5) + 1/2(1,20) + 1(6,4) + 1(6,1) + 1/2(6,16) + 1(10,1)

Spectrum 330; b1=(2,4), b2=(3,8); singlets = 1

1(1,1) + 1/2(15,4) + 6(4,4) + 3(4,1) + 1(4,5) + 4(1,10) + 2(1,4) + 7(1,5) + 1/2(1,20) + 1(6,5) + 1/2(6,16) + 1(10,1)

Spectrum 331; b1=(2,4), b2=(3,8); singlets = 10

10(1,1) + 1/2(15,4) + 2(4,10) + 3(4,4) + 23/2(1,4) + 1(1,14) + 3/2(1,16) + 1/2(6,4) + 1(6,1) + 2(6,5) + 1(10,1)

Spectrum 332; b1=(2,4), b2=(3,8); singlets = 16

16(1,1) + 1/2(15,4) + 4(4,4) + 1(4,16) + 5(1,10) + 5/2(1,4) + 2(1,5) + 1/2(1,16) + 2(6,4) + 1(6,5) + 1(10,1)

Spectrum 333; b1=(2,4), b2=(3,8); singlets = 15

15(1,1) + 1/2(15,4) + 2(4,10) + 2(4,1) + 2(4,5) + 1(1,10) + 17/2(1,4) + 1(1,5) + 1(1,14) + 1(1,16) + 2(6,4) + 1(6,5) + 1(10,1)

Spectrum 334; b1=(2,4), b2=(3,8); singlets = 17

17(1,1) + 1/2(15,4) + 2(4,10) + 3(4,4) + 1(1,10) + 11/2(1,4) + 3(1,5) + 1(1,14) + 1(1,16) + 1/2(6,4) + 1(6,1) + 2(6,5) + 1(10,1)

Spectrum 335; b1=(2,4), b2=(3,8); singlets = 11

11(1,1) + 1/2(15,4) + 2(4,4) + 3(4,1) + 1(4,5) + 1(4,16) + 6(1,10) + 9/2(1,4) + 1(1,5) + 2(6,4) + 1(6,5) + 1(10,1)

Spectrum 336; b1=(2,4), b2=(3,8); singlets = 11

11(1,1) + 1/2(15,4) + 6(4,4) + 3(4,1) + 1(4,5) + 6(1,10) + 1/2(1,4) + 1(1,5) + 1(1,16) + 1(6,5) + 1/2(6,16) + 1(10,1)

Spectrum 337; b1=(2,4), b2=(3,8); singlets = 9

9(1,1) + 1/2(15,4) + 1(4,4) + 3(4,1) + 5(4,5) + 7(1,4) + 3(1,5) + 2(1,16) + 1/2(1,20) + 1(6,10) + 1/2(6,4) + 1(6,1) + 1(10,1)

Spectrum 338; b1=(2,4), b2=(3,8); singlets = 22

22(1,1) + 1/2(15,4) + 2(4,10) + 2(4,1) + 2(4,5) + 2(1,10) + 5/2(1,4) + 4(1,5) + 1(1,14) + 1/2(1,16) + 2(6,4) + 1(6,5) + 1(10,1)

Spectrum 339; b1=(2,4), b2=(3,8); singlets = 12

12(1,1) + 1/2(15,4) + 2(4,10) + 1(4,4) + 3(4,1) + 1(4,5) + 2(1,10) + 15/2(1,4) + 2(1,5) + 1(1,14) + 1/2(1,16) + 1/2(6,4) + 1(6,1) + 2(6,5) + 1(10,1)

Spectrum 340; b1=(2,4), b2=(3,8); singlets = 6

6(1,1) + 1/2(15,4) + 4(4,4) + 6(4,1) + 2(4,5) + 7(1,10) + 5/2(1,4) + 1/2(1,16) + 1(6,5) + 1/2(6,16) + 1(10,1)

Spectrum 341; b1=(2,4), b2=(3,8); singlets = 5

5(1,1) + 1/2(15,4) + 2(4,1) + 6(4,5) + 7/2(1,4) + 11(1,5) + 1(1,20) + 1(6,10) + 3(6,1) + 1(10,1)

Spectrum 342; b1=(2,4), b2=(3,8); singlets = 16

16(1,1) + 1/2(15,4) + 1(4,4) + 3(4,1) + 5(4,5) + 1(1,10) + 1(1,4) + 6(1,5) + 3/2(1,16) + 1/2(1,20) + 1(6,10) + 1/2(6,4) + 1(6,1) + 1(10,1)

Spectrum 343; b1=(2,4), b2=(3,8); singlets = 8

8(1,1) + 1/2(15,4) + 2(4,1) + 6(4,5) + 1(1,10) + 8(1,4) + 2(1,5) + 3/2(1,16) + 1/2(1,20) + 1(6,10) + 3(6,1) + 1(10,1)

Spectrum 344; b1=(2,4), b2=(3,8); singlets = 19

19(1,1) + 1/2(15,4) + 2(4,10) + 1(4,4) + 3(4,1) + 1(4,5) + 3(1,10) + 3/2(1,4) + 5(1,5) + 1(1,14) + 1/2(6,4) + 1(6,1) + 2(6,5) + 1(10,1)

Spectrum 345; b1=(2,4), b2=(3,8); singlets = 11

11(1,1) + 1/2(15,4) + 2(4,10) + 2(4,1) + 2(4,5) + 3(1,10) + 17/2(1,4) + 1(1,5) + 1(1,14) + 3(6,1) + 2(6,5) + 1(10,1)

Spectrum 346; b1=(2,4), b2=(3,8); singlets = 19

19(1,1) + 1/2(15,4) + 1(4,4) + 3(4,1) + 5(4,5) + 5(1,10) + 3/2(1,4) + 1(1,5) + 1(1,14) + 1(6,10) + 1/2(6,4) + 1(6,1) + 1(10,1)

Spectrum 347; b1=(2,4), b2=(3,8); singlets = 3

3(1,1) + 1/2(15,4) + 1(4,10) + 2(4,4) + 4(4,1) + 2(4,5) + 1(1,10) + 1/2(1,4) + 13(1,5) + 1(1,14) + 1(6,10) + 3(6,1) + 1(10,1)

Spectrum 348; b1=(2,4), b2=(3,8); singlets = 29

29(1,1) + 1/2(15,4) + 2(4,10) + 2(4,1) + 2(4,5) + 1/2(1,4) + 3(1,5) + 3(1,16) + 2(6,4) + 1(6,5) + 1(10,1)

Spectrum 349; b1=(2,4), b2=(3,8); singlets = 19

19(1,1) + 1/2(15,4) + 2(4,10) + 1(4,4) + 3(4,1) + 1(4,5) + 11/2(1,4) + 1(1,5) + 3(1,16) + 1/2(6,4) + 1(6,1) + 2(6,5) + 1(10,1)

Spectrum 350; b1=(2,4), b2=(3,8); singlets = 3

3(1,1) + 2(15,1) + 1(4,10) + 4(4,4) + 1(4,1) + 1(4,5) + 4(1,10) + 4(1,4) + 7(1,5) + 1(6,5) + 1/2(6,16) + 1(10,1)

Spectrum 351; b1=(2,4), b2=(3,8); singlets = 15

15(1,1) + 1/2(15,4) + 2(4,10) + 2(4,1) + 2(4,5) + 2(1,4) + 9(1,5) + 1(1,16) + 1/2(1,20) + 3(6,1) + 2(6,5) + 1(10,1)

Spectrum 352; b1=(2,4), b2=(3,8); singlets = 15

15(1,1) + 1/2(15,4) + 2(4,1) + 6(4,5) + 2(1,10) + 2(1,4) + 5(1,5) + 1(1,16) + 1/2(1,20) + 1(6,10) + 3(6,1) + 1(10,1)

Spectrum 353; b1=(2,4), b2=(3,8); singlets = 18

18(1,1) + 1/2(15,4) + 2(4,10) + 2(4,1) + 2(4,5) + 1(1,10) + 13/2(1,4) + 5/2(1,16) + 3(6,1) + 2(6,5) + 1(10,1)

Spectrum 354; b1=(2,4), b2=(3,8); singlets = 25

25(1,1) + 1/2(15,4) + 2(4,10) + 2(4,1) + 2(4,5) + 2(1,10) + 1/2(1,4) + 3(1,5) + 2(1,16) + 3(6,1) + 2(6,5) + 1(10,1)

Spectrum 355; b1=(2,4), b2=(3,8); singlets = 5

5(1,1) + 1/2(15,4) + 1(4,10) + 7(4,1) + 3(4,5) + 1/2(1,4) + 11(1,5) + 2(1,16) + 1(6,10) + 3(6,1) + 1(10,1)

Spectrum 356; b1=(2,4), b2=(3,8); singlets = 1

1(1,1) + 1/2(15,4) + 1(15,1) + 3(4,4) + 4(4,5) + 13(1,4) + 2(1,5) + 1(1,14) + 1/2(1,16) + 1/2(1,20) + 1(6,10) + 1/2(6,4)

Spectrum 357; b1=(2,4), b2=(3,8); singlets = 8

8(1,1) + 1/2(15,4) + 1(15,1) + 3(4,4) + 4(4,5) + 1(1,10) + 7(1,4) + 5(1,5) + 1(1,14) + 1/2(1,20) + 1(6,10) + 1/2(6,4)

Spectrum 358; b1=(2,4), b2=(3,8); singlets = 4

4(1,1) + 1/2(15,4) + 1(15,1) + 4(4,4) + 1(4,1) + 3(4,5) + 4(1,10) + 4(1,4) + 5(1,5) + 1/2(1,20) + 1(6,4) + 1/2(6,16)

Spectrum 359; b1=(2,4), b2=(3,8); singlets = 11

11(1,1) + 1/2(15,4) + 1(15,1) + 2(4,10) + 3(4,4) + 23/2(1,4) + 1(1,14) + 3/2(1,16) + 1/2(6,4) + 2(6,5)

Spectrum 360; b1=(2,4), b2=(3,8); singlets = 18

18(1,1) + 1/2(15,4) + 1(15,1) + 2(4,10) + 3(4,4) + 1(1,10) + 11/2(1,4) + 3(1,5) + 1(1,14) + 1(1,16) + 1/2(6,4) + 2(6,5)

Spectrum 361; b1=(2,4), b2=(3,8); singlets = 10

10(1,1) + 1/2(15,4) + 1(15,1) + 1(4,4) + 3(4,1) + 5(4,5) + 7(1,4) + 3(1,5) + 2(1,16) + 1/2(1,20) + 1(6,10) + 1/2(6,4)

Spectrum 362; b1=(2,4), b2=(3,8); singlets = 13

13(1,1) + 1/2(15,4) + 1(15,1) + 2(4,10) + 1(4,4) + 3(4,1) + 1(4,5) + 2(1,10) + 15/2(1,4) + 2(1,5) + 1(1,14) + 1/2(1,16) + 1/2(6,4) + 2(6,5)

Spectrum 363; b1=(2,4), b2=(3,8); singlets = 6

6(1,1) + 1/2(15,4) + 1(15,1) + 2(4,1) + 6(4,5) + 7/2(1,4) + 11(1,5) + 1(1,20) + 1(6,10) + 2(6,1)

Spectrum 364; b1=(2,4), b2=(3,8); singlets = 17

17(1,1) + 1/2(15,4) + 1(15,1) + 1(4,4) + 3(4,1) + 5(4,5) + 1(1,10) + 1(1,4) + 6(1,5) + 3/2(1,16) + 1/2(1,20) + 1(6,10) + 1/2(6,4)

Spectrum 365; b1=(2,4), b2=(3,8); singlets = 9

9(1,1) + 1/2(15,4) + 1(15,1) + 2(4,1) + 6(4,5) + 1(1,10) + 8(1,4) + 2(1,5) + 3/2(1,16) + 1/2(1,20) + 1(6,10) + 2(6,1)

Spectrum 366; b1=(2,4), b2=(3,8); singlets = 20

20(1,1) + 1/2(15,4) + 1(15,1) + 2(4,10) + 1(4,4) + 3(4,1) + 1(4,5) + 3(1,10) + 3/2(1,4) + 5(1,5) + 1(1,14) + 1/2(6,4) + 2(6,5)

$$33(\mathbf{1},\mathbf{1}) + 1(\mathbf{15},\mathbf{1}) + 7(\mathbf{4},\mathbf{4}) + 15(\mathbf{4},\mathbf{1}) + 11(\mathbf{1},\mathbf{4}) + 1/2(\mathbf{1},\mathbf{16}) + 1/2(\mathbf{6},\mathbf{4}) + 1(\mathbf{6},\mathbf{1}) + 1(\mathbf{6},\mathbf{5}) + 1(20',\mathbf{1})$$

Spectrum 401; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 50
50(1,1) + 1(15,1) + 6(4,4) + 14(4,1) + 1(4,5) + 1(1,10) + 5(1,5) + 2(6,4) + 1(20',1)
Spectrum 402; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 54
54(1,1) + 1(15,1) + 8(4,4) + 1(4,1) + 2(4,5) + 1(1,10) + 4(1,4) + 1(1,5) + 8(6,1) + 1(20',1)
Spectrum 403; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 50
50(1,1) + 1(15,1) + 7(4,4) + 5(4,1) + 2(4,5) + 1(1,10) + 5(1,4) + 1(1,5) + 1/2(6,4) + 6(6,1) + 1(20',1)
Spectrum 404; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 46
46(1,1) + 1(15,1) + 6(4,4) + 9(4,1) + 2(4,5) + 1(1,10) + 6(1,4) + 1(1,5) + 1(6,4) + 4(6,1) + 1(20',1)
Spectrum 405; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 42
42(1,1) + 1(15,1) + 5(4,4) + 13(4,1) + 2(4,5) + 1(1,10) + 7(1,4) + 1(1,5) + 3/2(6,4) + 2(6,1) + 1(20',1)
Spectrum 406; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 38
38(1,1) + 1(15,1) + 4(4,4) + 17(4,1) + 2(4,5) + 1(1,10) + 8(1,4) + 1(1,5) + 2(6,4) + 1(20',1)
Spectrum 407; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 44
44(1,1) + 1(15,1) + 8(4,4) + 11(4,1) + 1(1,10) + 4(1,4) + 3(1,5) + 3(6,1) + 1(6,5) + 1(20',1)
Spectrum 408; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 40
40(1,1) + 1(15,1) + 7(4,4) + 15(4,1) + 1(1,10) + 5(1,4) + 3(1,5) + 1/2(6,4) + 1(6,1) + 1(6,5) + 1(20',1)
Spectrum 409; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 54
54(1,1) + 6(4,4) + 8(4,1) + 6(1,5) + 1/2(1,16) + 3(6,4) + 3(6,1) + 3(10,1)
Spectrum 410; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 50
50(1,1) + 5(4,4) + 12(4,1) + 1(1,4) + 6(1,5) + 1/2(1,16) + 7/2(6,4) + 1(6,1) + 3(10,1)
Spectrum 411; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 50
50(1,1) + 6(4,4) + 3(4,1) + 1(4,5) + 6(1,4) + 2(1,5) + 1/2(1,16) + 2(6,4) + 7(6,1) + 3(10,1)
Spectrum 412; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 46
46(1,1) + 5(4,4) + 7(4,1) + 1(4,5) + 7(1,4) + 2(1,5) + 1/2(1,16) + 5/2(6,4) + 5(6,1) + 3(10,1)
Spectrum 413; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 42
42(1,1) + 4(4,4) + 11(4,1) + 1(4,5) + 8(1,4) + 2(1,5) + 1/2(1,16) + 3(6,4) + 3(6,1) + 3(10,1)
Spectrum 414; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 38
38(1,1) + 3(4,4) + 15(4,1) + 1(4,5) + 9(1,4) + 2(1,5) + 1/2(1,16) + 7/2(6,4) + 1(6,1) + 3(10,1)
Spectrum 415; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 44
44(1,1) + 8(4,4) + 10(1,4) + 1/2(1,16) + 10(6,1) + 1(6,5) + 3(10,1)
Spectrum 416; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 40
40(1,1) + 7(4,4) + 4(4,1) + 11(1,4) + 1/2(1,16) + 1/2(6,4) + 8(6,1) + 1(6,5) + 3(10,1)
Spectrum 417; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 36
36(1,1) + 6(4,4) + 8(4,1) + 12(1,4) + 1/2(1,16) + 1(6,4) + 6(6,1) + 1(6,5) + 3(10,1)
Spectrum 418; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 32
32(1,1) + 5(4,4) + 12(4,1) + 13(1,4) + 1/2(1,16) + 3/2(6,4) + 4(6,1) + 1(6,5) + 3(10,1)
Spectrum 419; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 28
28(1,1) + 4(4,4) + 16(4,1) + 14(1,4) + 1/2(1,16) + 2(6,4) + 2(6,1) + 1(6,5) + 3(10,1)
Spectrum 420; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 24
24(1,1) + 3(4,4) + 20(4,1) + 15(1,4) + 1/2(1,16) + 5/2(6,4) + 1(6,5) + 3(10,1)
Spectrum 421; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 57
57(1,1) + 6(4,4) + 3(4,1) + 1(4,5) + 1(1,10) + 5(1,5) + 2(6,4) + 7(6,1) + 3(10,1)
Spectrum 422; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 53
53(1,1) + 5(4,4) + 7(4,1) + 1(4,5) + 1(1,10) + 1(1,4) + 5(1,5) + 5/2(6,4) + 5(6,1) + 3(10,1)
Spectrum 423; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 49
49(1,1) + 4(4,4) + 11(4,1) + 1(4,5) + 1(1,10) + 2(1,4) + 5(1,5) + 3(6,4) + 3(6,1) + 3(10,1)
Spectrum 424; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 45
45(1,1) + 3(4,4) + 15(4,1) + 1(4,5) + 1(1,10) + 3(1,4) + 5(1,5) + 7/2(6,4) + 1(6,1) + 3(10,1)
Spectrum 425; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 49
49(1,1) + 5(4,4) + 2(4,1) + 2(4,5) + 1(1,10) + 7(1,4) + 1(1,5) + 3/2(6,4) + 9(6,1) + 3(10,1)
Spectrum 426; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 45
45(1,1) + 4(4,4) + 6(4,1) + 2(4,5) + 1(1,10) + 8(1,4) + 1(1,5) + 2(6,4) + 7(6,1) + 3(10,1)
Spectrum 427; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 41
41(1,1) + 3(4,4) + 10(4,1) + 2(4,5) + 1(1,10) + 9(1,4) + 1(1,5) + 5/2(6,4) + 5(6,1) + 3(10,1)
Spectrum 428; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 37
37(1,1) + 2(4,4) + 14(4,1) + 2(4,5) + 1(1,10) + 10(1,4) + 1(1,5) + 3(6,4) + 3(6,1) + 3(10,1)
Spectrum 429; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 33
33(1,1) + 1(4,4) + 18(4,1) + 2(4,5) + 1(1,10) + 11(1,4) + 1(1,5) + 7/2(6,4) + 1(6,1) + 3(10,1)
Spectrum 430; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 51
51(1,1) + 8(4,4) + 1(1,10) + 4(1,4) + 3(1,5) + 10(6,1) + 1(6,5) + 3(10,1)
Spectrum 431; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 47
47(1,1) + 7(4,4) + 4(4,1) + 1(1,10) + 5(1,4) + 3(1,5) + 1/2(6,4) + 8(6,1) + 1(6,5) + 3(10,1)
Spectrum 432; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 43
43(1,1) + 6(4,4) + 8(4,1) + 1(1,10) + 6(1,4) + 3(1,5) + 1(6,4) + 6(6,1) + 1(6,5) + 3(10,1)
Spectrum 433; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 39
39(1,1) + 5(4,4) + 12(4,1) + 1(1,10) + 7(1,4) + 3(1,5) + 3/2(6,4) + 4(6,1) + 1(6,5) + 3(10,1)
Spectrum 434; $b1=(2,4)$, $b2=(3/2,3)$; singlets = 35
35(1,1) + 4(4,4) + 16(4,1) + 1(1,10) + 8(1,4) + 3(1,5) + 2(6,4) + 2(6,1) + 1(6,5) + 3(10,1)

Spectrum 435: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 51
 $31(1,1) + 3(4,4) + 20(4,1) + 1(1,10) + 9(1,4) + 3(1,5) + 5/2(6,4) + 1(6,5) + 3(10,1)$
Spectrum 436: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 55
 $55(1,1) + 1(15,1) + 6(4,4) + 8(4,1) + 6(1,5) + 1/2(1,16) + 3(6,4) + 2(6,1) + 2(10,1)$
Spectrum 437: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 51
 $51(1,1) + 1(15,1) + 5(4,4) + 12(4,1) + 1(1,4) + 6(1,5) + 1/2(1,16) + 7/2(6,4) + 2(10,1)$
Spectrum 438: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 51
 $51(1,1) + 1(15,1) + 6(4,4) + 3(4,1) + 1(4,5) + 6(1,4) + 2(1,5) + 1/2(1,16) + 2(6,4) + 6(6,1) + 2(10,1)$
Spectrum 439: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 47
 $47(1,1) + 1(15,1) + 5(4,4) + 7(4,1) + 1(4,5) + 7(1,4) + 2(1,5) + 1/2(1,16) + 5/2(6,4) + 4(6,1) + 2(10,1)$
Spectrum 440: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 43
 $43(1,1) + 1(15,1) + 4(4,4) + 11(4,1) + 1(4,5) + 8(1,4) + 2(1,5) + 1/2(1,16) + 3(6,4) + 2(6,1) + 2(10,1)$
Spectrum 441: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 39
 $39(1,1) + 1(15,1) + 3(4,4) + 15(4,1) + 1(4,5) + 9(1,4) + 2(1,5) + 1/2(1,16) + 7/2(6,4) + 2(10,1)$
Spectrum 442: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 45
 $45(1,1) + 1(15,1) + 8(4,4) + 10(1,4) + 1/2(1,16) + 9(6,1) + 1(6,5) + 2(10,1)$
Spectrum 443: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 41
 $41(1,1) + 1(15,1) + 7(4,4) + 4(4,1) + 11(1,4) + 1/2(1,16) + 1/2(6,4) + 7(6,1) + 1(6,5) + 2(10,1)$
Spectrum 444: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 37
 $37(1,1) + 1(15,1) + 6(4,4) + 8(4,1) + 12(1,4) + 1/2(1,16) + 1(6,4) + 5(6,1) + 1(6,5) + 2(10,1)$
Spectrum 445: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 33
 $33(1,1) + 1(15,1) + 5(4,4) + 12(4,1) + 13(1,4) + 1/2(1,16) + 3/2(6,4) + 3(6,1) + 1(6,5) + 2(10,1)$
Spectrum 446: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 29
 $29(1,1) + 1(15,1) + 4(4,4) + 16(4,1) + 14(1,4) + 1/2(1,16) + 2(6,4) + 1(6,1) + 1(6,5) + 2(10,1)$
Spectrum 447: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 58
 $58(1,1) + 1(15,1) + 6(4,4) + 3(4,1) + 1(4,5) + 1(1,10) + 5(1,5) + 2(6,4) + 6(6,1) + 2(10,1)$
Spectrum 448: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 54
 $54(1,1) + 1(15,1) + 5(4,4) + 7(4,1) + 1(4,5) + 1(1,10) + 1(1,4) + 5(1,5) + 5/2(6,4) + 4(6,1) + 2(10,1)$
Spectrum 449: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 50
 $50(1,1) + 1(15,1) + 4(4,4) + 11(4,1) + 1(4,5) + 1(1,10) + 2(1,4) + 5(1,5) + 3(6,4) + 2(6,1) + 2(10,1)$
Spectrum 450: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 46
 $46(1,1) + 1(15,1) + 3(4,4) + 15(4,1) + 1(4,5) + 1(1,10) + 3(1,4) + 5(1,5) + 7/2(6,4) + 2(10,1)$
Spectrum 451: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 50
 $50(1,1) + 1(15,1) + 5(4,4) + 2(4,1) + 2(4,5) + 1(1,10) + 7(1,4) + 1(1,5) + 3/2(6,4) + 8(6,1) + 2(10,1)$
Spectrum 452: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 46
 $46(1,1) + 1(15,1) + 4(4,4) + 6(4,1) + 2(4,5) + 1(1,10) + 8(1,4) + 1(1,5) + 2(6,4) + 6(6,1) + 2(10,1)$
Spectrum 453: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 42
 $42(1,1) + 1(15,1) + 3(4,4) + 10(4,1) + 2(4,5) + 1(1,10) + 9(1,4) + 1(1,5) + 5/2(6,4) + 4(6,1) + 2(10,1)$
Spectrum 454: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 38
 $38(1,1) + 1(15,1) + 2(4,4) + 14(4,1) + 2(4,5) + 1(1,10) + 10(1,4) + 1(1,5) + 3(6,4) + 2(6,1) + 2(10,1)$
Spectrum 455: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 34
 $34(1,1) + 1(15,1) + 1(4,4) + 18(4,1) + 2(4,5) + 1(1,10) + 11(1,4) + 1(1,5) + 7/2(6,4) + 2(10,1)$
Spectrum 456: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 52
 $52(1,1) + 1(15,1) + 8(4,4) + 1(1,10) + 4(1,4) + 3(1,5) + 9(6,1) + 1(6,5) + 2(10,1)$
Spectrum 457: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 48
 $48(1,1) + 1(15,1) + 7(4,4) + 4(4,1) + 1(1,10) + 5(1,4) + 3(1,5) + 1/2(6,4) + 7(6,1) + 1(6,5) + 2(10,1)$
Spectrum 458: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 44
 $44(1,1) + 1(15,1) + 6(4,4) + 8(4,1) + 1(1,10) + 6(1,4) + 3(1,5) + 1(6,4) + 5(6,1) + 1(6,5) + 2(10,1)$
Spectrum 459: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 40
 $40(1,1) + 1(15,1) + 5(4,4) + 12(4,1) + 1(1,10) + 7(1,4) + 3(1,5) + 3/2(6,4) + 3(6,1) + 1(6,5) + 2(10,1)$
Spectrum 460: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 36
 $36(1,1) + 1(15,1) + 4(4,4) + 16(4,1) + 1(1,10) + 8(1,4) + 3(1,5) + 2(6,4) + 1(6,1) + 1(6,5) + 2(10,1)$
Spectrum 461: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 56
 $56(1,1) + 2(15,1) + 6(4,4) + 8(4,1) + 6(1,5) + 1/2(1,16) + 3(6,4) + 1(6,1) + 1(10,1)$
Spectrum 462: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 52
 $52(1,1) + 2(15,1) + 6(4,4) + 3(4,1) + 1(4,5) + 6(1,4) + 2(1,5) + 1/2(1,16) + 2(6,4) + 5(6,1) + 1(10,1)$
Spectrum 463: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 48
 $48(1,1) + 2(15,1) + 5(4,4) + 7(4,1) + 1(4,5) + 7(1,4) + 2(1,5) + 1/2(1,16) + 5/2(6,4) + 3(6,1) + 1(10,1)$
Spectrum 464: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 44
 $44(1,1) + 2(15,1) + 4(4,4) + 11(4,1) + 1(4,5) + 8(1,4) + 2(1,5) + 1/2(1,16) + 3(6,4) + 1(6,1$

Spectrum 503: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 41
 $41(1,1) + 5/2(15,4) + 6(4,4) + 3(4,1) + 1(4,5) + 1(1,10) + 9/2(1,4) + 5(1,5) + 13(6,1) + 1(10,1)$
Spectrum 504: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 37
 $37(1,1) + 1/2(15,4) + 5(4,4) + 7(4,1) + 1(4,5) + 1(1,10) + 11/2(1,4) + 5(1,5) + 1/2(6,4) + 11(6,1) + 1(10,1)$
Spectrum 505: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 33
 $33(1,1) + 1/2(15,4) + 4(4,4) + 11(4,1) + 1(4,5) + 1(1,10) + 13/2(1,4) + 5(1,5) + 1(6,4) + 9(6,1) + 1(10,1)$
Spectrum 506: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 29
 $29(1,1) + 1/2(15,4) + 3(4,4) + 15(4,1) + 1(4,5) + 1(1,10) + 15/2(1,4) + 5(1,5) + 3/2(6,4) + 7(6,1) + 1(10,1)$
Spectrum 507: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 25
 $25(1,1) + 1/2(15,4) + 2(4,4) + 19(4,1) + 1(4,5) + 1(1,10) + 17/2(1,4) + 5(1,5) + 2(6,4) + 5(6,1) + 1(10,1)$
Spectrum 508: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 21
 $21(1,1) + 1/2(15,4) + 1(4,4) + 23(4,1) + 1(4,5) + 1(1,10) + 19/2(1,4) + 5(1,5) + 5/2(6,4) + 3(6,1) + 1(10,1)$
Spectrum 509: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 17
 $17(1,1) + 1/2(15,4) + 27(4,1) + 1(4,5) + 1(1,10) + 21/2(1,4) + 5(1,5) + 3(6,4) + 1(6,1) + 1(10,1)$
Spectrum 510: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 29
 $29(1,1) + 1/2(15,4) + 4(4,4) + 6(4,1) + 2(4,5) + 1(1,10) + 25/2(1,4) + 1(1,5) + 13(6,1) + 1(10,1)$
Spectrum 511: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 25
 $25(1,1) + 1/2(15,4) + 3(4,4) + 10(4,1) + 2(4,5) + 1(1,10) + 27/2(1,4) + 1(1,5) + 1/2(6,4) + 11(6,1) + 1(10,1)$
Spectrum 512: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 21
 $21(1,1) + 1/2(15,4) + 2(4,4) + 14(4,1) + 2(4,5) + 1(1,10) + 29/2(1,4) + 1(1,5) + 1(6,4) + 9(6,1) + 1(10,1)$
Spectrum 513: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 17
 $17(1,1) + 1/2(15,4) + 1(4,4) + 18(4,1) + 2(4,5) + 1(1,10) + 31/2(1,4) + 1(1,5) + 3/2(6,4) + 7(6,1) + 1(10,1)$
Spectrum 514: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 13
 $13(1,1) + 1/2(15,4) + 22(4,1) + 2(4,5) + 1(1,10) + 33/2(1,4) + 1(1,5) + 2(6,4) + 5(6,1) + 1(10,1)$
Spectrum 515: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 19
 $19(1,1) + 1/2(15,4) + 4(4,4) + 16(4,1) + 1(1,10) + 25/2(1,4) + 3(1,5) + 8(6,1) + 1(6,5) + 1(10,1)$
Spectrum 516: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 15
 $15(1,1) + 1/2(15,4) + 3(4,4) + 20(4,1) + 1(1,10) + 27/2(1,4) + 3(1,5) + 1/2(6,4) + 6(6,1) + 1(6,5) + 1(10,1)$
Spectrum 517: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 11
 $11(1,1) + 1/2(15,4) + 2(4,4) + 24(4,1) + 1(1,10) + 29/2(1,4) + 3(1,5) + 1(6,4) + 4(6,1) + 1(6,5) + 1(10,1)$
Spectrum 518: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 7
 $7(1,1) + 1/2(15,4) + 1(4,4) + 28(4,1) + 1(1,10) + 31/2(1,4) + 3(1,5) + 3/2(6,4) + 2(6,1) + 1(6,5) + 1(10,1)$
Spectrum 519: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 3
 $3(1,1) + 1/2(15,4) + 32(4,1) + 1(1,10) + 33/2(1,4) + 3(1,5) + 2(6,4) + 1(6,5) + 1(10,1)$
Spectrum 520: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 57
 $57(1,1) + 3(15,1) + 6(4,4) + 8(4,1) + 6(1,5) + 1/2(1,16) + 3(6,4)$
Spectrum 521: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 53
 $53(1,1) + 3(15,1) + 6(4,4) + 3(4,1) + 1(4,5) + 6(1,4) + 2(1,5) + 1/2(1,16) + 2(6,4) + 4(6,1)$
Spectrum 522: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 49
 $49(1,1) + 3(15,1) + 5(4,4) + 7(4,1) + 1(4,5) + 7(1,4) + 2(1,5) + 1/2(1,16) + 5/2(6,4) + 2(6,1)$
Spectrum 523: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 45
 $45(1,1) + 3(15,1) + 4(4,4) + 11(4,1) + 1(4,5) + 8(1,4) + 2(1,5) + 1/2(1,16) + 3(6,4)$
Spectrum 524: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 47
 $47(1,1) + 3(15,1) + 8(4,4) + 10(1,4) + 1/2(1,16) + 7(6,1) + 1(6,5)$
Spectrum 525: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 43
 $43(1,1) + 3(15,1) + 7(4,4) + 4(4,1) + 11(1,4) + 1/2(1,16) + 1/2(6,4) + 5(6,1) + 1(6,5)$
Spectrum 526: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 39
 $39(1,1) + 3(15,1) + 6(4,4) + 8(4,1) + 12(1,4) + 1/2(1,16) + 1(6,4) + 3(6,1) + 1(6,5)$
Spectrum 527: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 35
 $35(1,1) + 3(15,1) + 5(4,4) + 12(4,1) + 13(1,4) + 1/2(1,16) + 3/2(6,4) + 1(6,1) + 1(6,5)$
Spectrum 528: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 47
 $47(1,1) + 1/2(15,4) + 1(15,1) + 8(4,4) + 5/2(1,4) + 6(1,5) + 1/2(1,16) + 12(6,1)$
Spectrum 529: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 43
 $43(1,1) + 1/2(15,4) + 1(15,1) + 7(4,4) + 4(4,1) + 7/2(1,4) + 6(1,5) + 1/2(1,16) + 1/2(6,4) + 10(6,1)$
Spectrum 530: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 39
 $39(1,1) + 1/2(15,4) + 1(15,1) + 6(4,4) + 8(4,1) + 9/2(1,4) + 6(1,5) + 1/2(1,16) + 1(6,4) + 8(6,1)$
Spectrum 531: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 35
 $35(1,1) + 1/2(15,4) + 1(15,1) + 5(4,4) + 12(4,1) + 11/2(1,4) + 6(1,5) + 1/2(1,16) + 3/2(6,4) + 6(6,1)$
Spectrum 532: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 31
 $31(1,1) + 1/2(15,4) + 1(15,1) + 4(4,4) + 16(4,1) + 13/2(1,4) + 6(1,5) + 1/2(1,16)$

Spectrum 537: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 27
 $27(1,1) + 1/2(15,4) + 1(15,1) + 4(4,4) + 11(4,1) + 1(4,5) + 25/2(1,4) + 2(1,5) + 1/2(1,16) + 1(6,4) + 8(6,1)$
Spectrum 538: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 23
 $23(1,1) + 1/2(15,4) + 1(15,1) + 3(4,4) + 15(4,1) + 1(4,5) + 27/2(1,4) + 2(1,5) + 1/2(1,16) + 3/2(6,4) + 6(6,1)$
Spectrum 539: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 19
 $19(1,1) + 1/2(15,4) + 1(15,1) + 2(4,4) + 19(4,1) + 1(4,5) + 29/2(1,4) + 2(1,5) + 1/2(1,16) + 2(6,4) + 4(6,1)$
Spectrum 540: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 15
 $15(1,1) + 1/2(15,4) + 1(15,1) + 1(4,4) + 23(4,1) + 1(4,5) + 31/2(1,4) + 2(1,5) + 1/2(1,16) + 5/2(6,4) + 2(6,1)$
Spectrum 541: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 11
 $11(1,1) + 1/2(15,4) + 1(15,1) + 27(4,1) + 1(4,5) + 33/2(1,4) + 2(1,5) + 1/2(1,16) + 3(6,4)$
Spectrum 542: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 13
 $13(1,1) + 1/2(15,4) + 1(15,1) + 4(4,4) + 16(4,1) + 37/2(1,4) + 1/2(1,16) + 7(6,1) + 1(6,5)$
Spectrum 543: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 9
 $9(1,1) + 1/2(15,4) + 1(15,1) + 3(4,4) + 20(4,1) + 39/2(1,4) + 1/2(1,16) + 1/2(6,4) + 5(6,1) + 1(6,5)$
Spectrum 544: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 5
 $5(1,1) + 1/2(15,4) + 1(15,1) + 2(4,4) + 24(4,1) + 41/2(1,4) + 1/2(1,16) + 1(6,4) + 3(6,1) + 1(6,5)$
Spectrum 545: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 1
 $1(1,1) + 1/2(15,4) + 1(15,1) + 1(4,4) + 28(4,1) + 43/2(1,4) + 1/2(1,16) + 3/2(6,4) + 1(6,1) + 1(6,5)$
Spectrum 546: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 60
 $60(1,1) + 3(15,1) + 6(4,4) + 3(4,1) + 1(4,5) + 1(1,10) + 5(1,5) + 2(6,4) + 4(6,1)$
Spectrum 547: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 56
 $56(1,1) + 3(15,1) + 5(4,4) + 7(4,1) + 1(4,5) + 1(1,10) + 1(1,4) + 5(1,5) + 5/2(6,4) + 2(6,1)$
Spectrum 548: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 52
 $52(1,1) + 3(15,1) + 4(4,4) + 11(4,1) + 1(4,5) + 1(1,10) + 2(1,4) + 5(1,5) + 3(6,4)$
Spectrum 549: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 52
 $52(1,1) + 3(15,1) + 5(4,4) + 2(4,1) + 2(4,5) + 1(1,10) + 7(1,4) + 1(1,5) + 3/2(6,4) + 6(6,1)$
Spectrum 550: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 48
 $48(1,1) + 3(15,1) + 4(4,4) + 6(4,1) + 2(4,5) + 1(1,10) + 8(1,4) + 1(1,5) + 2(6,4) + 4(6,1)$
Spectrum 551: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 44
 $44(1,1) + 3(15,1) + 3(4,4) + 10(4,1) + 2(4,5) + 1(1,10) + 9(1,4) + 1(1,5) + 5/2(6,4) + 2(6,1)$
Spectrum 552: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 40
 $40(1,1) + 3(15,1) + 2(4,4) + 14(4,1) + 2(4,5) + 1(1,10) + 10(1,4) + 1(1,5) + 3(6,4)$
Spectrum 553: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 54
 $54(1,1) + 3(15,1) + 8(4,4) + 1(1,10) + 4(1,4) + 3(1,5) + 7(6,1) + 1(6,5)$
Spectrum 554: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 50
 $50(1,1) + 3(15,1) + 7(4,4) + 4(4,1) + 1(1,10) + 5(1,4) + 3(1,5) + 1/2(6,4) + 5(6,1) + 1(6,5)$
Spectrum 555: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 46
 $46(1,1) + 3(15,1) + 6(4,4) + 8(4,1) + 1(1,10) + 6(1,4) + 3(1,5) + 1(6,4) + 3(6,1) + 1(6,5)$
Spectrum 556: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 42
 $42(1,1) + 3(15,1) + 5(4,4) + 12(4,1) + 1(1,10) + 7(1,4) + 3(1,5) + 3/2(6,4) + 1(6,1) + 1(6,5)$
Spectrum 557: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 38
 $38(1,1) + 1/2(15,4) + 1(15,1) + 4(4,4) + 16(4,1) + 1(1,10) + 1/2(1,4) + 9(1,5) + 2(6,4) + 4(6,1)$
Spectrum 558: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 34
 $34(1,1) + 1/2(15,4) + 1(15,1) + 3(4,4) + 20(4,1) + 1(1,10) + 3/2(1,4) + 9(1,5) + 5/2(6,4) + 2(6,1)$
Spectrum 559: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 30
 $30(1,1) + 1/2(15,4) + 1(15,1) + 2(4,4) + 24(4,1) + 1(1,10) + 5/2(1,4) + 9(1,5) + 3(6,4)$
Spectrum 560: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 42
 $42(1,1) + 1/2(15,4) + 1(15,1) + 6(4,4) + 3(4,1) + 1(4,5) + 1(1,10) + 9/2(1,4) + 5(1,5) + 12(6,1)$
Spectrum 561: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 38
 $38(1,1) + 1/2(15,4) + 1(15,1) + 5(4,4) + 7(4,1) + 1(4,5) + 1(1,10) + 11/2(1,4) + 5(1,5) + 1/2(6,4) + 10(6,1)$
Spectrum 562: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 34
 $34(1,1) + 1/2(15,4) + 1(15,1) + 4(4,4) + 11(4,1) + 1(4,5) + 1(1,10) + 13/2(1,4) + 5(1,5) + 1(6,4) + 8(6,1)$
Spectrum 563: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 30
 $30(1,1) + 1/2(15,4) + 1(15,1) + 3(4,4) + 15(4,1) + 1(4,5) + 1(1,10) + 15/2(1,4) + 5(1,5) + 3/2(6,4) + 6(6,1)$
Spectrum 564: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 26
 $26(1,1) + 1/2(15,4) + 1(15,1) + 2(4,4) + 19(4,1) + 1(4,5) + 1(1,10) + 17/2(1,4) + 5(1,5) + 2(6,4) + 4(6,1)$
Spectrum 565: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 22
 $22(1,1) + 1/2(15,4) + 1(15,1) + 1(4,4) + 23(4,1) + 1(4,5) + 1(1,10) + 19/2(1,4) + 5(1,5) + 5/2(6,4) + 2(6,1)$
Spectrum 566: $b_1=(2,4)$, $b_2=(3/2,3)$; singlets = 18
 $18(1,1) + 1/2(15,4) + 1(15,1) + 27(4,1) + 1(4,5) + 1(1,10) + 21/2(1,4) + 5(1,5) + 3(6,$

Spectrum 571; b1=(2,4), b2=(3/2,3); singlets = 14

14(1,1) + 1/2(15,4) + 1(15,1) + 22(4,1) + 2(4,5) + 1(1,10) + 33/2(1,4) + 1(1,5) + 2(6,4) + 4(6,1)

Spectrum 572; b1=(2,4), b2=(3/2,3); singlets = 20

20(1,1) + 1/2(15,4) + 1(15,1) + 4(4,4) + 16(4,1) + 1(1,10) + 25/2(1,4) + 3(1,5) + 7(6,1) + 1(6,5)

Spectrum 573; b1=(2,4), b2=(3/2,3); singlets = 16

16(1,1) + 1/2(15,4) + 1(15,1) + 3(4,4) + 20(4,1) + 1(1,10) + 27/2(1,4) + 3(1,5) + 1/2(6,4) + 5(6,1) + 1(6,5)

Spectrum 574; b1=(2,4), b2=(3/2,3); singlets = 12

12(1,1) + 1/2(15,4) + 1(15,1) + 2(4,4) + 24(4,1) + 1(1,10) + 29/2(1,4) + 3(1,5) + 1(6,4) + 3(6,1) + 1(6,5)

Spectrum 575; b1=(2,4), b2=(3/2,3); singlets = 8

8(1,1) + 1/2(15,4) + 1(15,1) + 1(4,4) + 28(4,1) + 1(1,10) + 31/2(1,4) + 3(1,5) + 3/2(6,4) + 1(6,1) + 1(6,5)

A3 x C3

Spectrum 576; b1=(2,4), b2=(2,4); singlets = 29

29(1,1) + 3(4,6) + 1(4,14) + 3(1,21) + 1(6,1) + 1(6,14') + 3(10,1)

Spectrum 577; b1=(2,4), b2=(2,4); singlets = 29

29(1,1) + 4(4,1) + 2(4,14') + 3(1,21) + 1(6,1) + 1(6,14') + 3(10,1)

Spectrum 578; b1=(2,4), b2=(2,4); singlets = 30

30(1,1) + 1(15,1) + 3(4,6) + 1(4,14) + 3(1,21) + 1(6,14') + 2(10,1)

Spectrum 579; b1=(2,4), b2=(2,4); singlets = 30

30(1,1) + 1(15,1) + 4(4,1) + 2(4,14') + 3(1,21) + 1(6,14') + 2(10,1)

Spectrum 580; b1=(2,4), b2=(2,4); singlets = 9

9(1,1) + 1/2(15,6) + 3(4,6) + 1(4,14') + 1(1,21) + 11/2(1,6) + 1/2(1,64) + 5(6,1) + 1/2(6,14)

Spectrum 581; b1=(2,4), b2=(2,4); singlets = 3

3(1,1) + 1/2(15,6) + 2(4,6) + 6(4,1) + 1(4,14') + 1(1,21) + 13/2(1,6) + 1/2(1,64) + 1/2(6,6) + 2(6,1) + 1/2(6,14)

A3 x C4

Spectrum 582; b1=(2,4), b2=(1,0); singlets = 37

37(1,1) + 4(4,8) + 32(4,1) + 1(1,27) + 1(20,1)

Spectrum 583; b1=(2,4), b2=(1,0); singlets = 57

57(1,1) + 4(4,8) + 11(4,1) + 1(1,27) + 9(6,1) + 1(10,1) + 1(20',1)

Spectrum 584; b1=(2,4), b2=(1,0); singlets = 49

49(1,1) + 3(4,8) + 19(4,1) + 1(1,8) + 1(1,27) + 1/2(6,8) + 5(6,1) + 1(10,1) + 1(20',1)

Spectrum 585; b1=(2,4), b2=(1,0); singlets = 41

41(1,1) + 2(4,8) + 27(4,1) + 2(1,8) + 1(1,27) + 1(6,8) + 1(6,1) + 1(10,1) + 1(20',1)

Spectrum 586; b1=(2,4), b2=(1,0); singlets = 58

58(1,1) + 1(15,1) + 4(4,8) + 11(4,1) + 1(1,27) + 8(6,1) + 1(20',1)

Spectrum 587; b1=(2,4), b2=(1,0); singlets = 50

50(1,1) + 1(15,1) + 3(4,8) + 19(4,1) + 1(1,8) + 1(1,27) + 1/2(6,8) + 4(6,1) + 1(20',1)

Spectrum 588; b1=(2,4), b2=(1,0); singlets = 42

42(1,1) + 1(15,1) + 2(4,8) + 27(4,1) + 2(1,8) + 1(1,27) + 1(6,8) + 1(20',1)

Spectrum 589; b1=(2,4), b2=(1,0); singlets = 65

65(1,1) + 4(4,8) + 1(1,27) + 15(6,1) + 3(10,1)

Spectrum 590; b1=(2,4), b2=(1,0); singlets = 57

57(1,1) + 3(4,8) + 8(4,1) + 1(1,8) + 1(1,27) + 1/2(6,8) + 11(6,1) + 3(10,1)

Spectrum 591; b1=(2,4), b2=(1,0); singlets = 49

49(1,1) + 2(4,8) + 16(4,1) + 2(1,8) + 1(1,27) + 1(6,8) + 7(6,1) + 3(10,1)

Spectrum 592; b1=(2,4), b2=(1,0); singlets = 41

41(1,1) + 1(4,8) + 24(4,1) + 3(1,8) + 1(1,27) + 3/2(6,8) + 3(6,1) + 3(10,1)

Spectrum 593; b1=(2,4), b2=(1,0); singlets = 66

66(1,1) + 1(15,1) + 4(4,8) + 1(1,27) + 14(6,1) + 2(10,1)

Spectrum 594; b1=(2,4), b2=(1,0); singlets = 58

58(1,1) + 1(15,1) + 3(4,8) + 8(4,1) + 1(1,8) + 1(1,27) + 1/2(6,8) + 10(6,1) + 2(10,1)

Spectrum 595; b1=(2,4), b2=(1,0); singlets = 50

50(1,1) + 1(15,1) + 2(4,8) + 16(4,1) + 2(1,8) + 1(1,27) + 1(6,8) + 6(6,1) + 2(10,1)

Spectrum 596; b1=(2,4), b2=(1,0); singlets = 42

42(1,1) + 1(15,1) + 1(4,8) + 24(4,1) + 3(1,8) + 1(1,27) + 3/2(6,8) + 2(6,1) + 2(10,1)

Spectrum 597; b1=(2,4), b2=(1,0); singlets = 67

67(1,1) + 2(15,1) + 4(4,8) + 1(1,27) + 13(6,1) + 1(10,1)

Spectrum 598; b1=(2,4), b2=(1,0); singlets = 59

59(1,1) + 2(15,1) + 3(4,8) + 8(4,1) + 1(1,8) + 1(1,27) + 1/2(6,8) + 9(6,1) + 1(10,1)

Spectrum 599; b1=(2,4), b2=(1,0); singlets = 51

51(1,1) + 2(15,1) + 2(4,8) + 16(4,1) + 2(1,8) + 1(1,27) + 1(6,8) + 5(6,1) + 1(10,1)

Spectrum 600; b1=(2,4), b2=(1,0); singlets = 43

43(1,1) + 2(15,1) + 1(4,8) + 24(4,1) + 3(1,8) + 1(1,27) + 3/2(6,8) + 1(6,1) + 1(10,1)

Spectrum 601; b1=(2,4), b2=(1,0); singlets = 68

68(1,1) + 3(15,1) + 4(4,8) + 1(1,27) + 12(6,1)

Spectrum 602; b1=(2,4), b2=(1,0); singlets = 60

60(1,1) + 3(15,1) + 3(4,8) + 8(4,1) + 1(1,8) + 1(1,27) + 1/2(6,8) + 8(6,1)

Spectrum 603; b1=(2,4), b2=(1,0); singlets = 52

52(1,1) + 3(15,1) + 2(4,8) + 16(4,1) + 2(1,8) + 1(1,27) + 1(6,8) + 4(6,1)

Spectrum 604; b1=(2,4), b2=(1,0); singlets = 44

44(1,1) + 3(15,1) + 1(4,8) + 24(4,1) + 3(1,8) + 1(1,27) + 3/2(6,8)

A3 x D4

Spectrum 605; b1=(2,4), b2=(3,6); singlets = 8

8(1,1) + 2(4,28) + 8(4,1) + 2(1,28) + 1(20,1)

Spectrum 606; b1=(2,4), b2=(3/2,1); singlets = 28

28(1,1) + 1(4,8) + 27(4,1) + 1(4,8) + 3(1,8) + 1(1,8_ϕ) + 1(1,8_ϕ) + 1(6,1) + 1(6,8_ϕ) + 1(10,1) + 1(20',1)

Spectrum 607; b1=(2,4), b2=(3/2,1); singlets = 28

28(1,1) + 1(4,8) + 27(4,1) + 1(4,8) + 3(1,8) + 1(1,8_s) + 3(1,8_ϕ) + 1(6,1) + 1(6,8_s) + 1(10,1) + 1(20',1)

Spectrum 608; b1=(2,4), b2=(3/2,1); singlets = 28

28(1,1) + 27(4,1) + 1(4,8) + 1(4,8) + 1(1,8) + 1(1,8_ϕ) + 3(1,8_s) + 3(1,8_ϕ) + 1(6,8_ϕ) + 1(6,1) + 1(10,1) + 1(20',1)

Spectrum 609; b1=(2,4), b2=(3/2,1); singlets = 29

29(1,1) + 1(15,1) + 1(4,8) + 27(4,1) + 1(4,8) + 3(1,8_ϕ) + 3(1,8_s) + 1(1,8_ϕ) + 1(6,8_ϕ) + 1(20',1)

Spectrum 610; b1=(2,4), b2=(3/2,1); singlets = 29

29(1,1) + 1(15,1) + 1(4,8) + 27(4,1) + 1(4,8) + 3(1,8_ϕ) + 1(1,8_s) + 3(1,8_ϕ) + 1(6,8_s) + 1(20',1)

Spectrum 611; b1=(2,4), b2=(3/2,1); singlets = 29

29(1,1) + 1(15,1) + 27(4,1) + 1(4,8) + 1(4,8) + 1(1,8_ϕ) + 3(1,8_s) + 3(1,8_ϕ) + 1(6,8_ϕ) + 1(20',1)

Spectrum 612; b1=(2,4), b2=(3/2,1); singlets = 36

36(1,1) + 1(4,8) + 16(4,1) + 1(4,8) + 3(1,8) + 3(1,8_s) + 1(1,8_ϕ) + 7(6,1) + 1(6,8_ϕ) + 3(10,1)

Spectrum 613; b1=(2,4), b2=(3/2,1); singlets = 36

36(1,1) + 1(4,8) + 16(4,1) + 1(4,8) + 3(1,8) + 1(1,8_s) + 3(1,8_ϕ) + 7(6,1) + 1(6,8_s) + 3(10,1)

Spectrum 614; b1=(2,4), b2=(3/2,1); singlets = 36

36(1,1) + 16(4,1) + 1(4,8) + 1(4,8) + 1(4,8) + 1(1,8) + 3(1,8_s) + 3(1,8_ϕ) + 1(6,8_ϕ) + 7(6,1) + 3(10,1)

Spectrum 615; b1=(2,4), b2=(3/2,1); singlets = 37

37(1,1) + 1(15,1) + 1(4,8) + 16(4,1) + 1(4,8) + 3(1,8_ϕ) + 3(1,8_s) + 1(1,8_ϕ) + 6(6,1) + 1(6,8_ϕ) + 2(10,1)

Spectrum 616; b1=(2,4), b2=(3/2,1); singlets = 37

37(1,1) + 1(15,1) + 1(4,8) + 16(4,1) + 1(4,8) + 3(1,8_ϕ) + 1(1,8_s) + 3(1,8_ϕ) + 6(6,1) + 1(6,8_s) + 2(10,1)

Spectrum 617; b1=(2,4), b2=(3/2,1); singlets = 37

37(1,1) + 1(15,1) + 16(4,1) + 1(4,8) + 1(4,8) + 1(1,8_ϕ) + 3(1,8_s) + 3(1,8_ϕ) + 1(6,8_ϕ) + 6(6,1) + 2(10,1)

Spectrum 618; b1=(2,4), b2=(3/2,1); singlets = 38

38(1,1) + 2(15,1) + 1(4,8) + 16(4,1) + 1(4,8) + 3(1,8_ϕ) + 3(1,8_s) + 1(1,8_ϕ) + 5(6,1) + 1(6,8_ϕ) + 1(10,1)

Spectrum 619; b1=(2,4), b2=(3/2,1); singlets = 38

38(1,1) + 2(15,1) + 1(4,8) + 16(4,1) + 1(4,8) + 3(1,8_ϕ) + 1(1,8_s) + 3(1,8_ϕ) + 5(6,1) + 1(6,8_s) + 1(10,1)

Spectrum 620; b1=(2,4), b2=(3/2,1); singlets = 38

38(1,1) + 2(15,1) + 16(4,1) + 1(4,8) + 1(4,8) + 1(1,8_ϕ) + 3(1,8_s) + 3(1,8_ϕ) + 1(6,8_ϕ) + 5(6,1) + 1(10,1)

Spectrum 621; b1=(2,4), b2=(3/2,1); singlets = 39

39(1,1) + 3(15,1) + 1(4,8) + 16(4,1) + 1(4,8) + 3(1,8_ϕ) + 3(1,8_s) + 1(1,8_ϕ) + 4(6,1) + 1(6,8_ϕ)

Spectrum 622; b1=(2,4), b2=(3/2,1); singlets = 39

39(1,1) + 3(15,1) + 1(4,8) + 16(4,1) + 1(4,8) + 3(1,8_ϕ) + 1(1,8_s) + 3(1,8_ϕ) + 4(6,1) + 1(6,8_s)

Spectrum 623; b1=(2,4), b2=(3/2,1); singlets = 39

39(1,1) + 3(15,1) + 16(4,1) + 1(4,8_ϕ) + 1(4,8_ϕ) + 1(1,8_ϕ) + 3(1,8_s) + 3(1,8_ϕ) + 1(6,8_ϕ) + 4(6,1)

A3 x D5

Spectrum 624; b1=(2,4), b2=(1,-2); singlets = 44

44(1,1) + 64(4,1) + 2(1,10) + 1(20,1)

Spectrum 625; b1=(2,4), b2=(1,-2); singlets = 64

64(1,1) + 43(4,1) + 2(1,10) + 9(6,1) + 1(10,1) + 1(20',1)

Spectrum 626; b1=(2,4), b2=(1,-2); singlets = 65

65(1,1) + 1(15,1) + 43(4,1) + 2(1,10) + 8(6,1) + 1(20',1)

Spectrum 627; b1=(2,4), b2=(1,-2); singlets = 72

72(1,1) + 32(4,1) + 2(1,10) + 15(6,1) + 3(10,1)

Spectrum 628; b1=(2,4), b2=(1,-2); singlets = 73

73(1,1) + 1(15,1) + 32(4,1) + 2(1,10) + 14(6,1) + 2(10,1)

Spectrum 629; b1=(2,4), b2=(1,-2); singlets = 74

74(1,1) + 2(15,1) + 32(4,1) + 2(1,10) + 13(6,1) + 1(10,1)

Spectrum 630; b1=(2,4), b2=(1,-2); singlets = 75

75(1,1) + 3(15,1) + 32(4,1) + 2(1,10) + 12(6,1)

A3 x G2

Spectrum 631; b1=(7/2,9), b2=(2,4); singlets = 2

2(1,1) + 1(15,7) + 12(4,1) + 1(1,27) + 1(6,7) + 1(6,1) + 1(10,7) + 2(20,1)

Spectrum 632; b1=(7/2,9), b2=(2,4); singlets = 2

2(1,1) + 2(15,7) + 12(4,1) + 1(1,7) + 1(1,27) + 1(6,1) + 2(20,1)

Spectrum 633; b1=(2,4), b2=(7/2,9); singlets = 9

9(1,1) + 1(4,14) + 3(4,7) + 2(4,1) + 1(4,27) + 1(1,14) + 2(1,7) + 1(1,27) + 1(20,1)

Spectrum 634; b1=(2,4), b2=(7/2,9); singlets = 9

9(1,1) + 4(4,14) + 8(4,1) + 1(1,14) + 2(1,7) + 1(1,27) + 1(20,1)

Spectrum 635; b1=(2,4), b2=(7/2,9); singlets = 3

3(1,1) + 2(4,14) + 4(4,1) + 5(1,7) + 2(1,27) + 1(6,14) + 1(6,1) + 3(10,1)

Spectrum 636; b1=(2,4), b2=(7/2,9); singlets = 9

9(1,1) + 2(4,14) + 4(4,1) + 3(1,14) + 2(1,7) + 1(1,27) + 1(6,14) + 1(6,1) + 3(10,1)

Spectrum 637; b1=(2,4), b2=(7/2,9); singlets = 4

4(1,1) + 1(15,1) + 2(4,14) + 4(4,1) + 5(1,7) + 2(1,27) + 1(6,14) + 2(10,1)

Spectrum 638; b1=(2,4), b2=(7/2,9); singlets = 10

10(1,1) + 1(15,1) + 2(4,14) + 4(4,1) + 3(1,14) + 2(1,7) + 1(1,27) + 1(6,14) + 2(10,1)

Spectrum 639; b1=(2,4), b2=(2,4); singlets = 50

50(1,1) + 6(4,7) + 1(4,1) + 1(1,7) + 1(1,27) + 1(6,7) + 2(6,1) + 1(10,1) + 1(20,1)

Spectrum 640; b1=(2,4), b2=(2,4); singlets = 51

51(1,1) + 1(15,1) + 6(4,7) + 1(4,1) + 1(1,7) + 1(1,27) + 1(6,7) + 1(6,1) + 1(20,1)

Spectrum 641; b1=(2,4), b2=(2,4); singlets = 44

44(1,1) + 4(4,7) + 4(4,1) + 3(1,7) + 1(1,27) + 2(6,7) + 1(6,1) + 3(10,1)

Spectrum 642; b1=(2,4), b2=(2,4); singlets = 50

50(1,1) + 4(4,7) + 4(4,1) + 3(1,14) + 2(6,7) + 1(6,1) + 3(10,1)

Spectrum 643; b1=(2,4), b2=(2,4); singlets = 45

45(1,1) + 1(15,1) + 4(4,7) + 4(4,1) + 3(1,7) + 1(1,27) + 2(6,7) + 2(10,1)

Spectrum 644; b1=(2,4), b2=(2,4); singlets = 51

51(1,1) + 1(15,1) + 4(4,7) + 4(4,1) + 3(1,14) + 2(6,7) + 2(10,1)

A4 x A4

Spectrum 645; b1=(1,0), b2=(1,0); singlets = 140

140(1,1) + 16(5,1) + 16(1,5) + 2(1,10) + 2(10,1)

A4 x A5

Spectrum 646; b1=(5/2,5), b2=(3/2,1); singlets = 28

28(1,1) + 1(5,15) + 2(1,6) + 2(10,6) + 2(10,1) + 3(15,1) + 1(40,1)

Spectrum 647; b1=(5/2,5), b2=(3/2,1); singlets = 28

28(1,1) + 1(5,1) + 1(5,15) + 2(1,6) + 2(10,6) + 1(10,1) + 3(15,1) + 1(45,1)

Spectrum 648; b1=(5/2,5), b2=(3/2,1); singlets = 29

29(1,1) + 1(24,1) + 1(5,15) + 2(1,6) + 2(10,6) + 1(10,1) + 2(15,1) + 1(40,1)

Spectrum 649; b1=(5/2,5), b2=(3/2,1); singlets = 29

29(1,1) + 1(24,1) + 1(5,1) + 1(5,15) + 2(1,6) + 2(10,6) + 2(15,1) + 1(45,1)

Spectrum 650; b1=(5/2,5), b2=(3/2,1); singlets = 30

30(1,1) + 2(24,1) + 1(5,15) + 2(1,6) + 2(10,6) + 1(15,1) + 1(40,1)

Spectrum 651; b1=(1,0), b2=(3/2,1); singlets = 58

58(1,1) + 1(5,6) + 10(5,1) + 22(1,6) + 5/2(1,20) + 2(10,1)

Spectrum 652; b1=(1,0), b2=(3/2,1); singlets = 59

59(1,1) + 1(5,6) + 10(5,1) + 21(1,6) + 1(1,15) + 2(1,20) + 2(10,1)

Spectrum 653; b1=(1,0), b2=(3/2,1); singlets = 60

60(1,1) + 1(5,6) + 10(5,1) + 20(1,6) + 2(1,15) + 3/2(1,20) + 2(10,1)

Spectrum 654; b1=(1,0), b2=(3/2,1); singlets = 61

61(1,1) + 1(5,6) + 10(5,1) + 19(1,6) + 3(1,15) + 1(1,20) + 2(10,1)

Spectrum 655; b1=(1,0), b2=(3/2,1); singlets = 62

62(1,1) + 1(5,6) + 10(5,1) + 18(1,6) + 4(1,15) + 1/2(1,20) + 2(10,1)

Spectrum 656; b1=(1,0), b2=(3/2,1); singlets = 63

63(1,1) + 1(5,6) + 10(5,1) + 17(1,6) + 5(1,15) + 2(10,1)

A4 x A6

Spectrum 657; b1=(1,0), b2=(2,2); singlets = 16

16(1,1) + 2(5,7) + 2(5,1) + 13(1,7) + 1(1,28) + 3(1,35) + 2(10,1)

Spectrum 658; b1=(1,0), b2=(2,2); singlets = 23

23(1,1) + 2(5,7) + 2(5,1) + 8(1,7) + 3(1,21) + 1(1,28) + 2(1,35) + 2(10,1)

Spectrum 659; b1=(1,0), b2=(2,2); singlets = 24

24(1,1) + 2(5,7) + 2(5,1) + 1(1,48) + 8(1,7) + 2(1,21) + 2(1,35) + 2(10,1)

Spectrum 660; b1=(1,0), b2=(2,2); singlets = 30

30(1,1) + 2(5,7) + 2(5,1) + 3(1,7) + 6(1,21) + 1(1,28) + 1(1,35) + 2(10,1)

Spectrum 661; b1=(1,0), b2=(2,2); singlets = 31

31(1,1) + 2(5,7) + 2(5,1) + 1(1,48) + 3(1,7) + 5(1,21) + 1(1,35) + 2(10,1)

A4 x B3

Spectrum 662; b1=(1,0), b2=(1,0); singlets = 155

155(1,1) + 16(5,1) + 3(1,7) + 8(1,8) + 2(10,1)

A4 x B4

Spectrum 663; b1=(1,0), b2=(2,2); singlets = 49

$$49(\mathbf{1,1}) + 1(\mathbf{5,9}) + 7(\mathbf{5,1}) + 1(\mathbf{1,36}) + 3(\mathbf{1,9}) + 8(\mathbf{1,16}) + 2(\mathbf{10,1})$$

A4 x C2

Spectrum 664; b1=(5/2,5), b2=(3/2,3); singlets = 29

$$29(\mathbf{1,1}) + 1/2(\mathbf{24,4}) + 2(\mathbf{5,4}) + 2(\mathbf{5,1}) + 1(\mathbf{5,5}) + 1(\mathbf{1,5}) + 1/2(\mathbf{1,16}) + 2(\mathbf{10,4}) + 4(\mathbf{10,1}) + 1(\mathbf{15,1}) + 1(\mathbf{40,1})$$

Spectrum 665; b1=(5/2,5), b2=(3/2,3); singlets = 29

$$29(\mathbf{1,1}) + 1/2(\mathbf{24,4}) + 2(\mathbf{5,4}) + 3(\mathbf{5,1}) + 1(\mathbf{5,5}) + 1(\mathbf{1,5}) + 1/2(\mathbf{1,16}) + 2(\mathbf{10,4}) + 3(\mathbf{10,1}) + 1(\mathbf{15,1}) + 1(\mathbf{45,1})$$

Spectrum 666; b1=(5/2,5), b2=(3/2,3); singlets = 30

$$30(\mathbf{1,1}) + 1/2(\mathbf{24,4}) + 1(\mathbf{24,1}) + 2(\mathbf{5,4}) + 2(\mathbf{5,1}) + 1(\mathbf{5,5}) + 1(\mathbf{1,5}) + 1/2(\mathbf{1,16}) + 2(\mathbf{10,4}) + 3(\mathbf{10,1}) + 1(\mathbf{40,1})$$

Spectrum 667; b1=(5/2,5), b2=(3/2,3); singlets = 30

$$30(\mathbf{1,1}) + 1/2(\mathbf{24,4}) + 1(\mathbf{24,1}) + 2(\mathbf{5,4}) + 3(\mathbf{5,1}) + 1(\mathbf{5,5}) + 1(\mathbf{1,5}) + 1/2(\mathbf{1,16}) + 2(\mathbf{10,4}) + 2(\mathbf{10,1}) + 1(\mathbf{45,1})$$

Spectrum 668; b1=(1,0), b2=(3,8); singlets = 1

$$1(\mathbf{1,1}) + 1(\mathbf{5,10}) + 1(\mathbf{5,1}) + 1(\mathbf{5,5}) + 3(\mathbf{1,10}) + 34(\mathbf{1,4}) + 7(\mathbf{1,5}) + 1(\mathbf{1,14}) + 3/2(\mathbf{1,16}) + 2(\mathbf{10,1})$$

Spectrum 669; b1=(1,0), b2=(3,8); singlets = 1

$$1(\mathbf{1,1}) + 1(\mathbf{5,10}) + 1(\mathbf{5,1}) + 1(\mathbf{5,5}) + 38(\mathbf{1,4}) + 3(\mathbf{1,5}) + 9/2(\mathbf{1,16}) + 2(\mathbf{10,1})$$

Spectrum 670; b1=(1,0), b2=(3,8); singlets = 5

$$5(\mathbf{1,1}) + 1(\mathbf{5,10}) + 1(\mathbf{5,1}) + 1(\mathbf{5,5}) + 55/2(\mathbf{1,4}) + 15(\mathbf{1,5}) + 5/2(\mathbf{1,16}) + 1/2(\mathbf{1,20}) + 2(\mathbf{10,1})$$

Spectrum 671; b1=(1,0), b2=(3,8); singlets = 8

$$8(\mathbf{1,1}) + 1(\mathbf{5,10}) + 1(\mathbf{5,1}) + 1(\mathbf{5,5}) + 4(\mathbf{1,10}) + 28(\mathbf{1,4}) + 10(\mathbf{1,5}) + 1(\mathbf{1,14}) + 1(\mathbf{1,16}) + 2(\mathbf{10,1})$$

Spectrum 672; b1=(1,0), b2=(3,8); singlets = 8

$$8(\mathbf{1,1}) + 1(\mathbf{5,10}) + 1(\mathbf{5,1}) + 1(\mathbf{5,5}) + 1(\mathbf{1,10}) + 32(\mathbf{1,4}) + 6(\mathbf{1,5}) + 4(\mathbf{1,16}) + 2(\mathbf{10,1})$$

Spectrum 673; b1=(1,0), b2=(3,8); singlets = 9

$$9(\mathbf{1,1}) + 1(\mathbf{5,10}) + 1(\mathbf{5,1}) + 1(\mathbf{5,5}) + 17(\mathbf{1,4}) + 27(\mathbf{1,5}) + 1/2(\mathbf{1,16}) + 1(\mathbf{1,20}) + 2(\mathbf{10,1})$$

Spectrum 674; b1=(1,0), b2=(3,8); singlets = 12

$$12(\mathbf{1,1}) + 1(\mathbf{5,10}) + 1(\mathbf{5,1}) + 1(\mathbf{5,5}) + 1(\mathbf{1,10}) + 43/2(\mathbf{1,4}) + 18(\mathbf{1,5}) + 2(\mathbf{1,16}) + 1/2(\mathbf{1,20}) + 2(\mathbf{10,1})$$

Spectrum 675; b1=(1,0), b2=(3,8); singlets = 15

$$15(\mathbf{1,1}) + 1(\mathbf{5,10}) + 1(\mathbf{5,1}) + 1(\mathbf{5,5}) + 5(\mathbf{1,10}) + 22(\mathbf{1,4}) + 13(\mathbf{1,5}) + 1(\mathbf{1,14}) + 1/2(\mathbf{1,16}) + 2(\mathbf{10,1})$$

Spectrum 676; b1=(1,0), b2=(3,8); singlets = 15

$$15(\mathbf{1,1}) + 1(\mathbf{5,10}) + 1(\mathbf{5,1}) + 1(\mathbf{5,5}) + 2(\mathbf{1,10}) + 26(\mathbf{1,4}) + 9(\mathbf{1,5}) + 7/2(\mathbf{1,16}) + 2(\mathbf{10,1})$$

Spectrum 677; b1=(1,0), b2=(3,8); singlets = 16

$$16(\mathbf{1,1}) + 1(\mathbf{5,10}) + 1(\mathbf{5,1}) + 1(\mathbf{5,5}) + 1(\mathbf{1,10}) + 11(\mathbf{1,4}) + 30(\mathbf{1,5}) + 1(\mathbf{1,20}) + 2(\mathbf{10,1})$$

Spectrum 678; b1=(1,0), b2=(3,8); singlets = 19

$$19(\mathbf{1,1}) + 1(\mathbf{5,10}) + 1(\mathbf{5,1}) + 1(\mathbf{5,5}) + 2(\mathbf{1,10}) + 31/2(\mathbf{1,4}) + 21(\mathbf{1,5}) + 3/2(\mathbf{1,16}) + 1/2(\mathbf{1,20}) + 2(\mathbf{10,1})$$

Spectrum 679; b1=(1,0), b2=(3,8); singlets = 22

$$22(\mathbf{1,1}) + 1(\mathbf{5,10}) + 1(\mathbf{5,1}) + 1(\mathbf{5,5}) + 6(\mathbf{1,10}) + 16(\mathbf{1,4}) + 16(\mathbf{1,5}) + 1(\mathbf{1,14}) + 2(\mathbf{10,1})$$

Spectrum 680; b1=(1,0), b2=(3,8); singlets = 22

$$22(\mathbf{1,1}) + 1(\mathbf{5,10}) + 1(\mathbf{5,1}) + 1(\mathbf{5,5}) + 3(\mathbf{1,10}) + 20(\mathbf{1,4}) + 12(\mathbf{1,5}) + 3(\mathbf{1,16}) + 2(\mathbf{10,1})$$

Spectrum 681; b1=(1,0), b2=(3,8); singlets = 26

$$26(\mathbf{1,1}) + 1(\mathbf{5,10}) + 1(\mathbf{5,1}) + 1(\mathbf{5,5}) + 3(\mathbf{1,10}) + 19/2(\mathbf{1,4}) + 24(\mathbf{1,5}) + 1(\mathbf{1,16}) + 1/2(\mathbf{1,20}) + 2(\mathbf{10,1})$$

Spectrum 682; b1=(1,0), b2=(3,8); singlets = 29

$$29(\mathbf{1,1}) + 1(\mathbf{5,10}) + 1(\mathbf{5,1}) + 1(\mathbf{5,5}) + 4(\mathbf{1,10}) + 14(\mathbf{1,4}) + 15(\mathbf{1,5}) + 5/2(\mathbf{1,16}) + 2(\mathbf{10,1})$$

Spectrum 683; b1=(1,0), b2=(3,8); singlets = 33

$$33(\mathbf{1,1}) + 1(\mathbf{5,10}) + 1(\mathbf{5,1}) + 1(\mathbf{5,5}) + 4(\mathbf{1,10}) + 7/2(\mathbf{1,4}) + 27(\mathbf{1,5}) + 1/2(\mathbf{1,16}) + 1/2(\mathbf{1,20}) + 2(\mathbf{10,1})$$

Spectrum 684; b1=(1,0), b2=(3,8); singlets = 36

$$36(\mathbf{1,1}) + 1(\mathbf{5,10}) + 1(\mathbf{5,1}) + 1(\mathbf{5,5}) + 5(\mathbf{1,10}) + 8(\mathbf{1,4}) + 18(\mathbf{1,5}) + 2(\mathbf{1,16}) + 2(\mathbf{10,1})$$

Spectrum 685; b1=(1,0), b2=(3,8); singlets = 43

$$43(\mathbf{1,1}) + 1(\mathbf{5,10}) + 1(\mathbf{5,1}) + 1(\mathbf{5,5}) + 6(\mathbf{1,10}) + 2(\mathbf{1,4}) + 21(\mathbf{1,5}) + 3/2(\mathbf{1,16}) + 2(\mathbf{10,1})$$

Spectrum 686; b1=(1,0), b2=(3/2,3); singlets = 94

$$94(\mathbf{1,1}) + 3(\mathbf{5,4}) + 4(\mathbf{5,1}) + 27(\mathbf{1,4}) + 6(\mathbf{1,5}) + 1/2(\mathbf{1,16}) + 2(\mathbf{10,1})$$

Spectrum 687; b1=(1,0), b2=(3/2,3); singlets = 79

$$79(\mathbf{1,1}) + 1(\mathbf{5,4}) + 7(\mathbf{5,1}) + 1(\mathbf{5,5}) + 37(\mathbf{1,4}) + 1(\mathbf{1,5}) + 1/2(\mathbf{1,16}) + 2(\mathbf{10,1})$$

Spectrum 688; b1=(1,0), b2=(3/2,3); singlets = 101

$$101(\mathbf{1,1}) + 3(\mathbf{5,4}) + 4(\mathbf{5,1}) + 1(\mathbf{1,10}) + 21(\mathbf{1,4}) + 9(\mathbf{1,5}) + 2(\mathbf{10,1})$$

Spectrum 689; b1=(1,0), b2=(3/2,3); singlets = 86

$$86(\mathbf{1,1}) + 1(\mathbf{5,4}) + 7(\mathbf{5,1}) + 1(\mathbf{5,5}) + 1(\mathbf{1,10}) + 31(\mathbf{1,4}) + 4(\mathbf{1,5}) + 2(\mathbf{10,1})$$

A4 x C3

Spectrum 690; b1=(1,0), b2=(2,4); singlets = 11

$$11(\mathbf{1,1}) + 2(\mathbf{5,1}) + 1(\mathbf{5,14'}) + 1(\mathbf{1,21}) + 47/2(\mathbf{1,6}) + 5/2(\mathbf{1,14}) + 1/2(\mathbf{1,64}) + 2(\mathbf{10,1})$$

Spectrum 691; b1=(1,0), b2=(2,4); singlets = 13

$$13(\mathbf{1,1}) + 2(\mathbf{5,1}) + 1(\mathbf{5,14'}) + 1(\mathbf{1,21}) + 22(\mathbf{1,6}) + 1(\mathbf{1,14'}) + 2(\mathbf{1,14}) + 1/2(\mathbf{1,64}) + 2(\mathbf{10,1})$$

Spectrum 692; b1=(1,0), b2=(2,4); singlets = 15

$$15(\mathbf{1,1}) + 2(\mathbf{5,1}) + 1(\mathbf{5,14'}) + 1(\mathbf{1,21}) + 41/2(\mathbf{1,6}) + 2(\mathbf{1,14'}) + 3/2(\mathbf{1,14}) + 1/2(\mathbf{1,64}) + 2(\mathbf{10,1})$$

Spectrum 693; b1=(1,0), b2=(2,4); singlets = 17

$$17(\mathbf{1,1}) + 2(\mathbf{5,1}) + 1(\mathbf{5,14'}) + 1(\mathbf{1,21}) + 19(\mathbf{1,6}) + 3(\mathbf{1,14'}) + 1(\mathbf{1,14}) + 1/2(\mathbf{1,64}) + 2(\mathbf{10,1})$$

Spectrum 694; b1=(1,0), b2=(2,4); singlets = 19

19(1,1) + 2(5,1) + 1(5,14') + 1(1,21) + 35/2(1,6) + 4(1,14') + 1/2(1,14) + 1/2(1,64) + 2(10,1)

Spectrum 695; b1=(1,0), b2=(2,4); singlets = 21

21(1,1) + 2(5,1) + 1(5,14') + 1(1,21) + 16(1,6) + 5(1,14') + 1/2(1,64) + 2(10,1)

Spectrum 696; b1=(1,0), b2=(2,4); singlets = 33

33(1,1) + 2(5,1) + 1(5,14') + 3(1,21) + 27/2(1,6) + 9/2(1,14) + 2(10,1)

Spectrum 697; b1=(1,0), b2=(2,4); singlets = 35

35(1,1) + 2(5,1) + 1(5,14') + 3(1,21) + 12(1,6) + 1(1,14') + 4(1,14) + 2(10,1)

Spectrum 698; b1=(1,0), b2=(2,4); singlets = 37

37(1,1) + 2(5,1) + 1(5,14') + 3(1,21) + 21/2(1,6) + 2(1,14') + 7/2(1,14) + 2(10,1)

Spectrum 699; b1=(1,0), b2=(2,4); singlets = 39

39(1,1) + 2(5,1) + 1(5,14') + 3(1,21) + 9(1,6) + 3(1,14') + 3(1,14) + 2(10,1)

Spectrum 700; b1=(1,0), b2=(2,4); singlets = 41

41(1,1) + 2(5,1) + 1(5,14') + 3(1,21) + 15/2(1,6) + 4(1,14') + 5/2(1,14) + 2(10,1)

Spectrum 701; b1=(1,0), b2=(2,4); singlets = 43

43(1,1) + 2(5,1) + 1(5,14') + 3(1,21) + 6(1,6) + 5(1,14') + 2(1,14) + 2(10,1)

Spectrum 702; b1=(1,0), b2=(2,4); singlets = 45

45(1,1) + 2(5,1) + 1(5,14') + 3(1,21) + 9/2(1,6) + 6(1,14') + 3/2(1,14) + 2(10,1)

Spectrum 703; b1=(1,0), b2=(2,4); singlets = 47

47(1,1) + 2(5,1) + 1(5,14') + 3(1,21) + 3(1,6) + 7(1,14') + 1(1,14) + 2(10,1)

Spectrum 704; b1=(1,0), b2=(2,4); singlets = 49

49(1,1) + 2(5,1) + 1(5,14') + 3(1,21) + 3/2(1,6) + 8(1,14') + 1/2(1,14) + 2(10,1)

Spectrum 705; b1=(1,0), b2=(2,4); singlets = 51

51(1,1) + 2(5,1) + 1(5,14') + 3(1,21) + 9(1,14') + 2(10,1)

A4 x C4

Spectrum 706; b1=(1,0), b2=(1,0); singlets = 85

85(1,1) + 16(5,1) + 16(1,8) + 1(1,27) + 2(10,1)

A4 x C5

Spectrum 707; b1=(1,0), b2=(3/2,1); singlets = 14

14(1,1) + 1(5,10) + 6(5,1) + 5(1,10) + 4(1,44) + 2(10,1)

A4 x D5

Spectrum 708; b1=(5/2,5), b2=(1,-2); singlets = 20

20(1,1) + 15(5,1) + 2(1,10) + 14(10,1) + 3(15,1) + 1(40,1)

Spectrum 709; b1=(5/2,5), b2=(1,-2); singlets = 20

20(1,1) + 16(5,1) + 2(1,10) + 13(10,1) + 3(15,1) + 1(45,1)

Spectrum 710; b1=(5/2,5), b2=(1,-2); singlets = 21

21(1,1) + 1(24,1) + 15(5,1) + 2(1,10) + 13(10,1) + 2(15,1) + 1(40,1)

Spectrum 711; b1=(5/2,5), b2=(1,-2); singlets = 21

21(1,1) + 1(24,1) + 16(5,1) + 2(1,10) + 12(10,1) + 2(15,1) + 1(45,1)

Spectrum 712; b1=(5/2,5), b2=(1,-2); singlets = 22

22(1,1) + 2(24,1) + 15(5,1) + 2(1,10) + 12(10,1) + 1(15,1) + 1(40,1)

Spectrum 713; b1=(5/2,5), b2=(1,-2); singlets = 22

22(1,1) + 2(24,1) + 16(5,1) + 2(1,10) + 11(10,1) + 1(15,1) + 1(45,1)

Spectrum 714; b1=(5/2,5), b2=(1,-2); singlets = 23

23(1,1) + 3(24,1) + 15(5,1) + 2(1,10) + 11(10,1) + 1(40,1)

Spectrum 715; b1=(5/2,5), b2=(1,-2); singlets = 23

23(1,1) + 3(24,1) + 16(5,1) + 2(1,10) + 10(10,1) + 1(45,1)

A4 x G2

Spectrum 716; b1=(1,0), b2=(2,4); singlets = 66

66(1,1) + 2(5,7) + 2(5,1) + 21(1,7) + 1(1,27) + 2(10,1)

Spectrum 717; b1=(1,0), b2=(2,4); singlets = 72

72(1,1) + 2(5,7) + 2(5,1) + 3(1,14) + 18(1,7) + 2(10,1)

A5 x A5

Spectrum 718; b1=(3/2,1), b2=(3/2,1); singlets = 24

24(1,1) + 3(6,6) + 9(6,1) + 9(1,6) + 5/2(1,20) + 5/2(20,1)

Spectrum 719; b1=(3/2,1), b2=(3/2,1); singlets = 25

25(1,1) + 3(6,6) + 9(6,1) + 8(1,6) + 1(1,15) + 2(1,20) + 5/2(20,1)

Spectrum 720; b1=(3/2,1), b2=(3/2,1); singlets = 26

26(1,1) + 3(6,6) + 9(6,1) + 7(1,6) + 2(1,15) + 3/2(1,20) + 5/2(20,1)

Spectrum 721; b1=(3/2,1), b2=(3/2,1); singlets = 27

27(1,1) + 3(6,6) + 9(6,1) + 6(1,6) + 3(1,15) + 1(1,20) + 5/2(20,1)

Spectrum 722; b1=(3/2,1), b2=(3/2,1); singlets = 28

Spectrum 756; $b_1=(3/2,1)$, $b_2=(3,8)$; singlets = 6

Spectrum 757; $b_1=(3/2,1)$, $b_2=(3,8)$; singlets = 13

Spectrum 758; $b_1=(3/2,1)$, $b_2=(3,8)$; singlets = 3

Spectrum 759; $b1=(3/2,1)$, $b2=(3,8)$; singlets = 10

Spectrum 760; $b_1=(3/2,1)$, $b_2=(3,8)$; singlets = 17

Spectrum 761; $b_1=(3/2,1)$, $b_2=(3,8)$; singlets = 27

Spectrum 762; $b_1=(3/2,1)$, $b_2=(3,8)$; singlets = 23

Spectrum 763; $b_1=(3/2,1)$, $b_2=(3,8)$; singlets = 3

Spectrum 764; $b_1=(3/2,1)$, $b_2=(3,8)$; singlets = 6

Spectrum 765: $b_1=(3/2.1)$, $b_2=(3.8)$; singlets = 13

Spectrum 766; $b_1=(3/2,1)$, $b_2=(3/2,3)$; singlets = 66

Spectrum 767: $b_1=(3/2.1)$, $b_2=(3/2.3)$; singlets = 48

Spectrum 768: $b_1=(3/2.1)$, $b_2=(3/2.3)$; singlets = 73

Spectrum 769: $b_1=(3/2.1)$, $b_2=(3/2.3)$: singlets = 55

Spectrum 770: $b_1=(3/2.1)$, $b_2=(3/2.3)$: singlets = 67

Spectrum 771: $b_1 \equiv (3/2, 1)$, $b_2 \equiv (3/2, 3)$: singlets $\equiv 49$

Spectrum 772: $b_1=(3/2,1)$, $b_2=(3/2,3)$; singlets = 74

Spectrum 773: $b_1=(3/2, 1)$, $b_2=(3/2, 3)$: singlets = 56

Spectrum 774: $h_1 \equiv (3/2, 1)$, $h_2 \equiv (3/2, 3)$: singlets $\equiv 68$

Spectrum 775: $b_1 \equiv (3/2, 1)$, $b_2 \equiv (3/2, 3)$: singlets $\equiv 50$

Spectrum 776: $h_1=(3/2\ 1)$ $h_2=(3/2\ 3)$: singlets = 75

Spectrum 777: $h1=(3/2\ 1)$ $h2=(3/2\ 3)$: singlets = 57

Spectrum 778: $h1=(3/2\ 1)$, $h2=(3/2\ 3)$: singlets = 69

Spectrum 779: $h_1=(3/2\ 1)$ $h_2=(3/2\ 3)$: singlets = 51

Spectrum 780: $h_1 = (3/2 \ 1)$ $h_2 = (3/2 \ 3)$; singlets = 76

Spectrum 781: $h_1=(3/2\ 1)$ $h_2=(3/2\ 3)$: singlets = 58

Spectrum 782: $h1=(3/2\ 1)$ $h2=(3/2\ 3)$: singlets = 52

Spectrum 783: $h1=(3/2\ 1)$ $h2=(3/2\ 3)$: singlets = 34

Spectrum 784: $h_1 = (3/2 \ 1)$ $h_2 = (3/2 \ 3)$; singlets = 16

Spectrum 785: $h_1 = (3/2, 1)$, $h_2 = (3/2, 3)$; singlets = 58

Spectrum 786: $h_1 = (3/2 \ 1)$, $h_2 = (3/2 \ 3)$; singlets = 41

Spectrum 787: $h1=(3/2\ 1)$ $h2=(3/2\ 3)$: singlets = 23

Spectrum 788: $h1=-(3/2\ 1)$ $h3=-(3/2\ 3)$; singlets = 52

Spectrum 780: $b_1 = (3/2, 1)$, $b_3 = (3/2, 3)$; singlets = 3F

35(1,1) + 2(6,4) + 14(6,1) + 15(1,4) + 6(1,5) + 1/2(1,16) + 1(15,4) + 1(15,1)

Spectrum 790; b1=(3/2,1), b2=(3/2,3); singlets = 17

17(1,1) + 17(6,1) + 1(6,5) + 27(1,4) + 1/2(1,16) + 1(15,4) + 1(15,1)

Spectrum 791; b1=(3/2,1), b2=(3/2,3); singlets = 60

60(1,1) + 4(6,4) + 1(6,1) + 1(6,5) + 1(1,10) + 12(1,4) + 3(1,5) + 5(15,1)

Spectrum 792; b1=(3/2,1), b2=(3/2,3); singlets = 42

42(1,1) + 2(6,4) + 14(6,1) + 1(1,10) + 9(1,4) + 9(1,5) + 1(15,4) + 1(15,1)

Spectrum 793; b1=(3/2,1), b2=(3/2,3); singlets = 24

24(1,1) + 17(6,1) + 1(6,5) + 1(1,10) + 21(1,4) + 3(1,5) + 1(15,4) + 1(15,1)

A5 x C4

Spectrum 794; b1=(3/2,1), b2=(1,0); singlets = 21

21(1,1) + 1(6,8) + 19(6,1) + 10(1,8) + 1(1,27) + 5/2(20,1)

Spectrum 795; b1=(3/2,1), b2=(1,0); singlets = 22

22(1,1) + 1(6,8) + 18(6,1) + 10(1,8) + 1(1,27) + 1(15,1) + 2(20,1)

Spectrum 796; b1=(3/2,1), b2=(1,0); singlets = 23

23(1,1) + 1(6,8) + 17(6,1) + 10(1,8) + 1(1,27) + 2(15,1) + 3/2(20,1)

Spectrum 797; b1=(3/2,1), b2=(1,0); singlets = 24

24(1,1) + 1(6,8) + 16(6,1) + 10(1,8) + 1(1,27) + 3(15,1) + 1(20,1)

Spectrum 798; b1=(3/2,1), b2=(1,0); singlets = 25

25(1,1) + 1(6,8) + 15(6,1) + 10(1,8) + 1(1,27) + 4(15,1) + 1/2(20,1)

Spectrum 799; b1=(3/2,1), b2=(1,0); singlets = 26

26(1,1) + 1(6,8) + 14(6,1) + 10(1,8) + 1(1,27) + 5(15,1)

A5 x D4

Spectrum 800; b1=(3,6), b2=(3,6); singlets = 4

4(1,1) + 1/2(20,28) + 1(56,1)

A5 x D5

Spectrum 801; b1=(3,6), b2=(1,-2); singlets = 3

3(1,1) + 3(6,1) + 2(1,10) + 11/2(20,1) + 4(21,1) + 1(105,1)

Spectrum 802; b1=(3,6), b2=(1,-2); singlets = 4

4(1,1) + 2(6,1) + 2(1,10) + 1(15,1) + 5(20,1) + 4(21,1) + 1(105,1)

Spectrum 803; b1=(3,6), b2=(1,-2); singlets = 5

5(1,1) + 1(6,1) + 2(1,10) + 2(15,1) + 9/2(20,1) + 4(21,1) + 1(105,1)

Spectrum 804; b1=(3,6), b2=(1,-2); singlets = 5

5(1,1) + 1(35,1) + 2(6,1) + 2(1,10) + 5(20,1) + 3(21,1) + 1(105,1)

Spectrum 805; b1=(3,6), b2=(1,-2); singlets = 6

6(1,1) + 2(1,10) + 3(15,1) + 4(20,1) + 4(21,1) + 1(105,1)

Spectrum 806; b1=(3,6), b2=(1,-2); singlets = 6

6(1,1) + 1(35,1) + 1(6,1) + 2(1,10) + 1(15,1) + 9/2(20,1) + 3(21,1) + 1(105,1)

Spectrum 807; b1=(3,6), b2=(1,-2); singlets = 6

6(1,1) + 2(6,1) + 2(1,10) + 6(20,1) + 2(21,1) + 2(70,1)

Spectrum 808; b1=(3,6), b2=(1,-2); singlets = 6

6(1,1) + 3(6,1) + 2(1,10) + 5(20,1) + 2(21,1) + 1(70,1) + 1(84,1)

Spectrum 809; b1=(3,6), b2=(1,-2); singlets = 6

6(1,1) + 4(6,1) + 2(1,10) + 4(20,1) + 2(21,1) + 2(84,1)

Spectrum 810; b1=(3,6), b2=(1,-2); singlets = 7

7(1,1) + 1(35,1) + 2(1,10) + 2(15,1) + 4(20,1) + 3(21,1) + 1(105,1)

Spectrum 811; b1=(3,6), b2=(1,-2); singlets = 7

7(1,1) + 1(6,1) + 2(1,10) + 1(15,1) + 11/2(20,1) + 2(21,1) + 2(70,1)

Spectrum 812; b1=(3,6), b2=(1,-2); singlets = 7

7(1,1) + 2(35,1) + 1(6,1) + 2(1,10) + 9/2(20,1) + 2(21,1) + 1(105,1)

Spectrum 813; b1=(3,6), b2=(1,-2); singlets = 7

7(1,1) + 2(6,1) + 2(1,10) + 1(15,1) + 9/2(20,1) + 2(21,1) + 1(70,1) + 1(84,1)

Spectrum 814; b1=(3,6), b2=(1,-2); singlets = 7

7(1,1) + 3(6,1) + 2(1,10) + 1(15,1) + 7/2(20,1) + 2(21,1) + 2(84,1)

Spectrum 815; b1=(3,6), b2=(1,-2); singlets = 8

8(1,1) + 2(1,10) + 2(15,1) + 5(20,1) + 2(21,1) + 2(70,1)

Spectrum 816; b1=(3,6), b2=(1,-2); singlets = 8

8(1,1) + 2(35,1) + 2(1,10) + 1(15,1) + 4(20,1) + 2(21,1) + 1(105,1)

Spectrum 817; b1=(3,6), b2=(1,-2); singlets = 8

8(1,1) + 1(6,1) + 2(1,10) + 2(15,1) + 4(20,1) + 2(21,1) + 1(70,1) + 1(84,1)

Spectrum 818; b1=(3,6), b2=(1,-2); singlets = 8

8(1,1) + 1(35,1) + 1(6,1) + 2(1,10) + 11/2(20,1) + 1(21,1) + 2(70,1)

Spectrum 819; b1=(3,6), b2=(1,-2); singlets = 8

8(1,1) + 2(6,1) + 2(1,10) + 2(15,1) + 3(20,1) + 2(21,1) + 2(84,1)

Spectrum 820; b1=(3,6), b2=(1,-2); singlets = 8

$8(\mathbf{1,1}) + 1(\mathbf{35,1}) + 2(\mathbf{6,1}) + 2(\mathbf{1,10}) + 9/2(\mathbf{20,1}) + 1(\mathbf{21,1}) + 1(\mathbf{70,1}) + 1(\mathbf{84,1})$
Spectrum 821; b1=(3,6), b2=(1,-2); singlets = 8
 $8(\mathbf{1,1}) + 1(\mathbf{35,1}) + 3(\mathbf{6,1}) + 2(\mathbf{1,10}) + 7/2(\mathbf{20,1}) + 1(\mathbf{21,1}) + 2(\mathbf{84,1})$
Spectrum 822; b1=(3,6), b2=(1,-2); singlets = 9
 $9(\mathbf{1,1}) + 2(\mathbf{1,10}) + 3(\mathbf{15,1}) + 7/2(\mathbf{20,1}) + 2(\mathbf{21,1}) + 1(\mathbf{70,1}) + 1(\mathbf{84,1})$
Spectrum 823; b1=(3,6), b2=(1,-2); singlets = 9
 $9(\mathbf{1,1}) + 1(\mathbf{35,1}) + 2(\mathbf{1,10}) + 1(\mathbf{15,1}) + 5(\mathbf{20,1}) + 1(\mathbf{21,1}) + 2(\mathbf{70,1})$
Spectrum 824; b1=(3,6), b2=(1,-2); singlets = 9
 $9(\mathbf{1,1}) + 3(\mathbf{35,1}) + 2(\mathbf{1,10}) + 4(\mathbf{20,1}) + 1(\mathbf{21,1}) + 1(\mathbf{105,1})$
Spectrum 825; b1=(3,6), b2=(1,-2); singlets = 9
 $9(\mathbf{1,1}) + 1(\mathbf{6,1}) + 2(\mathbf{1,10}) + 3(\mathbf{15,1}) + 5/2(\mathbf{20,1}) + 2(\mathbf{21,1}) + 2(\mathbf{84,1})$
Spectrum 826; b1=(3,6), b2=(1,-2); singlets = 9
 $9(\mathbf{1,1}) + 1(\mathbf{35,1}) + 1(\mathbf{6,1}) + 2(\mathbf{1,10}) + 1(\mathbf{15,1}) + 4(\mathbf{20,1}) + 1(\mathbf{21,1}) + 1(\mathbf{70,1}) + 1(\mathbf{84,1})$
Spectrum 827; b1=(3,6), b2=(1,-2); singlets = 9
 $9(\mathbf{1,1}) + 1(\mathbf{35,1}) + 2(\mathbf{6,1}) + 2(\mathbf{1,10}) + 1(\mathbf{15,1}) + 3(\mathbf{20,1}) + 1(\mathbf{21,1}) + 2(\mathbf{84,1})$
Spectrum 828; b1=(3,6), b2=(1,-2); singlets = 10
 $10(\mathbf{1,1}) + 2(\mathbf{1,10}) + 4(\mathbf{15,1}) + 2(\mathbf{20,1}) + 2(\mathbf{21,1}) + 2(\mathbf{84,1})$
Spectrum 829; b1=(3,6), b2=(1,-2); singlets = 10
 $10(\mathbf{1,1}) + 1(\mathbf{35,1}) + 2(\mathbf{1,10}) + 2(\mathbf{15,1}) + 7/2(\mathbf{20,1}) + 1(\mathbf{21,1}) + 1(\mathbf{70,1}) + 1(\mathbf{84,1})$
Spectrum 830; b1=(3,6), b2=(1,-2); singlets = 10
 $10(\mathbf{1,1}) + 2(\mathbf{35,1}) + 2(\mathbf{1,10}) + 5(\mathbf{20,1}) + 2(\mathbf{70,1})$
Spectrum 831; b1=(3,6), b2=(1,-2); singlets = 10
 $10(\mathbf{1,1}) + 1(\mathbf{35,1}) + 1(\mathbf{6,1}) + 2(\mathbf{1,10}) + 2(\mathbf{15,1}) + 5/2(\mathbf{20,1}) + 1(\mathbf{21,1}) + 2(\mathbf{84,1})$
Spectrum 832; b1=(3,6), b2=(1,-2); singlets = 10
 $10(\mathbf{1,1}) + 2(\mathbf{35,1}) + 1(\mathbf{6,1}) + 2(\mathbf{1,10}) + 4(\mathbf{20,1}) + 1(\mathbf{70,1}) + 1(\mathbf{84,1})$
Spectrum 833; b1=(3,6), b2=(1,-2); singlets = 10
 $10(\mathbf{1,1}) + 2(\mathbf{35,1}) + 2(\mathbf{6,1}) + 2(\mathbf{1,10}) + 3(\mathbf{20,1}) + 2(\mathbf{84,1})$
Spectrum 834; b1=(3,6), b2=(1,-2); singlets = 11
 $11(\mathbf{1,1}) + 1(\mathbf{35,1}) + 2(\mathbf{1,10}) + 3(\mathbf{15,1}) + 2(\mathbf{20,1}) + 1(\mathbf{21,1}) + 2(\mathbf{84,1})$
Spectrum 835; b1=(3,6), b2=(1,-2); singlets = 11
 $11(\mathbf{1,1}) + 2(\mathbf{35,1}) + 2(\mathbf{1,10}) + 1(\mathbf{15,1}) + 7/2(\mathbf{20,1}) + 1(\mathbf{70,1}) + 1(\mathbf{84,1})$
Spectrum 836; b1=(3,6), b2=(1,-2); singlets = 11
 $11(\mathbf{1,1}) + 2(\mathbf{35,1}) + 1(\mathbf{6,1}) + 2(\mathbf{1,10}) + 1(\mathbf{15,1}) + 5/2(\mathbf{20,1}) + 2(\mathbf{84,1})$
Spectrum 837; b1=(3,6), b2=(1,-2); singlets = 12
 $12(\mathbf{1,1}) + 2(\mathbf{35,1}) + 2(\mathbf{1,10}) + 2(\mathbf{15,1}) + 2(\mathbf{20,1}) + 2(\mathbf{84,1})$

A6 x B3

Spectrum 838; b1=(2,2), b2=(1,0); singlets = 17
 $17(\mathbf{1,1}) + 15(\mathbf{7,1}) + 1(\mathbf{7,8}) + 3(\mathbf{1,7}) + 1(\mathbf{1,8}) + 1(\mathbf{28,1}) + 3(\mathbf{35,1})$
Spectrum 839; b1=(2,2), b2=(1,0); singlets = 24
 $24(\mathbf{1,1}) + 10(\mathbf{7,1}) + 1(\mathbf{7,8}) + 3(\mathbf{1,7}) + 1(\mathbf{1,8}) + 3(\mathbf{21,1}) + 1(\mathbf{28,1}) + 2(\mathbf{35,1})$
Spectrum 840; b1=(2,2), b2=(1,0); singlets = 25
 $25(\mathbf{1,1}) + 1(\mathbf{48,1}) + 10(\mathbf{7,1}) + 1(\mathbf{7,8}) + 3(\mathbf{1,7}) + 1(\mathbf{1,8}) + 2(\mathbf{21,1}) + 2(\mathbf{35,1})$
Spectrum 841; b1=(2,2), b2=(1,0); singlets = 31
 $31(\mathbf{1,1}) + 5(\mathbf{7,1}) + 1(\mathbf{7,8}) + 3(\mathbf{1,7}) + 1(\mathbf{1,8}) + 6(\mathbf{21,1}) + 1(\mathbf{28,1}) + 1(\mathbf{35,1})$
Spectrum 842; b1=(2,2), b2=(1,0); singlets = 32
 $32(\mathbf{1,1}) + 1(\mathbf{48,1}) + 5(\mathbf{7,1}) + 1(\mathbf{7,8}) + 3(\mathbf{1,7}) + 1(\mathbf{1,8}) + 5(\mathbf{21,1}) + 1(\mathbf{35,1})$
Spectrum 843; b1=(2,2), b2=(1,0); singlets = 38
 $38(\mathbf{1,1}) + 1(\mathbf{7,8}) + 3(\mathbf{1,7}) + 1(\mathbf{1,8}) + 9(\mathbf{21,1}) + 1(\mathbf{28,1})$
Spectrum 844; b1=(2,2), b2=(1,0); singlets = 39
 $39(\mathbf{1,1}) + 1(\mathbf{48,1}) + 1(\mathbf{7,8}) + 3(\mathbf{1,7}) + 1(\mathbf{1,8}) + 8(\mathbf{21,1})$

A6 x C2

Spectrum 845; b1=(2,2), b2=(3/2,3); singlets = 36
 $36(\mathbf{1,1}) + 2(\mathbf{7,4}) + 1(\mathbf{7,5}) + 1(\mathbf{1,10}) + 1(\mathbf{1,4}) + 2(\mathbf{1,5}) + 1(\mathbf{21,4}) + 2(\mathbf{21,1}) + 1(\mathbf{28,1}) + 1(\mathbf{35,1})$
Spectrum 846; b1=(2,2), b2=(3/2,3); singlets = 37
 $37(\mathbf{1,1}) + 1(\mathbf{48,1}) + 2(\mathbf{7,4}) + 1(\mathbf{7,5}) + 1(\mathbf{1,10}) + 1(\mathbf{1,4}) + 2(\mathbf{1,5}) + 1(\mathbf{21,4}) + 1(\mathbf{21,1}) + 1(\mathbf{35,1})$

A6 x C4

Spectrum 847; b1=(2,2), b2=(1,0); singlets = 3
 $3(\mathbf{1,1}) + 2(\mathbf{7,8}) + 7(\mathbf{7,1}) + 2(\mathbf{1,8}) + 1(\mathbf{1,27}) + 1(\mathbf{28,1}) + 3(\mathbf{35,1})$
Spectrum 848; b1=(2,2), b2=(1,0); singlets = 10
 $10(\mathbf{1,1}) + 2(\mathbf{7,8}) + 2(\mathbf{7,1}) + 2(\mathbf{1,8}) + 1(\mathbf{1,27}) + 3(\mathbf{21,1}) + 1(\mathbf{28,1}) + 2(\mathbf{35,1})$
Spectrum 849; b1=(2,2), b2=(1,0); singlets = 11
 $11(\mathbf{1,1}) + 1(\mathbf{48,1}) + 2(\mathbf{7,8}) + 2(\mathbf{7,1}) + 2(\mathbf{1,8}) + 1(\mathbf{1,27}) + 2(\mathbf{21,1}) + 2(\mathbf{35,1})$

A6 x G2

Spectrum 850; b1=(2,2), b2=(2,4); singlets = 19

19(1,1) + 1(7,7) + 1(7,1) + 3(1,7) + 1(1,27) + 1(21,7) + 2(21,1) + 1(28,1)

Spectrum 851; b1=(2,2), b2=(2,4); singlets = 25

25(1,1) + 1(7,7) + 1(7,1) + 3(1,14) + 1(21,7) + 2(21,1) + 1(28,1)

Spectrum 852; b1=(2,2), b2=(2,4); singlets = 20

20(1,1) + 1(48,1) + 1(7,7) + 1(7,1) + 3(1,7) + 1(1,27) + 1(21,7) + 1(21,1)

Spectrum 853; b1=(2,2), b2=(2,4); singlets = 26

26(1,1) + 1(48,1) + 1(7,7) + 1(7,1) + 3(1,14) + 1(21,7) + 1(21,1)

B3 x B3

Spectrum 854; b1=(1,0), b2=(1,0); singlets = 170

170(1,1) + 3(7,1) + 3(1,7) + 8(1,8) + 8(8,1)

B3 x C2

Spectrum 855; b1=(5/2,5), b2=(3,8); singlets = 2

2(1,1) + 2(7,1) + 2(7,5) + 1/2(7,16) + 3(1,10) + 2(1,4) + 1/2(1,20) + 5/2(8,4) + 1/2(35,4)

Spectrum 856; b1=(5/2,5), b2=(3,8); singlets = 0

1/2(7,4) + 2(7,5) + 1/2(7,16) + 3(1,10) + 5/2(1,4) + 1/2(1,20) + 2(8,4) + 2(8,1) + 1/2(35,4)

Spectrum 857; b1=(5/2,5), b2=(3/2,3); singlets = 21

21(1,1) + 1/2(21,4) + 1(21,1) + 1(1,10) + 3/2(1,4) + 1(1,5) + 3(8,4) + 9(8,1) + 1(8,5) + 1(27,1)

Spectrum 858; b1=(5/2,5), b2=(3/2,3); singlets = 27

27(1,1) + 1/2(21,4) + 6(7,1) + 1(1,10) + 3/2(1,4) + 1(1,5) + 3(8,4) + 3(8,1) + 1(8,5) + 1(48,1)

Spectrum 859; b1=(5/2,5), b2=(3/2,3); singlets = 25

25(1,1) + 1/2(21,4) + 1/2(7,4) + 4(7,1) + 1(1,10) + 2(1,4) + 1(1,5) + 5/2(8,4) + 5(8,1) + 1(8,5) + 1(48,1)

Spectrum 860; b1=(5/2,5), b2=(3/2,3); singlets = 23

23(1,1) + 1/2(21,4) + 1(7,4) + 2(7,1) + 1(1,10) + 5/2(1,4) + 1(1,5) + 2(8,4) + 7(8,1) + 1(8,5) + 1(48,1)

Spectrum 861; b1=(5/2,5), b2=(3/2,3); singlets = 21

21(1,1) + 1/2(21,4) + 3/2(7,4) + 1(1,10) + 3(1,4) + 1(1,5) + 3/2(8,4) + 9(8,1) + 1(8,5) + 1(48,1)

Spectrum 862; b1=(5/2,5), b2=(3/2,3); singlets = 22

22(1,1) + 1/2(21,4) + 1(7,1) + 1(7,5) + 1(1,10) + 3/2(1,4) + 2(1,5) + 3(8,4) + 8(8,1) + 1(48,1)

Spectrum 863; b1=(5/2,5), b2=(3/2,3); singlets = 28

28(1,1) + 1/2(21,4) + 1(21,1) + 2(7,4) + 7(7,1) + 1(1,10) + 7/2(1,4) + 1(1,5) + 1(8,4) + 1(8,1) + 1(8,5) + 1(35,1)

Spectrum 864; b1=(5/2,5), b2=(3/2,3); singlets = 26

26(1,1) + 1/2(21,4) + 1(21,1) + 5/2(7,4) + 5(7,1) + 1(1,10) + 4(1,4) + 1(1,5) + 1/2(8,4) + 3(8,1) + 1(8,5) + 1(35,1)

Spectrum 865; b1=(5/2,5), b2=(3/2,3); singlets = 24

24(1,1) + 1/2(21,4) + 1(21,1) + 3(7,4) + 3(7,1) + 1(1,10) + 9/2(1,4) + 1(1,5) + 5(8,1) + 1(8,5) + 1(35,1)

Spectrum 866; b1=(5/2,5), b2=(3/2,3); singlets = 29

29(1,1) + 1/2(21,4) + 1(21,1) + 1/2(7,4) + 8(7,1) + 1(7,5) + 1(1,10) + 2(1,4) + 2(1,5) + 5/2(8,4) + 1(35,1)

Spectrum 867; b1=(5/2,5), b2=(3/2,3); singlets = 27

27(1,1) + 1/2(21,4) + 1(21,1) + 1(7,4) + 6(7,1) + 1(7,5) + 1(1,10) + 5/2(1,4) + 2(1,5) + 2(8,4) + 2(8,1) + 1(35,1)

Spectrum 868; b1=(5/2,5), b2=(3/2,3); singlets = 25

25(1,1) + 1/2(21,4) + 1(21,1) + 3/2(7,4) + 4(7,1) + 1(7,5) + 1(1,10) + 3(1,4) + 2(1,5) + 3/2(8,4) + 4(8,1) + 1(35,1)

Spectrum 869; b1=(5/2,5), b2=(3/2,3); singlets = 23

23(1,1) + 1/2(21,4) + 1(21,1) + 2(7,4) + 2(7,1) + 1(7,5) + 1(1,10) + 7/2(1,4) + 2(1,5) + 1(8,4) + 6(8,1) + 1(35,1)

Spectrum 870; b1=(5/2,5), b2=(3/2,3); singlets = 21

21(1,1) + 1/2(21,4) + 1(21,1) + 5/2(7,4) + 1(7,5) + 1(1,10) + 4(1,4) + 2(1,5) + 1/2(8,4) + 8(8,1) + 1(35,1)

Spectrum 871; b1=(5/2,5), b2=(3/2,3); singlets = 22

22(1,1) + 1(21,4) + 2(21,1) + 10(7,1) + 1(1,4) + 6(1,5) + 1/2(1,16) + 5/2(8,4)

Spectrum 872; b1=(5/2,5), b2=(3/2,3); singlets = 20

20(1,1) + 1(21,4) + 2(21,1) + 1/2(7,4) + 8(7,1) + 3/2(1,4) + 6(1,5) + 1/2(1,16) + 2(8,4) + 2(8,1)

Spectrum 873; b1=(5/2,5), b2=(3/2,3); singlets = 18

18(1,1) + 1(21,4) + 2(21,1) + 1(7,4) + 6(7,1) + 2(1,4) + 6(1,5) + 1/2(1,16) + 3/2(8,4) + 4(8,1)

Spectrum 874; b1=(5/2,5), b2=(3/2,3); singlets = 16

16(1,1) + 1(21,4) + 2(21,1) + 3/2(7,4) + 4(7,1) + 5/2(1,4) + 6(1,5) + 1/2(1,16) + 1(8,4) + 6(8,1)

Spectrum 875; b1=(5/2,5), b2=(3/2,3); singlets = 14

14(1,1) + 1(21,4) + 2(21,1) + 2(7,4) + 2(7,1) + 3(1,4) + 6(1,5) + 1/2(1,16) + 1/2(8,4) + 8(8,1)

Spectrum 876; b1=(5/2,5), b2=(3/2,3); singlets = 12

12(1,1) + 1(21,4) + 2(21,1) + 5/2(7,4) + 7/2(1,4) + 6(1,5) + 1/2(1,16) + 10(8,1)

Spectrum 877; b1=(5/2,5), b2=(3/2,3); singlets = 35

35(1,1) + 1/2(21,4) + 4(21,1) + 2(7,4) + 2(7,1) + 1(1,10) + 7/2(1,4) + 1(1,5) + 1(8,4) + 1(8,1) + 1(8,5)

Spectrum 878; b1=(5/2,5), b2=(3/2,3); singlets = 33

33(1,1) + 1/2(21,4) + 4(21,1) + 5/2(7,4) + 1(1,10) + 4(1,4) + 1(1,5) + 1/2(8,4) + 3(8,1) + 1(8,5)

Spectrum 879; b1=(5/2,5), b2=(3/2,3); singlets = 36

36(1,1) + 1/2(21,4) + 4(21,1) + 1/2(7,4) + 3(7,1) + 1(7,5) + 1(1,10) + 2(1,4) + 2(1,5) + 5/2(8,4)

Spectrum 880; b1=(5/2,5), b2=(3/2,3); singlets = 34

34(1,1) + 1/2(21,4) + 4(21,1) + 1(7,4) + 1(7,1) + 1(7,5) + 1(1,10) + 5/2(1,4) + 2(1,5) + 2(8,4) + 2(8,1)

Spectrum 881; b1=(5/2,5), b2=(3/2,3); singlets = 5

5(1,1) + 1(21,4) + 2(21,1) + 10(7,1) + 1(1,10) + 11(1,4) + 1(1,5) + 1/2(8,4) + 3(8,1) + 1(8,5)

Spectrum 882; $b_1=(5/2,5)$, $b_2=(3/2,3)$; singlets = 3
 $3(1,1) + 1(21,4) + 2(21,1) + 1/2(7,4) + 8(7,1) + 1(1,10) + 23/2(1,4) + 1(1,5) + 5(8,1) + 1(8,5)$
Spectrum 883; $b_1=(5/2,5)$, $b_2=(3/2,3)$; singlets = 0
 $1(21,4) + 2(21,1) + 5(7,1) + 1(7,5) + 1(1,10) + 11(1,4) + 2(1,5) + 1/2(8,4) + 8(8,1)$
Spectrum 884; $b_1=(1,0)$, $b_2=(9/2,13)$; singlets = 0
 $1/2(7,4) + 1(7,1) + 8(1,10) + 25/2(1,4) + 7(1,5) + 3(1,14) + 1/2(1,16) + 2(1,20) + 1/2(8,16)$
Spectrum 885; $b_1=(1,0)$, $b_2=(9/2,13)$; singlets = 0
 $1/2(7,4) + 1(7,1) + 5(1,10) + 33/2(1,4) + 3(1,5) + 2(1,14) + 7/2(1,16) + 2(1,20) + 1/2(8,16)$
Spectrum 886; $b_1=(1,0)$, $b_2=(9/2,13)$; singlets = 1
 $1(1,1) + 1/2(7,4) + 1(7,1) + 4(1,10) + 3/2(1,4) + 24(1,5) + 2(1,14) + 3(1,20) + 1/2(8,16)$
Spectrum 887; $b_1=(1,0)$, $b_2=(9/2,13)$; singlets = 1
 $1(1,1) + 1/2(7,4) + 1(7,1) + 1(1,10) + 11/2(1,4) + 20(1,5) + 1(1,14) + 3(1,16) + 3(1,20) + 1/2(8,16)$
Spectrum 888; $b_1=(1,0)$, $b_2=(9/2,13)$; singlets = 2
 $2(1,1) + 1/2(7,4) + 1(7,1) + 10(1,10) + 4(1,4) + 14(1,5) + 1(1,14) + 1/2(1,16) + 1/2(1,20) + 1(1,35) + 1/2(8,16)$
Spectrum 889; $b_1=(1,0)$, $b_2=(9/2,13)$; singlets = 2
 $2(1,1) + 1/2(7,4) + 1(7,1) + 13(1,10) + 7(1,4) + 5(1,5) + 5/2(1,16) + 1/2(1,20) + 1/2(1,40) + 1/2(8,16)$
Spectrum 890; $b_1=(1,0)$, $b_2=(9/2,13)$; singlets = 2
 $2(1,1) + 1/2(7,4) + 1(7,1) + 7(1,10) + 8(1,4) + 10(1,5) + 7/2(1,16) + 1/2(1,20) + 1(1,35) + 1/2(8,16)$
Spectrum 891; $b_1=(1,0)$, $b_2=(9/2,13)$; singlets = 4
 $4(1,1) + 1/2(7,4) + 1(7,1) + 5(1,10) + 6(1,4) + 15(1,5) + 2(1,14) + 3/2(1,16) + 5/2(1,20) + 1/2(8,16)$
Spectrum 892; $b_1=(1,0)$, $b_2=(9/2,13)$; singlets = 4
 $4(1,1) + 1/2(7,4) + 1(7,1) + 2(1,10) + 10(1,4) + 11(1,5) + 1(1,14) + 9/2(1,16) + 5/2(1,20) + 1/2(8,16)$
Spectrum 893; $b_1=(1,0)$, $b_2=(9/2,13)$; singlets = 5
 $5(1,1) + 1/2(7,4) + 1(7,1) + 17(1,10) + 15/2(1,4) + 1(1,14) + 1(1,16) + 1/2(1,40) + 1/2(8,16)$
Spectrum 894; $b_1=(1,0)$, $b_2=(9/2,13)$; singlets = 5
 $5(1,1) + 1/2(7,4) + 1(7,1) + 11(1,10) + 17/2(1,4) + 5(1,5) + 1(1,14) + 2(1,16) + 1(1,35) + 1/2(8,16)$
Spectrum 895; $b_1=(1,0)$, $b_2=(9/2,13)$; singlets = 5
 $5(1,1) + 1/2(7,4) + 1(7,1) + 8(1,10) + 25/2(1,4) + 1(1,5) + 5(1,16) + 1(1,35) + 1/2(8,16)$
Spectrum 896; $b_1=(1,0)$, $b_2=(9/2,13)$; singlets = 7
 $7(1,1) + 1/2(7,4) + 1(7,1) + 9(1,10) + 13/2(1,4) + 10(1,5) + 3(1,14) + 2(1,20) + 1/2(8,16)$
Spectrum 897; $b_1=(1,0)$, $b_2=(9/2,13)$; singlets = 7
 $7(1,1) + 1/2(7,4) + 1(7,1) + 6(1,10) + 21/2(1,4) + 6(1,5) + 2(1,14) + 3(1,16) + 2(1,20) + 1/2(8,16)$
Spectrum 898; $b_1=(1,0)$, $b_2=(9/2,13)$; singlets = 7
 $7(1,1) + 1/2(7,4) + 1(7,1) + 3(1,10) + 29/2(1,4) + 2(1,5) + 1(1,14) + 6(1,16) + 2(1,20) + 1/2(8,16)$
Spectrum 899; $b_1=(1,0)$, $b_2=(9/2,13)$; singlets = 9
 $9(1,1) + 1/2(7,4) + 1(7,1) + 14(1,10) + 1(1,4) + 8(1,5) + 2(1,16) + 1/2(1,20) + 1/2(1,40) + 1/2(8,16)$
Spectrum 900; $b_1=(1,0)$, $b_2=(9/2,13)$; singlets = 9
 $9(1,1) + 1/2(7,4) + 1(7,1) + 8(1,10) + 2(1,4) + 13(1,5) + 3(1,16) + 1/2(1,20) + 1(1,35) + 1/2(8,16)$
Spectrum 901; $b_1=(1,0)$, $b_2=(9/2,13)$; singlets = 10
 $10(1,1) + 1/2(7,4) + 1(7,1) + 10(1,10) + 11(1,4) + 1(1,5) + 3(1,14) + 3/2(1,16) + 3/2(1,20) + 1/2(8,16)$
Spectrum 902; $b_1=(1,0)$, $b_2=(9/2,13)$; singlets = 11
 $11(1,1) + 1/2(7,4) + 1(7,1) + 6(1,10) + 18(1,5) + 2(1,14) + 1(1,16) + 5/2(1,20) + 1/2(8,16)$
Spectrum 903; $b_1=(1,0)$, $b_2=(9/2,13)$; singlets = 11
 $11(1,1) + 1/2(7,4) + 1(7,1) + 3(1,10) + 4(1,4) + 14(1,5) + 1(1,14) + 4(1,16) + 5/2(1,20) + 1/2(8,16)$
Spectrum 904; $b_1=(1,0)$, $b_2=(9/2,13)$; singlets = 11
 $11(1,1) + 1/2(7,4) + 1(7,1) + 8(1,4) + 10(1,5) + 7(1,16) + 5/2(1,20) + 1/2(8,16)$
Spectrum 905; $b_1=(1,0)$, $b_2=(9/2,13)$; singlets = 12
 $12(1,1) + 1/2(7,4) + 1(7,1) + 18(1,10) + 3/2(1,4) + 3(1,5) + 1(1,14) + 1/2(1,16) + 1/2(1,40) + 1/2(8,16)$
Spectrum 906; $b_1=(1,0)$, $b_2=(9/2,13)$; singlets = 12
 $12(1,1) + 1/2(7,4) + 1(7,1) + 12(1,10) + 5/2(1,4) + 8(1,5) + 1(1,14) + 3/2(1,16) + 1(1,35) + 1/2(8,16)$
Spectrum 907; $b_1=(1,0)$, $b_2=(9/2,13)$; singlets = 12
 $12(1,1) + 1/2(7,4) + 1(7,1) + 9(1,10) + 13/2(1,4) + 4(1,5) + 9/2(1,16) + 1(1,35) + 1/2(8,16)$
Spectrum 908; $b_1=(1,0)$, $b_2=(9/2,13)$; singlets = 14
 $14(1,1) + 1/2(7,4) + 1(7,1) + 7(1,10) + 9/2(1,4) + 9(1,5) + 2(1,14) + 5/2(1,16) + 2(1,20) + 1/2(8,16)$
Spectrum 909; $b_1=(1,0)$, $b_2=(9/2,13)$; singlets = 14
 $14(1,1) + 1/2(7,4) + 1(7,1) + 4(1,10) + 17/2(1,4) + 5(1,5) + 1(1,14) + 11/2(1,16) + 2(1,20) + 1/2(8,16)$
Spectrum 910; $b_1=(1,0)$, $b_2=(9/2,13)$; singlets = 14
 $14(1,1) + 1/2(7,4) + 1(7,1) + 1(1,10) + 25/2(1,4) + 1(1,5) + 17/2(1,16) + 2(1,20) + 1/2(8,16)$
Spectrum 911; $b_1=(1,0)$, $b_2=(9/2,13)$; singlets = 17
 $17(1,1) + 1/2(7,4) + 1(7,1) +$

Spectrum 916; b1=(1,0), b2=(9/2,13); singlets = 21

21(1,1) + 1/2(7,4) + 1(7,1) + 2(1,10) + 13/2(1,4) + 4(1,5) + 8(1,16) + 2(1,20) + 1/2(8,16)

Spectrum 917; b1=(1,0), b2=(9/2,13); singlets = 24

24(1,1) + 1/2(7,4) + 1(7,1) + 9(1,10) + 3(1,4) + 3(1,5) + 2(1,14) + 7/2(1,16) + 3/2(1,20) + 1/2(8,16)

Spectrum 918; b1=(1,0), b2=(9/2,13); singlets = 28

28(1,1) + 1/2(7,4) + 1(7,1) + 3(1,10) + 1/2(1,4) + 7(1,5) + 15/2(1,16) + 2(1,20) + 1/2(8,16)

Spectrum 919; b1=(1,0), b2=(9/2,13); singlets = 31

31(1,1) + 1/2(7,4) + 1(7,1) + 7(1,10) + 1(1,4) + 2(1,5) + 1(1,14) + 6(1,16) + 3/2(1,20) + 1/2(8,16)

Spectrum 920; b1=(1,0), b2=(3/2,3); singlets = 97

97(1,1) + 3(7,1) + 30(1,4) + 6(1,5) + 1/2(1,16) + 3/2(8,4) + 2(8,1)

Spectrum 921; b1=(1,0), b2=(3/2,3); singlets = 95

95(1,1) + 1/2(7,4) + 1(7,1) + 61/2(1,4) + 6(1,5) + 1/2(1,16) + 1(8,4) + 4(8,1)

Spectrum 922; b1=(1,0), b2=(3/2,3); singlets = 104

104(1,1) + 3(7,1) + 1(1,10) + 24(1,4) + 9(1,5) + 3/2(8,4) + 2(8,1)

Spectrum 923; b1=(1,0), b2=(3/2,3); singlets = 102

102(1,1) + 1/2(7,4) + 1(7,1) + 1(1,10) + 49/2(1,4) + 9(1,5) + 1(8,4) + 4(8,1)

B3 x C4

Spectrum 924; b1=(1,0), b2=(1,0); singlets = 100

100(1,1) + 3(7,1) + 16(1,8) + 1(1,27) + 8(8,1)

B3 x C5

Spectrum 925; b1=(1,0), b2=(3/2,1); singlets = 19

19(1,1) + 3(7,1) + 6(1,10) + 4(1,44) + 1/2(8,10) + 3(8,1)

B3 x D5

Spectrum 926; b1=(5/2,5), b2=(1,-2); singlets = 15

15(1,1) + 5(7,1) + 2(1,10) + 26(8,1) + 1(27,1) + 1(35,1)

Spectrum 927; b1=(5/2,5), b2=(1,-2); singlets = 22

22(1,1) + 3(21,1) + 2(1,10) + 26(8,1) + 1(27,1)

Spectrum 928; b1=(5/2,5), b2=(1,-2); singlets = 28

28(1,1) + 2(21,1) + 6(7,1) + 2(1,10) + 20(8,1) + 1(48,1)

Spectrum 929; b1=(5/2,5), b2=(1,-2); singlets = 30

30(1,1) + 20(7,1) + 2(1,10) + 10(8,1) + 2(35,1)

Spectrum 930; b1=(5/2,5), b2=(1,-2); singlets = 37

37(1,1) + 3(21,1) + 15(7,1) + 2(1,10) + 10(8,1) + 1(35,1)

Spectrum 931; b1=(5/2,5), b2=(1,-2); singlets = 44

44(1,1) + 6(21,1) + 10(7,1) + 2(1,10) + 10(8,1)

B3 x G2

Spectrum 932; b1=(1,0), b2=(2,4); singlets = 67

67(1,1) + 3(7,1) + 23(1,7) + 1(1,27) + 1(8,7) + 1(8,1)

Spectrum 933; b1=(1,0), b2=(2,4); singlets = 73

73(1,1) + 3(7,1) + 3(1,14) + 20(1,7) + 1(8,7) + 1(8,1)

B4 x C2

Spectrum 934; b1=(2,2), b2=(3/2,3); singlets = 48

48(1,1) + 1(36,1) + 3/2(9,4) + 2(9,1) + 9/2(1,4) + 6(1,5) + 1/2(1,16) + 3/2(16,4) + 2(16,1)

Spectrum 935; b1=(2,2), b2=(3/2,3); singlets = 44

44(1,1) + 1(36,1) + 1/2(9,4) + 6(9,1) + 11/2(1,4) + 6(1,5) + 1/2(1,16) + 2(16,4)

Spectrum 936; b1=(2,2), b2=(3/2,3); singlets = 32

32(1,1) + 1(36,1) + 1/2(9,4) + 1(9,1) + 1(9,5) + 1(1,10) + 31/2(1,4) + 1(16,4) + 4(16,1)

B4 x C4

Spectrum 937; b1=(2,2), b2=(1,0); singlets = 21

21(1,1) + 1(36,1) + 1(9,8) + 7(1,8) + 1(1,27) + 8(16,1)

Spectrum 938; b1=(2,2), b2=(1,0); singlets = 13

13(1,1) + 1(36,1) + 8(9,1) + 8(1,8) + 1(1,27) + 1/2(16,8) + 4(16,1)

B4 x D5

Spectrum 939; b1=(7/2,7), b2=(1,-2); singlets = 11

11(1,1) + 2(36,1) + 1(9,1) + 2(1,10) + 1(16,1) + 1(84,1) + 1(128,1)

Spectrum 940; b1=(7/2,7), b2=(1,-2); singlets = 11

11(1,1) + 2(36,1) + 1(9,1) + 2(1,10) + 1(16,1) + 1(44,1) + 2(84,1)

B4 x G2

Spectrum 941; b1=(2,2), b2=(2,4); singlets = 35

35(1,1) + 1(36,1) + 1(9,7) + 1(9,1) + 6(1,7) + 1(1,27) + 1(16,7) + 1(16,1)

Spectrum 942; b1=(2,2), b2=(2,4); singlets = 41

41(1,1) + 1(36,1) + 1(9,7) + 1(9,1) + 3(1,14) + 3(1,7) + 1(16,7) + 1(16,1)

C2 x C2

Spectrum 943; b1=(6,18), b2=(3/2,3); singlets = 2

2(1,1) + 1/2(10,4) + 1(4,4) + 1(1,10) + 1(1,5) + 1/2(5,4) + 1/2(14,4) + 4(16,1) + 1/2(16,5) + 2(20,1) + 1/2(35,4) + 1(35,1)

Spectrum 944; b1=(6,18), b2=(3/2,3); singlets = 10

10(1,1) + 3/2(4,1) + 3/2(4,5) + 1(1,4) + 1/2(1,16) + 3/2(14,4) + 3(14,1) + 1(16,1) + 1(20,4) + 3(20,1)

Spectrum 945; b1=(9/2,13), b2=(3/2,3); singlets = 0

5/2(10,4) + 5(10,1) + 5(4,1) + 3/2(4,5) + 1/2(1,4) + 1/2(1,16) + 1/2(5,4) + 4(5,1) + 1(14,4) + 1(14,1) + 3/2(20,1)

Spectrum 946; b1=(9/2,13), b2=(3/2,3); singlets = 0

5/2(10,4) + 2(10,1) + 9(4,1) + 3/2(4,5) + 1/2(1,4) + 1/2(1,16) + 1/2(5,4) + 1(14,4) + 3(16,1) + 3/2(20,1)

Spectrum 947; b1=(9/2,13), b2=(3/2,3); singlets = 4

4(1,1) + 1(10,4) + 4(10,1) + 17/2(4,1) + 3/2(4,5) + 1/2(1,16) + 1(5,1) + 2(16,4) + 1(16,1) + 1(35,1)

Spectrum 948; b1=(9/2,13), b2=(3/2,3); singlets = 1

1(1,1) + 3/2(10,4) + 3(10,1) + 1(4,4) + 6(4,1) + 1(1,5) + 1/2(1,16) + 5(5,1) + 1(5,5) + 1/2(14,4) + 1(14,1) + 1(16,4) + 2(20,1)

Spectrum 949; b1=(9/2,13), b2=(3/2,3); singlets = 1

1(1,1) + 3/2(10,4) + 1(4,4) + 10(4,1) + 1(1,5) + 1/2(1,16) + 1(5,1) + 1(5,5) + 1/2(14,4) + 1(16,4) + 3(16,1) + 2(20,1)

Spectrum 950; b1=(9/2,13), b2=(3/2,3); singlets = 0

1/2(10,4) + 3(4,1) + 1(1,10) + 1(1,5) + 3(5,4) + 11(5,1) + 1(14,1) + 1(16,4) + 1/2(16,5) + 3(20,1)

Spectrum 951; b1=(9/2,13), b2=(3/2,3); singlets = 1

1(1,1) + 1/2(10,4) + 6(10,1) + 11/2(4,1) + 1(1,10) + 1(1,5) + 3(5,4) + 1(5,1) + 1(16,4) + 1/2(16,1) + 1/2(16,5) + 1/2(20,1) + 1(35,1)

Spectrum 952; b1=(9/2,13), b2=(3/2,3); singlets = 7

7(1,1) + 5/2(10,4) + 3(10,1) + 3(4,1) + 3/2(4,5) + 1/2(1,4) + 1/2(1,16) + 1/2(5,4) + 3(5,1) + 1(14,4) + 5/2(16,1) + 3/2(20,1)

Spectrum 953; b1=(9/2,13), b2=(3/2,3); singlets = 5

5(1,1) + 1/2(10,4) + 1(10,1) + 15/2(4,1) + 1(1,5) + 1/2(1,16) + 1(5,4) + 5(5,1) + 1(5,5) + 1(14,1) + 2(16,4) + 5/2(20,1)

Spectrum 954; b1=(9/2,13), b2=(3/2,3); singlets = 3

3(1,1) + 1/2(10,4) + 1(10,1) + 15/2(4,1) + 1(1,10) + 1(1,5) + 3(5,4) + 2(5,1) + 1(14,1) + 1(16,4) + 3/2(16,1) + 1/2(16,5) + 5/2(20,1)

Spectrum 955; b1=(9/2,13), b2=(3/2,3); singlets = 0

7/2(10,4) + 1(10,1) + 2(4,1) + 1(1,10) + 1(1,4) + 1(1,5) + 12(5,1) + 2(14,1) + 1/2(16,5) + 1(35,1)

Spectrum 956; b1=(9/2,13), b2=(3/2,3); singlets = 0

7/2(10,4) + 4(10,1) + 5(4,1) + 1(1,10) + 1(1,4) + 1(1,5) + 3(5,1) + 1(14,1) + 2(16,1) + 1/2(16,5) + 1/2(40,1)

Spectrum 957; b1=(9/2,13), b2=(3/2,3); singlets = 11

11(1,1) + 1(10,4) + 5(10,1) + 5/2(4,1) + 3/2(4,5) + 1/2(1,16) + 4(5,1) + 2(16,4) + 1/2(16,1) + 1(35,1)

Spectrum 958; b1=(9/2,13), b2=(3/2,3); singlets = 0

1(10,4) + 5(10,1) + 1(4,4) + 6(4,1) + 1(1,10) + 1(1,4) + 1(1,5) + 1(5,4) + 1(16,4) + 2(16,1) + 1/2(16,5) + 1(35,1)

Spectrum 959; b1=(9/2,13), b2=(3/2,3); singlets = 6

6(1,1) + 2(10,4) + 5(10,1) + 1/2(1,4) + 1(1,5) + 1/2(1,16) + 3/2(5,4) + 6(5,1) + 1(5,5) + 1(14,4) + 1/2(16,1) + 2(20,1)

Spectrum 960; b1=(9/2,13), b2=(3/2,3); singlets = 6

6(1,1) + 3/2(10,4) + 4(10,1) + 1(4,4) + 1(1,10) + 1(1,5) + 2(5,4) + 5(5,1) + 1/2(14,4) + 1(14,1) + 1(16,1) + 1/2(16,5) + 2(20,1)

Spectrum 961; b1=(9/2,13), b2=(3/2,3); singlets = 0

3/2(10,4) + 4(10,1) + 4(4,1) + 1(1,10) + 3/2(1,4) + 1(1,5) + 5/2(5,4) + 3(5,1) + 1/2(14,4) + 1(14,1) + 1(16,1) + 1/2(16,5) + 2(20,1)

Spectrum 962; b1=(9/2,13), b2=(3/2,3); singlets = 8

8(1,1) + 3/2(10,4) + 1(10,1) + 1(4,4) + 4(4,1) + 1(1,5) + 1/2(1,16) + 4(5,1) + 1(5,5) + 1/2(14,4) + 1(16,4) + 5/2(16,1) + 2(20,1)

Spectrum 963; b1=(9/2,13), b2=(3/2,3); singlets = 2

2(1,1) + 3/2(10,4) + 1(10,1) + 8(4,1) + 3/2(1,4) + 1(1,5) + 1/2(1,16) + 1/2(5,4) + 2(5,1) + 1(5,5) + 1/2(14,4) + 1(16,4) + 5/2(16,1) + 2(20,1)

Spectrum 964; b1=(9/2,13), b2=(3/2,3); singlets = 2

2(1,1) + 1(10,4) + 3(10,1) + 1(4,4) + 4(4,1) + 1(1,10) + 1(1,4) + 1(1,5) + 1(5,4) + 5(5,1) + 2(14,1) + 1(16,4) + 1/2(16,5) + 2(20,1)

Spectrum 965; b1=(9/2,13), b2=(3/2,3); singlets = 6

6(1,1) + 3/2(10,4) + 1(10,1) + 1(4,4) + 4(4,1) + 1(1,10) + 1(1,5) + 2(5,4) + 1(5,1) + 1/2(14,4) + 4(16,1) + 1/2(16,5) + 2(20,1)

Spectrum 966; b1=(9/2,13), b2=(3/2,3); singlets = 13

13(1,1) + 1(10,4) + 9/2(4,1) + 3/2(4,5) + 1/2(1,16) + 5(5,1) + 1(14,1) + 2(16,4) + 3/2(16,1) + 2(20,1)

Spectrum 967; b1=(9/2,13), b2=(3/2,3); singlets = 2

2(1,1) + 1(10,4) + 1(4,4) + 8(4,1) + 1(1,10) + 1(1,4) + 1(1,5) + 1(5,4) + 1(5,1) + 1(14,1) + 1(16,4) + 3(16,1) + 1/2(16,5) + 2(20,1)

Spectrum 968; b1=(9/2,13), b2=(3/2,3); singlets = 5

5(1,1) + 2(10,4) + 3(10,1) + 2(4,4) + 1/2(4,1) + 1(1,10) + 1(1,4) + 1(1,5) + 4(5,1) + 1/2(14,4) + 1(14,1) + 5/2(16,1) + 1/2(16,5) + 3/2(20,1)

Spectrum 969; b1=(9/2,13), b2=(3/2,3); singlets = 16

16(1,1) + 1(10,4) + 4(10,1) + 5(4,1) + 3/2(4,5) + 1/2(1,16) + 2(14,1) + 2(16,4) + 3/2(20,1)

Spectrum 970; b1=(9/2,13), b2=(3/2,3); singlets = 5

5(1,1) + 2(10,4) + 2(4,4) + 9/2(4,1) + 1(1,10) + 1(1,4) + 1(1,5) + 1/2(14,4) + 11/2(16,1) + 1/2(16,5) + 3/2(20,1)

Spectrum 971; b1=(9/2,13), b2=(3/2,3); singlets = 10

10(1,1) + 1/2(10,4) + 2(10,1) + 3/2(4,1) + 1(1,10) + 1(1,5) + 3(5,4) + 5(5,1) + 1(14,1) + 1(16,4) + 1(16,1) + 1/2(16,5) + 5/2(20,1)

Spectrum 972; b1=(9/2,13), b2=(3/2,3); singlets = 0

1(10,4) + 3/2(4,1) + 1(1,10) + 5/2(1,4) + 1(1,5) + 3/2(5,4) + 11(5,1) + 1(14,1) + 1(16,4) + 1(16,1) + 1/2(16,5) + 5/2(20,1)

Spectrum 973; b1=(9/2,13), b2=(3/2,3); singlets = 7

7(1,1) + 7/2(10,4) + 2(10,1) + 3(4,1) + 1(1,10) + 1(1,4) + 1(1,5) + 2(5,1) + 9/2(16,1) + 1/2(16,5) + 1/2(40,1)

$$5(1,1) + 1(10,4) + 1(10,1) + 2(4,10) + 2(4,4) + 2(4,5) + 2(1,4) + 1(1,5) + 1/2(1,16) + 1/2(5,16) + 2(16,1) + 1/2(16,5)$$

$$3(1,1) + 1/2(10,4) + 39/2(4,1) + 1(4,5) + 1(1,10) + 1(1,4) + 6(5,4) + 6(5,1) + 1(5,5) + 3/2(20,1)$$

Spectrum 1076; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 9
 $9(1,1) + 3/2(10,4) + 1(10,1) + 4(4,4) + 19(4,1) + 1(4,5) + 1(1,10) + 1(5,4) + 1(5,5) + 1(14,1) + 2(16,1)$
Spectrum 1077; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 3
 $3(1,1) + 4(10,1) + 6(4,4) + 25/2(4,1) + 3/2(4,5) + 2(1,4) + 1/2(1,16) + 5(5,1) + 1(16,4) + 1(16,1)$
Spectrum 1078; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 4
 $4(1,1) + 4(10,1) + 5(4,4) + 14(4,1) + 2(4,5) + 1(1,10) + 1(1,5) + 5(5,1) + 1(16,4) + 1(16,1)$
Spectrum 1079; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 4
 $4(1,1) + 4(10,1) + 9(4,4) + 8(4,1) + 1(1,10) + 1(1,5) + 5(5,1) + 5/2(16,1) + 1/2(16,5)$
Spectrum 1080; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 10
 $10(1,1) + 1(10,4) + 3(10,1) + 3(4,4) + 35/2(4,1) + 1(1,5) + 1/2(1,16) + 4(5,4) + 1(5,5) + 1(14,1) + 1/2(20,1)$
Spectrum 1081; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 3
 $3(1,1) + 3/2(10,4) + 1(10,1) + 1(4,4) + 41/2(4,1) + 1(4,5) + 1/2(1,4) + 2(1,5) + 1/2(1,16) + 9/2(5,4) + 3(5,1) + 1(14,1) + 1/2(20,1)$
Spectrum 1082; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 15
 $15(1,1) + 3/2(10,4) + 1(10,1) + 3(4,4) + 10(4,1) + 3/2(4,5) + 1/2(1,16) + 3(5,4) + 9(5,1) + 1(14,1) + 1/2(20,1)$
Spectrum 1083; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 9
 $9(1,1) + 3/2(10,4) + 1(10,1) + 2(4,4) + 14(4,1) + 3/2(4,5) + 3/2(1,4) + 1/2(1,16) + 7/2(5,4) + 7(5,1) + 1(14,1) + 1/2(20,1)$
Spectrum 1084; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 3
 $3(1,1) + 3/2(10,4) + 1(10,1) + 1(4,4) + 18(4,1) + 3/2(4,5) + 3(1,4) + 1/2(1,16) + 4(5,4) + 5(5,1) + 1(14,1) + 1/2(20,1)$
Spectrum 1085; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 6
 $6(1,1) + 3/2(10,4) + 1(10,1) + 4(4,4) + 27/2(4,1) + 1(1,4) + 1(1,5) + 1/2(1,16) + 2(5,4) + 8(5,1) + 1(5,5) + 1(14,1) + 1/2(20,1)$
Spectrum 1086; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 0
 $3/2(10,4) + 1(10,1) + 3(4,4) + 35/2(4,1) + 5/2(1,4) + 1(1,5) + 1/2(1,16) + 5/2(5,4) + 6(5,1) + 1(5,5) + 1(14,1) + 1/2(20,1)$
Spectrum 1087; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 11
 $11(1,1) + 1(10,4) + 3(10,1) + 2(4,4) + 33/2(4,1) + 1(4,5) + 1(1,10) + 1/2(1,4) + 7/2(5,4) + 2(5,1) + 1(5,5) + 1(14,1) + 1/2(20,1)$
Spectrum 1088; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 5
 $5(1,1) + 1(10,4) + 3(10,1) + 1(4,4) + 41/2(4,1) + 1(4,5) + 1(1,10) + 2(1,4) + 4(5,4) + 1(5,5) + 1(14,1) + 1/2(20,1)$
Spectrum 1089; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 4
 $4(1,1) + 3/2(10,4) + 1(10,1) + 39/2(4,1) + 2(4,5) + 1(1,10) + 1(1,4) + 1(1,5) + 4(5,4) + 5(5,1) + 1(14,1) + 1/2(20,1)$
Spectrum 1090; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 1
 $1(1,1) + 3/2(10,4) + 1(10,1) + 2(4,4) + 19(4,1) + 1/2(4,5) + 1(1,10) + 1/2(1,4) + 2(1,5) + 5/2(5,4) + 6(5,1) + 1(5,5) + 1(14,1) + 1/2(20,1)$
Spectrum 1091; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 13
 $13(1,1) + 3/2(10,4) + 1(10,1) + 4(4,4) + 17/2(4,1) + 1(4,5) + 1(1,10) + 1(5,4) + 12(5,1) + 1(5,5) + 1(14,1) + 1/2(20,1)$
Spectrum 1092; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 7
 $7(1,1) + 3/2(10,4) + 1(10,1) + 3(4,4) + 25/2(4,1) + 1(4,5) + 1(1,10) + 3/2(1,4) + 3/2(5,4) + 10(5,1) + 1(5,5) + 1(14,1) + 1/2(20,1)$
Spectrum 1093; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 1
 $1(1,1) + 3/2(10,4) + 1(10,1) + 2(4,4) + 33/2(4,1) + 1(4,5) + 1(1,10) + 3(1,4) + 2(5,4) + 8(5,1) + 1(5,5) + 1(14,1) + 1/2(20,1)$
Spectrum 1094; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 2
 $2(1,1) + 4(10,1) + 8(4,4) + 3/2(4,1) + 1(1,10) + 3/2(1,4) + 1(1,5) + 1/2(5,4) + 15(5,1) + 1/2(16,1) + 1/2(16,5) + 1/2(20,1)$
Spectrum 1095; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 18
 $18(1,1) + 3/2(10,4) + 2(10,1) + 3(4,4) + 29/2(4,1) + 3/2(4,5) + 1/2(1,16) + 3(5,4) + 1(14,1) + 3/2(16,1)$
Spectrum 1096; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 8
 $8(1,1) + 2(10,4) + 3(4,4) + 17(4,1) + 1(4,5) + 2(1,5) + 1/2(1,16) + 2(5,4) + 4(5,1) + 1(14,1) + 3/2(16,1)$
Spectrum 1097; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 2
 $2(1,1) + 2(10,4) + 2(4,4) + 21(4,1) + 1(4,5) + 3/2(1,4) + 2(1,5) + 1/2(1,16) + 5/2(5,4) + 2(5,1) + 1(14,1) + 3/2(16,1)$
Spectrum 1098; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 14
 $14(1,1) + 2(10,4) + 4(4,4) + 21/2(4,1) + 3/2(4,5) + 1(1,4) + 1/2(1,16) + 1(5,4) + 8(5,1) + 1(14,1) + 3/2(16,1)$
Spectrum 1099; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 8
 $8(1,1) + 2(10,4) + 3(4,4) + 29/2(4,1) + 3/2(4,5) + 5/2(1,4) + 1/2(1,16) + 3/2(5,4) + 6(5,1) + 1(14,1) + 3/2(16,1)$
Spectrum 1100; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 2
 $2(1,1) + 2(10,4) + 2(4,4) + 37/2(4,1) + 3/2(4,5) + 4(1,4) + 1/2(1,16) + 2(5,4) + 4(5,1) + 1(14,1) + 3/2(16,1)$
Spectrum 1101; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 5
 $5(1,1) + 2(10,4) + 5(4,4) + 14(4,1) + 2(1,4) + 1(1,5) + 1/2(1,16) + 7(5,1) + 1(5,5) + 1(14,1) + 3/2(16,1)$
Spectrum 1102; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 16
 $16(1,1) + 3/2(10,4) + 2(10,1) + 4(4,4) + 13(4,1) + 1(4,5) + 1(1,10) + 1(5,4) + 3(5,1) + 1(5,5)$

$0(1,1) + 5(10,1) + 6(4,4) + 13/2(4,1) + 3/2(4,5) + 2(1,4) + 1/2(1,16) + 8(5,1) + 1(16,4) + 1/2(16,1)$
 Spectrum 1111; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 4
 $4(1,1) + 5(10,1) + 5(4,4) + 21/2(4,1) + 3/2(4,5) + 7/2(1,4) + 1/2(1,16) + 1/2(5,4) + 6(5,1) + 1(16,4) + 1/2(16,1)$
 Spectrum 1112; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 11
 $11(1,1) + 5(10,1) + 5(4,4) + 8(4,1) + 2(4,5) + 1(1,10) + 1(1,5) + 8(5,1) + 1(16,4) + 1/2(16,1)$
 Spectrum 1113; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 5
 $5(1,1) + 5(10,1) + 4(4,4) + 12(4,1) + 2(4,5) + 1(1,10) + 3/2(1,4) + 1(1,5) + 1/2(5,4) + 6(5,1) + 1(16,4) + 1/2(16,1)$
 Spectrum 1114; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 11
 $11(1,1) + 5(10,1) + 9(4,4) + 2(4,1) + 1(1,10) + 1(1,5) + 8(5,1) + 2(16,1) + 1/2(16,5)$
 Spectrum 1115; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 5
 $5(1,1) + 5(10,1) + 8(4,4) + 6(4,1) + 1(1,10) + 3/2(1,4) + 1(1,5) + 1/2(5,4) + 6(5,1) + 2(16,1) + 1/2(16,5)$
 Spectrum 1116; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 17
 $17(1,1) + 1(10,4) + 1(4,4) + 23/2(4,1) + 3/2(4,5) + 1/2(1,4) + 1/2(1,16) + 11/2(5,4) + 7(5,1) + 1(16,1) + 1(20,1)$
 Spectrum 1117; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 11
 $11(1,1) + 1(10,4) + 31/2(4,1) + 3/2(4,5) + 2(1,4) + 1/2(1,16) + 6(5,4) + 5(5,1) + 1(16,1) + 1(20,1)$
 Spectrum 1118; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 14
 $14(1,1) + 1(10,4) + 3(4,4) + 11(4,1) + 1(1,5) + 1/2(1,16) + 4(5,4) + 8(5,1) + 1(5,5) + 1(16,1) + 1(20,1)$
 Spectrum 1119; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 8
 $8(1,1) + 1(10,4) + 2(4,4) + 15(4,1) + 3/2(1,4) + 1(1,5) + 1/2(1,16) + 9/2(5,4) + 6(5,1) + 1(5,5) + 1(16,1) + 1(20,1)$
 Spectrum 1120; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 2
 $2(1,1) + 1(10,4) + 1(4,4) + 19(4,1) + 3(1,4) + 1(1,5) + 1/2(1,16) + 5(5,4) + 4(5,1) + 1(5,5) + 1(16,1) + 1(20,1)$
 Spectrum 1121; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 13
 $13(1,1) + 1/2(10,4) + 2(10,1) + 18(4,1) + 1(4,5) + 1(1,10) + 1(1,4) + 6(5,4) + 1(5,5) + 1(16,1) + 1(20,1)$
 Spectrum 1122; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 3
 $3(1,1) + 1(10,4) + 41/2(4,1) + 1/2(4,5) + 1(1,10) + 1(1,4) + 2(1,5) + 5(5,4) + 4(5,1) + 1(5,5) + 1(16,1) + 1(20,1)$
 Spectrum 1123; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 15
 $15(1,1) + 1(10,4) + 2(4,4) + 10(4,1) + 1(4,5) + 1(1,10) + 1/2(1,4) + 7/2(5,4) + 10(5,1) + 1(5,5) + 1(16,1) + 1(20,1)$
 Spectrum 1124; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 9
 $9(1,1) + 1(10,4) + 1(4,4) + 14(4,1) + 1(4,5) + 1(1,10) + 2(1,4) + 4(5,4) + 8(5,1) + 1(5,5) + 1(16,1) + 1(20,1)$
 Spectrum 1125; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 3
 $3(1,1) + 1(10,4) + 18(4,1) + 1(4,5) + 1(1,10) + 7/2(1,4) + 9/2(5,4) + 6(5,1) + 1(5,5) + 1(16,1) + 1(20,1)$
 Spectrum 1126; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 18
 $18(1,1) + 1(10,4) + 1(10,1) + 2(4,4) + 29/2(4,1) + 1(4,5) + 1(1,10) + 1/2(1,4) + 7/2(5,4) + 1(5,1) + 1(5,5) + 5/2(16,1) + 1/2(20,1)$
 Spectrum 1127; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 25
 $25(1,1) + 3/2(10,4) + 3(10,1) + 3(4,4) + 17/2(4,1) + 3/2(4,5) + 1/2(1,16) + 3(5,4) + 3(5,1) + 1(14,1) + 1(16,1)$
 Spectrum 1128; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 19
 $19(1,1) + 3/2(10,4) + 3(10,1) + 2(4,4) + 25/2(4,1) + 3/2(4,5) + 3/2(1,4) + 1/2(1,16) + 7/2(5,4) + 1(5,1) + 1(14,1) + 1(16,1)$
 Spectrum 1129; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 16
 $16(1,1) + 3/2(10,4) + 3(10,1) + 4(4,4) + 12(4,1) + 1(1,4) + 1(1,5) + 1/2(1,16) + 2(5,4) + 2(5,1) + 1(5,5) + 1(14,1) + 1(16,1)$
 Spectrum 1130; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 10
 $10(1,1) + 3/2(10,4) + 3(10,1) + 3(4,4) + 16(4,1) + 5/2(1,4) + 1(1,5) + 1/2(1,16) + 5/2(5,4) + 1(5,5) + 1(14,1) + 1(16,1)$
 Spectrum 1131; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 3
 $3(1,1) + 2(10,4) + 1(10,1) + 1(4,4) + 43/2(4,1) + 1/2(4,5) + 1/2(1,4) + 4(1,5) + 1/2(1,16) + 7/2(5,4) + 1(5,1) + 1(14,1) + 1(16,1)$
 Spectrum 1132; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 15
 $15(1,1) + 2(10,4) + 1(10,1) + 3(4,4) + 11(4,1) + 1(4,5) + 2(1,5) + 1/2(1,16) + 2(5,4) + 7(5,1) + 1(14,1) + 1(16,1)$
 Spectrum 1133; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 9
 $9(1,1) + 2(10,4) + 1(10,1) + 2(4,4) + 15(4,1) + 1(4,5) + 3/2(1,4) + 2(1,5) + 1/2(1,16) + 5/2(5,4) + 5(5,1) + 1(14,1) + 1(16,1)$
 Spectrum 1134; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 3
 $3(1,1) + 2(10,4) + 1(10,1) + 1(4,4) + 25/2(4,1) + 3/2(4,5) + 4(1,4) + 1/2(1,16) + 3(5,4) + 3(5,1) + 1(14,1) + 1(16,1)$
 Spectrum 1135; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 21
 $21(1,1) + 2(10,4) + 1(10,1) + 4(4,4) + 9/2(4,1) + 3/2(4,5) + 1(1,4) + 1/2(1,16) + 1(5,4) + 11(5,1) + 1(14,1) + 1(16,1)$
 Spectrum 1136; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 15
 $15(1,1) + 2(10,4) + 1(10,1) + 3(4,4) + 17/2(4,1) + 3/2(4,5) + 5/2(1,4) + 1/2(1,16) + 3/2(5,4) + 9(5,1) + 1(14,1) + 1(16,1)$
 Spectrum 1137; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 9
 $9(1,1) + 2(10,4) + 1$

Spectrum 1178; b1={3,8}, b2={3/2,3}; singlets = 18

18(1,1) + 1(10,4) + 1(10,1) + 19/2(4,1) + 3/2(4,5) + 2(1,4) + 1/2(1,16) + 6(5,4) + 8(5,1) + 1/2(16,1) + 1(20,1)

Spectrum 1179; b1={3,8}, b2={3/2,3}; singlets = 21

21(1,1) + 1(10,4) + 1(10,1) + 3(4,4) + 5(4,1) + 1(1,5) + 1/2(1,16) + 4(5,4) + 11(5,1) + 1(5,5) + 1/2(16,1) + 1(20,1)

Spectrum 1180; b1={3,8}, b2={3/2,3}; singlets = 15

15(1,1) + 1(10,4) + 1(10,1) + 2(4,4) + 9(4,1) + 3/2(1,4) + 1(1,5) + 1/2(1,16) + 9/2(5,4) + 9(5,1) + 1(5,5) + 1/2(16,1) + 1(20,1)

Spectrum 1181; b1={3,8}, b2={3/2,3}; singlets = 9

9(1,1) + 1(10,4) + 1(10,1) + 1(4,4) + 13(4,1) + 3(1,4) + 1(1,5) + 1/2(1,16) + 5(5,4) + 7(5,1) + 1(5,5) + 1/2(16,1) + 1(20,1)

Spectrum 1182; b1={3,8}, b2={3/2,3}; singlets = 3

3(1,1) + 1(10,4) + 1(10,1) + 17(4,1) + 9/2(1,4) + 1(1,5) + 1/2(1,16) + 11/2(5,4) + 5(5,1) + 1(5,5) + 1/2(16,1) + 1(20,1)

Spectrum 1183; b1={3,8}, b2={3/2,3}; singlets = 20

20(1,1) + 1/2(10,4) + 3(10,1) + 12(4,1) + 1(4,5) + 1(1,10) + 1(1,4) + 6(5,4) + 3(5,1) + 1(5,5) + 1/2(16,1) + 1(20,1)

Spectrum 1184; b1={3,8}, b2={3/2,3}; singlets = 10

10(1,1) + 1(10,4) + 1(10,1) + 29/2(4,1) + 1/2(4,5) + 1(1,10) + 1(1,4) + 2(1,5) + 5(5,4) + 7(5,1) + 1(5,5) + 1/2(16,1) + 1(20,1)

Spectrum 1185; b1={3,8}, b2={3/2,3}; singlets = 22

22(1,1) + 1(10,4) + 1(10,1) + 2(4,4) + 4(4,1) + 1(4,5) + 1(1,10) + 1/2(1,4) + 7/2(5,4) + 13(5,1) + 1(5,5) + 1/2(16,1) + 1(20,1)

Spectrum 1186; b1={3,8}, b2={3/2,3}; singlets = 16

16(1,1) + 1(10,4) + 1(10,1) + 1(4,4) + 8(4,1) + 1(4,5) + 1(1,10) + 2(1,4) + 4(5,4) + 11(5,1) + 1(5,5) + 1/2(16,1) + 1(20,1)

Spectrum 1187; b1={3,8}, b2={3/2,3}; singlets = 10

10(1,1) + 1(10,4) + 1(10,1) + 12(4,1) + 1(4,5) + 1(1,10) + 7/2(1,4) + 9/2(5,4) + 9(5,1) + 1(5,5) + 1/2(16,1) + 1(20,1)

Spectrum 1188; b1={3,8}, b2={3/2,3}; singlets = 27

27(1,1) + 1(10,4) + 2(10,1) + 1(4,4) + 10(4,1) + 3/2(4,5) + 1/2(1,4) + 1/2(1,16) + 11/2(5,4) + 1(5,1) + 2(16,1) + 1/2(20,1)

Spectrum 1189; b1={3,8}, b2={3/2,3}; singlets = 24

24(1,1) + 1(10,4) + 2(10,1) + 3(4,4) + 19/2(4,1) + 1(1,5) + 1/2(1,16) + 4(5,4) + 2(5,1) + 1(5,5) + 2(16,1) + 1/2(20,1)

Spectrum 1190; b1={3,8}, b2={3/2,3}; singlets = 18

18(1,1) + 1(10,4) + 2(10,1) + 2(4,4) + 27/2(4,1) + 3/2(1,4) + 1(1,5) + 1/2(1,16) + 9/2(5,4) + 1(5,5) + 2(16,1) + 1/2(20,1)

Spectrum 1191; b1={3,8}, b2={3/2,3}; singlets = 17

17(1,1) + 3/2(10,4) + 1(4,4) + 25/2(4,1) + 1(4,5) + 1/2(1,4) + 2(1,5) + 1/2(1,16) + 9/2(5,4) + 5(5,1) + 2(16,1) + 1/2(20,1)

Spectrum 1192; b1={3,8}, b2={3/2,3}; singlets = 11

11(1,1) + 3/2(10,4) + 33/2(4,1) + 1(4,5) + 2(1,4) + 2(1,5) + 1/2(1,16) + 5(5,4) + 3(5,1) + 2(16,1) + 1/2(20,1)

Spectrum 1193; b1={3,8}, b2={3/2,3}; singlets = 29

29(1,1) + 3/2(10,4) + 3(4,4) + 2(4,1) + 3/2(4,5) + 1/2(1,16) + 3(5,4) + 11(5,1) + 2(16,1) + 1/2(20,1)

Spectrum 1194; b1={3,8}, b2={3/2,3}; singlets = 23

23(1,1) + 3/2(10,4) + 2(4,4) + 6(4,1) + 3/2(4,5) + 3/2(1,4) + 1/2(1,16) + 7/2(5,4) + 9(5,1) + 2(16,1) + 1/2(20,1)

Spectrum 1195; b1={3,8}, b2={3/2,3}; singlets = 17

17(1,1) + 3/2(10,4) + 1(4,4) + 10(4,1) + 3/2(4,5) + 3(1,4) + 1/2(1,16) + 4(5,4) + 7(5,1) + 2(16,1) + 1/2(20,1)

Spectrum 1196; b1={3,8}, b2={3/2,3}; singlets = 11

11(1,1) + 3/2(10,4) + 14(4,1) + 3/2(4,5) + 9/2(1,4) + 1/2(1,16) + 9/2(5,4) + 5(5,1) + 2(16,1) + 1/2(20,1)

Spectrum 1197; b1={3,8}, b2={3/2,3}; singlets = 20

20(1,1) + 3/2(10,4) + 4(4,4) + 11/2(4,1) + 1(1,4) + 1(1,5) + 1/2(1,16) + 2(5,4) + 10(5,1) + 1(5,5) + 2(16,1) + 1/2(20,1)

Spectrum 1198; b1={3,8}, b2={3/2,3}; singlets = 14

14(1,1) + 3/2(10,4) + 3(4,4) + 19/2(4,1) + 5/2(1,4) + 1(1,5) + 1/2(1,16) + 5/2(5,4) + 8(5,1) + 1(5,5) + 2(16,1) + 1/2(20,1)

Spectrum 1199; b1={3,8}, b2={3/2,3}; singlets = 8

8(1,1) + 3/2(10,4) + 2(4,4) + 27/2(4,1) + 4(1,4) + 1(1,5) + 1/2(1,16) + 3(5,4) + 6(5,1) + 1(5,5) + 2(16,1) + 1/2(20,1)

Spectrum 1200; b1={3,8}, b2={3/2,3}; singlets = 2

2(1,1) + 3/2(10,4) + 1(4,4) + 35/2(4,1) + 11/2(1,4) + 1(1,5) + 1/2(1,16) + 7/2(5,4) + 4(5,1) + 1(5,5) + 2(16,1) + 1/2(20,1)

Spectrum 1201; b1={3,8}, b2={3/2,3}; singlets = 25

25(1,1) + 1(10,4) + 2(10,1) + 2(4,4) + 17/2(4,1) + 1(4,5) + 1(1,10) + 1/2(1,4) + 7/2(5,4) + 4(5,1) + 1(5,5) + 2(16,1) + 1/2(20,1)

Spectrum 1202; b1={3,8}, b2={3/2,3}; singlets = 19

19(1,1) + 1(10,4) + 2(10,1) + 1(4,4) + 25/2(4,1) + 1(4,5) + 1(1,10) + 2(1,4) + 4(5,4) + 2(5,1) + 1(5,5) + 2(16,1) + 1/2(20,1)

Spectrum 1203; b1={3,8}, b2={3/2,3}; singlets = 13

13(1,1) + 1(10,4) + 2(10,1) + 33/2(4,1) + 1(4,5) + 1(1,10) + 7/2(1,4) + 9/2(5,4) + 1(5,5) + 2(16,1) + 1/2(20,1)

Spectrum 1204; b1={3,8}, b2={3/2,3}; singlets = 18

18(1,1) + 3/2(10,4) + 23/2(4,1) + 2(4,5) + 1(1,10) + 1(1,4) + 1(1,5) + 4(5,4) + 7(5,1) + 2(16,1) + 1/2(20,1)

Spectrum 1205; b1={3,8}, b2={3/2,3}; singlets = 3

3(1,1) + 3/2(10,4) + 43/2(4,1) + 1(1,10) + 1(1,4) + 4(1,5) + 4(5,4) + 2(5,1) + 1(5,5) + 2(16,1) + 1/2(20,1)

Spectrum 1206; b1={3,8}, b2={3/2,3}; singlets = 15

15(1,1) + 3/2(10,4) + 2(4,4) + 11(4,1) + 1/2(4,5) + 1(1,10) + 1/2(1,4) + 2(1,5) + 5/2(5,4) + 8(5,1) + 1(5,5) + 2(16,1) + 1/2(20,1)

Spectrum 1207; b1={3,8}, b2={3/2,3}; singlets = 9

9(1,1) + 3/2(10,4) + 1(4,4) + 15(4,1) + 1/2(4,5) + 1(1,10) + 2(1,4) + 2(1,5) + 3(5,4) + 6(5,1) + 1(5,5) + 2(16,1) + 1/2(20,1)

Spectrum 1208; b1={3,8}, b2={3/2,3}; singlets = 3

3(1,1) + 3/2(10,4) + 19(4,1) + 1/2(4,5) + 1(1,10) + 7/2(1,4) + 2(1,5) + 7/2(5,4) + 4(5,1) + 1(5,5) + 2(16,1) + 1/2(20,1)

Spectrum 1209; b1={3,8}, b2={3/2,3}; singlets = 27

27(1,1) + 3/2(10,4) + 4(4,4) + 1/2(4,1) + 1(4,5) + 1(1,10) + 1(5,4) + 14(5,1) + 1(5,5) + 2(16,1) + 1/2(20,1)

Spectrum 1210; b1={3,8}, b2={3/2,3}; singlets = 21

21(1,1) + 3/2(10,4) + 3(4,4) + 9/2(4,1) + 1(4,5) + 1(1,10) + 3/2(1,4) + 3/2(5,4) + 12(5,1) + 1(5,5) + 2(16,1) + 1/2(20,1)

Spectrum 1211; b1={3,8}, b2={3/2,3}; singlets = 15

15(1,1) + 3/2(10,4) + 2(4,4) + 17/2(4,1) + 1(4,5) + 1(1,10) + 3(1,4) + 2(5,4) + 10(5,1) + 1(5,5) + 2(16,1) + 1/2(20,1)

Spectrum 1212; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 9
 $9(1,1) + 3/2(10,4) + 1(4,4) + 25/2(4,1) + 1(4,5) + 1(1,10) + 9/2(1,4) + 5/2(5,4) + 8(5,1) + 1(5,5) + 2(16,1) + 1/2(20,1)$
Spectrum 1213; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 3
 $3(1,1) + 3/2(10,4) + 33/2(4,1) + 1(4,5) + 1(1,10) + 6(1,4) + 3(5,4) + 6(5,1) + 1(5,5) + 2(16,1) + 1/2(20,1)$
Spectrum 1214; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 20
 $20(1,1) + 3/2(10,4) + 4(10,1) + 1(4,4) + 13(4,1) + 1(4,5) + 1/2(1,4) + 2(1,5) + 1/2(1,16) + 9/2(5,4) + 1(14,1) + 1/2(16,1)$
Spectrum 1215; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 32
 $32(1,1) + 3/2(10,4) + 4(10,1) + 3(4,4) + 5/2(4,1) + 3/2(4,5) + 1/2(1,16) + 3(5,4) + 6(5,1) + 1(14,1) + 1/2(16,1)$
Spectrum 1216; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 26
 $26(1,1) + 3/2(10,4) + 4(10,1) + 2(4,4) + 13/2(4,1) + 3/2(4,5) + 3/2(1,4) + 1/2(1,16) + 7/2(5,4) + 4(5,1) + 1(14,1) + 1/2(16,1)$
Spectrum 1217; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 20
 $20(1,1) + 3/2(10,4) + 4(10,1) + 1(4,4) + 21/2(4,1) + 3/2(4,5) + 3(1,4) + 1/2(1,16) + 4(5,4) + 2(5,1) + 1(14,1) + 1/2(16,1)$
Spectrum 1218; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 14
 $14(1,1) + 3/2(10,4) + 4(10,1) + 29/2(4,1) + 3/2(4,5) + 9/2(1,4) + 1/2(1,16) + 9/2(5,4) + 1(14,1) + 1/2(16,1)$
Spectrum 1219; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 23
 $23(1,1) + 3/2(10,4) + 4(10,1) + 4(4,4) + 6(4,1) + 1(1,4) + 1(1,5) + 1/2(1,16) + 2(5,4) + 5(5,1) + 1(5,5) + 1(14,1) + 1/2(16,1)$
Spectrum 1220; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 17
 $17(1,1) + 3/2(10,4) + 4(10,1) + 3(4,4) + 10(4,1) + 5/2(1,4) + 1(1,5) + 1/2(1,16) + 5/2(5,4) + 3(5,1) + 1(5,5) + 1(14,1) + 1/2(16,1)$
Spectrum 1221; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 11
 $11(1,1) + 3/2(10,4) + 4(10,1) + 2(4,4) + 14(4,1) + 4(1,4) + 1(1,5) + 1/2(1,16) + 3(5,4) + 1(5,1) + 1(5,5) + 1(14,1) + 1/2(16,1)$
Spectrum 1222; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 10
 $10(1,1) + 2(10,4) + 2(10,1) + 1(4,4) + 31/2(4,1) + 1/2(4,5) + 1/2(1,4) + 4(1,5) + 1/2(1,16) + 7/2(5,4) + 4(5,1) + 1(14,1) + 1/2(16,1)$
Spectrum 1223; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 4
 $4(1,1) + 2(10,4) + 2(10,1) + 39/2(4,1) + 1/2(4,5) + 2(1,4) + 4(1,5) + 1/2(1,16) + 4(5,4) + 2(5,1) + 1(14,1) + 1/2(16,1)$
Spectrum 1224; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 22
 $22(1,1) + 2(10,4) + 2(10,1) + 3(4,4) + 5(4,1) + 1(4,5) + 2(1,5) + 1/2(1,16) + 2(5,4) + 10(5,1) + 1(14,1) + 1/2(16,1)$
Spectrum 1225; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 16
 $16(1,1) + 2(10,4) + 2(10,1) + 2(4,4) + 9(4,1) + 1(4,5) + 3/2(1,4) + 2(1,5) + 1/2(1,16) + 5/2(5,4) + 8(5,1) + 1(14,1) + 1/2(16,1)$
Spectrum 1226; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 10
 $10(1,1) + 2(10,4) + 2(10,1) + 1(4,4) + 13(4,1) + 1(4,5) + 3(1,4) + 2(1,5) + 1/2(1,16) + 3(5,4) + 6(5,1) + 1(14,1) + 1/2(16,1)$
Spectrum 1227; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 4
 $4(1,1) + 2(10,4) + 2(10,1) + 17(4,1) + 1(4,5) + 9/2(1,4) + 2(1,5) + 1/2(1,16) + 7/2(5,4) + 4(5,1) + 1(14,1) + 1/2(16,1)$
Spectrum 1228; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 22
 $22(1,1) + 2(10,4) + 2(10,1) + 3(4,4) + 5/2(4,1) + 3/2(4,5) + 5/2(1,4) + 1/2(1,16) + 3/2(5,4) + 12(5,1) + 1(14,1) + 1/2(16,1)$
Spectrum 1229; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 16
 $16(1,1) + 2(10,4) + 2(10,1) + 2(4,4) + 13/2(4,1) + 3/2(4,5) + 4(1,4) + 1/2(1,16) + 2(5,4) + 10(5,1) + 1(14,1) + 1/2(16,1)$
Spectrum 1230; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 10
 $10(1,1) + 2(10,4) + 2(10,1) + 1(4,4) + 21/2(4,1) + 3/2(4,5) + 11/2(1,4) + 1/2(1,16) + 5/2(5,4) + 8(5,1) + 1(14,1) + 1/2(16,1)$
Spectrum 1231; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 4
 $4(1,1) + 2(10,4) + 2(10,1) + 29/2(4,1) + 3/2(4,5) + 7(1,4) + 1/2(1,16) + 3(5,4) + 6(5,1) + 1(14,1) + 1/2(16,1)$
Spectrum 1232; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 19
 $19(1,1) + 2(10,4) + 2(10,1) + 5(4,4) + 2(4,1) + 2(1,4) + 1(1,5) + 1/2(1,16) + 13(5,1) + 1(5,5) + 1(14,1) + 1/2(16,1)$
Spectrum 1233; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 13
 $13(1,1) + 2(10,4) + 2(10,1) + 4(4,4) + 6(4,1) + 7/2(1,4) + 1(1,5) + 1/2(1,16) + 1/2(5,4) + 11(5,1) + 1(5,5) + 1(14,1) + 1/2(16,1)$
Spectrum 1234; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 7
 $7(1,1) + 2(10,4) + 2(10,1) + 3(4,4) + 10(4,1) + 5(1,4) + 1(1,5) + 1/2(1,16) + 1(5,4) + 9(5,1) + 1(5,5) + 1(14,1) + 1/2(16,1)$
Spectrum 1235; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 1
 $1(1,1) + 2(10,4) + 2(10,1) + 2(4,4) + 14(4,1) + 13/2(1,4) + 1(1,5) + 1/2(1,16) + 3/2(5,4) + 7(5,1) + 1(5,5) + 1(14,1) + 1/2(16,1)$
Spectrum 1236; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 0
 $5/2(10,4) + 1(4,4) + 18(4,1) + 1/2(1,4) + 6(1,5) + 1/2(1,16) + 5/2(5,4) + 8(5,1) + 1(14,1) + 1/2(16,1)$
Spectrum 1237; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 12
 $12(1,1) + 5/2(10,4) + 3(4,4) + 15/2(4,1) + 1/2(4,5) + 4(1,5) + 1/2(1,16) + 1(5,4) + 14(5,1) + 1(14,$

Spectrum 1246; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 0
 $5/2(10,4) + 1(4,4) + 21/2(4,1) + 3/2(4,5) + 8(1,4) + 1/2(1,16) + 1(5,4) + 14(5,1) + 1(14,1) + 1/2(16,1)$
Spectrum 1247; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 21
 $21(1,1) + 3/2(10,4) + 4(10,1) + 12(4,1) + 2(4,5) + 1(1,10) + 1(1,4) + 1(1,5) + 4(5,4) + 2(5,1) + 1(14,1) + 1/2(16,1)$
Spectrum 1248; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 18
 $18(1,1) + 3/2(10,4) + 4(10,1) + 2(4,4) + 23/2(4,1) + 1/2(4,5) + 1(1,10) + 1/2(1,4) + 2(1,5) + 5/2(5,4) + 3(5,1) + 1(5,5) + 1(14,1) + 1/2(16,1)$
Spectrum 1249; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 12
 $12(1,1) + 3/2(10,4) + 4(10,1) + 1(4,4) + 31/2(4,1) + 1/2(4,5) + 1(1,10) + 2(1,4) + 2(1,5) + 3(5,4) + 1(5,1) + 1(5,5) + 1(14,1) + 1/2(16,1)$
Spectrum 1250; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 30
 $30(1,1) + 3/2(10,4) + 4(10,1) + 4(4,4) + 1(4,1) + 1(4,5) + 1(1,10) + 1(5,4) + 9(5,1) + 1(5,5) + 1(14,1) + 1/2(16,1)$
Spectrum 1251; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 24
 $24(1,1) + 3/2(10,4) + 4(10,1) + 3(4,4) + 5(4,1) + 1(4,5) + 1(1,10) + 3/2(1,4) + 3/2(5,4) + 7(5,1) + 1(5,5) + 1(14,1) + 1/2(16,1)$
Spectrum 1252; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 18
 $18(1,1) + 3/2(10,4) + 4(10,1) + 2(4,4) + 9(4,1) + 1(4,5) + 1(1,10) + 3(1,4) + 2(5,4) + 5(5,1) + 1(5,5) + 1(14,1) + 1/2(16,1)$
Spectrum 1253; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 12
 $12(1,1) + 3/2(10,4) + 4(10,1) + 1(4,4) + 13(4,1) + 1(4,5) + 1(1,10) + 9/2(1,4) + 5/2(5,4) + 3(5,1) + 1(5,5) + 1(14,1) + 1/2(16,1)$
Spectrum 1254; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 6
 $6(1,1) + 3/2(10,4) + 4(10,1) + 17(4,1) + 1(4,5) + 1(1,10) + 6(1,4) + 3(5,4) + 1(5,1) + 1(5,5) + 1(14,1) + 1/2(16,1)$
Spectrum 1255; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 11
 $11(1,1) + 2(10,4) + 2(10,1) + 29/2(4,1) + 3/2(4,5) + 1(1,10) + 1(1,4) + 3(1,5) + 3(5,4) + 6(5,1) + 1(14,1) + 1/2(16,1)$
Spectrum 1256; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 23
 $23(1,1) + 2(10,4) + 2(10,1) + 2(4,4) + 4(4,1) + 2(4,5) + 1(1,10) + 1/2(1,4) + 1(1,5) + 3/2(5,4) + 12(5,1) + 1(14,1) + 1/2(16,1)$
Spectrum 1257; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 17
 $17(1,1) + 2(10,4) + 2(10,1) + 1(4,4) + 8(4,1) + 2(4,5) + 1(1,10) + 2(1,4) + 1(1,5) + 2(5,4) + 10(5,1) + 1(14,1) + 1/2(16,1)$
Spectrum 1258; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 11
 $11(1,1) + 2(10,4) + 2(10,1) + 12(4,1) + 2(4,5) + 1(1,10) + 7/2(1,4) + 1(1,5) + 5/2(5,4) + 8(5,1) + 1(14,1) + 1/2(16,1)$
Spectrum 1259; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 8
 $8(1,1) + 2(10,4) + 2(10,1) + 2(4,4) + 14(4,1) + 1(1,10) + 1/2(1,4) + 4(1,5) + 3/2(5,4) + 7(5,1) + 1(5,5) + 1(14,1) + 1/2(16,1)$
Spectrum 1260; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 2
 $2(1,1) + 2(10,4) + 2(10,1) + 1(4,4) + 18(4,1) + 1(1,10) + 2(1,4) + 4(1,5) + 2(5,4) + 5(5,1) + 1(5,5) + 1(14,1) + 1/2(16,1)$
Spectrum 1261; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 20
 $20(1,1) + 2(10,4) + 2(10,1) + 4(4,4) + 7/2(4,1) + 1/2(4,5) + 1(1,10) + 2(1,5) + 13(5,1) + 1(5,5) + 1(14,1) + 1/2(16,1)$
Spectrum 1262; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 14
 $14(1,1) + 2(10,4) + 2(10,1) + 3(4,4) + 15/2(4,1) + 1/2(4,5) + 1(1,10) + 3/2(1,4) + 2(1,5) + 1/2(5,4) + 11(5,1) + 1(5,5) + 1(14,1) + 1/2(16,1)$
Spectrum 1263; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 8
 $8(1,1) + 2(10,4) + 2(10,1) + 2(4,4) + 23/2(4,1) + 1/2(4,5) + 1(1,10) + 3(1,4) + 2(1,5) + 1(5,4) + 9(5,1) + 1(5,5) + 1(14,1) + 1/2(16,1)$
Spectrum 1264; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 2
 $2(1,1) + 2(10,4) + 2(10,1) + 1(4,4) + 31/2(4,1) + 1/2(4,5) + 1(1,10) + 9/2(1,4) + 2(1,5) + 3/2(5,4) + 7(5,1) + 1(5,5) + 1(14,1) + 1/2(16,1)$
Spectrum 1265; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 14
 $14(1,1) + 2(10,4) + 2(10,1) + 3(4,4) + 5(4,1) + 1(4,5) + 1(1,10) + 4(1,4) + 13(5,1) + 1(5,5) + 1(14,1) + 1/2(16,1)$
Spectrum 1266; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 8
 $8(1,1) + 2(10,4) + 2(10,1) + 2(4,4) + 9(4,1) + 1(4,5) + 1(1,10) + 11/2(1,4) + 1/2(5,4) + 11(5,1) + 1(5,5) + 1(14,1) + 1/2(16,1)$
Spectrum 1267; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 2
 $2(1,1) + 2(10,4) + 2(10,1) + 1(4,4) + 13(4,1) + 1(4,5) + 1(1,10) + 7(1,4) + 1(5,4) + 9(5,1) + 1(5,5) + 1(14,1) + 1/2(16,1)$
Spectrum 1268; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 1
 $1(1,1) + 5/2(10,4) + 17(4,1) + 1(4,5) + 1(1,10) + 1(1,4) + 5(1,5) + 2(5,4) + 10(5,1) + 1(14,1) + 1/2(16,1)$
Spectrum 1269; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 13
 $13(1,1) + 5/2(10,4) + 2(4,4) + 13/2(4,1) + 3/2(4,5) + 1(1,10) + 1/2(1,4) + 3(1,5) + 1/2(5,4) + 16(5,1) + 1(14,1) + 1/2(16,1)$
Spectrum 1270; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 7
 $7(1,1) + 5/2(10,4) + 1(4,4) + 21/2(4,1) + 3/2(4,5) + 1(1,10) + 2(1,4) + 3(1,5) + 1(5,4) + 14(5,1) + 1(14,1) + 1/2(16,1)$
Spectrum 1271; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 1
 $1(1,1) + 5/2(10,4) + 29/2(4,1) + 3/2(4,5) +$

Spectrum 1280; b1={3,8}, b2={3/2,3}; singlets = 18

18(1,1) + 3/2(10,4) + 1(10,1) + 2(4,4) + 13(4,1) + 1(4,5) + 1(1,10) + 3(1,4) + 2(5,4) + 1(5,1) + 1(5,5) + 7/2(16,1)

Spectrum 1281; b1={3,8}, b2={3/2,3}; singlets = 7

7(1,1) + 7(10,1) + 6(4,4) + 2(4,1) + 1(1,10) + 9/2(1,4) + 1(1,5) + 3/2(5,4) + 8(5,1) + 1(16,1) + 1/2(16,5)

Spectrum 1282; b1={3,8}, b2={3/2,3}; singlets = 1

1(1,1) + 7(10,1) + 5(4,4) + 6(4,1) + 1(1,10) + 6(1,4) + 1(1,5) + 2(5,4) + 6(5,1) + 1(16,1) + 1/2(16,5)

Spectrum 1283; b1={3,8}, b2={3/2,3}; singlets = 26

26(1,1) + 1/2(10,4) + 4(10,1) + 1(4,4) + 7(4,1) + 1/2(1,4) + 1(1,5) + 1/2(1,16) + 13/2(5,4) + 4(5,1) + 1(5,5) + 1(20,1)

Spectrum 1284; b1={3,8}, b2={3/2,3}; singlets = 20

20(1,1) + 1/2(10,4) + 4(10,1) + 11(4,1) + 2(1,4) + 1(1,5) + 1/2(1,16) + 7(5,4) + 2(5,1) + 1(5,5) + 1(20,1)

Spectrum 1285; b1={3,8}, b2={3/2,3}; singlets = 25

25(1,1) + 1(10,4) + 2(10,1) + 7/2(4,1) + 3/2(4,5) + 2(1,4) + 1/2(1,16) + 6(5,4) + 11(5,1) + 1(20,1)

Spectrum 1286; b1={3,8}, b2={3/2,3}; singlets = 22

22(1,1) + 1(10,4) + 2(10,1) + 2(4,4) + 3(4,1) + 3/2(1,4) + 1(1,5) + 1/2(1,16) + 9/2(5,4) + 12(5,1) + 1(5,5) + 1(20,1)

Spectrum 1287; b1={3,8}, b2={3/2,3}; singlets = 16

16(1,1) + 1(10,4) + 2(10,1) + 1(4,4) + 7(4,1) + 3(1,4) + 1(1,5) + 1/2(1,16) + 5(5,4) + 10(5,1) + 1(5,5) + 1(20,1)

Spectrum 1288; b1={3,8}, b2={3/2,3}; singlets = 10

10(1,1) + 1(10,4) + 2(10,1) + 11(4,1) + 9/2(1,4) + 1(1,5) + 1/2(1,16) + 11/2(5,4) + 8(5,1) + 1(5,5) + 1(20,1)

Spectrum 1289; b1={3,8}, b2={3/2,3}; singlets = 21

21(1,1) + 3/2(10,4) + 1(4,4) + 2(4,1) + 1(4,5) + 1/2(1,4) + 2(1,5) + 1/2(1,16) + 9/2(5,4) + 17(5,1) + 1(20,1)

Spectrum 1290; b1={3,8}, b2={3/2,3}; singlets = 15

15(1,1) + 3/2(10,4) + 6(4,1) + 1(4,5) + 2(1,4) + 2(1,5) + 1/2(1,16) + 5(5,4) + 15(5,1) + 1(20,1)

Spectrum 1291; b1={3,8}, b2={3/2,3}; singlets = 15

15(1,1) + 3/2(10,4) + 7/2(4,1) + 3/2(4,5) + 9/2(1,4) + 1/2(1,16) + 9/2(5,4) + 17(5,1) + 1(20,1)

Spectrum 1292; b1={3,8}, b2={3/2,3}; singlets = 12

12(1,1) + 3/2(10,4) + 2(4,4) + 3(4,1) + 4(1,4) + 1(1,5) + 1/2(1,16) + 3(5,4) + 18(5,1) + 1(5,5) + 1(20,1)

Spectrum 1293; b1={3,8}, b2={3/2,3}; singlets = 6

6(1,1) + 3/2(10,4) + 1(4,4) + 7(4,1) + 11/2(1,4) + 1(1,5) + 1/2(1,16) + 7/2(5,4) + 16(5,1) + 1(5,5) + 1(20,1)

Spectrum 1294; b1={3,8}, b2={3/2,3}; singlets = 0

3/2(10,4) + 11(4,1) + 7(1,4) + 1(1,5) + 1/2(1,16) + 4(5,4) + 14(5,1) + 1(5,5) + 1(20,1)

Spectrum 1295; b1={3,8}, b2={3/2,3}; singlets = 27

27(1,1) + 1/2(10,4) + 4(10,1) + 6(4,1) + 1(4,5) + 1(1,10) + 1(1,4) + 6(5,4) + 6(5,1) + 1(5,5) + 1(20,1)

Spectrum 1296; b1={3,8}, b2={3/2,3}; singlets = 17

17(1,1) + 1(10,4) + 2(10,1) + 17/2(4,1) + 1/2(4,5) + 1(1,10) + 1(1,4) + 2(1,5) + 5(5,4) + 10(5,1) + 1(5,5) + 1(20,1)

Spectrum 1297; b1={3,8}, b2={3/2,3}; singlets = 23

23(1,1) + 1(10,4) + 2(10,1) + 1(4,4) + 2(4,1) + 1(4,5) + 1(1,10) + 2(1,4) + 4(5,4) + 14(5,1) + 1(5,5) + 1(20,1)

Spectrum 1298; b1={3,8}, b2={3/2,3}; singlets = 17

17(1,1) + 1(10,4) + 2(10,1) + 6(4,1) + 1(4,5) + 1(1,10) + 7/2(1,4) + 9/2(5,4) + 12(5,1) + 1(5,5) + 1(20,1)

Spectrum 1299; b1={3,8}, b2={3/2,3}; singlets = 22

22(1,1) + 3/2(10,4) + 1(4,1) + 2(4,5) + 1(1,10) + 1(1,4) + 1(1,5) + 4(5,4) + 19(5,1) + 1(20,1)

Spectrum 1300; b1={3,8}, b2={3/2,3}; singlets = 7

7(1,1) + 3/2(10,4) + 11(4,1) + 1(1,10) + 1(1,4) + 4(1,5) + 4(5,4) + 14(5,1) + 1(5,5) + 1(20,1)

Spectrum 1301; b1={3,8}, b2={3/2,3}; singlets = 19

19(1,1) + 3/2(10,4) + 2(4,4) + 1/2(4,1) + 1/2(4,5) + 1(1,10) + 1/2(1,4) + 2(1,5) + 5/2(5,4) + 20(5,1) + 1(5,5) + 1(20,1)

Spectrum 1302; b1={3,8}, b2={3/2,3}; singlets = 13

13(1,1) + 3/2(10,4) + 1(4,4) + 9/2(4,1) + 1/2(4,5) + 1(1,10) + 2(1,4) + 2(1,5) + 3(5,4) + 18(5,1) + 1(5,5) + 1(20,1)

Spectrum 1303; b1={3,8}, b2={3/2,3}; singlets = 7

7(1,1) + 3/2(10,4) + 17/2(4,1) + 1/2(4,5) + 1(1,10) + 7/2(1,4) + 2(1,5) + 7/2(5,4) + 16(5,1) + 1(5,5) + 1(20,1)

Spectrum 1304; b1={3,8}, b2={3/2,3}; singlets = 13

13(1,1) + 3/2(10,4) + 1(4,4) + 2(4,1) + 1(4,5) + 1(1,10) + 9/2(1,4) + 5/2(5,4) + 20(5,1) + 1(5,5) + 1(20,1)

Spectrum 1305; b1={3,8}, b2={3/2,3}; singlets = 7

7(1,1) + 3/2(10,4) + 6(4,1) + 1(4,5) + 1(1,10) + 6(1,4) + 3(5,4) + 18(5,1) + 1(5,5) + 1(20,1)

Spectrum 1306; b1={3,8}, b2={3/2,3}; singlets = 34

34(1,1) + 1(10,4) + 3(10,1) + 1(4,4) + 4(4,1) + 3/2(4,5) + 1/2(1,4) + 1/2(1,16) + 11/2(5,4) + 4(5,1) + 3/2(16,1) + 1/2(20,1)

Spectrum 1307; b1={3,8}, b2={3/2,3}; singlets = 28

28(1,1) + 1(10,4) + 3(10,1) + 8(4,1) + 3/2(4,5) + 2(1,4) + 1/2(1,16) + 6(5,4) + 2(5,1) + 3/2(16,1) + 1/2(20,1)

Spectrum 1308; b1={3,8}, b2={3/2,3}; singlets = 31

31(1,1) + 1(10,4) + 3(10,1) + 3(4,4) + 7/2(4,1) + 1(1,5) + 1/2(1,16) + 4(5,4) + 5(5,1) + 1(5,5) + 3/2(16,1) + 1/2(20,1)

Spectrum 1309; b1={3,8}, b2={3/2,3}; singlets = 25

25(1,1) + 1(10,4) + 3(10,1) + 2(4,4) + 15/2(4,1) + 3/2(1,4) + 1(1,5) + 1/2(1,16) + 9/2(5,4) + 3(5,1) + 1(5,5) + 3/2(16,1) + 1/2(20,1)

Spectrum 1310; b1={3,8}, b2={3/2,3}; singlets = 19

19(1,1) + 1(10,4) + 3(10,1) + 1(4,4) + 23/2(4,1) + 3(1,4) + 1(1,5) + 1/2(1,16) + 5(5,4) + 1(5,1) + 1(5,5) + 3/2(16,1) + 1/2(20,1)

Spectrum 1311; b1={3,8}, b2={3/2,3}; singlets = 24

24(1,1) + 3/2(10,4) + 1(10,1) + 1(4,4) + 13/2(4,1) + 1(4,5) + 1/2(1,4) + 2(1,5) + 1/2(1,16) + 9/2(5,4) + 8(5,1) + 3/2(16,1) + 1/2(20,1)

Spectrum 1312; b1={3,8}, b2={3/2,3}; singlets = 18

18(1,1) + 3/2(10,4) + 1(10,1) + 21/2(4,1) + 1(4,5) + 2(1,4) + 2(1,5) + 1/2(1,16) + 5(5,4) + 6(5,1) + 3/2(16,1) + 1/2(20,1)

Spectrum 1313; b1={3,8}, b2={3/2,3}; singlets = 30

30(1,1) + 3/2(10,4) + 1(10,1) + 2(4,4) + 3/2(4,5) + 3/2(1,4) + 1/2(1,16) + 7/2(5,4) + 12(5,1) + 3/2(16,1) + 1/2(20,1)

Spectrum 1348; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 23
 $23(1,1) + 2(10,4) + 3(10,1) + 2(4,4) + 1/2(4,1) + 3/2(4,5) + 4(1,4) + 1/2(1,16) + 2(5,4) + 13(5,1) + 1(14,1)$
Spectrum 1349; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 17
 $17(1,1) + 2(10,4) + 3(10,1) + 1(4,4) + 9/2(4,1) + 3/2(4,5) + 11/2(1,4) + 1/2(1,16) + 5/2(5,4) + 11(5,1) + 1(14,1)$
Spectrum 1350; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 11
 $11(1,1) + 2(10,4) + 3(10,1) + 17/2(4,1) + 3/2(4,5) + 7(1,4) + 1/2(1,16) + 3(5,4) + 9(5,1) + 1(14,1)$
Spectrum 1351; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 20
 $20(1,1) + 2(10,4) + 3(10,1) + 4(4,4) + 7/2(1,4) + 1(1,5) + 1/2(1,16) + 1/2(5,4) + 14(5,1) + 1(5,5) + 1(14,1)$
Spectrum 1352; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 14
 $14(1,1) + 2(10,4) + 3(10,1) + 3(4,4) + 4(4,1) + 5(1,4) + 1(1,5) + 1/2(1,16) + 1(5,4) + 12(5,1) + 1(5,5) + 1(14,1)$
Spectrum 1353; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 8
 $8(1,1) + 2(10,4) + 3(10,1) + 2(4,4) + 8(4,1) + 13/2(1,4) + 1(1,5) + 1/2(1,16) + 3/2(5,4) + 10(5,1) + 1(5,5) + 1(14,1)$
Spectrum 1354; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 2
 $2(1,1) + 2(10,4) + 3(10,1) + 1(4,4) + 12(4,1) + 8(1,4) + 1(1,5) + 1/2(1,16) + 2(5,4) + 8(5,1) + 1(5,5) + 1(14,1)$
Spectrum 1355; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 7
 $7(1,1) + 5/2(10,4) + 1(10,1) + 1(4,4) + 12(4,1) + 1/2(1,4) + 6(1,5) + 1/2(1,16) + 5/2(5,4) + 11(5,1) + 1(14,1)$
Spectrum 1356; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 1
 $1(1,1) + 5/2(10,4) + 1(10,1) + 16(4,1) + 2(1,4) + 6(1,5) + 1/2(1,16) + 3(5,4) + 9(5,1) + 1(14,1)$
Spectrum 1357; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 19
 $19(1,1) + 5/2(10,4) + 1(10,1) + 3(4,4) + 3/2(4,1) + 1/2(4,5) + 4(1,5) + 1/2(1,16) + 1(5,4) + 17(5,1) + 1(14,1)$
Spectrum 1358; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 13
 $13(1,1) + 5/2(10,4) + 1(10,1) + 2(4,4) + 11/2(4,1) + 1/2(4,5) + 3/2(1,4) + 4(1,5) + 1/2(1,16) + 3/2(5,4) + 15(5,1) + 1(14,1)$
Spectrum 1359; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 7
 $7(1,1) + 5/2(10,4) + 1(10,1) + 1(4,4) + 19/2(4,1) + 1/2(4,5) + 3(1,4) + 4(1,5) + 1/2(1,16) + 2(5,4) + 13(5,1) + 1(14,1)$
Spectrum 1360; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 1
 $1(1,1) + 5/2(10,4) + 1(10,1) + 27/2(4,1) + 1/2(4,5) + 9/2(1,4) + 4(1,5) + 1/2(1,16) + 5/2(5,4) + 11(5,1) + 1(14,1)$
Spectrum 1361; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 13
 $13(1,1) + 5/2(10,4) + 1(10,1) + 2(4,4) + 3(4,1) + 1(4,5) + 4(1,4) + 2(1,5) + 1/2(1,16) + 1(5,4) + 17(5,1) + 1(14,1)$
Spectrum 1362; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 7
 $7(1,1) + 5/2(10,4) + 1(10,1) + 1(4,4) + 7(4,1) + 1(4,5) + 11/2(1,4) + 2(1,5) + 1/2(1,16) + 3/2(5,4) + 15(5,1) + 1(14,1)$
Spectrum 1363; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 1
 $1(1,1) + 5/2(10,4) + 1(10,1) + 11(4,1) + 1(4,5) + 7(1,4) + 2(1,5) + 1/2(1,16) + 2(5,4) + 13(5,1) + 1(14,1)$
Spectrum 1364; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 13
 $13(1,1) + 5/2(10,4) + 1(10,1) + 2(4,4) + 1/2(4,1) + 3/2(4,5) + 13/2(1,4) + 1/2(1,16) + 1/2(5,4) + 19(5,1) + 1(14,1)$
Spectrum 1365; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 7
 $7(1,1) + 5/2(10,4) + 1(10,1) + 1(4,4) + 9/2(4,1) + 3/2(4,5) + 8(1,4) + 1/2(1,16) + 1(5,4) + 17(5,1) + 1(14,1)$
Spectrum 1366; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 1
 $1(1,1) + 5/2(10,4) + 1(10,1) + 17/2(4,1) + 3/2(4,5) + 19/2(1,4) + 1/2(1,16) + 3/2(5,4) + 15(5,1) + 1(14,1)$
Spectrum 1367; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 35
 $35(1,1) + 1(10,4) + 7(10,1) + 2(4,4) + 3(4,1) + 1(4,5) + 1(1,10) + 1/2(1,4) + 7/2(5,4) + 2(5,1) + 1(5,5) + 1(14,1)$
Spectrum 1368; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 29
 $29(1,1) + 1(10,4) + 7(10,1) + 1(4,4) + 7(4,1) + 1(4,5) + 1(1,10) + 2(1,4) + 4(5,4) + 1(5,5) + 1(14,1)$
Spectrum 1369; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 28
 $28(1,1) + 3/2(10,4) + 5(10,1) + 6(4,1) + 2(4,5) + 1(1,10) + 1(1,4) + 1(1,5) + 4(5,4) + 5(5,1) + 1(14,1)$
Spectrum 1370; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 13
 $13(1,1) + 3/2(10,4) + 5(10,1) + 16(4,1) + 1(1,10) + 1(1,4) + 4(1,5) + 4(5,4) + 1(5,5) + 1(14,1)$
Spectrum 1371; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 25
 $25(1,1) + 3/2(10,4) + 5(10,1) + 2(4,4) + 11/2(4,1) + 1/2(4,5) + 1(1,10) + 1/2(1,4) + 2(1,5) + 5/2(5,4) + 6(5,1) + 1(5,5) + 1(14,1)$
Spectrum 1372; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 19
 $19(1,1) + 3/2(10,4) + 5(10,1) + 1(4,4) + 19/2(4,1) + 1/2(4,5) + 1(1,10) + 2(1,4) + 2(1,5) + 3(5,4) + 4(5,1) + 1(5,5) + 1(14,1)$
Spectrum 1373; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 13
 $13(1,1) + 3/2(10,4) + 5(10,1) + 27/2(4,1) + 1/2(4,5) + 1(1,10) + 7/2(1,4) + 2(1,5) + 7/2(5,4) + 2(5,1) + 1(5,5) + 1(14,1)$
Spectrum 1374; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 25
 $25(1,1) + 3/2(10,4) + 5(10,1) + 2(4,4) + 3(4,1) + 1(4,5) + 1(1,10) + 3(1,4) + 2(5,4) + 8(5,1) + 1(5,5) + 1(14,1)$
Spectrum 1375; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 19
 $19(1,1) + 3/2(10,4) + 5(10,1) + 1(4,4) + 7(4,1) + 1(4,5) + 1(1,10) + 9/2(1,4) + 5/2(5,4) + 6(5,1) + 1(5,5) + 1(14,1)$
Spectrum 1376; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 13
 $13(1,1) + 3/2(10,4) + 5(10,1) + 11(4,1) + 1(4,5) + 1(1,10) + 6(1,4) + 3(5,4) + 4(5,1) + 1(5,5) + 1(14,1)$
Spectrum 1377; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 18
 $18(1,1) + 2(10,4) + 3(10,1) + 17/2(4,1) + 3/2(4,5) + 1(1,10) + 1(1,4) + 3(1,5) + 3(5,4) + 9(5,1) + 1(14,1)$
Spectrum 1378; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 24
 $24(1,1) + 2(10,4) + 3(10,1) + 1(4,4) + 2(4,1) + 2(4,5) + 1(1,10) + 2(1,4) + 1(1,5) + 2(5,4) + 13(5,1) + 1(14,1)$
Spectrum 1379; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 18
 $18(1,1) + 2(10,4) + 3(10,1) + 6(4,1) + 2(4,5) + 1(1,10) + 7/2(1,4) + 1(1,5) + 5/2(5,4) + 11(5,1) + 1(14,1)$
Spectrum 1380; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 15
 $15(1,1) + 2(10,4) + 3(10,1) + 2(4,4) + 8(4,1) + 1(1,10) + 1/2(1,4) + 4(1,5) + 3/2(5,4) + 10(5,1) + 1(5,5) + 1(14,1)$
Spectrum 1381; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 9
 $9(1,1) + 2(10,4) + 3(10,1) + 1(4,4) + 12(4,1) + 1(1,10) + 2(1,4) + 4(1,5) + 2(5,4) + 8(5,1) + 1(5,5) + 1(14,1)$

Spectrum 1382; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 3
 $3(1,1) + 2(10,4) + 3(10,1) + 16(4,1) + 1(1,10) + 7/2(1,4) + 4(1,5) + 5/2(5,4) + 6(5,1) + 1(5,5) + 1(14,1)$
Spectrum 1383; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 21
 $21(1,1) + 2(10,4) + 3(10,1) + 3(4,4) + 3/2(4,1) + 1/2(4,5) + 1(1,10) + 3/2(1,4) + 2(1,5) + 1/2(5,4) + 14(5,1) + 1(5,5) + 1(14,1)$
Spectrum 1384; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 15
 $15(1,1) + 2(10,4) + 3(10,1) + 2(4,4) + 11/2(4,1) + 1/2(4,5) + 1(1,10) + 3(1,4) + 2(1,5) + 1(5,4) + 12(5,1) + 1(5,5) + 1(14,1)$
Spectrum 1385; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 9
 $9(1,1) + 2(10,4) + 3(10,1) + 1(4,4) + 19/2(4,1) + 1/2(4,5) + 1(1,10) + 9/2(1,4) + 2(1,5) + 3/2(5,4) + 10(5,1) + 1(5,5) + 1(14,1)$
Spectrum 1386; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 3
 $3(1,1) + 2(10,4) + 3(10,1) + 27/2(4,1) + 1/2(4,5) + 1(1,10) + 6(1,4) + 2(1,5) + 2(5,4) + 8(5,1) + 1(5,5) + 1(14,1)$
Spectrum 1387; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 15
 $15(1,1) + 2(10,4) + 3(10,1) + 2(4,4) + 3(4,1) + 1(4,5) + 1(1,10) + 11/2(1,4) + 1/2(5,4) + 14(5,1) + 1(5,5) + 1(14,1)$
Spectrum 1388; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 9
 $9(1,1) + 2(10,4) + 3(10,1) + 1(4,4) + 7(4,1) + 1(4,5) + 1(1,10) + 7(1,4) + 1(5,4) + 12(5,1) + 1(5,5) + 1(14,1)$
Spectrum 1389; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 3
 $3(1,1) + 2(10,4) + 3(10,1) + 11(4,1) + 1(4,5) + 1(1,10) + 17/2(1,4) + 3/2(5,4) + 10(5,1) + 1(5,5) + 1(14,1)$
Spectrum 1390; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 8
 $8(1,1) + 5/2(10,4) + 1(10,1) + 11(4,1) + 1(4,5) + 1(1,10) + 1(1,4) + 5(1,5) + 2(5,4) + 13(5,1) + 1(14,1)$
Spectrum 1391; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 20
 $20(1,1) + 5/2(10,4) + 1(10,1) + 2(4,4) + 1/2(4,1) + 3/2(4,5) + 1(1,10) + 1/2(1,4) + 3(1,5) + 1/2(5,4) + 19(5,1) + 1(14,1)$
Spectrum 1392; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 14
 $14(1,1) + 5/2(10,4) + 1(10,1) + 1(4,4) + 9/2(4,1) + 3/2(4,5) + 1(1,10) + 2(1,4) + 3(1,5) + 1(5,4) + 17(5,1) + 1(14,1)$
Spectrum 1393; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 8
 $8(1,1) + 5/2(10,4) + 1(10,1) + 17/2(4,1) + 3/2(4,5) + 1(1,10) + 7/2(1,4) + 3(1,5) + 3/2(5,4) + 15(5,1) + 1(14,1)$
Spectrum 1394; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 14
 $14(1,1) + 5/2(10,4) + 1(10,1) + 1(4,4) + 2(4,1) + 2(4,5) + 1(1,10) + 9/2(1,4) + 1(1,5) + 1/2(5,4) + 19(5,1) + 1(14,1)$
Spectrum 1395; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 8
 $8(1,1) + 5/2(10,4) + 1(10,1) + 6(4,1) + 2(4,5) + 1(1,10) + 6(1,4) + 1(1,5) + 1(5,4) + 17(5,1) + 1(14,1)$
Spectrum 1396; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 5
 $5(1,1) + 5/2(10,4) + 1(10,1) + 2(4,4) + 8(4,1) + 1(1,10) + 3(1,4) + 4(1,5) + 16(5,1) + 1(5,5) + 1(14,1)$
Spectrum 1397; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 39
 $39(1,1) + 3/2(10,4) + 2(10,1) + 3(4,4) + 1/2(4,1) + 3/2(4,5) + 1/2(1,16) + 3(5,4) + 5(5,1) + 3(16,1)$
Spectrum 1398; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 33
 $33(1,1) + 3/2(10,4) + 2(10,1) + 2(4,4) + 9/2(4,1) + 3/2(4,5) + 3/2(1,4) + 1/2(1,16) + 7/2(5,4) + 3(5,1) + 3(16,1)$
Spectrum 1399; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 27
 $27(1,1) + 3/2(10,4) + 2(10,1) + 1(4,4) + 17/2(4,1) + 3/2(4,5) + 3(1,4) + 1/2(1,16) + 4(5,4) + 1(5,1) + 3(16,1)$
Spectrum 1400; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 30
 $30(1,1) + 3/2(10,4) + 2(10,1) + 4(4,4) + 4(4,1) + 1(1,4) + 1(1,5) + 1/2(1,16) + 2(5,4) + 4(5,1) + 1(5,5) + 3(16,1)$
Spectrum 1401; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 24
 $24(1,1) + 3/2(10,4) + 2(10,1) + 3(4,4) + 8(4,1) + 5/2(1,4) + 1(1,5) + 1/2(1,16) + 5/2(5,4) + 2(5,1) + 1(5,5) + 3(16,1)$
Spectrum 1402; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 18
 $18(1,1) + 3/2(10,4) + 2(10,1) + 2(4,4) + 12(4,1) + 4(1,4) + 1(1,5) + 1/2(1,16) + 3(5,4) + 1(5,5) + 3(16,1)$
Spectrum 1403; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 17
 $17(1,1) + 2(10,4) + 1(4,4) + 27/2(4,1) + 1/2(4,5) + 1/2(1,4) + 4(1,5) + 1/2(1,16) + 7/2(5,4) + 3(5,1) + 3(16,1)$
Spectrum 1404; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 11
 $11(1,1) + 2(10,4) + 35/2(4,1) + 1/2(4,5) + 2(1,4) + 4(1,5) + 1/2(1,16) + 4(5,4) + 1(5,1) + 3(16,1)$
Spectrum 1405; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 29
 $29(1,1) + 2(10,4) + 3(4,4) + 3(4,1) + 1(4,5) + 2(1,5) + 1/2(1,16) + 2(5,4) + 9(5,1) + 3(16,1)$
Spectrum 1406; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 23
 $23(1,1) + 2(10,4) + 2(4,4) + 7(4,1) + 1(4,5) + 3/2(1,4) + 2(1,5) + 1/2(1,16) + 5/2(5,4) + 7(5,1) + 3(16,1)$
Spectrum 1407; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 17
 $17(1,1) + 2(10,4) + 1(4,4) + 11(4,1) + 1(4,5) + 3(1,4) + 2(1,5) + 1/2(1,16) + 3(5,4) + 5(5,1) + 3(16,1)$
Spectrum 1408; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 11
 $11(1,1) + 2(10,4) + 15(4,1) + 1(4,5) + 9/2(1,4) + 2(1,5) + 1/2(1,16) + 7/2(5,4) + 3(5,1) + 3(16,1)$
Spectrum 1409; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 29
 $29(1,1) + 2(10,4) + 3(4,4) + 1/2(4,1) + 3/2(4,5) + 5/2(1,4) + 1/2(1,16) + 3/2(5,4) + 11(5,1) + 3(16,1)$
Spectrum 1410; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 23
 $23(1,1) + 2(10,4) + 2(4,4) + 9/2(4,1) + 3/2(4,5) + 4(1,4) + 1/2(1,16) + 2(5,4) + 9(5,1) + 3(16,1)$
Spectrum 1411; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 17
 $17(1,1) + 2(10,4) + 1(4,4) + 17/2(4,1) + 3/2(4,5) + 11/2(1,4) + 1/2(1,16) + 5/2(5,4) + 7(5,1) + 3(16,1)$
Spectrum 1412; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 11
 $11(1,1) + 2(10,4) + 25/2(4,1) + 3/2(4,5) + 7(1,4) + 1/2(1,16) + 3(5,4) + 5(5,1) + 3(16,1)$
Spectrum 1413; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 26
 $26(1,1) + 2(10,4) + 5(4,4) + 2(1,4) + 1(1,5) + 1/2(1,16) + 12(5,1) + 1(5,5) + 3(16,1)$
Spectrum 1414; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 20
 $20(1,1) + 2(10,4) + 4(4,4) + 4(4,1) + 7/2(1,4) + 1(1,5) + 1/2(1,16) + 1/2(5,4) + 10(5,1) + 1(5,5) + 3(16,1)$
Spectrum 1415; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 14
 $14(1,1) + 2(10,4) + 3(4,4) + 8(4,1) + 5(1,4) + 1(1,5) + 1/2(1,16) + 1(5,4) + 8(5,1) + 1(5,5) + 3(16,1)$

Spectrum 1416; $b1=(3,8)$, $b2=(2/2,3)$; singlets = 8
 $8(1,1) + 2(10,4) + 2(4,4) + 12(4,1) + 13/2(1,4) + 1(1,5) + 1/2(1,16) + 3/2(5,4) + 6(5,1) + 1(5,5) + 3(16,1)$
Spectrum 1417; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 2
 $2(1,1) + 2(10,4) + 1(4,4) + 16(4,1) + 8(1,4) + 1(1,5) + 1/2(1,16) + 2(5,4) + 4(5,1) + 1(5,5) + 3(16,1)$
Spectrum 1418; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 28
 $28(1,1) + 3/2(10,4) + 2(10,1) + 10(4,1) + 2(4,5) + 1(1,10) + 1(1,4) + 1(1,5) + 4(5,4) + 1(5,1) + 3(16,1)$
Spectrum 1419; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 25
 $25(1,1) + 3/2(10,4) + 2(10,1) + 2(4,4) + 19/2(4,1) + 1/2(4,5) + 1(1,10) + 1/2(1,4) + 2(1,5) + 5/2(5,4) + 2(5,1) + 1(5,5) + 3(16,1)$
Spectrum 1420; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 19
 $19(1,1) + 3/2(10,4) + 2(10,1) + 1(4,4) + 27/2(4,1) + 1/2(4,5) + 1(1,10) + 2(1,4) + 2(1,5) + 3(5,4) + 1(5,5) + 3(16,1)$
Spectrum 1421; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 31
 $31(1,1) + 3/2(10,4) + 2(10,1) + 3(4,4) + 3(4,1) + 1(4,5) + 1(1,10) + 3/2(1,4) + 3/2(5,4) + 6(5,1) + 1(5,5) + 3(16,1)$
Spectrum 1422; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 25
 $25(1,1) + 3/2(10,4) + 2(10,1) + 2(4,4) + 7(4,1) + 1(4,5) + 1(1,10) + 3(1,4) + 2(5,4) + 4(5,1) + 1(5,5) + 3(16,1)$
Spectrum 1423; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 19
 $19(1,1) + 3/2(10,4) + 2(10,1) + 1(4,4) + 11(4,1) + 1(4,5) + 1(1,10) + 9/2(1,4) + 5/2(5,4) + 2(5,1) + 1(5,5) + 3(16,1)$
Spectrum 1424; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 13
 $13(1,1) + 3/2(10,4) + 2(10,1) + 15(4,1) + 1(4,5) + 1(1,10) + 6(1,4) + 3(5,4) + 1(5,5) + 3(16,1)$
Spectrum 1425; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 18
 $18(1,1) + 2(10,4) + 25/2(4,1) + 3/2(4,5) + 1(1,10) + 1(1,4) + 3(1,5) + 3(5,4) + 5(5,1) + 3(16,1)$
Spectrum 1426; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 30
 $30(1,1) + 2(10,4) + 2(4,4) + 2(4,1) + 2(4,5) + 1(1,10) + 1/2(1,4) + 1(1,5) + 3/2(5,4) + 11(5,1) + 3(16,1)$
Spectrum 1427; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 24
 $24(1,1) + 2(10,4) + 1(4,4) + 6(4,1) + 2(4,5) + 1(1,10) + 2(1,4) + 1(1,5) + 2(5,4) + 9(5,1) + 3(16,1)$
Spectrum 1428; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 18
 $18(1,1) + 2(10,4) + 10(4,1) + 2(4,5) + 1(1,10) + 7/2(1,4) + 1(1,5) + 5/2(5,4) + 7(5,1) + 3(16,1)$
Spectrum 1429; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 15
 $15(1,1) + 2(10,4) + 2(4,4) + 12(4,1) + 1(1,10) + 1/2(1,4) + 4(1,5) + 3/2(5,4) + 6(5,1) + 1(5,5) + 3(16,1)$
Spectrum 1430; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 9
 $9(1,1) + 2(10,4) + 1(4,4) + 16(4,1) + 1(1,10) + 2(1,4) + 4(1,5) + 2(5,4) + 4(5,1) + 1(5,5) + 3(16,1)$
Spectrum 1431; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 3
 $3(1,1) + 2(10,4) + 20(4,1) + 1(1,10) + 7/2(1,4) + 4(1,5) + 5/2(5,4) + 2(5,1) + 1(5,5) + 3(16,1)$
Spectrum 1432; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 27
 $27(1,1) + 2(10,4) + 4(4,4) + 3/2(4,1) + 1/2(4,5) + 1(1,10) + 2(1,5) + 12(5,1) + 1(5,5) + 3(16,1)$
Spectrum 1433; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 21
 $21(1,1) + 2(10,4) + 3(4,4) + 11/2(4,1) + 1/2(4,5) + 1(1,10) + 3/2(1,4) + 2(1,5) + 1/2(5,4) + 10(5,1) + 1(5,5) + 3(16,1)$
Spectrum 1434; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 15
 $15(1,1) + 2(10,4) + 2(4,4) + 19/2(4,1) + 1/2(4,5) + 1(1,10) + 3(1,4) + 2(1,5) + 1(5,4) + 8(5,1) + 1(5,5) + 3(16,1)$
Spectrum 1435; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 9
 $9(1,1) + 2(10,4) + 1(4,4) + 27/2(4,1) + 1/2(4,5) + 1(1,10) + 9/2(1,4) + 2(1,5) + 3/2(5,4) + 6(5,1) + 1(5,5) + 3(16,1)$
Spectrum 1436; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 3
 $3(1,1) + 2(10,4) + 35/2(4,1) + 1/2(4,5) + 1(1,10) + 6(1,4) + 2(1,5) + 2(5,4) + 4(5,1) + 1(5,5) + 3(16,1)$
Spectrum 1437; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 21
 $21(1,1) + 2(10,4) + 3(4,4) + 3(4,1) + 1(4,5) + 1(1,10) + 4(1,4) + 12(5,1) + 1(5,5) + 3(16,1)$
Spectrum 1438; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 15
 $15(1,1) + 2(10,4) + 2(4,4) + 7(4,1) + 1(4,5) + 1(1,10) + 11/2(1,4) + 1/2(5,4) + 10(5,1) + 1(5,5) + 3(16,1)$
Spectrum 1439; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 9
 $9(1,1) + 2(10,4) + 1(4,4) + 11(4,1) + 1(4,5) + 1(1,10) + 7(1,4) + 1(5,4) + 8(5,1) + 1(5,5) + 3(16,1)$
Spectrum 1440; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 3
 $3(1,1) + 2(10,4) + 15(4,1) + 1(4,5) + 1(1,10) + 17/2(1,4) + 3/2(5,4) + 6(5,1) + 1(5,5) + 3(16,1)$
Spectrum 1441; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 8
 $8(1,1) + 8(10,1) + 5(4,4) + 1(1,10) + 6(1,4) + 1(1,5) + 2(5,4) + 9(5,1) + 1/2(16,1) + 1/2(16,5)$
Spectrum 1442; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 2
 $2(1,1) + 8(10,1) + 4(4,4) + 4(4,1) + 1(1,10) + 15/2(1,4) + 1(1,5) + 5/2(5,4) + 7(5,1) + 1/2(16,1) + 1/2(16,5)$
Spectrum 1443; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 35
 $35(1,1) + 1(10,4) + 4(10,1) + 2(4,1) + 3/2(4,5) + 2(1,4) + 1/2(1,16) + 6(5,4) + 5(5,1) + 1(16,1) + 1/2(20,1)$
Spectrum 1444; $b1=(3,8)$, $b2=(3/2,3)$; singlets = 32
 $32(1,1) + 1(10,4) +$

22(1,1) + 3/2(10,4) + 2(10,1) + 2(4,1) + 4(1,4) + 1(1,5) + 1/2(1,16) + 3(5,4) + 12(5,1) + 1(5,5) + 1(16,1) + 1/2(20,1)
Spectrum 1451; b1=(3,8), b2=(3/2,3); singlets = 16
16(1,1) + 3/2(10,4) + 2(10,1) + 1(4,4) + 11/2(4,1) + 11/2(1,4) + 1(1,5) + 1/2(1,16) + 7/2(5,4) + 10(5,1) + 1(5,5) + 1(16,1) + 1/2(20,1)
Spectrum 1452; b1=(3,8), b2=(3/2,3); singlets = 10
10(1,1) + 3/2(10,4) + 2(10,1) + 19/2(4,1) + 7(1,4) + 1(1,5) + 1/2(1,16) + 4(5,4) + 8(5,1) + 1(5,5) + 1(16,1) + 1/2(20,1)
Spectrum 1453; b1=(3,8), b2=(3/2,3); singlets = 21
21(1,1) + 2(10,4) + 1(4,4) + 3(4,1) + 1/2(4,5) + 1/2(1,4) + 4(1,5) + 1/2(1,16) + 7/2(5,4) + 15(5,1) + 1(16,1) + 1/2(20,1)
Spectrum 1454; b1=(3,8), b2=(3/2,3); singlets = 15
15(1,1) + 2(10,4) + 7(4,1) + 1/2(4,5) + 2(1,4) + 4(1,5) + 1/2(1,16) + 4(5,4) + 13(5,1) + 1(16,1) + 1/2(20,1)
Spectrum 1455; b1=(3,8), b2=(3/2,3); singlets = 21
21(1,1) + 2(10,4) + 1(4,4) + 1/2(4,1) + 1(4,5) + 3(1,4) + 2(1,5) + 1/2(1,16) + 3(5,4) + 17(5,1) + 1(16,1) + 1/2(20,1)
Spectrum 1456; b1=(3,8), b2=(3/2,3); singlets = 15
15(1,1) + 2(10,4) + 9/2(4,1) + 1(4,5) + 9/2(1,4) + 2(1,5) + 1/2(1,16) + 7/2(5,4) + 15(5,1) + 1(16,1) + 1/2(20,1)
Spectrum 1457; b1=(3,8), b2=(3/2,3); singlets = 15
15(1,1) + 2(10,4) + 2(4,1) + 3/2(4,5) + 7(1,4) + 1/2(1,16) + 3(5,4) + 17(5,1) + 1(16,1) + 1/2(20,1)
Spectrum 1458; b1=(3,8), b2=(3/2,3); singlets = 12
12(1,1) + 2(10,4) + 2(4,4) + 3/2(4,1) + 13/2(1,4) + 1(1,5) + 1/2(1,16) + 3/2(5,4) + 18(5,1) + 1(5,5) + 1(16,1) + 1/2(20,1)
Spectrum 1459; b1=(3,8), b2=(3/2,3); singlets = 6
6(1,1) + 2(10,4) + 1(4,4) + 11/2(4,1) + 8(1,4) + 1(1,5) + 1/2(1,16) + 2(5,4) + 16(5,1) + 1(5,5) + 1(16,1) + 1/2(20,1)
Spectrum 1460; b1=(3,8), b2=(3/2,3); singlets = 0
2(10,4) + 19/2(4,1) + 19/2(1,4) + 1(1,5) + 1/2(1,16) + 5/2(5,4) + 14(5,1) + 1(5,5) + 1(16,1) + 1/2(20,1)
Spectrum 1461; b1=(3,8), b2=(3/2,3); singlets = 37
37(1,1) + 1/2(10,4) + 6(10,1) + 9/2(4,1) + 1(4,5) + 1(1,10) + 1(1,4) + 6(5,4) + 1(5,5) + 1(16,1) + 1/2(20,1)
Spectrum 1462; b1=(3,8), b2=(3/2,3); singlets = 27
27(1,1) + 1(10,4) + 4(10,1) + 7(4,1) + 1/2(4,5) + 1(1,10) + 1(1,4) + 2(1,5) + 5(5,4) + 4(5,1) + 1(5,5) + 1(16,1) + 1/2(20,1)
Spectrum 1463; b1=(3,8), b2=(3/2,3); singlets = 33
33(1,1) + 1(10,4) + 4(10,1) + 1(4,4) + 1/2(4,1) + 1(4,5) + 1(1,10) + 2(1,4) + 4(5,4) + 8(5,1) + 1(5,5) + 1(16,1) + 1/2(20,1)
Spectrum 1464; b1=(3,8), b2=(3/2,3); singlets = 27
27(1,1) + 1(10,4) + 4(10,1) + 9/2(4,1) + 1(4,5) + 1(1,10) + 7/2(1,4) + 9/2(5,4) + 6(5,1) + 1(5,5) + 1(16,1) + 1/2(20,1)
Spectrum 1465; b1=(3,8), b2=(3/2,3); singlets = 17
17(1,1) + 3/2(10,4) + 2(10,1) + 19/2(4,1) + 1(1,10) + 1(1,4) + 4(1,5) + 4(5,4) + 8(5,1) + 1(5,5) + 1(16,1) + 1/2(20,1)
Spectrum 1466; b1=(3,8), b2=(3/2,3); singlets = 23
23(1,1) + 3/2(10,4) + 2(10,1) + 1(4,4) + 3(4,1) + 1/2(4,5) + 1(1,10) + 2(1,4) + 2(1,5) + 3(5,4) + 12(5,1) + 1(5,5) + 1(16,1) + 1/2(20,1)
Spectrum 1467; b1=(3,8), b2=(3/2,3); singlets = 17
17(1,1) + 3/2(10,4) + 2(10,1) + 7(4,1) + 1/2(4,5) + 1(1,10) + 7/2(1,4) + 2(1,5) + 7/2(5,4) + 10(5,1) + 1(5,5) + 1(16,1) + 1/2(20,1)
Spectrum 1468; b1=(3,8), b2=(3/2,3); singlets = 23
23(1,1) + 3/2(10,4) + 2(10,1) + 1(4,4) + 1/2(4,1) + 1(4,5) + 1(1,10) + 9/2(1,4) + 5/2(5,4) + 14(5,1) + 1(5,5) + 1(16,1) + 1/2(20,1)
Spectrum 1469; b1=(3,8), b2=(3/2,3); singlets = 17
17(1,1) + 3/2(10,4) + 2(10,1) + 9/2(4,1) + 1(4,5) + 1(1,10) + 6(1,4) + 3(5,4) + 12(5,1) + 1(5,5) + 1(16,1) + 1/2(20,1)
Spectrum 1470; b1=(3,8), b2=(3/2,3); singlets = 22
22(1,1) + 2(10,4) + 2(4,1) + 3/2(4,5) + 1(1,10) + 1(1,4) + 3(1,5) + 3(5,4) + 17(5,1) + 1(16,1) + 1/2(20,1)
Spectrum 1471; b1=(3,8), b2=(3/2,3); singlets = 19
19(1,1) + 2(10,4) + 2(4,4) + 3/2(4,1) + 1(1,10) + 1/2(1,4) + 4(1,5) + 3/2(5,4) + 18(5,1) + 1(5,5) + 1(16,1) + 1/2(20,1)
Spectrum 1472; b1=(3,8), b2=(3/2,3); singlets = 13
13(1,1) + 2(10,4) + 1(4,4) + 11/2(4,1) + 1(1,10) + 2(1,4) + 4(1,5) + 2(5,4) + 16(5,1) + 1(5,5) + 1(16,1) + 1/2(20,1)
Spectrum 1473; b1=(3,8), b2=(3/2,3); singlets = 7
7(1,1) + 2(10,4) + 19/2(4,1) + 1(1,10) + 7/2(1,4) + 4(1,5) + 5/2(5,4) + 14(5,1) + 1(5,5) + 1(16,1) + 1/2(20,1)
Spectrum 1474; b1=(3,8), b2=(3/2,3); singlets = 13
13(1,1) + 2(10,4) + 1(4,4) + 3(4,1) + 1/2(4,5) + 1(1,10) + 9/2(1,4) + 2(1,5) + 3/2(5,4) + 18(5,1) + 1(5,5) + 1(16,1) + 1/2(20,1)
Spectrum 1475; b1=(3,8), b2=(3/2,3); singlets = 7
7(1,1) + 2(10,4) + 7(4,1) + 1/2(4,5) + 1(1,10) + 6(1,4) + 2(1,5) + 2(5,4) + 16(5,1) + 1(5,5) + 1(16,1) + 1/2(20,1)
Spectrum 1476; b1=(3,8), b2=(3/2,3); singlets = 13
13(1,1) + 2(10,4) + 1(4,4) + 1/2(4,1) + 1(4,5) + 1(1,10) + 7(1,4) + 1(5,4) + 20(5,1) + 1(5,5) + 1(16,1) + 1/2(20,1)
Spectrum 1477; b1=(3,8), b2=(3/2,3); singlets = 7
7(1,1) + 2(10,4) + 9/2(4,1) + 1(4,5) + 1(1,10) + 17/2(1,4) + 3/2(5,4) + 18(5,1) + 1(5,5) + 1(16,1) + 1/2(20,1)
Spectrum 1478; b1=(3,8), b2=(3/2,3); singlets = 34
34(1,1) + 3/2(10,4) + 3(10,1) + 1(4,4) + 5(4,1) + 1(4,5) + 1/2(1,4) + 2(1,5) + 1/2(1,16) + 9/2(5,4) + 2(5,1) + 5/2(16,1)
Spectrum 1479; b1=(3,8), b2=(3/2,3); singlets = 28
28(1,1) + 3

$$19(1,1) + 3/2(10,4) + 3(10,1) + 1(4,4) + 10(4,1) + 11/2(1,4) + 1(1,5) + 1/2(1,16) + 7/2(5,4) + 1(5,1) + 1(5,5) + 5/2(16,1)$$
$$24(\mathbf{1}, \mathbf{1}) + 2(\mathbf{10}, \mathbf{4}) + 1(\mathbf{10}, \mathbf{1}) + 1(\mathbf{4}, \mathbf{4}) + 15/2(\mathbf{4}, \mathbf{1}) + 1/2(\mathbf{4}, \mathbf{5}) + 1/2(\mathbf{1}, \mathbf{4}) + 4(\mathbf{1}, \mathbf{5}) + 1/2(\mathbf{1}, \mathbf{16}) + 7/2(\mathbf{5}, \mathbf{4}) + 6(\mathbf{5}, \mathbf{1}) + 5/2(\mathbf{16}, \mathbf{1})$$
$$18(1,1) + 2(10,4) + 1(10,1) + 23/2(4,1) + 1/2(4,5) + 2(1,4) + 4(1,5) + 1/2(1,16) + 4(5,4) + 4(5,1) + 5/2(16,1)$$
$$30(\mathbf{1}, \mathbf{1}) + 2(\mathbf{10}, \mathbf{4}) + 1(\mathbf{10}, \mathbf{1}) + 2(\mathbf{4}, \mathbf{4}) + 1(\mathbf{4}, \mathbf{1}) + 1(\mathbf{4}, \mathbf{5}) + 3/2(\mathbf{1}, \mathbf{4}) + 2(\mathbf{1}, \mathbf{5}) + 1/2(\mathbf{1}, \mathbf{16}) + 5/2(\mathbf{5}, \mathbf{4}) + 10(\mathbf{5}, \mathbf{1}) + 5/2(\mathbf{16}, \mathbf{1})$$
$$24(\mathbf{1}, \mathbf{1}) + 2(\mathbf{10}, \mathbf{4}) + 1(\mathbf{10}, \mathbf{1}) + 1(\mathbf{4}, \mathbf{4}) + 5(\mathbf{4}, \mathbf{1}) + 1(\mathbf{4}, \mathbf{5}) + 3(\mathbf{1}, \mathbf{4}) + 2(\mathbf{1}, \mathbf{5}) + 1/2(\mathbf{1}, \mathbf{16}) + 3(\mathbf{5}, \mathbf{4}) + 8(\mathbf{5}, \mathbf{1}) + 5/2(\mathbf{16}, \mathbf{1})$$
$$18(1,1) + 2(10,4) + 1(10,1) + 9(4,1) + 1(4,5) + 9/2(1,4) + 2(1,5) + 1/2(1,16) + 7/2(5,4) + 6(5,1) + 5/2(16,1)$$
$$24(\mathbf{1}, \mathbf{1}) + 2(\mathbf{10}, \mathbf{4}) + 1(\mathbf{10}, \mathbf{1}) + 1(\mathbf{4}, \mathbf{4}) + 5/2(\mathbf{4}, \mathbf{1}) + 3/2(\mathbf{4}, \mathbf{5}) + 11/2(\mathbf{1}, \mathbf{4}) + 1/2(\mathbf{1}, \mathbf{16}) + 5/2(\mathbf{5}, \mathbf{4}) + 10(\mathbf{5}, \mathbf{1}) + 5/2(\mathbf{16}, \mathbf{1})$$
$$18(\mathbf{1,1}) + 2(\mathbf{10,4}) + 1(\mathbf{10,1}) + 13/2(\mathbf{4,1}) + 3/2(\mathbf{4,5}) + 7(\mathbf{1,4}) + 1/2(\mathbf{1,16}) + 3(\mathbf{5,4}) + 8(\mathbf{5,1}) + 5/2(\mathbf{16,1})$$
$$21(\mathbf{1}, \mathbf{1}) + 2(\mathbf{10}, \mathbf{4}) + 1(\mathbf{10}, \mathbf{1}) + 3(\mathbf{4}, \mathbf{4}) + 2(\mathbf{4}, \mathbf{1}) + 5(\mathbf{1}, \mathbf{4}) + 1(\mathbf{1}, \mathbf{5}) + 1/2(\mathbf{1}, \mathbf{16}) + 1(\mathbf{5}, \mathbf{4}) + 11(\mathbf{5}, \mathbf{1}) + 1(\mathbf{5}, \mathbf{5}) + 5/2(\mathbf{16}, \mathbf{1})$$
$$15(\mathbf{1},\mathbf{1}) + 2(\mathbf{10},\mathbf{4}) + 1(\mathbf{10},\mathbf{1}) + 2(\mathbf{4},\mathbf{4}) + 6(\mathbf{4},\mathbf{1}) + 13/2(\mathbf{1},\mathbf{4}) + 1(\mathbf{1},\mathbf{5}) + 1/2(\mathbf{1},\mathbf{16}) + 3/2(\mathbf{5},\mathbf{4}) + 9(\mathbf{5},\mathbf{1}) + 1(\mathbf{5},\mathbf{5}) + 5/2(\mathbf{16},\mathbf{1})$$
$$9(1,1) + 2(10,4) + 1(10,1) + 1(4,4) + 10(4,1) + 8(1,4) + 1(1,5) + 1/2(1,16) + 2(5,4) + 7(5,1) + 1(5,5) + 5/2(16,1)$$
$$3(1,1) + 2(10,4) + 1(10,1) + 14(4,1) + 19/2(1,4) + 1(1,5) + 1/2(1,16) + 5/2(5,4) + 5(5,1) + 1(5,5) + 5/2(16,1)$$

Spectrum 1496; $b_1=(3,8)$, $b_2=(3/2,3)$; singlets = 42

Spectrum 1497; $b_1=(3,8)$, $b_2=(3/2,3)$; singlets = 35

Spectrum 1498; $b_1=(3,8)$, $b_2=(3/2,3)$; singlets = 32

Spectrum 1499; $b_1=(3,8)$, $b_2=(3/2,3)$; singlets = 26

Spectrum 1500; $b_1=(3,8)$, $b_2=(3/2,3)$; singlets = 20

Spectrum 1501; $b_1=(3,8)$, $b_2=(3/2,3)$; singlets = 32

Spectrum 1502; $b_1=(3,8)$, $b_2=(3/2,3)$; singlets = 26

Spectrum 1503; $b_1=(3,8)$, $b_2=(3/2,3)$; singlets = 20

Spectrum 1504; $b_1=(3,8)$, $b_2=(3/2,3)$; singlets = 25

Spectrum 1505; $b_1=(3,8)$, $b_2=(3/2,3)$; singlets = 31

Spectrum 1506; $b_1=(3,8)$, $b_2=(3/2,3)$; singlets = 25

Spectrum 1507; $b_1=(3,8)$, $b_2=(3/2,3)$; singlets = 22

Spectrum 1508; $b_1=(3,8)$, $b_2=(3/2,3)$; singlets = 16

Spectrum 1509; $b_1=(3,8)$, $b_2=(3/2,3)$; singlets = 10

$$10(1,1) + 2(10,4) + 1(10,1) + 14(4,1) + 1(1,10) + 7/2(1,4) + 4(1,5) + 5/2(5,4) + 5(5,1) + 1(5,5) + 5/2(16,1)$$
$$22(\mathbf{1},\mathbf{1}) + 2(\mathbf{10},\mathbf{4}) + 1(\mathbf{10},\mathbf{1}) + 2(\mathbf{4},\mathbf{4}) + 7/2(\mathbf{4},\mathbf{1}) + 1/2(\mathbf{4},\mathbf{5}) + 1(\mathbf{1},\mathbf{10}) + 3(\mathbf{1},\mathbf{4}) + 2(\mathbf{1},\mathbf{5}) + 1(\mathbf{5},\mathbf{4}) + 11(\mathbf{5},\mathbf{1}) + 1(\mathbf{5},\mathbf{5}) + 5/2(\mathbf{16},\mathbf{1})$$
$$16(\mathbf{1}, \mathbf{1}) + 2(\mathbf{10}, \mathbf{4}) + 1(\mathbf{10}, \mathbf{1}) + 1(\mathbf{4}, \mathbf{4}) + 15/2(\mathbf{4}, \mathbf{1}) + 1/2(\mathbf{4}, \mathbf{5}) + 1(\mathbf{1}, \mathbf{10}) + 9/2(\mathbf{1}, \mathbf{4}) + 2(\mathbf{1}, \mathbf{5}) + 3/2(\mathbf{5}, \mathbf{4}) + 9(\mathbf{5}, \mathbf{1}) + 1(\mathbf{5}, \mathbf{5}) + 5/2(\mathbf{16}, \mathbf{1})$$
$$10(1,1) + 2(10,4) + 1(10,1) + 23/2(4,1) + 1/2(4,5) + 1(1,10) + 6(1,4) + 2(1,5) + 2(5,4) + 7(5,1) + 1(5,5) + 5/2(16,1)$$
$$22(\mathbf{1}, \mathbf{1}) + 2(\mathbf{10}, \mathbf{4}) + 1(\mathbf{10}, \mathbf{1}) + 2(\mathbf{4}, \mathbf{4}) + 1(\mathbf{4}, \mathbf{1}) + 1(\mathbf{4}, \mathbf{5}) + 1(\mathbf{1}, \mathbf{10}) + 11/2(\mathbf{1}, \mathbf{4}) + 1/2(\mathbf{5}, \mathbf{4}) + 13(\mathbf{5}, \mathbf{1}) + 1(\mathbf{5}, \mathbf{5}) + 5/2(\mathbf{16}, \mathbf{1})$$

Spectrum 1514; $b_1=(3,8)$, $b_2=(3/2,3)$; singlets = 16

Spectrum 1515; $b_1=(3,8)$, $b_2=(3/2,3)$; singlets = 10

Spectrum 1516; $b_1=(3,8)$, $b_2=(3/2,3)$; singlets = 3

$$3(1,1) + 9(10,1) + 3(4,4) + 2(4,1) + 1(1,10) + 9(1,4) + 1(1,5) + 3(5,4) + 8(5,1) + 1/2(16,5)$$
$$27(\mathbf{1},\mathbf{1}) + 1(\mathbf{10},\mathbf{4}) + 5(\mathbf{10},\mathbf{1}) + 7/2(\mathbf{4},\mathbf{1}) + 9/2(\mathbf{1},\mathbf{4}) + 1(\mathbf{1},\mathbf{5}) + 1/2(\mathbf{1},\mathbf{16}) + 11/2(\mathbf{5},\mathbf{4}) + 5(\mathbf{5},\mathbf{1}) + 1(\mathbf{5},\mathbf{5}) + 1/2(\mathbf{16},\mathbf{1}) + 1/2(\mathbf{20},\mathbf{1})$$

$$27(1.1) + 3/2(10.4) + 4(10.1) + 11/2(4.1) + 1/2(4.5) + 1(1.10) + 7/2(1.4) + 2(1.5) + 7/2(5.4) + 4(5.1) + 1(5.5) + 2(16.1)$$

Spectrum 1586; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 79
79(1,1) + 9(4,4) + 6(4,1) + 1(1,10) + 9(1,5) + 6(5,1) + 1/2(16,1)
Spectrum 1587; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 73
73(1,1) + 8(4,4) + 10(4,1) + 1(1,10) + 3/2(1,4) + 9(1,5) + 1/2(5,4) + 4(5,1) + 1/2(16,1)
Spectrum 1588; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 67
67(1,1) + 7(4,4) + 14(4,1) + 1(1,10) + 3(1,4) + 9(1,5) + 1(5,4) + 2(5,1) + 1/2(16,1)
Spectrum 1589; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 61
61(1,1) + 6(4,4) + 18(4,1) + 1(1,10) + 9/2(1,4) + 9(1,5) + 3/2(5,4) + 1/2(16,1)
Spectrum 1590; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 73
73(1,1) + 8(4,4) + 15/2(4,1) + 1/2(4,5) + 1(1,10) + 4(1,4) + 7(1,5) + 6(5,1) + 1/2(16,1)
Spectrum 1591; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 67
67(1,1) + 7(4,4) + 23/2(4,1) + 1/2(4,5) + 1(1,10) + 11/2(1,4) + 7(1,5) + 1/2(5,4) + 4(5,1) + 1/2(16,1)
Spectrum 1592; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 61
61(1,1) + 6(4,4) + 31/2(4,1) + 1/2(4,5) + 1(1,10) + 7(1,4) + 7(1,5) + 1(5,4) + 2(5,1) + 1/2(16,1)
Spectrum 1593; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 55
55(1,1) + 5(4,4) + 39/2(4,1) + 1/2(4,5) + 1(1,10) + 17/2(1,4) + 7(1,5) + 3/2(5,4) + 1/2(16,1)
Spectrum 1594; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 67
67(1,1) + 7(4,4) + 9(4,1) + 1(4,5) + 1(1,10) + 8(1,4) + 5(1,5) + 6(5,1) + 1/2(16,1)
Spectrum 1595; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 61
61(1,1) + 6(4,4) + 13(4,1) + 1(4,5) + 1(1,10) + 19/2(1,4) + 5(1,5) + 1/2(5,4) + 4(5,1) + 1/2(16,1)
Spectrum 1596; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 55
55(1,1) + 5(4,4) + 17(4,1) + 1(4,5) + 1(1,10) + 11(1,4) + 5(1,5) + 1(5,4) + 2(5,1) + 1/2(16,1)
Spectrum 1597; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 49
49(1,1) + 4(4,4) + 21(4,1) + 1(4,5) + 1(1,10) + 25/2(1,4) + 5(1,5) + 3/2(5,4) + 1/2(16,1)
Spectrum 1598; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 61
61(1,1) + 6(4,4) + 21/2(4,1) + 3/2(4,5) + 1(1,10) + 12(1,4) + 3(1,5) + 6(5,1) + 1/2(16,1)
Spectrum 1599; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 55
55(1,1) + 5(4,4) + 29/2(4,1) + 3/2(4,5) + 1(1,10) + 27/2(1,4) + 3(1,5) + 1/2(5,4) + 4(5,1) + 1/2(16,1)
Spectrum 1600; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 49
49(1,1) + 4(4,4) + 37/2(4,1) + 3/2(4,5) + 1(1,10) + 15(1,4) + 3(1,5) + 1(5,4) + 2(5,1) + 1/2(16,1)
Spectrum 1601; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 43
43(1,1) + 3(4,4) + 45/2(4,1) + 3/2(4,5) + 1(1,10) + 33/2(1,4) + 3(1,5) + 3/2(5,4) + 1/2(16,1)
Spectrum 1602; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 55
55(1,1) + 5(4,4) + 12(4,1) + 2(4,5) + 1(1,10) + 16(1,4) + 1(1,5) + 6(5,1) + 1/2(16,1)
Spectrum 1603; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 49
49(1,1) + 4(4,4) + 16(4,1) + 2(4,5) + 1(1,10) + 35/2(1,4) + 1(1,5) + 1/2(5,4) + 4(5,1) + 1/2(16,1)
Spectrum 1604; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 43
43(1,1) + 3(4,4) + 20(4,1) + 2(4,5) + 1(1,10) + 19(1,4) + 1(1,5) + 1(5,4) + 2(5,1) + 1/2(16,1)
Spectrum 1605; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 37
37(1,1) + 2(4,4) + 24(4,1) + 2(4,5) + 1(1,10) + 41/2(1,4) + 1(1,5) + 3/2(5,4) + 1/2(16,1)
Spectrum 1606; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 40
40(1,1) + 5(4,4) + 22(4,1) + 1(1,10) + 16(1,4) + 4(1,5) + 1(5,1) + 1(5,5) + 1/2(16,1)
Spectrum 1607; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 34
34(1,1) + 4(4,4) + 47/2(4,1) + 1/2(4,5) + 1(1,10) + 20(1,4) + 2(1,5) + 1(5,1) + 1(5,5) + 1/2(16,1)
Spectrum 1608; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 28
28(1,1) + 3(4,4) + 25(4,1) + 1(4,5) + 1(1,10) + 24(1,4) + 1(5,1) + 1(5,5) + 1/2(16,1)
Spectrum 1609; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 86
86(1,1) + 1(10,1) + 9(4,4) + 1(1,10) + 9(1,5) + 9(5,1)
Spectrum 1610; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 80
80(1,1) + 1(10,1) + 8(4,4) + 4(4,1) + 1(1,10) + 3/2(1,4) + 9(1,5) + 1/2(5,4) + 7(5,1)
Spectrum 1611; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 74
74(1,1) + 1(10,1) + 7(4,4) + 8(4,1) + 1(1,10) + 3(1,4) + 9(1,5) + 1(5,4) + 5(5,1)
Spectrum 1612; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 68
68(1,1) + 1(10,1) + 6(4,4) + 12(4,1) + 1(1,10) + 9/2(1,4) + 9(1,5) + 3/2(5,4) + 3(5,1)
Spectrum 1613; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 62
62(1,1) + 1(10,1) + 5(4,4) + 16(4,1) + 1(1,10) + 6(1,4) + 9(1,5) + 2(5,4) + 1(5,1)
Spectrum 1614; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 74
74(1,1) + 1(10,1) + 7(4,4) + 11/2(4,1) + 1/2(4,5) + 1(1,10) + 11/2(1,4) + 7(1,5) + 1/2(5,4) + 7(5,1)
Spectrum 1615; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 68
68(1,1) + 1(10,1) + 6(4,4) + 19/2(4,1) + 1/2(4,5) + 1(1,10) + 7(1,4) + 7(1,5) + 1(5,4) + 5(5,1)
Spectrum 1616; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 62
62(1,1) + 1(10,1) + 5(4,4) + 27/2(4,1) + 1/2(4,5) + 1(1,10) + 17/2(1,4) + 7(1,5) + 3/2(5,4) + 3(5,1)
Spectrum 1617; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 56
56(1,1) + 1(10,1) + 4(4,4) + 35/2(4,1) + 1/2(4,5) + 1(1,10) + 10(1,4) + 7(1,5) + 2(5,4) + 1(5,1)
Spectrum 1618; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 62
62(1,1) + 1(10,1) + 5(4,4) + 11(4,1) + 1(4,5) + 1(1,10) + 11(1,4) + 5(1,5) + 1(5,4) + 5(5,1)
Spectrum 1619; $b1=(3/2,3)$, $b2=(3/2,3)$; singlets = 56
56(1,1) + 1(10,1) + 4(4,4) + 15(4,1) + 1(4,5) + 1(1,10) + 25/2(1,4) + 5(1,5) + 3/2(5,4) + 3(5,1)

Spectrum 1620; b1={3/2,3}, b2={3/2,3}; singlets = 50

50(1,1) + 1(10,1) + 3(4,4) + 19(4,1) + 1(4,5) + 1(1,10) + 14(1,4) + 5(1,5) + 2(5,4) + 1(5,1)

Spectrum 1621; b1={3/2,3}, b2={3/2,3}; singlets = 50

50(1,1) + 1(10,1) + 1(10,1) + 3(4,4) + 33/2(4,1) + 3/2(4,5) + 1(1,10) + 33/2(1,4) + 3(1,5) + 3/2(5,4) + 3(5,1)

Spectrum 1622; b1={3/2,3}, b2={3/2,3}; singlets = 44

44(1,1) + 1(10,1) + 2(4,4) + 41/2(4,1) + 3/2(4,5) + 1(1,10) + 18(1,4) + 3(1,5) + 2(5,4) + 1(5,1)

Spectrum 1623; b1={3/2,3}, b2={3/2,3}; singlets = 38

38(1,1) + 1(10,1) + 1(4,4) + 22(4,1) + 2(4,5) + 1(1,10) + 22(1,4) + 1(1,5) + 2(5,4) + 1(5,1)

Spectrum 1624; b1={3/2,3}, b2={3/2,3}; singlets = 47

47(1,1) + 1(10,1) + 5(4,4) + 16(4,1) + 1(1,10) + 16(1,4) + 4(1,5) + 4(5,1) + 1(5,5)

Spectrum 1625; b1={3/2,3}, b2={3/2,3}; singlets = 41

41(1,1) + 1(10,1) + 4(4,4) + 20(4,1) + 1(1,10) + 35/2(1,4) + 4(1,5) + 1/2(5,4) + 2(5,1) + 1(5,5)

Spectrum 1626; b1={3/2,3}, b2={3/2,3}; singlets = 35

35(1,1) + 1(10,1) + 3(4,4) + 24(4,1) + 1(1,10) + 19(1,4) + 4(1,5) + 1(5,4) + 1(5,5)

Spectrum 1627; b1={3/2,3}, b2={3/2,3}; singlets = 35

35(1,1) + 1(10,1) + 3(4,4) + 43/2(4,1) + 1/2(4,5) + 1(1,10) + 43/2(1,4) + 2(1,5) + 1/2(5,4) + 2(5,1) + 1(5,5)

Spectrum 1628; b1={3/2,3}, b2={3/2,3}; singlets = 29

29(1,1) + 1(10,1) + 2(4,4) + 51/2(4,1) + 1/2(4,5) + 1(1,10) + 23(1,4) + 2(1,5) + 1(5,4) + 1(5,5)

Spectrum 1629; b1={3/2,3}, b2={3/2,3}; singlets = 23

23(1,1) + 1(10,1) + 1(4,4) + 27(4,1) + 1(4,5) + 1(1,10) + 27(1,4) + 1(5,4) + 1(5,5)

C2 x C3

Spectrum 1630; b1={3/2,3}, b2=(2,4); singlets = 15

15(1,1) + 3(4,6) + 3(4,1) + 1/2(4,14') + 1(4,14) + 1(1,21) + 11(1,6) + 1/2(1,64) + 1(5,6) + 1/2(16,1)

Spectrum 1631; b1={3/2,3}, b2=(2,4); singlets = 31

31(1,1) + 2(4,6) + 2(4,1) + 2(4,14') + 1(1,21) + 6(1,6) + 1(1,14) + 1/2(1,64) + 1(5,6) + 1/2(16,1)

Spectrum 1632; b1={3/2,3}, b2=(2,4); singlets = 33

33(1,1) + 2(4,6) + 2(4,1) + 2(4,14') + 1(1,21) + 9/2(1,6) + 1(1,14') + 1/2(1,14) + 1/2(1,64) + 1(5,6) + 1/2(16,1)

Spectrum 1633; b1={3/2,3}, b2=(2,4); singlets = 35

35(1,1) + 2(4,6) + 2(4,1) + 2(4,14') + 1(1,21) + 3(1,6) + 2(1,14') + 1/2(1,64) + 1(5,6) + 1/2(16,1)

Spectrum 1634; b1={3/2,3}, b2=(2,4); singlets = 24

24(1,1) + 1(4,6) + 1(4,1) + 5/2(4,14') + 1(1,21) + 19/2(1,6) + 1/2(1,64) + 1/2(5,6) + 3(5,1) + 1/2(16,1)

Spectrum 1635; b1={3/2,3}, b2=(2,4); singlets = 15

15(1,1) + 7(4,1) + 5/2(4,14') + 1(1,21) + 11(1,6) + 1/2(1,64) + 1(5,6) + 1/2(16,1)

Spectrum 1636; b1={3/2,3}, b2=(2,4); singlets = 8

8(1,1) + 1/2(4,21) + 2(4,6) + 11/2(4,1) + 1(4,14) + 1(1,21) + 21/2(1,6) + 3(1,14) + 1/2(5,6) + 3(5,1) + 1/2(16,1)

Spectrum 1637; b1={3/2,3}, b2=(2,4); singlets = 10

10(1,1) + 1/2(4,21) + 2(4,6) + 11/2(4,1) + 1(4,14) + 1(1,21) + 9(1,6) + 1(1,14') + 5/2(1,14) + 1/2(5,6) + 3(5,1) + 1/2(16,1)

Spectrum 1638; b1={3/2,3}, b2=(2,4); singlets = 1

1(1,1) + 1/2(4,21) + 1(4,6) + 23/2(4,1) + 1(4,14) + 1(1,21) + 21/2(1,6) + 1(1,14') + 5/2(1,14) + 1(5,6) + 1/2(16,1)

Spectrum 1639; b1={3/2,3}, b2=(2,4); singlets = 37

37(1,1) + 3(4,6) + 3(4,1) + 1/2(4,14') + 1(4,14) + 3(1,21) + 1(1,6) + 2(1,14) + 1(5,6) + 1/2(16,1)

Spectrum 1640; b1={3/2,3}, b2=(2,4); singlets = 12

12(1,1) + 1/2(4,21) + 2(4,6) + 11/2(4,1) + 1(4,14) + 1(1,21) + 15/2(1,6) + 2(1,14') + 2(1,14) + 1/2(5,6) + 3(5,1) + 1/2(16,1)

Spectrum 1641; b1={3/2,3}, b2=(2,4); singlets = 3

3(1,1) + 1/2(4,21) + 1(4,6) + 23/2(4,1) + 1(4,14) + 1(1,21) + 9(1,6) + 2(1,14') + 2(1,14) + 1(5,6) + 1/2(16,1)

Spectrum 1642; b1={3/2,3}, b2=(2,4); singlets = 1

1(1,1) + 1/2(4,21) + 1(4,6) + 9/2(4,1) + 1/2(4,14') + 1(4,14) + 1(1,21) + 14(1,6) + 2(1,14) + 6(5,1) + 1/2(16,1)

Spectrum 1643; b1={3/2,3}, b2=(2,4); singlets = 14

14(1,1) + 1/2(4,21) + 2(4,6) + 11/2(4,1) + 1(4,14) + 1(1,21) + 6(1,6) + 3(1,14') + 3/2(1,14) + 1/2(5,6) + 3(5,1) + 1/2(16,1)

Spectrum 1644; b1={3/2,3}, b2=(2,4); singlets = 5

5(1,1) + 1/2(4,21) + 1(4,6) + 23/2(4,1) + 1(4,14) + 1(1,21) + 15/2(1,6) + 3(1,14') + 3/2(1,14) + 1(5,6) + 1/2(16,1)

Spectrum 1645; b1={3/2,3}, b2=(2,4); singlets = 3

3(1,1) + 1/2(4,21) + 1(4,6) + 9/2(4,1) + 1/2(4,14') + 1(4,14) + 1(1,21) + 25/2(1,6) + 1(1,14') + 3/2(1,14) + 6(5,1) + 1/2(16,1)

Spectrum 1646; b1={3/2,3}, b2=(2,4); singlets = 30

30(1,1) + 2(4,6) + 2(4,1) + 1(4,14') + 1(4,14) + 3(1,21) + 9/2(1,6) + 1(1,14) + 1/2(5,6) + 3(5,1) + 1/2(16,1)

Spectrum 1647; b1={3/2,3}, b2=(2,4); singlets = 21

21(1,1) + 1(4,6) + 8(4,1) + 1(4,14') + 1(4,14) + 3(1,21) + 6(1,6) + 1(1,14) + 1(5,6) + 1/2(16,1)

Spectrum 1648; b1={3/2,3}, b2=(2,4); singlets = 31

31(1,1) + 1/2(4,21) + 2(4,6) + 11/2(4,1) + 1(4,14') + 1(1,21) + 2(1,6) + 5(1,14) + 1(5,6) + 1/2(16,1)

Spectrum 1649; b1={3/2,3}, b2=(2,4); singlets = 16

16(1,1) + 1/2(4,21) + 2(4,6) + 11/2(4,1) + 1(4,14) + 1(1,21) + 9/2(1,6) + 4(1,14') + 1(1,14) + 1/2(5,6) + 3(5,1) + 1/2(16,1)

Spectrum 1650; b1={3/2,3}, b2=(2,4); singlets = 7

7(1,1) + 1/2(4,21) + 1(4,6) + 23/2(4,1) + 1(4,14) + 1(1,21) + 6(1,6) + 4(1,14') + 1(1,14) + 1(5,6) + 1/2(16,1)

Spectrum 1651; b1={3/2,3}, b2=(2,4); singlets = 5

5(1,1) + 1/2(4,21) + 1(4,6) + 9/2(4,1) + 1/2(4,14') + 1(4,14) + 1(1,21) + 11(1,6) + 2(1,14') + 1(1,14) + 6(5,1) + 1/2(16,1)

Spectrum 1652; b1={3/2,3}, b2=(2,4); singlets = 32

32(1,1) + 2(4,6) + 2(4,1) + 1(4,14') + 1(4,14) + 3(1,21) + 3(1,6) + 1(1,14') + 1/2(1,14) + 1/2(5,6) + 3(5,1) + 1/2(16,1)

33(1,1) + 1(4,6) + 1(4,1) + 1(4,14) + 3(1,21) + 9/2(1,6) + 1(1,14') + 1/2(1,14) + 1(5,6) + 1/2(16,1)
Spectrum 1654; b1=(3/2,3), b2=(2,4); singlets = 33
33(1,1) + 1/2(4,21) + 2(4,6) + 11/2(4,1) + 1(4,14') + 1(1,21) + 1/2(1,6) + 1(1,14') + 9/2(1,14) + 1(5,6) + 1/2(16,1)
Spectrum 1655; b1=(3/2,3), b2=(2,4); singlets = 18
18(1,1) + 1/2(4,21) + 2(4,6) + 11/2(4,1) + 1(4,14) + 1(1,21) + 3(1,6) + 5(1,14') + 1/2(1,14) + 1/2(5,6) + 3(5,1) + 1/2(16,1)
Spectrum 1656; b1=(3/2,3), b2=(2,4); singlets = 9
9(1,1) + 1/2(4,21) + 1(4,6) + 23/2(4,1) + 1(4,14) + 1(1,21) + 9/2(1,6) + 5(1,14') + 1/2(1,14) + 1(5,6) + 1/2(16,1)
Spectrum 1657; b1=(3/2,3), b2=(2,4); singlets = 7
7(1,1) + 1/2(4,21) + 1(4,6) + 9/2(4,1) + 1/2(4,14') + 1(4,14) + 1(1,21) + 19/2(1,6) + 3(1,14') + 1/2(1,14) + 6(5,1) + 1/2(16,1)
Spectrum 1658; b1=(3/2,3), b2=(2,4); singlets = 34
34(1,1) + 2(4,6) + 2(4,1) + 1(4,14') + 1(4,14) + 3(1,21) + 3/2(1,6) + 2(1,14') + 1/2(5,6) + 3(5,1) + 1/2(16,1)
Spectrum 1659; b1=(3/2,3), b2=(2,4); singlets = 25
25(1,1) + 1(4,6) + 8(4,1) + 1(4,14') + 1(4,14) + 3(1,21) + 3(1,6) + 2(1,14') + 1(5,6) + 1/2(16,1)
Spectrum 1660; b1=(3/2,3), b2=(2,4); singlets = 23
23(1,1) + 1(4,6) + 1(4,1) + 3/2(4,14') + 1(4,14) + 3(1,21) + 8(1,6) + 6(5,1) + 1/2(16,1)
Spectrum 1661; b1=(3/2,3), b2=(2,4); singlets = 14
14(1,1) + 7(4,1) + 3/2(4,14') + 1(4,14) + 3(1,21) + 19/2(1,6) + 1/2(5,6) + 3(5,1) + 1/2(16,1)
Spectrum 1662; b1=(3/2,3), b2=(2,4); singlets = 24
24(1,1) + 1/2(4,21) + 1(4,6) + 9/2(4,1) + 3/2(4,14') + 1(1,21) + 11/2(1,6) + 4(1,14) + 1/2(5,6) + 3(5,1) + 1/2(16,1)
Spectrum 1663; b1=(3/2,3), b2=(2,4); singlets = 15
15(1,1) + 1/2(4,21) + 21/2(4,1) + 3/2(4,14') + 1(1,21) + 7(1,6) + 4(1,14) + 1(5,6) + 1/2(16,1)
Spectrum 1664; b1=(3/2,3), b2=(2,4); singlets = 20
20(1,1) + 1/2(4,21) + 2(4,6) + 11/2(4,1) + 1(4,14) + 1(1,21) + 3/2(1,6) + 6(1,14') + 1/2(5,6) + 3(5,1) + 1/2(16,1)
Spectrum 1665; b1=(3/2,3), b2=(2,4); singlets = 11
11(1,1) + 1/2(4,21) + 1(4,6) + 23/2(4,1) + 1(4,14) + 1(1,21) + 3(1,6) + 6(1,14') + 1(5,6) + 1/2(16,1)
Spectrum 1666; b1=(3/2,3), b2=(2,4); singlets = 9
9(1,1) + 1/2(4,21) + 1(4,6) + 9/2(4,1) + 1/2(4,14') + 1(4,14) + 1(1,21) + 8(1,6) + 4(1,14') + 6(5,1) + 1/2(16,1)
Spectrum 1667; b1=(3/2,3), b2=(2,4); singlets = 0
1/2(4,21) + 21/2(4,1) + 1/2(4,14') + 1(4,14) + 1(1,21) + 19/2(1,6) + 4(1,14') + 1/2(5,6) + 3(5,1) + 1/2(16,1)
Spectrum 1668; b1=(3/2,3), b2=(2,4); singlets = 26
26(1,1) + 1/2(4,21) + 1(4,6) + 9/2(4,1) + 3/2(4,14') + 1(1,21) + 4(1,6) + 1(1,14') + 7/2(1,14) + 1/2(5,6) + 3(5,1) + 1/2(16,1)
Spectrum 1669; b1=(3/2,3), b2=(2,4); singlets = 17
17(1,1) + 1/2(4,21) + 21/2(4,1) + 3/2(4,14') + 1(1,21) + 11/2(1,6) + 1(1,14') + 7/2(1,14) + 1(5,6) + 1/2(16,1)
Spectrum 1670; b1=(3/2,3), b2=(2,4); singlets = 28
28(1,1) + 1/2(4,21) + 1(4,6) + 9/2(4,1) + 3/2(4,14') + 1(1,21) + 5/2(1,6) + 2(1,14') + 3(1,14) + 1/2(5,6) + 3(5,1) + 1/2(16,1)
Spectrum 1671; b1=(3/2,3), b2=(2,4); singlets = 19
19(1,1) + 1/2(4,21) + 21/2(4,1) + 3/2(4,14') + 1(1,21) + 4(1,6) + 2(1,14') + 3(1,14) + 1(5,6) + 1/2(16,1)
Spectrum 1672; b1=(3/2,3), b2=(2,4); singlets = 17
17(1,1) + 1/2(4,21) + 7/2(4,1) + 2(4,14') + 1(1,21) + 9(1,6) + 3(1,14) + 6(5,1) + 1/2(16,1)
Spectrum 1673; b1=(3/2,3), b2=(2,4); singlets = 30
30(1,1) + 1/2(4,21) + 1(4,6) + 9/2(4,1) + 3/2(4,14') + 1(1,21) + 1(1,6) + 3(1,14') + 5/2(1,14) + 1/2(5,6) + 3(5,1) + 1/2(16,1)
Spectrum 1674; b1=(3/2,3), b2=(2,4); singlets = 21
21(1,1) + 1/2(4,21) + 21/2(4,1) + 3/2(4,14') + 1(1,21) + 5/2(1,6) + 3(1,14') + 5/2(1,14) + 1(5,6) + 1/2(16,1)
Spectrum 1675; b1=(3/2,3), b2=(2,4); singlets = 19
19(1,1) + 1/2(4,21) + 7/2(4,1) + 2(4,14') + 1(1,21) + 15/2(1,6) + 1(1,14') + 5/2(1,14) + 6(5,1) + 1/2(16,1)
Spectrum 1676; b1=(3/2,3), b2=(2,4); singlets = 37
37(1,1) + 7(4,1) + 5/2(4,14') + 3(1,21) + 1(1,6) + 2(1,14) + 1(5,6) + 1/2(16,1)
Spectrum 1677; b1=(3/2,3), b2=(2,4); singlets = 23
23(1,1) + 1/2(4,21) + 21/2(4,1) + 3/2(4,14') + 1(1,21) + 1(1,6) + 4(1,14') + 2(1,14) + 1(5,6) + 1/2(16,1)
Spectrum 1678; b1=(3/2,3), b2=(2,4); singlets = 21
21(1,1) + 1/2(4,21) + 7/2(4,1) + 2(4,14') + 1(1,21) + 6(1,6) + 2(1,14') + 2(1,14) + 6(5,1) + 1/2(16,1)
Spectrum 1679; b1=(3/2,3), b2=(2,4); singlets = 23
23(1,1) + 1/2(4,21) + 7/2(4,1) + 2(4,14') + 1(1,21) + 9/2(1,6) + 3(1,14') + 3/2(1,14) + 6(5,1) + 1/2(16,1)
Spectrum 1680; b1=(3/2,3), b2=(2,4); singlets = 39
39(1,1) + 3(4,14') + 3(1,21) + 3(1,6) + 1(1,14) + 6(5,1) + 1/2(16,1)
Spectrum 1681; b1=(3/2,3), b2=(2,4); singlets = 25
25(1,1) + 1/2(4,21) + 7/2(4,1) + 2(4,14') + 1(1,21) + 3(1,6) + 4(1,14') + 1(1,14) + 6(5,1) + 1/2(16,1)
Spectrum 1682; b1=(3/2,3), b2=(2,4); singlets = 41
41(1,1) + 3(4,14') + 3(1,21) + 3/2(1,6) + 1(1,14') + 1/2(1,14) + 6(5,1) + 1/2(16,1)
Spectrum 1683; b1=(3/2,3), b2=(2,4); singlets = 27
27(1,1) + 1/2(4,21) + 7/2(4,1) + 2(4,14') + 1(1,21) + 3/2(1,6) + 5(1,14') + 1/2(1,14) + 6(5,1) + 1/2(16,1)
Spectrum 1684; b1=(3/2,3), b2=(2,4); singlets = 43
43(1,1) + 3(4,14') + 3(1,21) + 2(1,14') + 6(5,1) + 1/2(16,1)
Spectrum 1685; b1=(3/2,3), b2=(2,4); singlets = 29
29(1,1) + 1/2(4,21) + 7/2(4

Spectrum 1687; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 3
 $3(1,1) + 1(10,1) + 3(4,6) + 4(4,1) + 1(4,14') + 1(1,21) + 13(1,6) + 1/2(1,14) + 1/2(1,64) + 2(5,1) + 1/2(5,14)$
Spectrum 1688; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 5
 $5(1,1) + 1(10,1) + 3(4,6) + 4(4,1) + 1(4,14') + 1(1,21) + 23/2(1,6) + 1(1,14') + 1/2(1,64) + 2(5,1) + 1/2(5,14)$
Spectrum 1689; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 29
 $29(1,1) + 1(10,1) + 1(4,6) + 2(4,1) + 2(4,14') + 1(1,21) + 15/2(1,6) + 1(1,14) + 1/2(1,64) + 3/2(5,6)$
Spectrum 1690; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 31
 $31(1,1) + 1(10,1) + 1(4,6) + 2(4,1) + 2(4,14') + 1(1,21) + 6(1,6) + 1(1,14') + 1/2(1,14) + 1/2(1,64) + 3/2(5,6)$
Spectrum 1691; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 33
 $33(1,1) + 1(10,1) + 1(4,6) + 2(4,1) + 2(4,14') + 1(1,21) + 9/2(1,6) + 2(1,14') + 1/2(1,64) + 3/2(5,6)$
Spectrum 1692; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 22
 $22(1,1) + 1(10,1) + 1(4,1) + 5/2(4,14') + 1(1,21) + 11(1,6) + 1/2(1,64) + 1(5,6) + 3(5,1)$
Spectrum 1693; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 6
 $6(1,1) + 1(10,1) + 1/2(4,21) + 1(4,6) + 11/2(4,1) + 1(4,14) + 1(1,21) + 12(1,6) + 3(1,14) + 1(5,6) + 3(5,1)$
Spectrum 1694; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 3
 $3(1,1) + 1(10,1) + 1/2(4,21) + 3(4,6) + 15/2(4,1) + 1(1,21) + 9(1,6) + 9/2(1,14) + 2(5,1) + 1/2(5,14)$
Spectrum 1695; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 8
 $8(1,1) + 1(10,1) + 1/2(4,21) + 1(4,6) + 11/2(4,1) + 1(4,14) + 1(1,21) + 21/2(1,6) + 1(1,14') + 5/2(1,14) + 1(5,6) + 3(5,1)$
Spectrum 1696; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 5
 $5(1,1) + 1(10,1) + 1/2(4,21) + 3(4,6) + 15/2(4,1) + 1(1,21) + 15/2(1,6) + 1(1,14') + 4(1,14) + 2(5,1) + 1/2(5,14)$
Spectrum 1697; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 35
 $35(1,1) + 1(10,1) + 2(4,6) + 3(4,1) + 1/2(4,14') + 1(4,14) + 3(1,21) + 5/2(1,6) + 2(1,14) + 3/2(5,6)$
Spectrum 1698; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 10
 $10(1,1) + 1(10,1) + 1/2(4,21) + 1(4,6) + 11/2(4,1) + 1(4,14) + 1(1,21) + 9(1,6) + 2(1,14') + 2(1,14) + 1(5,6) + 3(5,1)$
Spectrum 1699; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 1
 $1(1,1) + 1(10,1) + 1/2(4,21) + 23/2(4,1) + 1(4,14) + 1(1,21) + 21/2(1,6) + 2(1,14') + 2(1,14) + 3/2(5,6)$
Spectrum 1700; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 7
 $7(1,1) + 1(10,1) + 1/2(4,21) + 3(4,6) + 15/2(4,1) + 1(1,21) + 6(1,6) + 2(1,14') + 7/2(1,14) + 2(5,1) + 1/2(5,14)$
Spectrum 1701; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 37
 $37(1,1) + 1(10,1) + 2(4,6) + 3(4,1) + 1/2(4,14') + 1(4,14) + 3(1,21) + 1(1,6) + 1(1,14') + 3/2(1,14) + 3/2(5,6)$
Spectrum 1702; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 12
 $12(1,1) + 1(10,1) + 1/2(4,21) + 1(4,6) + 11/2(4,1) + 1(4,14) + 1(1,21) + 15/2(1,6) + 3(1,14') + 3/2(1,14) + 1(5,6) + 3(5,1)$
Spectrum 1703; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 3
 $3(1,1) + 1(10,1) + 1/2(4,21) + 23/2(4,1) + 1(4,14) + 1(1,21) + 9(1,6) + 3(1,14') + 3/2(1,14) + 3/2(5,6)$
Spectrum 1704; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 1
 $1(1,1) + 1(10,1) + 1/2(4,21) + 9/2(4,1) + 1/2(4,14') + 1(4,14) + 1(1,21) + 14(1,6) + 1(1,14') + 3/2(1,14) + 1/2(5,6) + 6(5,1)$
Spectrum 1705; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 9
 $9(1,1) + 1(10,1) + 1/2(4,21) + 3(4,6) + 15/2(4,1) + 1(1,21) + 9/2(1,6) + 3(1,14') + 3(1,14) + 2(5,1) + 1/2(5,14)$
Spectrum 1706; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 28
 $28(1,1) + 1(10,1) + 1(4,6) + 2(4,1) + 1(4,14') + 1(4,14) + 3(1,21) + 6(1,6) + 1(1,14) + 1(5,6) + 3(5,1)$
Spectrum 1707; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 19
 $19(1,1) + 1(10,1) + 8(4,1) + 1(4,14') + 1(4,14) + 3(1,21) + 15/2(1,6) + 1(1,14) + 3/2(5,6)$
Spectrum 1708; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 29
 $29(1,1) + 1(10,1) + 1/2(4,21) + 1(4,6) + 11/2(4,1) + 1(4,14') + 1(1,21) + 7/2(1,6) + 5(1,14) + 3/2(5,6)$
Spectrum 1709; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 14
 $14(1,1) + 1(10,1) + 1/2(4,21) + 1(4,6) + 11/2(4,1) + 1(4,14) + 1(1,21) + 6(1,6) + 4(1,14') + 1(1,14) + 1(5,6) + 3(5,1)$
Spectrum 1710; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 5
 $5(1,1) + 1(10,1) + 1/2(4,21) + 23/2(4,1) + 1(4,14) + 1(1,21) + 15/2(1,6) + 4(1,14') + 1(1,14) + 3/2(5,6)$
Spectrum 1711; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 3
 $3(1,1) + 1(10,1) + 1/2(4,21) + 9/2(4,1) + 1/2(4,14') + 1(4,14) + 1(1,21) + 25/2(1,6) + 2(1,14') + 1(1,14) + 1/2(5,6) + 6(5,1)$
Spectrum 1712; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 25
 $25(1,1) + 1(10,1) + 3(4,6) + 4(4,1) + 1(4,14') + 3(1,21) + 3(1,6) + 5/2(1,14) + 2(5,1) + 1/2(5,14)$
Spectrum 1713; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 11
 $11(1,1) + 1(10,1) + 1/2(4,21) + 3(4,6) + 15/2(4,1) + 1(1,21) + 3(1,6) + 4(1,14') + 5/2(1,14) + 2(5,1) + 1/2(5,14)$
Spectrum 1714; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 30
 $30(1,1) + 1(10,1) + 1(4,6) + 2(4,1) + 1(4,14') + 1(4,14) + 3(1,21) + 9/2(1,6) + 1(1,14') + 1/2(1,14) + 1(5,6) + 3(5,1)$
Spectrum 1715; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 21
 $21(1,1) + 1(10,1) + 8(4,1) + 1(4,14') + 1(4,14) + 3(1,21) + 6(1,6) + 1(1,14') + 1/2(1,14) + 3/2(5,6)$
Spectrum 1716; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 31
 $31(1,1) + 1(10,1) + 1/2(4,21) + 1(4,6) + 11/2(4,1) + 1(4,14') + 1(1,21) + 2(1,6) + 1(1,14') + 9/2(1,14) + 3/2(5,6)$
Spectrum 1717; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 16
 $16(1,1) + 1(10,1) + 1/2(4,21) + 1(4,6) + 11/2(4,1) + 1(4,14) + 1(1,21) + 9/2(1,6) + 5(1,14') + 1/2(1,14) + 1(5,6) + 3(5,1)$
Spectrum 1718; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 7
 $7(1,1) + 1(10,1) + 1/2(4,21) + 23/2(4,1) + 1(4,14) + 1(1,21) + 6(1,6) + 5(1,14') + 1/2(1,14) + 3/2(5,6)$
Spectrum 1719; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 5
 $5(1,1) + 1(10,1) + 1/2(4,21) + 9/2(4,1) + 1/2(4,14') + 1(4,14) + 1(1,21) + 11(1,6) + 3(1,14') + 1/2(1,14) + 1/2(5,6) + 6(5,1)$
Spectrum 1720; $b1=(3/2,3)$, $b2=(2,4)$; singlets = 27
 $27(1,1) + 1(10,1) + 3(4,6) + 4(4,1) + 1(4,14') + 3(1,21) + 3/2(1,6) + 1(1,14') + 2(1,14) + 2(5,1) + 1/2(5,14)$

Spectrum 1721; b1={3/2,3}, b2={2,4}; singlets = 13

13(1,1) + 1(10,1) + 1/2(4,21) + 3(4,6) + 15/2(4,1) + 1(1,21) + 3/2(1,6) + 5(1,14') + 2(1,14) + 2(5,1) + 1/2(5,14)

Spectrum 1722; b1={3/2,3}, b2={2,4}; singlets = 32

32(1,1) + 1(10,1) + 1(4,6) + 2(4,1) + 1(4,14') + 1(4,14) + 3(1,21) + 3(1,6) + 2(1,14') + 1(5,6) + 3(5,1)

Spectrum 1723; b1={3/2,3}, b2={2,4}; singlets = 23

23(1,1) + 1(10,1) + 8(4,1) + 1(4,14') + 1(4,14) + 3(1,21) + 9/2(1,6) + 2(1,14') + 3/2(5,6)

Spectrum 1724; b1={3/2,3}, b2={2,4}; singlets = 21

21(1,1) + 1(10,1) + 1(4,1) + 3/2(4,14') + 1(4,14) + 3(1,21) + 19/2(1,6) + 1/2(5,6) + 6(5,1)

Spectrum 1725; b1={3/2,3}, b2={2,4}; singlets = 33

33(1,1) + 1(10,1) + 1/2(4,21) + 1(4,6) + 11/2(4,1) + 1(4,14') + 1(1,21) + 1/2(1,6) + 2(1,14') + 4(1,14) + 3/2(5,6)

Spectrum 1726; b1={3/2,3}, b2={2,4}; singlets = 22

22(1,1) + 1(10,1) + 1/2(4,21) + 9/2(4,1) + 3/2(4,14') + 1(1,21) + 7(1,6) + 4(1,14) + 1(5,6) + 3(5,1)

Spectrum 1727; b1={3/2,3}, b2={2,4}; singlets = 18

18(1,1) + 1(10,1) + 1/2(4,21) + 1(4,6) + 11/2(4,1) + 1(4,14) + 1(1,21) + 3(1,6) + 6(1,14') + 1(5,6) + 3(5,1)

Spectrum 1728; b1={3/2,3}, b2={2,4}; singlets = 9

9(1,1) + 1(10,1) + 1/2(4,21) + 23/2(4,1) + 1(4,14) + 1(1,21) + 9/2(1,6) + 6(1,14') + 3/2(5,6)

Spectrum 1729; b1={3/2,3}, b2={2,4}; singlets = 7

7(1,1) + 1(10,1) + 1/2(4,21) + 9/2(4,1) + 1/2(4,14') + 1(4,14) + 1(1,21) + 19/2(1,6) + 4(1,14') + 1/2(5,6) + 6(5,1)

Spectrum 1730; b1={3/2,3}, b2={2,4}; singlets = 29

29(1,1) + 1(10,1) + 3(4,6) + 4(4,1) + 1(4,14') + 3(1,21) + 2(1,14') + 3/2(1,14) + 2(5,1) + 1/2(5,14)

Spectrum 1731; b1={3/2,3}, b2={2,4}; singlets = 9

9(1,1) + 1(10,1) + 1(4,6) + 9(4,1) + 3/2(4,14') + 3(1,21) + 8(1,6) + 3/2(1,14) + 2(5,1) + 1/2(5,14)

Spectrum 1732; b1={3/2,3}, b2={2,4}; singlets = 15

15(1,1) + 1(10,1) + 1/2(4,21) + 3(4,6) + 15/2(4,1) + 1(1,21) + 6(1,14') + 3/2(1,14) + 2(5,1) + 1/2(5,14)

Spectrum 1733; b1={3/2,3}, b2={2,4}; singlets = 24

24(1,1) + 1(10,1) + 1/2(4,21) + 9/2(4,1) + 3/2(4,14') + 1(1,21) + 11/2(1,6) + 1(1,14') + 7/2(1,14) + 1(5,6) + 3(5,1)

Spectrum 1734; b1={3/2,3}, b2={2,4}; singlets = 11

11(1,1) + 1(10,1) + 1(4,6) + 9(4,1) + 3/2(4,14') + 3(1,21) + 13/2(1,6) + 1(1,14') + 1(1,14) + 2(5,1) + 1/2(5,14)

Spectrum 1735; b1={3/2,3}, b2={2,4}; singlets = 26

26(1,1) + 1(10,1) + 1/2(4,21) + 9/2(4,1) + 3/2(4,14') + 1(1,21) + 4(1,6) + 2(1,14') + 3(1,14) + 1(5,6) + 3(5,1)

Spectrum 1736; b1={3/2,3}, b2={2,4}; singlets = 13

13(1,1) + 1(10,1) + 1(4,6) + 9(4,1) + 3/2(4,14') + 3(1,21) + 5(1,6) + 2(1,14') + 1/2(1,14) + 2(5,1) + 1/2(5,14)

Spectrum 1737; b1={3/2,3}, b2={2,4}; singlets = 28

28(1,1) + 1(10,1) + 1/2(4,21) + 9/2(4,1) + 3/2(4,14') + 1(1,21) + 5/2(1,6) + 3(1,14') + 5/2(1,14) + 1(5,6) + 3(5,1)

Spectrum 1738; b1={3/2,3}, b2={2,4}; singlets = 15

15(1,1) + 1(10,1) + 1(4,6) + 9(4,1) + 3/2(4,14') + 3(1,21) + 7/2(1,6) + 3(1,14') + 2(5,1) + 1/2(5,14)

Spectrum 1739; b1={3/2,3}, b2={2,4}; singlets = 1

1(1,1) + 1(10,1) + 1/2(4,21) + 1(4,6) + 25/2(4,1) + 1/2(4,14') + 1(1,21) + 7/2(1,6) + 7(1,14') + 2(5,1) + 1/2(5,14)

Spectrum 1740; b1={3/2,3}, b2={2,4}; singlets = 44

44(1,1) + 1(10,1) + 1(4,1) + 5/2(4,14') + 3(1,21) + 1(1,6) + 2(1,14) + 1(5,6) + 3(5,1)

Spectrum 1741; b1={3/2,3}, b2={2,4}; singlets = 30

30(1,1) + 1(10,1) + 1/2(4,21) + 9/2(4,1) + 3/2(4,14') + 1(1,21) + 1(1,6) + 4(1,14') + 2(1,14) + 1(5,6) + 3(5,1)

C2 x C4

Spectrum 1742; b1={3,8}, b2={1,0}; singlets = 5

5(1,1) + 1(10,8) + 2(10,1) + 22(4,1) + 1(1,8) + 1(1,27) + 1(5,8) + 10(5,1) + 1(14,1) + 1/2(16,1)

Spectrum 1743; b1={3,8}, b2={1,0}; singlets = 12

12(1,1) + 1(10,8) + 3(10,1) + 16(4,1) + 1(1,8) + 1(1,27) + 1(5,8) + 13(5,1) + 1(14,1)

Spectrum 1744; b1={3,8}, b2={1,0}; singlets = 12

12(1,1) + 1(10,8) + 20(4,1) + 1(1,8) + 1(1,27) + 1(5,8) + 9(5,1) + 3(16,1)

Spectrum 1745; b1={3,8}, b2={1,0}; singlets = 16

16(1,1) + 1(10,8) + 19/2(4,1) + 1(1,8) + 1(1,27) + 1(5,8) + 21(5,1) + 1(16,1) + 1/2(20,1)

Spectrum 1746; b1={3,8}, b2={1,0}; singlets = 19

19(1,1) + 1(10,8) + 1(10,1) + 14(4,1) + 1(1,8) + 1(1,27) + 1(5,8) + 12(5,1) + 5/2(16,1)

Spectrum 1747; b1={3,8}, b2={1,0}; singlets = 23

23(1,1) + 1(10,8) + 1(10,1) + 7/2(4,1) + 1(1,8) + 1(1,27) + 1(5,8) + 24(5,1) + 1/2(16,1) + 1/2(20,1)

Spectrum 1748; b1={3,8}, b2={1,0}; singlets = 26

26(1,1) + 1(10,8) + 2(10,1) + 8(4,1) + 1(1,8) + 1(1,27) + 1(5,8) + 15(5,1) + 2(16,1)

Spectrum 1749; b1={3,8}, b2={1,0}; singlets = 33

33(1,1) + 1(10,8) + 3(10,1) + 2(4,1) + 1(1,8) + 1(1,27) + 1(5,8) + 18(5,1) + 3/2(16,1)

Spectrum 1750; b1={3/2,3}, b2={1,0}; singlets = 75

75(1,1) + 3(4,8) + 18(4,1) + 4(1,8) + 1(1,27) + 6(5,1) + 1/2(16,1)

Spectrum 1751; b1={3/2,3}, b2={1,0}; singlets = 63

63(1,1) + 2(4,8) + 26(4,1) + 11/2(1,8) + 1(1,27) + 1/2(5,8) + 2(5,1) + 1/2(16,1)

Spectrum 1752; b1={3/2,3}, b2={1,0}; singlets = 82

82(1,1) + 1(10,1) + 3(4,8) + 12(4,1) + 4(1,8) + 1(1,27) + 9(5,1)

Spectrum 1753; b1={3/2,3}, b2={1,0}; singlets = 70

70(1,1) + 1(10,1) + 2(4,8) + 20(4,1) + 11/2(1,8) + 1(1,27) + 1/2(5,8) + 5(5,1)

Spectrum 1754; b1={3/2,3}, b2={1,0}; singlets = 58

58(1,1) + 1(10,1) + 1(4,8) + 28(4,1) + 7(1,8) + 1(1,27) + 1(5,8) + 1(5,1)

C2 x C5

Spectrum 1755; b1={3/2,3}, b2={3/2,1}; singlets = 26

26(1,1) + 2(4,10) + 1/2(4,44) + 2(1,10) + 2(1,44) + 6(5,1) + 1/2(16,1)

Spectrum 1756; b1={3/2,3}, b2={3/2,1}; singlets = 11

11(1,1) + 1(4,10) + 10(4,1) + 1/2(4,44) + 7/2(1,10) + 2(1,44) + 1/2(5,10) + 1(5,1) + 1/2(16,1)

Spectrum 1757; b1={3/2,3}, b2={3/2,1}; singlets = 18

18(1,1) + 1(10,1) + 1(4,10) + 4(4,1) + 1/2(4,44) + 7/2(1,10) + 2(1,44) + 1/2(5,10) + 4(5,1)

C2 x D4

Spectrum 1758; b1={3,8}, b2={3/2,1}; singlets = 3

3(1,1) + 1/2(4,8) + 49/2(4,1) + 1/2(4,8) + 1/2(4,8_v) + 1(5,8_v) + 5(5,1) + 1(5,8_s) + 1(5,8_c) + 1(16,1) + 3/2(20,1)

Spectrum 1759; b1={3,8}, b2={3/2,1}; singlets = 10

10(1,1) + 1(10,1) + 1/2(4,8) + 37/2(4,1) + 1/2(4,8_s) + 1/2(4,8_c) + 1(5,8_v) + 8(5,1) + 1(5,8_s) + 1(5,8_c) + 1/2(16,1) + 3/2(20,1)

Spectrum 1760; b1={3,8}, b2={3/2,1}; singlets = 17

17(1,1) + 2(10,1) + 1/2(4,8) + 25/2(4,1) + 1/2(4,8_s) + 1/2(4,8_c) + 1(5,8_v) + 11(5,1) + 1(5,8_s) + 1(5,8_c) + 3/2(20,1)

Spectrum 1761; b1={3,8}, b2={3/2,1}; singlets = 20

20(1,1) + 3(10,1) + 1/2(4,8) + 17(4,1) + 1/2(4,8_s) + 1/2(4,8_c) + 1(5,8_v) + 2(5,1) + 1(5,8_s) + 1(5,8_c) + 3/2(16,1) + 1(20,1)

Spectrum 1762; b1={3,8}, b2={3/2,1}; singlets = 27

27(1,1) + 4(10,1) + 1/2(4,8) + 11(4,1) + 1/2(4,8_s) + 1/2(4,8_c) + 1(5,8_v) + 5(5,1) + 1(5,8_s) + 1(5,8_c) + 1(16,1) + 1(20,1)

Spectrum 1763; b1={3,8}, b2={3/2,1}; singlets = 34

34(1,1) + 5(10,1) + 1/2(4,8) + 5(4,1) + 1/2(4,8) + 1/2(4,8_c) + 1(5,8_v) + 8(5,1) + 1(5,8_s) + 1(5,8_c) + 1/2(16,1) + 1(20,1)

Spectrum 1764; b1={3,8}, b2={3/2,1}; singlets = 44

44(1,1) + 7(10,1) + 1/2(4,8) + 7/2(4,1) + 1/2(4,8_s) + 1/2(4,8_c) + 1(5,8_v) + 2(5,1) + 1(5,8_s) + 1(5,8_c) + 3/2(16,1) + 1/2(20,1)

Spectrum 1765; b1={3/2,3}, b2={3,6}; singlets = 6

6(1,1) + 3/2(4,28) + 9(1,8) + 1(1,56) + 6(5,1) + 1/2(16,1)

Spectrum 1766; b1={3/2,3}, b2={3,6}; singlets = 6

6(1,1) + 3/2(4,28) + 9(1,8) + 1(1,56) + 6(5,1) + 1/2(16,1)

Spectrum 1767; b1={3/2,3}, b2={3,6}; singlets = 6

6(1,1) + 3/2(4,28) + 9(1,8) + 1(1,56) + 6(5,1) + 1/2(16,1)

Spectrum 1768; b1={3/2,3}, b2={3,6}; singlets = 22

22(1,1) + 3/2(4,28) + 4(1,28) + 6(5,1) + 1/2(16,1)

Spectrum 1769; b1={3/2,3}, b2={3,6}; singlets = 15

15(1,1) + 1(10,1) + 1(4,28) + 1(4,8) + 7(1,8) + 1(1,35_v) + 1(1,35_s) + 1(5,1) + 1(5,8_c)

Spectrum 1770; b1={3/2,3}, b2={3,6}; singlets = 15

15(1,1) + 1(10,1) + 1(4,28) + 1(4,8) + 7(1,8) + 1(1,35_v) + 1(1,35_c) + 1(5,1) + 1(5,8_s)

Spectrum 1771; b1={3/2,3}, b2={3,6}; singlets = 15

15(1,1) + 1(10,1) + 1(4,28) + 1(4,8) + 7(1,8) + 1(1,35_s) + 1(1,35_c) + 1(5,8_v) + 1(5,1)

Spectrum 1772; b1={3/2,3}, b2={3,6}; singlets = 29

29(1,1) + 1(10,1) + 1(4,28) + 1(4,8) + 2(1,28) + 1(1,56_c) + 1(5,1) + 1(5,8_c)

Spectrum 1773; b1={3/2,3}, b2={3,6}; singlets = 29

29(1,1) + 1(10,1) + 1(4,28) + 1(4,8) + 2(1,28) + 1(1,56_s) + 1(5,1) + 1(5,8_s)

Spectrum 1774; b1={3/2,3}, b2={3,6}; singlets = 29

29(1,1) + 1(10,1) + 1(4,28) + 1(4,8) + 2(1,28) + 1(1,56_v) + 1(5,8_v) + 1(5,1)

Spectrum 1775; b1={3/2,3}, b2={3/2,1}; singlets = 62

62(1,1) + 18(4,1) + 3/2(4,8) + 3/2(4,8) + 7(1,8_v) + 1(1,8_s) + 1(1,8_c) + 6(5,1) + 1/2(16,1)

Spectrum 1776; b1={3/2,3}, b2={3/2,1}; singlets = 62

62(1,1) + 1/2(4,8) + 18(4,1) + 1(4,8) + 3/2(4,8_c) + 5(1,8_v) + 3(1,8_s) + 1(1,8_c) + 6(5,1) + 1/2(16,1)

Spectrum 1777; b1={3/2,3}, b2={3/2,1}; singlets = 62

62(1,1) + 1/2(4,8) + 18(4,1) + 3/2(4,8) + 1(4,8_c) + 5(1,8_v) + 1(1,8_s) + 3(1,8_c) + 6(5,1) + 1/2(16,1)

Spectrum 1778; b1={3/2,3}, b2={3/2,1}; singlets = 62

62(1,1) + 1(4,8) + 18(4,1) + 1/2(4,8) + 3/2(4,8_c) + 3(1,8_v) + 5(1,8_s) + 1(1,8_c) + 6(5,1) + 1/2(16,1)

Spectrum 1779; b1={3/2,3}, b2={3/2,1}; singlets = 62

62(1,1) + 1(4,8) + 18(4,1) + 1(4,8) + 1(4,8_v) + 3(1,8_s) + 3(1,8_c) + 6(5,1) + 1/2(16,1)

Spectrum 1780; b1={3/2,3}, b2={3/2,1}; singlets = 62

62(1,1) + 1(4,8) + 18(4,1) + 3/2(4,8) + 1/2(4,8_c) + 3(1,8_v) + 1(1,8_s) + 5(1,8_c) + 6(5,1) + 1/2(16,1)

Spectrum 1781; b1={3/2,3}, b2={3/2,1}; singlets = 62

62(1,1) + 3/2(4,8) + 18(4,1) + 3/2(4,8) + 1(1,8_v) + 7(1,8_s) + 1(1,8_c) + 6(5,1) + 1/2(16,1)

Spectrum 1782; b1={3/2,3}, b2={3/2,1}; singlets = 62

62(1,1) + 3/2(4,8) + 18(4,1) + 1/2(4,8) + 1(4,8_c) + 1(1,8_v) + 5(1,8_s) + 3(1,8_c) + 6(5,1) + 1/2(16,1)

Spectrum 1783; b1={3/2,3}, b2={3/2,1}; singlets = 62

62(1,1) + 3/2(4,8) + 18(4,1) + 1(4,8) + 1/2(4,8_c) + 1(1,8_v) + 3(1,8_s) + 5(1,8_c) + 6(5,1) + 1/2(16,1)

Spectrum 1784; b1={3/2,3}, b2={3/2,1}; singlets = 62

62(1,1) + 3/2(4,8) + 18(4,1) + 3/2(4,8) + 1(1,8_v) + 1(1,8_s) + 7(1,8_c) + 6(5,1) + 1/2(16,1)

Spectrum 1785; b1={3/2,3}, b2={3/2,1}; singlets = 69

69(1,1) + 1(10,1) + 12(4,1) + 3/2(4,8_s) + 3/2(4,8_c) + 7(1,8_v) + 1(1,8_s) + 1(1,8_c) + 9(5,1)

Spectrum 1818; b1={3,8}, b2={2,4}; singlets = 9

9(1,1) + 1(10,7) + 1/2(4,14) + 17/2(4,1) + 1(1,14) + 3(1,7) + 3(5,7) + 5(5,1) + 3/2(16,1) + 1/2(20,1)

Spectrum 1819; b1={3,8}, b2={2,4}; singlets = 12

12(1,1) + 1(10,7) + 4(10,1) + 1/2(4,14) + 9(4,1) + 1(1,14) + 3(1,7) + 3(5,7) + 1(14,1)

Spectrum 1820; b1={3,8}, b2={2,4}; singlets = 17

17(1,1) + 1(10,7) + 1(10,1) + 19/2(4,1) + 1(1,7) + 1(1,27) + 4(5,7) + 1(5,1) + 1(16,1) + 1/2(20,1)

Spectrum 1821; b1={3,8}, b2={2,4}; singlets = 16

16(1,1) + 1(10,7) + 1(10,1) + 1/2(4,14) + 5/2(4,1) + 1(1,14) + 3(1,7) + 3(5,7) + 8(5,1) + 1(16,1) + 1/2(20,1)

Spectrum 1822; b1={3,8}, b2={2,4}; singlets = 24

24(1,1) + 1(10,7) + 2(10,1) + 7/2(4,1) + 1(1,7) + 1(1,27) + 4(5,7) + 4(5,1) + 1/2(16,1) + 1/2(20,1)

Spectrum 1823; b1={3,8}, b2={2,4}; singlets = 26

26(1,1) + 1(10,7) + 3(10,1) + 1/2(4,14) + 1(4,1) + 1(1,14) + 3(1,7) + 3(5,7) + 2(5,1) + 2(16,1)

Spectrum 1824; b1={3/2,3}, b2={7/2,9}; singlets = 1

1(1,1) + 1/2(4,14) + 1(4,7) + 1(4,1) + 1(4,27) + 6(1,14) + 7(1,7) + 6(5,1) + 1/2(16,1)

Spectrum 1825; b1={3/2,3}, b2={7/2,9}; singlets = 17

17(1,1) + 3(4,14) + 2(1,14) + 5(1,7) + 2(1,27) + 6(5,1) + 1/2(16,1)

Spectrum 1826; b1={3/2,3}, b2={7/2,9}; singlets = 23

23(1,1) + 3(4,14) + 5(1,14) + 2(1,7) + 1(1,27) + 6(5,1) + 1/2(16,1)

Spectrum 1827; b1={3/2,3}, b2={7/2,9}; singlets = 3

3(1,1) + 1(10,1) + 1(4,7) + 2(4,1) + 1(4,27) + 2(1,14) + 8(1,7) + 2(1,27) + 1(5,7) + 2(5,1)

Spectrum 1828; b1={3/2,3}, b2={7/2,9}; singlets = 9

9(1,1) + 1(10,1) + 1(4,7) + 2(4,1) + 1(4,27) + 5(1,14) + 5(1,7) + 1(1,27) + 1(5,7) + 2(5,1)

Spectrum 1829; b1={3/2,3}, b2={7/2,9}; singlets = 15

15(1,1) + 1(10,1) + 1(4,7) + 2(4,1) + 1(4,27) + 8(1,14) + 2(1,7) + 1(5,7) + 2(5,1)

Spectrum 1830; b1={3/2,3}, b2={7/2,9}; singlets = 25

25(1,1) + 1(10,1) + 5/2(4,14) + 1(4,1) + 1(1,14) + 3(1,7) + 3(1,27) + 1(5,7) + 2(5,1)

Spectrum 1831; b1={3/2,3}, b2={7/2,9}; singlets = 31

31(1,1) + 1(10,1) + 5/2(4,14) + 1(4,1) + 4(1,14) + 2(1,27) + 1(5,7) + 2(5,1)

Spectrum 1832; b1={3/2,3}, b2={2,4}; singlets = 58

58(1,1) + 6(4,7) + 7(1,7) + 1(1,27) + 6(5,1) + 1/2(16,1)

Spectrum 1833; b1={3/2,3}, b2={2,4}; singlets = 64

64(1,1) + 6(4,7) + 3(1,14) + 4(1,7) + 6(5,1) + 1/2(16,1)

Spectrum 1834; b1={3/2,3}, b2={2,4}; singlets = 36

36(1,1) + 1/2(4,14) + 4(4,7) + 7(4,1) + 1(1,14) + 12(1,7) + 6(5,1) + 1/2(16,1)

Spectrum 1835; b1={3/2,3}, b2={2,4}; singlets = 44

44(1,1) + 1(10,1) + 4(4,7) + 8(4,1) + 10(1,7) + 1(1,27) + 1(5,7) + 2(5,1)

Spectrum 1836; b1={3/2,3}, b2={2,4}; singlets = 50

50(1,1) + 1(10,1) + 4(4,7) + 8(4,1) + 3(1,14) + 7(1,7) + 1(5,7) + 2(5,1)

Spectrum 1837; b1={3/2,3}, b2={2,4}; singlets = 43

43(1,1) + 1(10,1) + 1/2(4,14) + 4(4,7) + 1(4,1) + 1(1,14) + 12(1,7) + 9(5,1)

Spectrum 1838; b1={3/2,3}, b2={2,4}; singlets = 22

22(1,1) + 1(10,1) + 1/2(4,14) + 2(4,7) + 15(4,1) + 1(1,14) + 15(1,7) + 1(5,7) + 2(5,1)

C3 x C3

Spectrum 1839; b1={2,4}, b2={2,4}; singlets = 18

18(1,1) + 3(21,1) + 3(1,21) + 1(14',14')

C3 x C4

Spectrum 1840; b1={2,4}, b2={1,0}; singlets = 4

4(1,1) + 1(21,1) + 19(6,1) + 2(1,8) + 1(1,27) + 1(14',8) + 1(14,1) + 1/2(64,1)

Spectrum 1841; b1={2,4}, b2={1,0}; singlets = 6

6(1,1) + 1(21,1) + 35/2(6,1) + 2(1,8) + 1(1,27) + 1(14',8) + 1(14',1) + 1/2(14,1) + 1/2(64,1)

Spectrum 1842; b1={2,4}, b2={1,0}; singlets = 8

8(1,1) + 1(21,1) + 16(6,1) + 2(1,8) + 1(1,27) + 1(14',8) + 2(14',1) + 1/2(64,1)

Spectrum 1843; b1={2,4}, b2={1,0}; singlets = 26

26(1,1) + 3(21,1) + 9(6,1) + 2(1,8) + 1(1,27) + 1(14',8) + 3(14,1)

Spectrum 1844; b1={2,4}, b2={1,0}; singlets = 28

28(1,1) + 3(21,1) + 15/2(6,1) + 2(1,8) + 1(1,27) + 1(14',8) + 1(14',1) + 5/2(14,1)

Spectrum 1845; b1={2,4}, b2={1,0}; singlets = 30

30(1,1) + 3(21,1) + 6(6,1) + 2(1,8) + 1(1,27) + 1(14',8) + 2(14',1) + 2(14,1)

Spectrum 1846; b1={2,4}, b2={1,0}; singlets = 32

32(1,1) + 3(21,1) + 9/2(6,1) + 2(1,8) + 1(1,27) + 1(14',8) + 3(14',1) + 3/2(14,1)

Spectrum 1847; b1={2,4}, b2={1,0}; singlets = 34

34(1,1) + 3(21,1) + 3(6,1) + 2(1,8) + 1(1,27) + 1(14',8) + 4(14',1) + 1(14,1)

Spectrum 1848; b1={2,4}, b2={1,0}; singlets = 36

36(1,1) + 3(21,1) + 3/2(6,1) + 2(1,8) + 1(1,27) + 1(14',8) + 5(14',1) + 1/2(14,1)

Spectrum 1849; b1={2,4}, b2={1,0}; singlets = 38

38(1,1) + 3(21,1) + 2(1,8) + 1(1,27) + 1(14',8) + 6(14',1)

[illegible]

59(**1,1**) + 3(**21,1**) + 3/2(**6,1**) + 2(**1,10**) + 13(**14',1**) + 1/2(**14,1**)

Spectrum 1917; b1=(2,4), b2=(1,-2); singlets = 61

61(**1,1**) + 3(**21,1**) + 2(**1,10**) + 14(**14',1**)

C3 x G2

Spectrum 1918; b1=(2,4), b2=(2,4); singlets = 33

33(**1,1**) + 3(**21,1**) + 3/2(**6,7**) + 1(**1,7**) + 1(**1,27**) + 1(**14',7**) + 1/2(**14,7**)

Spectrum 1919; b1=(2,4), b2=(2,4); singlets = 33

33(**1,1**) + 3(**21,1**) + 3(**1,7**) + 1(**1,27**) + 2(**14',7**)

Spectrum 1920; b1=(2,4), b2=(2,4); singlets = 39

39(**1,1**) + 3(**21,1**) + 3(**1,14**) + 2(**14',7**)

C4 x C4

Spectrum 1921; b1=(1,0), b2=(1,0); singlets = 30

30(**1,1**) + 16(**8,1**) + 16(**1,8**) + 1(**1,27**) + 1(**27,1**)

C4 x D4

Spectrum 1922; b1=(1,0), b2=(3,6); singlets = 1

1(**1,1**) + 1/2(**8,28**) + 2(**8,1**) + 2(**1,28**) + 9(**1,8**) + 1(**1,56_c**) + 1(**27,1**)

Spectrum 1923; b1=(1,0), b2=(3,6); singlets = 1

1(**1,1**) + 1/2(**8,28**) + 2(**8,1**) + 2(**1,28**) + 9(**1,8**) + 1(**1,56_s**) + 1(**27,1**)

Spectrum 1924; b1=(1,0), b2=(3,6); singlets = 1

1(**1,1**) + 1/2(**8,28**) + 2(**8,1**) + 2(**1,28**) + 9(**1,8**) + 1(**1,56_v**) + 1(**27,1**)

Spectrum 1925; b1=(1,0), b2=(3,6); singlets = 17

17(**1,1**) + 1/2(**8,28**) + 2(**8,1**) + 6(**1,28**) + 1(**27,1**)

Spectrum 1926; b1=(1,0), b2=(3/2,1); singlets = 49

49(**1,1**) + 12(**8,1**) + 1/2(**8,8**) + 7(**1,8**) + 7(**1,8**) + 3(**1,8_c**) + 1(**27,1**)

Spectrum 1927; b1=(1,0), b2=(3/2,1); singlets = 49

49(**1,1**) + 12(**8,1**) + 1/2(**8,8**) + 7(**1,8**) + 3(**1,8**) + 7(**1,8_c**) + 1(**27,1**)

Spectrum 1928; b1=(1,0), b2=(3/2,1); singlets = 49

49(**1,1**) + 1/2(**8,8_v**) + 12(**8,1**) + 3(**1,8_v**) + 7(**1,8_s**) + 7(**1,8_c**) + 1(**27,1**)

C4 x D5

Spectrum 1929; b1=(5/2,5), b2=(1,-2); singlets = 0

5/2(**8,1**) + 2(**1,10**) + 5(**48,1**) + 1/2(**120,1**)

Spectrum 1930; b1=(5/2,5), b2=(1,-2); singlets = 0

5(**8,1**) + 2(**1,10**) + 5/2(**48,1**) + 1(**160,1**)

Spectrum 1931; b1=(5/2,5), b2=(1,-2); singlets = 5

5(**1,1**) + 1(**8,1**) + 2(**1,10**) + 3(**27,1**) + 1(**42,1**) + 1/2(**48,1**) + 1(**160,1**)

Spectrum 1932; b1=(5/2,5), b2=(1,-2); singlets = 10

10(**1,1**) + 2(**1,10**) + 2(**27,1**) + 2(**48,1**) + 1(**160,1**)

C4 x G2

Spectrum 1933; b1=(1,0), b2=(2,4); singlets = 53

53(**1,1**) + 2(**8,7**) + 2(**8,1**) + 15(**1,7**) + 1(**1,27**) + 1(**27,1**)

Spectrum 1934; b1=(1,0), b2=(2,4); singlets = 59

59(**1,1**) + 2(**8,7**) + 2(**8,1**) + 3(**1,14**) + 12(**1,7**) + 1(**27,1**)

D4 x D5

Spectrum 1935; b1=(3,6), b2=(1,-2); singlets = 8

8(**1,1**) + 2(**28,1**) + 2(**1,10**) + 18(**8 ,1**) + 2(**56 ,1**)

Spectrum 1936; b1=(3,6), b2=(1,-2); singlets = 8

8(**1,1**) + 2(**28,1**) + 2(**1,10**) + 9(**8 ,1**) + 9(**8 ,1**) + 1(**56_c,1**) + 1(**56_s,1**)

Spectrum 1937; b1=(3,6), b2=(1,-2); singlets = 8

8(**1,1**) + 2(**28,1**) + 2(**1,10**) + 18(**8 ,1**) + 2(**56 ,1**)

Spectrum 1938; b1=(3,6), b2=(1,-2); singlets = 8

8(**1,1**) + 2(**28,1**) + 9(**8 ,1**) + 2(**1,10**) + 9(**8 ,1**) + 1(**56_c,1**) + 1(**56_v,1**)

Spectrum 1939; b1=(3,6), b2=(1,-2); singlets = 8

8(**1,1**) + 2(**28,1**) + 9(**8 ,1**) + 2(**1,10**) + 9(**8 ,1**) + 1(**56_s,1**) + 1(**56_v,1**)

Spectrum 1940; b1=(3,6), b2=(1,-2); singlets = 8

8(**1,1**) + 2(**28,1**) + 18(**8 ,1**) + 2(**1,10**) + 2(**56 ,1**)

Spectrum 1941; b1=(3,6), b2=(1,-2); singlets = 9

9(**1,1**) + 3(**28,1**) + 1(**8 ,1**) + 2(**1,10**) + 8(**8 ,1**) + 8(**8_c,1**) + 1(**35_v,1**) + 1(**56_v,1**)

Spectrum 1942; b1=(3,6), b2=(1,-2); singlets = 9

9(**1,1**) + 3(**28,1**) + 8(**8 ,1**) + 2(**1,10**) + 1(**8 ,1**) + 8(**8_c,1**) + 1(**35_s,1**) + 1(**56_s,1**)

Spectrum 1943; b1=(3,6), b2=(1,-2); singlets = 9

9(**1,1**) + 3(**28,1**) + 8(**8_v,1**) + 2(**1,10**) + 8(**8_s,1**) + 1(**8_c,1**) + 1(**35_c,1**) + 1(**56_c,1**)

Spectrum 1944; b1={3,6}, b2={1,-2}; singlets = 10
10(1,1) + 4(28,1) + 2(1,10) + 16(8 ,1) + 1(35 ,1) + 1(35_s,1)
Spectrum 1945; b1={3,6}, b2={1,-2}; singlets = 10
10(1,1) + 4(28,1) + 2(1,10) + 16(8 ,1) + 1(35 ,1) + 1(35_c,1)
Spectrum 1946; b1={3,6}, b2={1,-2}; singlets = 10
10(1,1) + 4(28,1) + 16(8 ,1) + 2(1,10) + 1(35 ,1) + 1(35_c,1)
Spectrum 1947; b1={3,6}, b2={1,-2}; singlets = 24
24(1,1) + 6(28,1) + 2(1,10) + 9(8 ,1) + 1(56 ,1)
Spectrum 1948; b1={3,6}, b2={1,-2}; singlets = 24
24(1,1) + 6(28,1) + 2(1,10) + 9(8 ,1) + 1(56 ,1)
Spectrum 1949; b1={3,6}, b2={1,-2}; singlets = 24
24(1,1) + 6(28,1) + 9(8 ,1) + 2(1,10) + 1(56 ,1)
Spectrum 1950; b1={3,6}, b2={1,-2}; singlets = 40
40(1,1) + 10(28,1) + 2(1,10)

D4 x G2

Spectrum 1951; b1={3,6}, b2={2,4}; singlets = 12
12(1,1) + 1(28,7) + 3(28,1) + 3(1,7) + 1(1,27)
Spectrum 1952; b1={3,6}, b2={2,4}; singlets = 18
18(1,1) + 1(28,7) + 3(28,1) + 3(1,14)
Spectrum 1953; b1={3/2,1}, b2={2,4}; singlets = 40
40(1,1) + 7(8 ,1) + 15(1,7) + 1(1,27) + 1(8 ,7) + 1(8_c,7)
Spectrum 1954; b1={3/2,1}, b2={2,4}; singlets = 40
40(1,1) + 1(8 ,7) + 15(1,7) + 1(1,27) + 7(8 ,1) + 1(8_c,7)
Spectrum 1955; b1={3/2,1}, b2={2,4}; singlets = 40
40(1,1) + 1(8 ,7) + 15(1,7) + 1(1,27) + 1(8 ,7) + 7(8_c,1)
Spectrum 1956; b1={3/2,1}, b2={2,4}; singlets = 46
46(1,1) + 7(8 ,1) + 3(1,14) + 12(1,7) + 1(8 ,7) + 1(8_c,7)
Spectrum 1957; b1={3/2,1}, b2={2,4}; singlets = 46
46(1,1) + 1(8 ,7) + 3(1,14) + 12(1,7) + 7(8 ,1) + 1(8_c,7)
Spectrum 1958; b1={3/2,1}, b2={2,4}; singlets = 46
46(1,1) + 1(8_v,7) + 3(1,14) + 12(1,7) + 1(8_s,7) + 7(8_c,1)

D5 x D5

Spectrum 1959; b1={4,8}, b2={1,-2}; singlets = 2
2(1,1) + 2(1,10) + 2(54,1) + 1(210,1)

D5 x G2

Spectrum 1960; b1={1,-2}, b2={2,4}; singlets = 76
76(1,1) + 2(10,1) + 31(1,7) + 1(1,27)
Spectrum 1961; b1={1,-2}, b2={2,4}; singlets = 82
82(1,1) + 2(10,1) + 3(1,14) + 28(1,7)

G2 x G2

Spectrum 1962; b1={7/2,9}, b2={2,4}; singlets = 7
7(1,1) + 2(14,7) + 5(7,1) + 3(1,7) + 1(1,27) + 2(27,1)
Spectrum 1963; b1={7/2,9}, b2={2,4}; singlets = 13
13(1,1) + 2(14,7) + 5(7,1) + 3(1,14) + 2(27,1)
Spectrum 1964; b1={7/2,9}, b2={2,4}; singlets = 13
13(1,1) + 2(14,7) + 3(14,1) + 2(7,1) + 3(1,7) + 1(1,27) + 1(27,1)
Spectrum 1965; b1={7/2,9}, b2={2,4}; singlets = 19
19(1,1) + 2(14,7) + 3(14,1) + 2(7,1) + 3(1,14) + 1(27,1)
Spectrum 1966; b1={2,4}, b2={2,4}; singlets = 48
48(1,1) + 4(7,7) + 3(7,1) + 3(1,7) + 1(1,27) + 1(27,1)
Spectrum 1967; b1={2,4}, b2={2,4}; singlets = 54
54(1,1) + 4(7,7) + 3(7,1) + 3(1,14) + 1(27,1)
Spectrum 1968; b1={2,4}, b2={2,4}; singlets = 60
60(1,1) + 3(14,1) + 4(7,7) + 3(1,14)

n = 3 detailed spectra

A3 x A3 x A3

Spectrum 1; branches = [(2,4), (2,4), (2,4)]; singlets = 24

24(1,1,1) + 4(4,4,4) + 1(1,1,20) + 1(1,20,1) + 1(20,1,1)

A3 x A3 x A4

Spectrum 2; branches = [(2,4), (2,4), (1,0)]; singlets = 24

24(1,1,1) + 1(4,4,5) + 11(4,4,1) + 2(1,1,10) + 1(1,20,1) + 1(20,1,1)

Spectrum 3; branches = [(2,4), (2,4), (1,0)]; singlets = 24

24(1,1,1) + 1(4,4,5) + 3(4,4,1) + 2(1,1,10) + 3(1,6,1) + 3(1,10,1) + 3(6,1,1) + 2(6,6,1) + 3(10,1,1)

Spectrum 4; branches = [(2,4), (2,4), (1,0)]; singlets = 25

25(1,1,1) + 1(4,4,5) + 3(4,4,1) + 1(1,15,1) + 2(1,1,10) + 2(1,6,1) + 2(1,10,1) + 3(6,1,1) + 2(6,6,1) + 3(10,1,1)

Spectrum 5; branches = [(2,4), (2,4), (1,0)]; singlets = 26

26(1,1,1) + 1(4,4,5) + 3(4,4,1) + 2(1,15,1) + 2(1,1,10) + 1(1,6,1) + 1(1,10,1) + 3(6,1,1) + 2(6,6,1) + 3(10,1,1)

Spectrum 6; branches = [(2,4), (2,4), (1,0)]; singlets = 27

27(1,1,1) + 1(4,4,5) + 3(4,4,1) + 3(1,15,1) + 2(1,1,10) + 3(6,1,1) + 2(6,6,1) + 3(10,1,1)

Spectrum 7; branches = [(2,4), (2,4), (1,0)]; singlets = 26

26(1,1,1) + 1(15,1,1) + 1(4,4,5) + 3(4,4,1) + 1(1,15,1) + 2(1,1,10) + 2(1,6,1) + 2(1,10,1) + 2(6,1,1) + 2(6,6,1) + 2(10,1,1)

Spectrum 8; branches = [(2,4), (2,4), (1,0)]; singlets = 27

27(1,1,1) + 1(15,1,1) + 1(4,4,5) + 3(4,4,1) + 2(1,15,1) + 2(1,1,10) + 1(1,6,1) + 1(1,10,1) + 2(6,1,1) + 2(6,6,1) + 2(10,1,1)

Spectrum 9; branches = [(2,4), (2,4), (1,0)]; singlets = 28

28(1,1,1) + 1(15,1,1) + 1(4,4,5) + 3(4,4,1) + 3(1,15,1) + 2(1,1,10) + 2(6,1,1) + 2(6,6,1) + 2(10,1,1)

Spectrum 10; branches = [(2,4), (2,4), (1,0)]; singlets = 28

28(1,1,1) + 2(15,1,1) + 1(4,4,5) + 3(4,4,1) + 2(1,15,1) + 2(1,1,10) + 1(1,6,1) + 1(1,10,1) + 1(6,1,1) + 2(6,6,1) + 1(10,1,1)

Spectrum 11; branches = [(2,4), (2,4), (1,0)]; singlets = 29

29(1,1,1) + 2(15,1,1) + 1(4,4,5) + 3(4,4,1) + 3(1,15,1) + 2(1,1,10) + 1(6,1,1) + 2(6,6,1) + 1(10,1,1)

Spectrum 12; branches = [(2,4), (2,4), (1,0)]; singlets = 30

30(1,1,1) + 3(15,1,1) + 1(4,4,5) + 3(4,4,1) + 3(1,15,1) + 2(1,1,10) + 2(6,6,1)

A3 x A3 x C4

Spectrum 13; branches = [(2,4), (2,4), (1,0)]; singlets = 17

17(1,1,1) + 1(4,4,8) + 8(4,4,1) + 1(1,1,27) + 1(1,20,1) + 1(20,1,1)

Spectrum 14; branches = [(2,4), (2,4), (1,0)]; singlets = 17

17(1,1,1) + 1(4,4,8) + 1(1,1,27) + 3(1,6,1) + 3(1,10,1) + 3(6,1,1) + 2(6,6,1) + 3(10,1,1)

Spectrum 15; branches = [(2,4), (2,4), (1,0)]; singlets = 18

18(1,1,1) + 1(4,4,8) + 1(1,15,1) + 1(1,1,27) + 2(1,6,1) + 2(1,10,1) + 3(6,1,1) + 2(6,6,1) + 3(10,1,1)

Spectrum 16; branches = [(2,4), (2,4), (1,0)]; singlets = 19

19(1,1,1) + 1(4,4,8) + 2(1,15,1) + 1(1,1,27) + 1(1,6,1) + 1(1,10,1) + 3(6,1,1) + 2(6,6,1) + 3(10,1,1)

Spectrum 17; branches = [(2,4), (2,4), (1,0)]; singlets = 20

20(1,1,1) + 1(4,4,8) + 3(1,15,1) + 1(1,1,27) + 3(6,1,1) + 2(6,6,1) + 3(10,1,1)

Spectrum 18; branches = [(2,4), (2,4), (1,0)]; singlets = 19

19(1,1,1) + 1(15,1,1) + 1(4,4,8) + 1(1,15,1) + 1(1,1,27) + 2(1,6,1) + 2(1,10,1) + 2(6,1,1) + 2(6,6,1) + 2(10,1,1)

Spectrum 19; branches = [(2,4), (2,4), (1,0)]; singlets = 20

20(1,1,1) + 1(15,1,1) + 1(4,4,8) + 2(1,15,1) + 1(1,1,27) + 1(1,6,1) + 1(1,10,1) + 2(6,1,1) + 2(6,6,1) + 2(10,1,1)

Spectrum 20; branches = [(2,4), (2,4), (1,0)]; singlets = 21

21(1,1,1) + 1(15,1,1) + 1(4,4,8) + 3(1,15,1) + 1(1,1,27) + 2(6,1,1) + 2(6,6,1) + 2(10,1,1)

Spectrum 21; branches = [(2,4), (2,4), (1,0)]; singlets = 21

21(1,1,1) + 2(15,1,1) + 1(4,4,8) + 2(1,15,1) + 1(1,1,27) + 1(1,6,1) + 1(1,10,1) + 1(6,1,1) + 2(6,6,1) + 1(10,1,1)

Spectrum 22; branches = [(2,4), (2,4), (1,0)]; singlets = 22

22(1,1,1) + 2(15,1,1) + 1(4,4,8) + 3(1,15,1) + 1(1,1,27) + 1(6,1,1) + 2(6,6,1) + 1(10,1,1)

Spectrum 23; branches = [(2,4), (2,4), (1,0)]; singlets = 23

23(1,1,1) + 3(15,1,1) + 1(4,4,8) + 3(1,15,1) + 1(1,1,27) + 2(6,6,1)

A3 x A3 x D5

Spectrum 24; branches = [(2,4), (2,4), (1,-2)]; singlets = 24

24(1,1,1) + 16(4,4,1) + 2(1,1,10) + 1(1,20,1) + 1(20,1,1)

Spectrum 25; branches = [(2,4), (2,4), (1,-2)]; singlets = 20

20(1,1,1) + 4(4,4,1) + 4(4,1,1) + 2(4,6,1) + 4(1,4,1) + 2(1,1,10) + 1(1,6,1) + 3(1,10,1) + 2(6,4,1) + 1(6,1,1) + 1(6,6,1) + 3(10,1,1)

Spectrum 26; branches = [(2,4), (2,4), (1,-2)]; singlets = 24

24(1,1,1) + 8(4,4,1) + 2(1,1,10) + 3(1,6,1) + 3(1,10,1) + 3(6,1,1) + 2(6,6,1) + 3(10,1,1)

Spectrum 27; branches = [(2,4), (2,4), (1,-2)]; singlets = 21

21(1,1,1) + 4(4,4,1) + 4(4,1,1) + 2(4,6,1) + 1(1,15,1) + 4(1,4,1) + 2(1,1,10) + 2(1,10,1) + 2(6,4,1) + 1(6,1,1) + 1(6,6,1) + 3(10,1,1)

Spectrum 28; branches = [(2,4), (2,4), (1,-2)]; singlets = 25

25(1,1,1) + 8(4,4,1) + 1(1,15,1) + 2(1,1,10) + 2(1,6,1) + 2(1,10,1) + 3(6,1,1) + 2(6,6,1) + 3(10,1,1)

Spectrum 29; branches = [(2,4), (2,4), (1,-2)]; singlets = 26

26(**1,1,1**) + 8(**4,4,1**) + 2(**1,15,1**) + 2(**1,1,10**) + 1(**1,6,1**) + 1(**1,10,1**) + 3(**6,1,1**) + 2(**6,6,1**) + 3(**10,1,1**)

Spectrum 30; branches = [(2,4), (2,4), (1,-2)]; singlets = 27

27(**1,1,1**) + 8(**4,4,1**) + 3(**1,15,1**) + 2(**1,1,10**) + 3(**6,1,1**) + 2(**6,6,1**) + 3(**10,1,1**)

Spectrum 31; branches = [(2,4), (2,4), (1,-2)]; singlets = 22

22(**1,1,1**) + 1(**15,1,1**) + 4(**4,4,1**) + 4(**4,1,1**) + 2(**4,6,1**) + 1(**1,15,1**) + 4(**1,4,1**) + 2(**1,1,10**) + 2(**1,10,1**) + 2(**6,4,1**) + 1(**6,6,1**) + 2(**10,1,1**)

Spectrum 32; branches = [(2,4), (2,4), (1,-2)]; singlets = 26

26(**1,1,1**) + 1(**15,1,1**) + 8(**4,4,1**) + 1(**1,15,1**) + 2(**1,1,10**) + 2(**1,6,1**) + 2(**1,10,1**) + 2(**6,1,1**) + 2(**6,6,1**) + 2(**10,1,1**)

Spectrum 33; branches = [(2,4), (2,4), (1,-2)]; singlets = 27

27(**1,1,1**) + 1(**15,1,1**) + 8(**4,4,1**) + 2(**1,15,1**) + 2(**1,1,10**) + 1(**1,6,1**) + 1(**1,10,1**) + 2(**6,1,1**) + 2(**6,6,1**) + 2(**10,1,1**)

Spectrum 34; branches = [(2,4), (2,4), (1,-2)]; singlets = 28

28(**1,1,1**) + 1(**15,1,1**) + 8(**4,4,1**) + 3(**1,15,1**) + 2(**1,1,10**) + 2(**6,1,1**) + 2(**6,6,1**) + 2(**10,1,1**)

Spectrum 35; branches = [(2,4), (2,4), (1,-2)]; singlets = 28

28(**1,1,1**) + 2(**15,1,1**) + 8(**4,4,1**) + 2(**1,15,1**) + 2(**1,1,10**) + 1(**1,6,1**) + 1(**1,10,1**) + 1(**6,1,1**) + 2(**6,6,1**) + 1(**10,1,1**)

Spectrum 36; branches = [(2,4), (2,4), (1,-2)]; singlets = 29

29(**1,1,1**) + 2(**15,1,1**) + 8(**4,4,1**) + 3(**1,15,1**) + 2(**1,1,10**) + 1(**6,1,1**) + 2(**6,6,1**) + 1(**10,1,1**)

Spectrum 37; branches = [(2,4), (2,4), (1,-2)]; singlets = 30

30(**1,1,1**) + 3(**15,1,1**) + 8(**4,4,1**) + 3(**1,15,1**) + 2(**1,1,10**) + 2(**6,6,1**)

A3 x A4 x A4

Spectrum 38; branches = [(2,4), (1,0), (1,0)]; singlets = 24

24(**1,1,1**) + 4(**4,5,1**) + 4(**4,1,5**) + 24(**4,1,1**) + 2(**1,1,10**) + 2(**1,10,1**) + 1(**20,1,1**)

Spectrum 39; branches = [(2,4), (1,0), (1,0)]; singlets = 44

44(**1,1,1**) + 4(**4,5,1**) + 4(**4,1,5**) + 3(**4,1,1**) + 2(**1,1,10**) + 2(**1,10,1**) + 9(**6,1,1**) + 1(**10,1,1**) + 1(**20',1,1**)

Spectrum 40; branches = [(2,4), (1,0), (1,0)]; singlets = 34

34(**1,1,1**) + 4(**4,5,1**) + 2(**4,1,5**) + 13(**4,1,1**) + 2(**1,1,5**) + 2(**1,1,10**) + 2(**1,10,1**) + 1(**6,1,5**) + 4(**6,1,1**) + 1(**10,1,1**) + 1(**20',1,1**)

Spectrum 41; branches = [(2,4), (1,0), (1,0)]; singlets = 45

45(**1,1,1**) + 1(**15,1,1**) + 4(**4,5,1**) + 4(**4,1,5**) + 3(**4,1,1**) + 2(**1,1,10**) + 2(**1,10,1**) + 8(**6,1,1**) + 1(**20',1,1**)

Spectrum 42; branches = [(2,4), (1,0), (1,0)]; singlets = 35

35(**1,1,1**) + 1(**15,1,1**) + 4(**4,5,1**) + 2(**4,1,5**) + 13(**4,1,1**) + 2(**1,1,5**) + 2(**1,1,10**) + 2(**1,10,1**) + 1(**6,1,5**) + 3(**6,1,1**) + 1(**20',1,1**)

Spectrum 43; branches = [(2,4), (1,0), (1,0)]; singlets = 42

42(**1,1,1**) + 4(**4,5,1**) + 2(**4,1,5**) + 2(**4,1,1**) + 2(**1,1,5**) + 2(**1,1,10**) + 2(**1,10,1**) + 1(**6,1,5**) + 10(**6,1,1**) + 3(**10,1,1**)

Spectrum 44; branches = [(2,4), (1,0), (1,0)]; singlets = 32

32(**1,1,1**) + 4(**4,5,1**) + 12(**4,1,1**) + 4(**1,1,5**) + 2(**1,1,10**) + 2(**1,10,1**) + 2(**6,1,5**) + 5(**6,1,1**) + 3(**10,1,1**)

Spectrum 45; branches = [(2,4), (1,0), (1,0)]; singlets = 32

32(**1,1,1**) + 2(**4,5,1**) + 2(**4,1,5**) + 12(**4,1,1**) + 2(**1,5,1**) + 2(**1,1,5**) + 2(**1,1,10**) + 2(**1,10,1**) + 1(**6,5,1**) + 1(**6,1,5**) + 5(**6,1,1**) + 3(**10,1,1**)

Spectrum 46; branches = [(2,4), (1,0), (1,0)]; singlets = 22

22(**1,1,1**) + 2(**4,5,1**) + 22(**4,1,1**) + 2(**1,5,1**) + 4(**1,1,5**) + 2(**1,1,10**) + 2(**1,10,1**) + 1(**6,5,1**) + 2(**6,1,5**) + 3(**10,1,1**)

Spectrum 47; branches = [(2,4), (1,0), (1,0)]; singlets = 43

43(**1,1,1**) + 1(**15,1,1**) + 4(**4,5,1**) + 2(**4,1,5**) + 2(**4,1,1**) + 2(**1,1,5**) + 2(**1,1,10**) + 2(**1,10,1**) + 1(**6,1,5**) + 9(**6,1,1**) + 2(**10,1,1**)

Spectrum 48; branches = [(2,4), (1,0), (1,0)]; singlets = 33

33(**1,1,1**) + 1(**15,1,1**) + 4(**4,5,1**) + 12(**4,1,1**) + 4(**1,1,5**) + 2(**1,1,10**) + 2(**1,10,1**) + 2(**6,1,5**) + 4(**6,1,1**) + 2(**10,1,1**)

Spectrum 49; branches = [(2,4), (1,0), (1,0)]; singlets = 33

33(**1,1,1**) + 1(**15,1,1**) + 2(**4,5,1**) + 2(**4,1,5**) + 12(**4,1,1**) + 2(**1,5,1**) + 2(**1,1,5**) + 2(**1,1,10**) + 2(**1,10,1**) + 1(**6,5,1**) + 1(**6,1,5**) + 4(**6,1,1**) + 2(**10,1,1**)

Spectrum 50; branches = [(2,4), (1,0), (1,0)]; singlets = 44

44(**1,1,1**) + 2(**15,1,1**) + 4(**4,5,1**) + 2(**4,1,5**) + 2(**4,1,1**) + 2(**1,1,5**) + 2(**1,1,10**) + 2(**1,10,1**) + 1(**6,1,5**) + 8(**6,1,1**) + 1(**10,1,1**)

Spectrum 51; branches = [(2,4), (1,0), (1,0)]; singlets = 34

34(**1,1,1**) + 2(**15,1,1**) + 4(**4,5,1**) + 12(**4,1,1**) + 4(**1,1,5**) + 2(**1,1,10**) + 2(**1,10,1**) + 2(**6,1,5**) + 3(**6,1,1**) + 1(**10,1,1**)

Spectrum 52; branches = [(2,4), (1,0), (1,0)]; singlets = 34

34(**1,1,1**) + 2(**15,1,1**) + 2(**4,5,1**) + 2(**4,1,5**) + 12(**4,1,1**) + 2(**1,5,1**) + 2(**1,1,5**) + 2(**1,1,10**) + 2(**1,10,1**) + 1(**6,5,1**) + 1(**6,1,5**) + 3(**6,1,1**) + 1(**10,1,1**)

Spectrum 53; branches = [(2,4), (1,0), (1,0)]; singlets = 45

45(**1,1,1**) + 3(**15,1,1**) + 4(**4,5,1**) + 2(**4,1,5**) + 2(**4,1,1**) + 2(**1,1,5**) + 2(**1,1,10**) + 2(**1,10,1**) + 1(**6,1,5**) + 7(**6,1,1**)

Spectrum 54; branches = [(2,4), (1,0), (1,0)]; singlets = 35

35(**1,1,1**) + 3(**15,1,1**) + 4(**4,5,1**) + 12(**4,1,1**) + 4(**1,1,5**) + 2(**1,1,10**) + 2(**1,10,1**) + 2(**6,1,5**) + 2(**6,1,1**)

Spectrum 55; branches = [(2,4), (1,0), (1,0)]; singlets = 35

35(**1,1,1**) + 3(**15,1,1**) + 2(**4,5,1**) + 2(**4,1,5**) + 12(**4,1,1**) + 2(**1,5,1**) + 2(**1,1,5**) + 2(**1,1,10**) + 2(**1,10,1**) + 1(**6,5,1**) + 1(**6,1,5**) + 2(**6,1,1**)

A3 x A4 x B3

Spectrum 56; branches = [(2,4), (1,0), (1,0)]; singlets = 23

23(**1,1,1**) + 4(**4,5,1**) + 28(**4,1,1**) + 2(**4,1,8**) + 3(**1,1,7**) + 2(**1,10,1**) + 1(**20,1,1**)

Spectrum 57; branches = [(2,4), (1,0), (1,0)]; singlets = 43

43(**1,1,1**) + 4(**4,5,1**) + 7(**4,1,1**) + 2(**4,1,8**) + 3(**1,1,7**) + 2(**1,10,1**) + 9(**6,1,1**) + 1(**10,1,1**) + 1(**20',1,1**)

Spectrum 58; branches = [(2,4), (1,0), (1,0)]; singlets = 27

27(**1,1,1**) + 4(**4,5,1**) + 23(**4,1,1**) + 3(**1,1,7**) + 2(**1,1,8**) + 2(**1,10,1**) + 1(**6,1,1**) + 1(**6,1,8**) + 1(**10,1,1**) + 1(**20',1,1**)

Spectrum 59; branches = [(2,4), (1,0), (1,0)]; singlets = 33

33(**1,1,1**) + 2(**4,5,1**) + 17(**4,1,1**) + 2(**4,1,8**) + 2(**1,5,1**) + 3(**1,1,7**) + 2(**1,10,1**) + 1(**6,5,1**) + 4(**6,1,1**) + 1(**10,1,1**) + 1(**20',1,1**)

Spectrum 60; branches = [(2,4), (1,0), (1,0)]; singlets = 44

44(**1,1,1**) + 1(**15,1,1**) + 4(**4,5,1**) + 7(**4,1,1**) + 2(**4,1,8**) + 3(**1,1,7**) + 2(**1,10,1**) + 8(**6,1,1**) + 1(**20',1,1**)

Spectrum 61; branches = [(2,4), (1,0), (1,0)]; singlets = 28

28(1,1,1) + 1(15,1,1) + 4(4,5,1) + 23(4,1,1) + 3(1,1,7) + 2(1,1,8) + 2(1,10,1) + 1(6,1,8) + 1(20',1,1)

Spectrum 62; branches = [(2,4), (1,0), (1,0)]; singlets = 34

34(1,1,1) + 1(15,1,1) + 2(4,5,1) + 17(4,1,1) + 2(4,1,8) + 2(1,5,1) + 3(1,1,7) + 2(1,10,1) + 1(6,5,1) + 3(6,1,1) + 1(20',1,1)

Spectrum 63; branches = [(2,4), (1,0), (1,0)]; singlets = 35

35(1,1,1) + 4(4,5,1) + 12(4,1,1) + 3(1,1,7) + 2(1,1,8) + 2(1,10,1) + 7(6,1,1) + 1(6,1,8) + 3(10,1,1)

Spectrum 64; branches = [(2,4), (1,0), (1,0)]; singlets = 41

41(1,1,1) + 2(4,5,1) + 6(4,1,1) + 2(4,1,8) + 2(1,5,1) + 3(1,1,7) + 2(1,10,1) + 1(6,5,1) + 10(6,1,1) + 3(10,1,1)

Spectrum 65; branches = [(2,4), (1,0), (1,0)]; singlets = 25

25(1,1,1) + 2(4,5,1) + 22(4,1,1) + 2(1,5,1) + 3(1,1,7) + 2(1,1,8) + 2(1,10,1) + 1(6,5,1) + 2(6,1,1) + 1(6,1,8) + 3(10,1,1)

Spectrum 66; branches = [(2,4), (1,0), (1,0)]; singlets = 31

31(1,1,1) + 16(4,1,1) + 2(4,1,8) + 4(1,5,1) + 3(1,1,7) + 2(1,10,1) + 2(6,5,1) + 5(6,1,1) + 3(10,1,1)

Spectrum 67; branches = [(2,4), (1,0), (1,0)]; singlets = 36

36(1,1,1) + 1(15,1,1) + 4(4,5,1) + 12(4,1,1) + 3(1,1,7) + 2(1,1,8) + 2(1,10,1) + 6(6,1,1) + 1(6,1,8) + 2(10,1,1)

Spectrum 68; branches = [(2,4), (1,0), (1,0)]; singlets = 42

42(1,1,1) + 1(15,1,1) + 2(4,5,1) + 6(4,1,1) + 2(4,1,8) + 2(1,5,1) + 3(1,1,7) + 2(1,10,1) + 1(6,5,1) + 9(6,1,1) + 2(10,1,1)

Spectrum 69; branches = [(2,4), (1,0), (1,0)]; singlets = 26

26(1,1,1) + 1(15,1,1) + 2(4,5,1) + 22(4,1,1) + 2(1,5,1) + 3(1,1,7) + 2(1,1,8) + 2(1,10,1) + 1(6,5,1) + 1(6,1,1) + 1(6,1,8) + 2(10,1,1)

Spectrum 70; branches = [(2,4), (1,0), (1,0)]; singlets = 32

32(1,1,1) + 1(15,1,1) + 16(4,1,1) + 2(4,1,8) + 4(1,5,1) + 3(1,1,7) + 2(1,10,1) + 2(6,5,1) + 4(6,1,1) + 2(10,1,1)

Spectrum 71; branches = [(2,4), (1,0), (1,0)]; singlets = 37

37(1,1,1) + 2(15,1,1) + 4(4,5,1) + 12(4,1,1) + 3(1,1,7) + 2(1,1,8) + 2(1,10,1) + 5(6,1,1) + 1(6,1,8) + 1(10,1,1)

Spectrum 72; branches = [(2,4), (1,0), (1,0)]; singlets = 43

43(1,1,1) + 2(15,1,1) + 2(4,5,1) + 6(4,1,1) + 2(4,1,8) + 2(1,5,1) + 3(1,1,7) + 2(1,10,1) + 1(6,5,1) + 8(6,1,1) + 1(10,1,1)

Spectrum 73; branches = [(2,4), (1,0), (1,0)]; singlets = 27

27(1,1,1) + 2(15,1,1) + 2(4,5,1) + 22(4,1,1) + 2(1,5,1) + 3(1,1,7) + 2(1,1,8) + 2(1,10,1) + 1(6,5,1) + 1(6,1,8) + 1(10,1,1)

Spectrum 74; branches = [(2,4), (1,0), (1,0)]; singlets = 33

33(1,1,1) + 2(15,1,1) + 16(4,1,1) + 2(4,1,8) + 4(1,5,1) + 3(1,1,7) + 2(1,10,1) + 2(6,5,1) + 3(6,1,1) + 1(10,1,1)

Spectrum 75; branches = [(2,4), (1,0), (1,0)]; singlets = 38

38(1,1,1) + 3(15,1,1) + 4(4,5,1) + 12(4,1,1) + 3(1,1,7) + 2(1,1,8) + 2(1,10,1) + 4(6,1,1) + 1(6,1,8)

Spectrum 76; branches = [(2,4), (1,0), (1,0)]; singlets = 44

44(1,1,1) + 3(15,1,1) + 2(4,5,1) + 6(4,1,1) + 2(4,1,8) + 2(1,5,1) + 3(1,1,7) + 2(1,10,1) + 1(6,5,1) + 7(6,1,1)

Spectrum 77; branches = [(2,4), (1,0), (1,0)]; singlets = 34

34(1,1,1) + 3(15,1,1) + 16(4,1,1) + 2(4,1,8) + 4(1,5,1) + 3(1,1,7) + 2(1,10,1) + 2(6,5,1) + 2(6,1,1)

A3 x A4 x C4

Spectrum 78; branches = [(2,4), (1,0), (1,0)]; singlets = 17

17(1,1,1) + 4(4,5,1) + 4(4,1,8) + 12(4,1,1) + 1(1,1,27) + 2(1,10,1) + 1(20,1,1)

Spectrum 79; branches = [(2,4), (1,0), (1,0)]; singlets = 21

21(1,1,1) + 4(4,5,1) + 2(4,1,8) + 7(4,1,1) + 2(1,1,8) + 1(1,1,27) + 2(1,10,1) + 1(6,1,8) + 1(6,1,1) + 1(10,1,1) + 1(20',1,1)

Spectrum 80; branches = [(2,4), (1,0), (1,0)]; singlets = 27

27(1,1,1) + 2(4,5,1) + 4(4,1,8) + 1(4,1,1) + 2(1,5,1) + 1(1,1,27) + 2(1,10,1) + 1(6,5,1) + 4(6,1,1) + 1(10,1,1) + 1(20',1,1)

Spectrum 81; branches = [(2,4), (1,0), (1,0)]; singlets = 19

19(1,1,1) + 2(4,5,1) + 3(4,1,8) + 9(4,1,1) + 2(1,5,1) + 1(1,1,8) + 1(1,1,27) + 2(1,10,1) + 1(6,5,1) + 1/2(6,1,8) + 1(10,1,1) + 1(20',1,1)

Spectrum 82; branches = [(2,4), (1,0), (1,0)]; singlets = 22

22(1,1,1) + 1(15,1,1) + 4(4,5,1) + 2(4,1,8) + 7(4,1,1) + 2(1,1,8) + 1(1,1,27) + 2(1,10,1) + 1(6,1,8) + 1(20',1,1)

Spectrum 83; branches = [(2,4), (1,0), (1,0)]; singlets = 28

28(1,1,1) + 1(15,1,1) + 2(4,5,1) + 4(4,1,8) + 1(4,1,1) + 2(1,5,1) + 1(1,1,27) + 2(1,10,1) + 1(6,5,1) + 3(6,1,1) + 1(20',1,1)

Spectrum 84; branches = [(2,4), (1,0), (1,0)]; singlets = 21

21(1,1,1) + 4(4,5,1) + 1(4,1,8) + 4(4,1,1) + 3(1,1,8) + 1(1,1,27) + 2(1,10,1) + 3/2(6,1,8) + 3(6,1,1) + 3(10,1,1)

Spectrum 85; branches = [(2,4), (1,0), (1,0)]; singlets = 19

19(1,1,1) + 2(4,5,1) + 2(4,1,8) + 6(4,1,1) + 2(1,5,1) + 2(1,1,8) + 1(1,1,27) + 2(1,10,1) + 1(6,5,1) + 1(6,1,8) + 2(6,1,1) + 3(10,1,1)

Spectrum 86; branches = [(2,4), (1,0), (1,0)]; singlets = 25

25(1,1,1) + 4(4,1,8) + 4(1,5,1) + 1(1,1,27) + 2(1,10,1) + 2(6,5,1) + 5(6,1,1) + 3(10,1,1)

Spectrum 87; branches = [(2,4), (1,0), (1,0)]; singlets = 17

17(1,1,1) + 3(4,1,8) + 8(4,1,1) + 4(1,5,1) + 1(1,1,8) + 1(1,1,27) + 2(1,10,1) + 2(6,5,1) + 1/2(6,1,8) + 1(6,1,1) + 3(10,1,1)

Spectrum 88; branches = [(2,4), (1,0), (1,0)]; singlets = 22

22(1,1,1) + 1(15,1,1) + 4(4,5,1) + 1(4,1,8) + 4(4,1,1) + 3(1,1,8) + 1(1,1,27) + 2(1,10,1) + 3/2(6,1,8) + 2(6,1,1) + 2(10,1,1)

Spectrum 89; branches = [(2,4), (1,0), (1,0)]; singlets = 20

20(1,1,1) + 1(15,1,1) + 2(4,5,1) + 2(4,1,8) + 6(4,1,1) + 2(1,5,1) + 2(1,1,8) + 1(1,1,27) + 2(1,10,1) + 1(6,5,1) + 1(6,1,8) + 1(6,1,1) + 2(10,1,1)

Spectrum 90; branches = [(2,4), (1,0), (1,0)]; singlets = 26

26(1,1,1) + 1(15,1,1) + 4(4,1,8) + 4(1,5,1) + 1(1,1,27) + 2(1,10,1) + 2(6,5,1) + 4(6,1,1) + 2(10,1,1)

Spectrum 91; branches = [(2,4), (1,0), (1,0)]; singlets = 18

18(1,1,1) + 1(15,1,1) + 3(4,1,8) + 8(4,1,1) + 4(1,5,1) + 1(1,1,8) + 1(1,1,27) + 2(1,10,1) + 2(6,5,1) + 1/2(6,1,8) + 2(10,1,1)

Spectrum 92; branches = [(2,4), (1,0), (1,0)]; singlets = 23

23(1,1,1) + 2(15,1,1) + 4(4,5,1) + 1(4,1,8) + 4(4,1,1) + 3(1,1,8) + 1(1,1,27) + 2(1,10,1) + 3/2(6,1,8) + 1(6,1,1) + 1(10,1,1)

Spectrum 93; branches = [(2,4), (1,0), (1,0)]; singlets = 21

21(1,1,1) + 2(15,1,1) + 2(4,5,1) + 2(4,1,8) + 6(4,1,1) + 2(1,5,1) + 2(1,1,8) + 1(1,1,27) + 2(1,10,1) + 1(6,5,1) + 1(6,1,8) + 1(10,1,1)

Spectrum 94; branches = [(2,4), (1,0), (1,0)]; singlets = 27

27(**1,1,1**) + 2(**15,1,1**) + 4(**4,1,8**) + 4(**1,5,1**) + 1(**1,1,27**) + 2(**1,10,1**) + 2(**6,5,1**) + 3(**6,1,1**) + 1(**10,1,1**)

Spectrum 95; branches = [(2,4), (1,0), (1,0)]; singlets = 24

24(**1,1,1**) + 3(**15,1,1**) + 4(**4,5,1**) + 1(**4,1,8**) + 4(**4,1,1**) + 3(**1,1,8**) + 1(**1,1,27**) + 2(**1,10,1**) + 3/2(**6,1,8**)

Spectrum 96; branches = [(2,4), (1,0), (1,0)]; singlets = 28

28(**1,1,1**) + 3(**15,1,1**) + 4(**4,1,8**) + 4(**1,5,1**) + 1(**1,1,27**) + 2(**1,10,1**) + 2(**6,5,1**) + 2(**6,1,1**)

A3 x B3 x B3

Spectrum 97; branches = [(2,4), (1,0), (1,0)]; singlets = 22

22(**1,1,1**) + 32(**4,1,1**) + 2(**4,1,8**) + 2(**4,8,1**) + 3(**1,7,1**) + 3(**1,1,7**) + 1(**20,1,1**)

Spectrum 98; branches = [(2,4), (1,0), (1,0)]; singlets = 42

42(**1,1,1**) + 11(**4,1,1**) + 2(**4,1,8**) + 2(**4,8,1**) + 3(**1,7,1**) + 3(**1,1,7**) + 9(**6,1,1**) + 1(**10,1,1**) + 1(**20',1,1**)

Spectrum 99; branches = [(2,4), (1,0), (1,0)]; singlets = 26

26(**1,1,1**) + 27(**4,1,1**) + 2(**4,8,1**) + 3(**1,7,1**) + 3(**1,1,7**) + 2(**1,1,8**) + 1(**6,1,1**) + 1(**6,1,8**) + 1(**10,1,1**) + 1(**20',1,1**)

Spectrum 100; branches = [(2,4), (1,0), (1,0)]; singlets = 43

43(**1,1,1**) + 1(**15,1,1**) + 11(**4,1,1**) + 2(**4,1,8**) + 2(**4,8,1**) + 3(**1,7,1**) + 3(**1,1,7**) + 8(**6,1,1**) + 1(**20',1,1**)

Spectrum 101; branches = [(2,4), (1,0), (1,0)]; singlets = 27

27(**1,1,1**) + 1(**15,1,1**) + 27(**4,1,1**) + 2(**4,8,1**) + 3(**1,7,1**) + 3(**1,1,7**) + 2(**1,1,8**) + 1(**6,1,8**) + 1(**20',1,1**)

Spectrum 102; branches = [(2,4), (1,0), (1,0)]; singlets = 50

50(**1,1,1**) + 2(**4,1,8**) + 2(**4,8,1**) + 3(**1,7,1**) + 3(**1,1,7**) + 15(**6,1,1**) + 3(**10,1,1**)

Spectrum 103; branches = [(2,4), (1,0), (1,0)]; singlets = 34

34(**1,1,1**) + 16(**4,1,1**) + 2(**4,8,1**) + 3(**1,7,1**) + 3(**1,1,7**) + 2(**1,1,8**) + 7(**6,1,1**) + 1(**6,1,8**) + 3(**10,1,1**)

Spectrum 104; branches = [(2,4), (1,0), (1,0)]; singlets = 51

51(**1,1,1**) + 1(**15,1,1**) + 2(**4,1,8**) + 2(**4,8,1**) + 3(**1,7,1**) + 3(**1,1,7**) + 14(**6,1,1**) + 2(**10,1,1**)

Spectrum 105; branches = [(2,4), (1,0), (1,0)]; singlets = 35

35(**1,1,1**) + 1(**15,1,1**) + 16(**4,1,1**) + 2(**4,8,1**) + 3(**1,7,1**) + 3(**1,1,7**) + 2(**1,1,8**) + 6(**6,1,1**) + 1(**6,1,8**) + 2(**10,1,1**)

Spectrum 106; branches = [(2,4), (1,0), (1,0)]; singlets = 52

52(**1,1,1**) + 2(**15,1,1**) + 2(**4,1,8**) + 2(**4,8,1**) + 3(**1,7,1**) + 3(**1,1,7**) + 13(**6,1,1**) + 1(**10,1,1**)

Spectrum 107; branches = [(2,4), (1,0), (1,0)]; singlets = 36

36(**1,1,1**) + 2(**15,1,1**) + 16(**4,1,1**) + 2(**4,8,1**) + 3(**1,7,1**) + 3(**1,1,7**) + 2(**1,1,8**) + 5(**6,1,1**) + 1(**6,1,8**) + 1(**10,1,1**)

Spectrum 108; branches = [(2,4), (1,0), (1,0)]; singlets = 53

53(**1,1,1**) + 3(**15,1,1**) + 2(**4,1,8**) + 2(**4,8,1**) + 3(**1,7,1**) + 3(**1,1,7**) + 12(**6,1,1**)

Spectrum 109; branches = [(2,4), (1,0), (1,0)]; singlets = 37

37(**1,1,1**) + 3(**15,1,1**) + 16(**4,1,1**) + 2(**4,8,1**) + 3(**1,7,1**) + 3(**1,1,7**) + 2(**1,1,8**) + 4(**6,1,1**) + 1(**6,1,8**)

A3 x B3 x C4

Spectrum 110; branches = [(2,4), (1,0), (1,0)]; singlets = 16

16(**1,1,1**) + 4(**4,1,8**) + 16(**4,1,1**) + 2(**4,8,1**) + 3(**1,7,1**) + 1(**1,1,27**) + 1(**20,1,1**)

Spectrum 111; branches = [(2,4), (1,0), (1,0)]; singlets = 28

28(**1,1,1**) + 3(**4,1,8**) + 3(**4,1,1**) + 2(**4,8,1**) + 3(**1,7,1**) + 1(**1,1,8**) + 1(**1,1,27**) + 1/2(**6,1,8**) + 5(**6,1,1**) + 1(**10,1,1**) + 1(**20',1,1**)

Spectrum 112; branches = [(2,4), (1,0), (1,0)]; singlets = 20

20(**1,1,1**) + 2(**4,1,8**) + 11(**4,1,1**) + 2(**4,8,1**) + 3(**1,7,1**) + 2(**1,1,8**) + 1(**1,1,27**) + 1(**6,1,8**) + 1(**6,1,1**) + 1(**10,1,1**) + 1(**20',1,1**)

Spectrum 113; branches = [(2,4), (1,0), (1,0)]; singlets = 20

20(**1,1,1**) + 4(**4,1,8**) + 11(**4,1,1**) + 3(**1,7,1**) + 1(**1,1,27**) + 2(**1,8,1**) + 1(**6,1,1**) + 1(**6,8,1**) + 1(**10,1,1**) + 1(**20',1,1**)

Spectrum 114; branches = [(2,4), (1,0), (1,0)]; singlets = 29

29(**1,1,1**) + 1(**15,1,1**) + 3(**4,1,8**) + 3(**4,1,1**) + 2(**4,8,1**) + 3(**1,7,1**) + 1(**1,1,8**) + 1(**1,1,27**) + 1/2(**6,1,8**) + 4(**6,1,1**) + 1(**20',1,1**)

Spectrum 115; branches = [(2,4), (1,0), (1,0)]; singlets = 21

21(**1,1,1**) + 1(**15,1,1**) + 2(**4,1,8**) + 11(**4,1,1**) + 2(**4,8,1**) + 3(**1,7,1**) + 2(**1,1,8**) + 1(**1,1,27**) + 1(**6,1,8**) + 1(**20',1,1**)

Spectrum 116; branches = [(2,4), (1,0), (1,0)]; singlets = 21

21(**1,1,1**) + 1(**15,1,1**) + 4(**4,1,8**) + 11(**4,1,1**) + 3(**1,7,1**) + 1(**1,1,27**) + 2(**1,8,1**) + 1(**6,8,1**) + 1(**20',1,1**)

Spectrum 117; branches = [(2,4), (1,0), (1,0)]; singlets = 28

28(**1,1,1**) + 2(**4,1,8**) + 2(**4,8,1**) + 3(**1,7,1**) + 2(**1,1,8**) + 1(**1,1,27**) + 1(**6,1,8**) + 7(**6,1,1**) + 3(**10,1,1**)

Spectrum 118; branches = [(2,4), (1,0), (1,0)]; singlets = 20

20(**1,1,1**) + 1(**4,1,8**) + 8(**4,1,1**) + 2(**4,8,1**) + 3(**1,7,1**) + 3(**1,1,8**) + 1(**1,1,27**) + 3/2(**6,1,8**) + 3(**6,1,1**) + 3(**10,1,1**)

Spectrum 119; branches = [(2,4), (1,0), (1,0)]; singlets = 28

28(**1,1,1**) + 4(**4,1,8**) + 3(**1,7,1**) + 1(**1,1,27**) + 2(**1,8,1**) + 7(**6,1,1**) + 1(**6,8,1**) + 3(**10,1,1**)

Spectrum 120; branches = [(2,4), (1,0), (1,0)]; singlets = 20

20(**1,1,1**) + 3(**4,1,8**) + 8(**4,1,1**) + 3(**1,7,1**) + 1(**1,1,8**) + 1(**1,1,27**) + 2(**1,8,1**) + 1/2(**6,1,8**) + 3(**6,1,1**) + 1(**6,8,1**) + 3(**10,1,1**)

Spectrum 121; branches = [(2,4), (1,0), (1,0)]; singlets = 29

29(**1,1,1**) + 1(**15,1,1**) + 2(**4,1,8**) + 2(**4,8,1**) + 3(**1,7,1**) + 2(**1,1,8**) + 1(**1,1,27**) + 1(**6,1,8**) + 6(**6,1,1**) + 2(**10,1,1**)

Spectrum 122; branches = [(2,4), (1,0), (1,0)]; singlets = 21

21(**1,1,1**) + 1(**15,1,1**) + 1(**4,1,8**) + 8(**4,1,1**) + 2(**4,8,1**) + 3(**1,7,1**) + 3(**1,1,8**) + 1(**1,1,27**) + 3/2(**6,1,8**) + 2(**6,1,1**) + 2(**10,1,1**)

Spectrum 123; branches = [(2,4), (1,0), (1,0)]; singlets = 29

29(**1,1,1**) + 1(**15,1,1**) + 4(**4,1,8**) + 3(**1,7,1**) + 1(**1,1,27**) + 2(**1,8,1**) + 6(**6,1,1**) + 1(**6,8,1**) + 2(**10,1,1**)

Spectrum 124; branches = [(2,4), (1,0), (1,0)]; singlets = 21

21(**1,1,1**) + 1(**15,1,1**) + 3(**4,1,8**) + 8(**4,1,1**) + 3(**1,7,1**) + 1(**1,1,8**) + 1(**1,1,27**) + 2(**1,8,1**) + 1/2(**6,1,8**) + 2(**6,1,1**) + 1(**6,8,1**) + 2(**10,1,1**)

Spectrum 125; branches = [(2,4), (1,0), (1,0)]; singlets = 30

30(**1,1,1**) + 2(**15,1,1**) + 2(**4,1,8**) + 2(**4,8,1**) + 3(**1,7,1**) + 2(**1,1,8**) + 1(**1,1,27**) + 1(**6,1,8**) + 5(**6,1,1**) + 1(**10,1,1**)

Spectrum 126; branches = [(2,4), (1,0), (1,0)]; singlets = 22

22(1,1,1) + 2(15,1,1) + 1(4,1,8) + 8(4,1,1) + 2(4,8,1) + 3(1,7,1) + 3(1,1,8) + 1(1,1,27) + 3/2(6,1,8) + 1(6,1,1) + 1(10,1,1)

Spectrum 127; branches = [(2,4), (1,0), (1,0)]; singlets = 30

30(1,1,1) + 2(15,1,1) + 4(4,1,8) + 3(1,7,1) + 1(1,1,27) + 2(1,8,1) + 5(6,1,1) + 1(6,8,1) + 1(10,1,1)

Spectrum 128; branches = [(2,4), (1,0), (1,0)]; singlets = 22

22(1,1,1) + 2(15,1,1) + 3(4,1,8) + 8(4,1,1) + 3(1,7,1) + 1(1,1,8) + 1(1,1,27) + 2(1,8,1) + 1/2(6,1,8) + 1(6,1,1) + 1(6,8,1) + 1(10,1,1)

Spectrum 129; branches = [(2,4), (1,0), (1,0)]; singlets = 31

31(1,1,1) + 3(15,1,1) + 2(4,1,8) + 2(4,8,1) + 3(1,7,1) + 2(1,1,8) + 1(1,1,27) + 1(6,1,8) + 4(6,1,1)

Spectrum 130; branches = [(2,4), (1,0), (1,0)]; singlets = 23

23(1,1,1) + 3(15,1,1) + 1(4,1,8) + 8(4,1,1) + 2(4,8,1) + 3(1,7,1) + 3(1,1,8) + 1(1,1,27) + 3/2(6,1,8)

Spectrum 131; branches = [(2,4), (1,0), (1,0)]; singlets = 31

31(1,1,1) + 3(15,1,1) + 4(4,1,8) + 3(1,7,1) + 1(1,1,27) + 2(1,8,1) + 4(6,1,1) + 1(6,8,1)

Spectrum 132; branches = [(2,4), (1,0), (1,0)]; singlets = 23

23(1,1,1) + 3(15,1,1) + 3(4,1,8) + 8(4,1,1) + 3(1,7,1) + 1(1,1,8) + 1(1,1,27) + 2(1,8,1) + 1/2(6,1,8) + 1(6,8,1)

A3 x B3 x D5

Spectrum 133; branches = [(2,4), (5/2,5), (1,-2)]; singlets = 8

8(1,1,1) + 1(4,21,1) + 2(4,7,1) + 1(4,8,1) + 2(1,21,1) + 2(1,7,1) + 2(1,1,10) + 1(6,1,1) + 1(6,8,1) + 1(10,1,1) + 1(20',1,1)

Spectrum 134; branches = [(2,4), (5/2,5), (1,-2)]; singlets = 9

9(1,1,1) + 1(15,1,1) + 1(4,21,1) + 2(4,7,1) + 1(4,8,1) + 2(1,21,1) + 2(1,7,1) + 2(1,1,10) + 1(6,8,1) + 1(20',1,1)

Spectrum 135; branches = [(2,4), (5/2,5), (1,-2)]; singlets = 2

2(1,1,1) + 1(4,21,1) + 3(4,1,1) + 1(4,8,1) + 2(1,21,1) + 4(1,7,1) + 2(1,1,10) + 1(6,7,1) + 1(6,8,1) + 3(10,1,1)

A3 x C2 x C2

Spectrum 136; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 17

17(1,1,1) + 2(4,4,4) + 2(4,4,1) + 9(4,1,1) + 2(4,1,5) + 1(4,5,1) + 2(1,4,1) + 1(1,1,10) + 3/2(1,1,4) + 1(1,1,5) + 1/2(1,5,4) + 1/2(1,16,1) + 1(20,1,1)

Spectrum 137; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 18

18(1,1,1) + 2(4,4,4) + 12(4,1,1) + 2(4,1,5) + 2(4,5,1) + 1(1,10,1) + 1(1,4,4) + 1(1,1,10) + 1(1,1,5) + 1(1,5,1) + 1(20,1,1)

Spectrum 138; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 16

16(1,1,1) + 2(4,4,4) + 1(4,4,1) + 2(4,1,1) + 1(4,5,1) + 3(1,4,1) + 15/2(1,1,4) + 1/2(1,1,16) + 1/2(1,5,4) + 1/2(1,16,1) + 1/2(6,4,1) + 2(6,1,1) + 1(6,1,5) + 1(10,1,1) + 1(20',1,1)

Spectrum 139; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 23

23(1,1,1) + 2(4,4,4) + 1(4,4,1) + 2(4,1,1) + 1(4,5,1) + 3(1,4,1) + 1(1,1,10) + 3/2(1,1,4) + 3(1,1,5) + 1/2(1,5,4) + 1/2(1,16,1) + 1/2(6,4,1) + 2(6,1,1) + 1(6,1,5) + 1(10,1,1) + 1(20',1,1)

Spectrum 140; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 23

23(1,1,1) + 2(4,4,4) + 1(4,4,1) + 2(4,1,1) + 1(4,5,1) + 1/2(1,4,1) + 1/2(1,4,5) + 1(1,1,10) + 4(1,1,4) + 1(1,1,5) + 2(1,5,1) + 1/2(1,16,1) + 1/2(6,4,1) + 2(6,1,1) + 1(6,1,5) + 1(10,1,1) + 1(20',1,1)

Spectrum 141; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 32

32(1,1,1) + 2(4,4,4) + 1(4,1,1) + 1(4,1,5) + 1(4,5,1) + 1(1,4,4) + 2(1,1,5) + 1/2(1,1,16) + 2(1,5,1) + 1/2(1,16,1) + 1(6,4,1) + 1(6,1,4) + 1(6,1,1) + 1(10,1,1) + 1(20',1,1)

Spectrum 142; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 26

26(1,1,1) + 2(4,4,4) + 1(4,1,1) + 1(4,1,5) + 1(4,5,1) + 4(1,4,1) + 3/2(1,1,4) + 2(1,1,5) + 1/2(1,1,16) + 1/2(1,5,4) + 1/2(1,16,1) + 1(6,4,1) + 1(6,1,4) + 1(6,1,1) + 1(10,1,1) + 1(20',1,1)

Spectrum 143; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 18

18(1,1,1) + 2(4,4,4) + 6(4,1,1) + 1(4,5,1) + 1(1,4,4) + 6(1,1,4) + 1/2(1,1,16) + 2(1,5,1) + 1/2(1,16,1) + 1(6,4,1) + 1(6,1,5) + 1(10,1,1) + 1(20',1,1)

Spectrum 144; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 12

12(1,1,1) + 2(4,4,4) + 6(4,1,1) + 1(4,5,1) + 4(1,4,1) + 15/2(1,1,4) + 1/2(1,1,16) + 1/2(1,5,4) + 1/2(1,16,1) + 1(6,4,1) + 1(6,1,5) + 1(10,1,1) + 1(20',1,1)

Spectrum 145; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 25

25(1,1,1) + 2(4,4,4) + 6(4,1,1) + 1(4,5,1) + 1(1,4,4) + 1(1,1,10) + 3(1,1,5) + 2(1,5,1) + 1/2(1,16,1) + 1(6,4,1) + 1(6,1,5) + 1(10,1,1) + 1(20',1,1)

Spectrum 146; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 19

19(1,1,1) + 2(4,4,4) + 6(4,1,1) + 1(4,5,1) + 4(1,4,1) + 1(1,1,10) + 3/2(1,1,4) + 3(1,1,5) + 1/2(1,5,4) + 1/2(1,16,1) + 1(6,4,1) + 1(6,1,5) + 1(10,1,1) + 1(20',1,1)

Spectrum 147; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 19

19(1,1,1) + 2(4,4,4) + 6(4,1,1) + 1(4,5,1) + 3/2(1,4,1) + 1/2(1,4,5) + 1(1,1,10) + 4(1,1,4) + 1(1,1,5) + 2(1,5,1) + 1/2(1,16,1) + 1(6,4,1) + 1(6,1,5) + 1(10,1,1) + 1(20',1,1)

Spectrum 148; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 21

21(1,1,1) + 2(4,4,4) + 1(4,1,1) + 2(4,1,5) + 1(1,4,4) + 6(1,4,1) + 1(1,1,10) + 1(1,1,5) + 1/2(1,16,1) + 4(6,1,1) + 1(6,5,1) + 1(10,1,1) + 1(20',1,1)

Spectrum 149; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 28

28(1,1,1) + 2(4,4,4) + 1(4,1,1) + 2(4,5,1) + 1(1,10,1) + 1(1,4,4) + 1(1,1,10) + 3(1,1,5) + 1(1,5,1) + 4(6,1,1) + 1(6,1,5) + 1(10,1,1) + 1(20',1,1)

Spectrum 150; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 22

22(1,1,1) + 2(4,4,4) + 1(4,1,1) + 2(4,5,1) + 1(1,10,1) + 3/2(1,4,1) + 1/2(1,4,5) + 1(1,1,10) + 4(1,1,4) + 1(1,1,5) + 1(1,5,1) + 4(6,1,1) + 1(6,1,5) + 1(10,1,1) + 1(20',1,1)

Spectrum 151; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 17

17(1,1,1) + 1(15,1,1) + 2(4,4,4) + 1(4,4,1) + 2(4,1,1) + 1(4,5,1) + 3(1,4,1) + 15/2(1,1,4) + 1/2(1,1,16) + 1/2(1,5,4) + 1/2(1,16,1) + 1/2(6,4,1) + 1(6,1,1) + 1(6,1,5) + 1(20',1,1)

Spectrum 152; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 24

24(1,1,1) + 1(15,1,1) + 2(4,4,4) + 1(4,4,1) + 2(4,1,1) + 1(4,5,1) + 3(1,4,1) + 1(1,1,10) + 3/2(1,1,4) + 3(1,1,5) + 1/2(1,5,4) + 1/2(1,16,1) + 1/2(6,4,1) + 1(6,1,1) + 1(6,1,5) + 1(20',1,1)

Spectrum 153; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 24

24(1,1,1) + 1(15,1,1) + 2(4,4,4) + 1(4,4,1) + 2(4,1,1) + 1(4,5,1) + 1/2(1,4,1) + 1/2(1,4,5) + 1(1,1,10) + 4(1,1,4) + 1(1,1,5) + 2(1,5,1) + 1/2(1,16,1) + 1/2(6,4,1) + 1(6,1,1) + 1(6,1,5) + 1(20',1,1)

Spectrum 154; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 33

33(1,1,1) + 1(15,1,1) + 2(4,4,4) + 1(4,1,1) + 1(4,1,5) + 1(4,5,1) + 1(1,4,4) + 2(1,1,5) + 1/2(1,1,16) + 2(1,5,1) + 1/2(1,16,1) + 1(6,4,1) + 1(6,1,4) + 1(20',1,1)

Spectrum 155; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 27

27(1,1,1) + 1(15,1,1) + 2(4,4,4) + 1(4,1,1) + 1(4,1,5) + 1(4,5,1) + 4(1,4,1) + 3/2(1,1,4) + 2(1,1,5) + 1/2(1,1,16) + 1/2(1,5,4) + 1/2(1,16,1) + 1(6,4,1) + 1(6,1,4) + 1(20',1,1)

Spectrum 156; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 22

22(1,1,1) + 1(15,1,1) + 2(4,4,4) + 1(4,1,1) + 2(4,1,5) + 1(1,4,4) + 6(1,4,1) + 1(1,1,10) + 1(1,1,5) + 1/2(1,16,1) + 3(6,1,1) + 1(6,5,1) + 1(20',1,1)

Spectrum 157; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 29

29(1,1,1) + 1(15,1,1) + 2(4,4,4) + 1(4,1,1) + 2(4,5,1) + 1(1,10,1) + 1(1,4,4) + 1(1,1,10) + 3(1,1,5) + 1(1,5,1) + 3(6,1,1) + 1(6,1,5) + 1(20',1,1)

Spectrum 158; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 23

23(1,1,1) + 1(15,1,1) + 2(4,4,4) + 1(4,1,1) + 2(4,5,1) + 1(1,10,1) + 3/2(1,4,1) + 1/2(1,4,5) + 1(1,1,10) + 4(1,1,4) + 1(1,1,5) + 1(1,5,1) + 3(6,1,1) + 1(6,1,5) + 1(20',1,1)

Spectrum 159; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 12

12(1,1,1) + 2(4,4,4) + 1(1,4,4) + 6(1,4,1) + 6(1,1,4) + 1/2(1,1,16) + 1/2(1,16,1) + 5(6,1,1) + 1(6,1,5) + 1(6,5,1) + 3(10,1,1)

Spectrum 160; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 19

19(1,1,1) + 2(4,4,4) + 1(1,4,4) + 6(1,4,1) + 1(1,1,10) + 3(1,1,5) + 1/2(1,16,1) + 5(6,1,1) + 1(6,1,5) + 1(6,5,1) + 3(10,1,1)

Spectrum 161; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 13

13(1,1,1) + 2(4,4,4) + 15/2(1,4,1) + 1/2(1,4,5) + 1(1,1,10) + 4(1,1,4) + 1(1,1,5) + 1/2(1,16,1) + 5(6,1,1) + 1(6,1,5) + 1(6,5,1) + 3(10,1,1)

Spectrum 162; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 26

26(1,1,1) + 2(4,4,4) + 1(1,10,1) + 1(1,4,4) + 1(1,1,10) + 3(1,1,5) + 3(1,5,1) + 5(6,1,1) + 1(6,1,5) + 1(6,5,1) + 3(10,1,1)

Spectrum 163; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 20

20(1,1,1) + 2(4,4,4) + 1(1,10,1) + 4(1,4,1) + 1(1,1,10) + 3/2(1,1,4) + 3(1,1,5) + 1/2(1,5,4) + 1(1,5,1) + 5(6,1,1) + 1(6,1,5) + 1(6,5,1) + 3(10,1,1)

Spectrum 164; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 13

13(1,1,1) + 1(15,1,1) + 2(4,4,4) + 1(1,4,4) + 6(1,4,1) + 6(1,1,4) + 1/2(1,1,16) + 1/2(1,16,1) + 4(6,1,1) + 1(6,1,5) + 1(6,5,1) + 2(10,1,1)

Spectrum 165; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 20

20(1,1,1) + 1(15,1,1) + 2(4,4,4) + 1(1,4,4) + 6(1,4,1) + 1(1,1,10) + 3(1,1,5) + 1/2(1,16,1) + 4(6,1,1) + 1(6,1,5) + 1(6,5,1) + 2(10,1,1)

Spectrum 166; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 14

14(1,1,1) + 1(15,1,1) + 2(4,4,4) + 15/2(1,4,1) + 1/2(1,4,5) + 1(1,1,10) + 4(1,1,4) + 1(1,1,5) + 1/2(1,16,1) + 4(6,1,1) + 1(6,1,5) + 1(6,5,1) + 2(10,1,1)

Spectrum 167; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 27

27(1,1,1) + 1(15,1,1) + 2(4,4,4) + 1(1,10,1) + 1(1,4,4) + 1(1,1,10) + 3(1,1,5) + 3(1,5,1) + 4(6,1,1) + 1(6,1,5) + 1(6,5,1) + 2(10,1,1)

Spectrum 168; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 21

21(1,1,1) + 1(15,1,1) + 2(4,4,4) + 1(1,10,1) + 4(1,4,1) + 1(1,1,10) + 3/2(1,1,4) + 3(1,1,5) + 1/2(1,5,4) + 1(1,5,1) + 4(6,1,1) + 1(6,1,5) + 1(6,5,1) + 2(10,1,1)

Spectrum 169; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 14

14(1,1,1) + 2(15,1,1) + 2(4,4,4) + 1(1,4,4) + 6(1,4,1) + 6(1,1,4) + 1/2(1,1,16) + 1/2(1,16,1) + 3(6,1,1) + 1(6,1,5) + 1(6,5,1) + 1(10,1,1)

Spectrum 170; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 8

8(1,1,1) + 1/2(15,1,4) + 2(4,4,4) + 15/2(1,4,1) + 1/2(1,4,5) + 5/2(1,1,4) + 4(1,1,5) + 1/2(1,1,16) + 1/2(1,16,1) + 8(6,1,1) + 1(6,5,1) + 1(10,1,1)

Spectrum 171; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 21

21(1,1,1) + 2(15,1,1) + 2(4,4,4) + 1(1,4,4) + 6(1,4,1) + 1(1,1,10) + 3(1,1,5) + 1/2(1,16,1) + 3(6,1,1) + 1(6,1,5) + 1(6,5,1) + 1(10,1,1)

Spectrum 172; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 15

15(1,1,1) + 2(15,1,1) + 2(4,4,4) + 15/2(1,4,1) + 1/2(1,4,5) + 1(1,1,10) + 4(1,1,4) + 1(1,1,5) + 1/2(1,16,1) + 3(6,1,1) + 1(6,1,5) + 1(6,5,1) + 1(10,1,1)

Spectrum 173; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 15

15(1,1,1) + 1/2(15,4,1) + 2(4,4,4) + 5/2(1,4,1) + 1(1,1,10) + 3/2(1,1,4) + 3(1,1,5) + 1/2(1,5,4) + 4(1,5,1) + 1/2(1,16,1) + 8(6,1,1) + 1(6,1,5) + 1(10,1,1)

Spectrum 174; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 15

15(1,1,1) + 1/2(15,4,1) + 2(4,4,4) + 1/2(1,4,5) + 1(1,1,10) + 4(1,1,4) + 1(1,1,5) + 6(1,5,1) + 1/2(1,16,1) + 8(6,1,1) + 1(6,1,5) + 1(10,1,1)

Spectrum 175; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 28

28(1,1,1) + 2(15,1,1) + 2(4,4,4) + 1(1,10,1) + 1(1,4,4) + 1(1,1,10) + 3(1,1,5) + 3(1,5,1) + 3(6,1,1) + 1(6,1,5) + 1(6,5,1) + 1(10,1,1)

Spectrum 176; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 22

22(1,1,1) + 2(15,1,1) + 2(4,4,4) + 1(1,10,1) + 4(1,4,1) + 1(1,1,10) + 3/2(1,1,4) + 3(1,1,5) + 1/2(1,5,4) + 1(1,5,1) + 3(6,1,1) + 1(6,1,5) + 1(6,5,1) + 1(10,1,1)

Spectrum 177; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 15

15(1,1,1) + 3(15,1,1) + 2(4,4,4) + 1(1,4,4) + 6(1,4,1) + 6(1,1,4) + 1/2(1,1,16) + 1/2(1,16,1) + 2(6,1,1) + 1(6,1,5) + 1(6,5,1)

Spectrum 178; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 9

9(1,1,1) + 1/2(15,1,4) + 1(15,1,1) + 2(4,4,4) + 15/2(1,4,1) + 1/2(1,4,5) + 5/2(1,1,4) + 4(1,1,5) + 1/2(1,1,16) + 1/2(1,16,1) + 7(6,1,1) + 1(6,5,1)

Spectrum 179; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 22

22(1,1,1) + 3(15,1,1) + 2(4,4,4) + 1(1,4,4) + 6(1,4,1) + 1(1,1,10) + 3(1,1,5) + 1/2(1,16,1) + 2(6,1,1) + 1(6,1,5) + 1(6,5,1)

Spectrum 180; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 16

16(1,1,1) + 3(15,1,1) + 2(4,4,4) + 15/2(1,4,1) + 1/2(1,4,5) + 1(1,1,10) + 4(1,1,4) + 1(1,1,5) + 1/2(1,16,1) + 2(6,1,1) + 1(6,1,5) + 1(6,5,1)

Spectrum 181; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 16

16(1,1,1) + 1/2(15,4,1) + 1(15,1,1) + 2(4,4,4) + 5/2(1,4,1) + 1(1,1,10) + 3/2(1,1,4) + 3(1,1,5) + 1/2(1,5,4) + 4(1,5,1) + 1/2(1,16,1) + 7(6,1,1) + 1(6,1,5)

Spectrum 182; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 16

16(1,1,1) + 1/2(15,4,1) + 1(15,1,1) + 2(4,4,4) + 1/2(1,4,5) + 1(1,1,10) + 4(1,1,4) + 1(1,1,5) + 6(1,5,1) + 1/2(1,16,1) + 7(6,1,1) + 1(6,1,5)

Spectrum 183; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 29

29(1,1,1) + 3(15,1,1) + 2(4,4,4) + 1(1,10,1) + 1(1,4,4) + 1(1,1,10) + 3(1,1,5) + 3(1,5,1) + 2(6,1,1) + 1(6,1,5) + 1(6,5,1)

Spectrum 184; branches = [(2,4), (3/2,3), (3/2,3)]; singlets = 23

23(1,1,1) + 3(15,1,1) + 2(4,4,4) + 1(1,10,1) + 4(1,4,1) + 1(1,1,10) + 3/2(1,1,4) + 3(1,1,5) + 1/2(1,5,4) + 1(1,5,1) + 2(6,1,1) + 1(6,1,5) + 1(6,5,1)

A3 x C2 x D5

Spectrum 185; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 18

18(1,1,1) + 10(4,4,1) + 19(4,1,1) + 1(4,5,1) + 2(1,4,1) + 2(1,1,10) + 2(1,5,1) + 1/2(1,16,1) + 1(20,1,1)

Spectrum 186; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 13

13(1,1,1) + 8(4,4,1) + 22(4,1,1) + 2(4,5,1) + 1(1,10,1) + 4(1,4,1) + 2(1,1,10) + 1(1,5,1) + 1(20,1,1)

Spectrum 187; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 34

34(1,1,1) + 9(4,4,1) + 2(4,1,1) + 1(4,5,1) + 3(1,4,1) + 2(1,1,10) + 2(1,5,1) + 1/2(1,16,1) + 1/2(6,4,1) + 7(6,1,1) + 1(10,1,1) + 1(20',1,1)

Spectrum 188; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 30

30(1,1,1) + 8(4,4,1) + 6(4,1,1) + 1(4,5,1) + 4(1,4,1) + 2(1,1,10) + 2(1,5,1) + 1/2(1,16,1) + 1(6,4,1) + 5(6,1,1) + 1(10,1,1) + 1(20',1,1)

Spectrum 189; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 26

26(1,1,1) + 7(4,4,1) + 10(4,1,1) + 1(4,5,1) + 5(1,4,1) + 2(1,1,10) + 2(1,5,1) + 1/2(1,16,1) + 3/2(6,4,1) + 3(6,1,1) + 1(10,1,1) + 1(20',1,1)

Spectrum 190; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 22

22(1,1,1) + 6(4,4,1) + 14(4,1,1) + 1(4,5,1) + 6(1,4,1) + 2(1,1,10) + 2(1,5,1) + 1/2(1,16,1) + 2(6,4,1) + 1(6,1,1) + 1(10,1,1) + 1(20',1,1)

Spectrum 191; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 16

Spectrum 192; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 12

Spectrum 193; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 8

Spectrum 194; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 29

Spectrum 195; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 33

Spectrum 196; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 29

Spectrum 197; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 25

Spectrum 198: branches = [(2,4), (3/2,3), (1,-2)]; singlets = 21

Spectrum 199; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 17

Spectrum 200; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 23

Spectrum 201: branches = [(2,4), (3/2,3), (1,-2)]; singlets = 19

Spectrum 202; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 15

Spectrum 203: branches = [(2,4), (3/2,3), (1,-2)]; singlets = 35

Spectrum 204: branches = [(2,4), (3/2,3), (1,-2)]; singlets = 31

Spectrum 205: branches = [(2,4), (3/2,3), (1,-2)]; singlets = 27

Spectrum 206: branches = [(2,4), (3/2,3), (1,-2)]: singlets = 23

Spectrum 207: branches $\equiv [(2,4), (3/2,3), (1,-2)]$; singlets $\equiv 17$

Spectrum 208: branches = [(2,4), (3/2,3), (1,-2)]: singlets = 13

Spectrum 209: branches = [(2,4), (3/2,3), (1,-2)]: singlets = 30

Spectrum 210: branches = [(2, 4), (3/2, 3), (1, -2)]: singlets = 34

Spectrum 211: branches $\equiv [(2,4), (3/2,3), (1,-2)]$; singlets $\equiv 30$

Spectrum 212: branches = [(2, 4), (3/2, 3), (1, -2)]; singlets = 26

Spectrum 213: branches = [(2,4), (3/2,3), (1,-2)]: singlets = 22

Spectrum 214: branches = [(2, 4), (3/2, 3), (1, -2)]; singlets = 18

Spectrum 215: branches = [(2, 4), (3/2, 3), (1, -2)]; singlets = 24

Spectrum 216: branches = [(2 4) (3/2 3) (1 -2)]: singlets = 20

Spectrum 217: branches = [(2 4) (3/2 3) (1 -2)]; singlets = 34

Spectrum 218: branches = [(2 4) (3/2 3) (1 -2)]: singlets = 30

Spectrum 219: branches = [(2 4) (3/2 3) (1 -2)]: singlets = 30

Spectrum 220: branches = [(2 4) (3/2 3) (1 -2)]: singlets = 26

Spectrum 221: branches = [(2 4) (3/2 3) (1 -2)]: singlets = 22

Spectrum 222: branches = [(2 4) (3/2 3) (1 -2)]: singlets = 18

Spectrum 223: branches = [(2 4) (3/2 3) (1 -2)]: singlets = 24

Spectrum 234: branches = $[2, 4], [3/2, 3], [1, -3]$; singlets = 30

Spectrum 22F: branches = [2, 4], [3/2, 3], [1, 3]; singlets = 16

Spectrum 225, branches – [(2,4), (3/2,3), (1,-2)], singlets – 10

1(1,1,1) + 6(4,4,1) + 8(4,1,1) + 12(1,1,1) + 2(1,1,10) + 1/2(1,16,1) + 1(6,4,1) + 6(6,1,1) + 1(6,5,1) + 3(10,1,1)
Spectrum 226; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 12
12(1,1,1) + 5(4,4,1) + 12(4,1,1) + 13(1,4,1) + 2(1,1,10) + 1/2(1,16,1) + 3/2(6,4,1) + 4(6,1,1) + 1(6,5,1) + 3(10,1,1)
Spectrum 227; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 8
8(1,1,1) + 4(4,4,1) + 16(4,1,1) + 14(1,4,1) + 2(1,1,10) + 1/2(1,16,1) + 2(6,4,1) + 2(6,1,1) + 1(6,5,1) + 3(10,1,1)
Spectrum 228; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 4
4(1,1,1) + 3(4,4,1) + 20(4,1,1) + 15(1,4,1) + 2(1,1,10) + 1/2(1,16,1) + 5/2(6,4,1) + 1(6,5,1) + 3(10,1,1)
Spectrum 229; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 37
37(1,1,1) + 6(4,4,1) + 3(4,1,1) + 1(4,5,1) + 1(1,10,1) + 2(1,1,10) + 5(1,5,1) + 2(6,4,1) + 7(6,1,1) + 3(10,1,1)
Spectrum 230; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 33
33(1,1,1) + 5(4,4,1) + 7(4,1,1) + 1(4,5,1) + 1(1,10,1) + 1(1,4,1) + 2(1,1,10) + 5(1,5,1) + 5/2(6,4,1) + 5(6,1,1) + 3(10,1,1)
Spectrum 231; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 29
29(1,1,1) + 4(4,4,1) + 11(4,1,1) + 1(4,5,1) + 1(1,10,1) + 2(1,4,1) + 2(1,1,10) + 5(1,5,1) + 3(6,4,1) + 3(6,1,1) + 3(10,1,1)
Spectrum 232; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 25
25(1,1,1) + 3(4,4,1) + 15(4,1,1) + 1(4,5,1) + 1(1,10,1) + 3(1,4,1) + 2(1,1,10) + 5(1,5,1) + 7/2(6,4,1) + 1(6,1,1) + 3(10,1,1)
Spectrum 233; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 29
29(1,1,1) + 5(4,4,1) + 2(4,1,1) + 2(4,5,1) + 1(1,10,1) + 7(1,4,1) + 2(1,1,10) + 1(1,5,1) + 3/2(6,4,1) + 9(6,1,1) + 3(10,1,1)
Spectrum 234; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 25
25(1,1,1) + 4(4,4,1) + 6(4,1,1) + 2(4,5,1) + 1(1,10,1) + 8(1,4,1) + 2(1,1,10) + 1(1,5,1) + 2(6,4,1) + 7(6,1,1) + 3(10,1,1)
Spectrum 235; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 21
21(1,1,1) + 3(4,4,1) + 10(4,1,1) + 2(4,5,1) + 1(1,10,1) + 9(1,4,1) + 2(1,1,10) + 1(1,5,1) + 5/2(6,4,1) + 5(6,1,1) + 3(10,1,1)
Spectrum 236; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 17
17(1,1,1) + 2(4,4,1) + 14(4,1,1) + 2(4,5,1) + 1(1,10,1) + 10(1,4,1) + 2(1,1,10) + 1(1,5,1) + 3(6,4,1) + 3(6,1,1) + 3(10,1,1)
Spectrum 237; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 13
13(1,1,1) + 1(4,4,1) + 18(4,1,1) + 2(4,5,1) + 1(1,10,1) + 11(1,4,1) + 2(1,1,10) + 1(1,5,1) + 7/2(6,4,1) + 1(6,1,1) + 3(10,1,1)
Spectrum 238; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 31
31(1,1,1) + 8(4,4,1) + 1(1,10,1) + 4(1,4,1) + 2(1,1,10) + 3(1,5,1) + 10(6,1,1) + 1(6,5,1) + 3(10,1,1)
Spectrum 239; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 27
27(1,1,1) + 7(4,4,1) + 4(4,1,1) + 1(1,10,1) + 5(1,4,1) + 2(1,1,10) + 3(1,5,1) + 1/2(6,4,1) + 8(6,1,1) + 1(6,5,1) + 3(10,1,1)
Spectrum 240; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 23
23(1,1,1) + 6(4,4,1) + 8(4,1,1) + 1(1,10,1) + 6(1,4,1) + 2(1,1,10) + 3(1,5,1) + 1(6,4,1) + 6(6,1,1) + 1(6,5,1) + 3(10,1,1)
Spectrum 241; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 19
19(1,1,1) + 5(4,4,1) + 12(4,1,1) + 1(1,10,1) + 7(1,4,1) + 2(1,1,10) + 3(1,5,1) + 3/2(6,4,1) + 4(6,1,1) + 1(6,5,1) + 3(10,1,1)
Spectrum 242; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 15
15(1,1,1) + 4(4,4,1) + 16(4,1,1) + 1(1,10,1) + 8(1,4,1) + 2(1,1,10) + 3(1,5,1) + 2(6,4,1) + 2(6,1,1) + 1(6,5,1) + 3(10,1,1)
Spectrum 243; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 11
11(1,1,1) + 3(4,4,1) + 20(4,1,1) + 1(1,10,1) + 9(1,4,1) + 2(1,1,10) + 3(1,5,1) + 5/2(6,4,1) + 1(6,5,1) + 3(10,1,1)
Spectrum 244; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 35
35(1,1,1) + 1(15,1,1) + 6(4,4,1) + 8(4,1,1) + 2(1,1,10) + 6(1,5,1) + 1/2(1,16,1) + 3(6,4,1) + 2(6,1,1) + 2(10,1,1)
Spectrum 245; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 31
31(1,1,1) + 1(15,1,1) + 5(4,4,1) + 12(4,1,1) + 1(1,4,1) + 2(1,1,10) + 6(1,5,1) + 1/2(1,16,1) + 7/2(6,4,1) + 2(10,1,1)
Spectrum 246; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 31
31(1,1,1) + 1(15,1,1) + 6(4,4,1) + 3(4,1,1) + 1(4,5,1) + 6(1,4,1) + 2(1,1,10) + 2(1,5,1) + 1/2(1,16,1) + 2(6,4,1) + 6(6,1,1) + 2(10,1,1)
Spectrum 247; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 27
27(1,1,1) + 1(15,1,1) + 5(4,4,1) + 7(4,1,1) + 1(4,5,1) + 7(1,4,1) + 2(1,1,10) + 2(1,5,1) + 1/2(1,16,1) + 5/2(6,4,1) + 4(6,1,1) + 2(10,1,1)
Spectrum 248; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 23
23(1,1,1) + 1(15,1,1) + 4(4,4,1) + 11(4,1,1) + 1(4,5,1) + 8(1,4,1) + 2(1,1,10) + 2(1,5,1) + 1/2(1,16,1) + 3(6,4,1) + 2(6,1,1) + 2(10,1,1)
Spectrum 249; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 19
19(1,1,1) + 1(15,1,1) + 3(4,4,1) + 15(4,1,1) + 1(4,5,1) + 9(1,4,1) + 2(1,1,10) + 2(1,5,1) + 1/2(1,16,1) + 7/2(6,4,1) + 2(10,1,1)
Spectrum 250; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 25
25(1,1,1) + 1(15,1,1) + 8(4,4,1) + 10(1,4,1) + 2(1,1,10) + 1/2(1,16,1) + 9(6,1,1) + 1(6,5,1) + 2(10,1,1)
Spectrum 251; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 21
21(1,1,1) + 1(15,1,1) + 7(4,4,1) + 4(4,1,1) + 11(1,4,1) + 2(1,1,10) + 1/2(1,16,1) + 1/2(6,4,1) + 7(6,1,1) + 1(6,5,1) + 2(10,1,1)
Spectrum 252; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 17
17(1,1,1) + 1(15,1,1) + 6(4,4,1) + 8(4,1,1) + 12(1,4,1) + 2(1,1,10) + 1/2(1,16,1) + 1(6,4,1) + 5(6,1,1) + 1(6,5,1) + 2(10,1,1)
Spectrum

1(1,1,1) + 1(15,1,1) + 5(4,4,1) + 2(4,1,1) + 2(4,5,1) + 1(1,10,1) + 7(1,4,1) + 2(1,1,10) + 1(1,5,1) + 3/2(6,4,1) + 8(6,1,1) + 2(10,1,1)
Spectrum 260; branches = [[2,4], (3/2,3), (1,-2)]; singlets = 26
26(1,1,1) + 1(15,1,1) + 4(4,4,1) + 6(4,1,1) + 2(4,5,1) + 1(1,10,1) + 8(1,4,1) + 2(1,1,10) + 1(1,5,1) + 2(6,4,1) + 6(6,1,1) + 2(10,1,1)
Spectrum 261; branches = [[2,4], (3/2,3), (1,-2)]; singlets = 22
22(1,1,1) + 1(15,1,1) + 3(4,4,1) + 10(4,1,1) + 2(4,5,1) + 1(1,10,1) + 9(1,4,1) + 2(1,1,10) + 1(1,5,1) + 5/2(6,4,1) + 4(6,1,1) + 2(10,1,1)
Spectrum 262; branches = [[2,4], (3/2,3), (1,-2)]; singlets = 18
18(1,1,1) + 1(15,1,1) + 2(4,4,1) + 14(4,1,1) + 2(4,5,1) + 1(1,10,1) + 10(1,4,1) + 2(1,1,10) + 1(1,5,1) + 3(6,4,1) + 2(6,1,1) + 2(10,1,1)
Spectrum 263; branches = [[2,4], (3/2,3), (1,-2)]; singlets = 14
14(1,1,1) + 1(15,1,1) + 1(4,4,1) + 18(4,1,1) + 2(4,5,1) + 1(1,10,1) + 11(1,4,1) + 2(1,1,10) + 1(1,5,1) + 7/2(6,4,1) + 2(10,1,1)
Spectrum 264; branches = [[2,4], (3/2,3), (1,-2)]; singlets = 32
32(1,1,1) + 1(15,1,1) + 8(4,4,1) + 1(1,10,1) + 4(1,4,1) + 2(1,1,10) + 3(1,5,1) + 9(6,1,1) + 1(6,5,1) + 2(10,1,1)
Spectrum 265; branches = [[2,4], (3/2,3), (1,-2)]; singlets = 28
28(1,1,1) + 1(15,1,1) + 7(4,4,1) + 4(4,1,1) + 1(1,10,1) + 5(1,4,1) + 2(1,1,10) + 3(1,5,1) + 1/2(6,4,1) + 7(6,1,1) + 1(6,5,1) + 2(10,1,1)
Spectrum 266; branches = [[2,4], (3/2,3), (1,-2)]; singlets = 24
24(1,1,1) + 1(15,1,1) + 6(4,4,1) + 8(4,1,1) + 1(1,10,1) + 6(1,4,1) + 2(1,1,10) + 3(1,5,1) + 1(6,4,1) + 5(6,1,1) + 1(6,5,1) + 2(10,1,1)
Spectrum 267; branches = [[2,4], (3/2,3), (1,-2)]; singlets = 20
20(1,1,1) + 1(15,1,1) + 5(4,4,1) + 12(4,1,1) + 1(1,10,1) + 7(1,4,1) + 2(1,1,10) + 3(1,5,1) + 3/2(6,4,1) + 3(6,1,1) + 1(6,5,1) + 2(10,1,1)
Spectrum 268; branches = [[2,4], (3/2,3), (1,-2)]; singlets = 16
16(1,1,1) + 1(15,1,1) + 4(4,4,1) + 16(4,1,1) + 1(1,10,1) + 8(1,4,1) + 2(1,1,10) + 3(1,5,1) + 2(6,4,1) + 1(6,1,1) + 1(6,5,1) + 2(10,1,1)
Spectrum 269; branches = [[2,4], (3/2,3), (1,-2)]; singlets = 36
36(1,1,1) + 2(15,1,1) + 6(4,4,1) + 8(4,1,1) + 2(1,1,10) + 6(1,5,1) + 1/2(1,16,1) + 3(6,4,1) + 1(6,1,1) + 1(10,1,1)
Spectrum 270; branches = [[2,4], (3/2,3), (1,-2)]; singlets = 32
32(1,1,1) + 2(15,1,1) + 6(4,4,1) + 3(4,1,1) + 1(4,5,1) + 6(1,4,1) + 2(1,1,10) + 2(1,5,1) + 1/2(1,16,1) + 2(6,4,1) + 5(6,1,1) + 1(10,1,1)
Spectrum 271; branches = [[2,4], (3/2,3), (1,-2)]; singlets = 28
28(1,1,1) + 2(15,1,1) + 5(4,4,1) + 7(4,1,1) + 1(4,5,1) + 7(1,4,1) + 2(1,1,10) + 2(1,5,1) + 1/2(1,16,1) + 5/2(6,4,1) + 3(6,1,1) + 1(10,1,1)
Spectrum 272; branches = [[2,4], (3/2,3), (1,-2)]; singlets = 24
24(1,1,1) + 2(15,1,1) + 4(4,4,1) + 11(4,1,1) + 1(4,5,1) + 8(1,4,1) + 2(1,1,10) + 2(1,5,1) + 1/2(1,16,1) + 3(6,4,1) + 1(6,1,1) + 1(10,1,1)
Spectrum 273; branches = [[2,4], (3/2,3), (1,-2)]; singlets = 26
26(1,1,1) + 2(15,1,1) + 8(4,4,1) + 10(1,4,1) + 2(1,1,10) + 1/2(1,16,1) + 8(6,1,1) + 1(6,5,1) + 1(10,1,1)
Spectrum 274; branches = [[2,4], (3/2,3), (1,-2)]; singlets = 22
22(1,1,1) + 2(15,1,1) + 7(4,4,1) + 4(4,1,1) + 11(1,4,1) + 2(1,1,10) + 1/2(1,16,1) + 1/2(6,4,1) + 6(6,1,1) + 1(6,5,1) + 1(10,1,1)
Spectrum 275; branches = [[2,4], (3/2,3), (1,-2)]; singlets = 18
18(1,1,1) + 2(15,1,1) + 6(4,4,1) + 8(4,1,1) + 12(1,4,1) + 2(1,1,10) + 1/2(1,16,1) + 1(6,4,1) + 4(6,1,1) + 1(6,5,1) + 1(10,1,1)
Spectrum 276; branches = [[2,4], (3/2,3), (1,-2)]; singlets = 14
14(1,1,1) + 2(15,1,1) + 5(4,4,1) + 12(4,1,1) + 13(1,4,1) + 2(1,1,10) + 1/2(1,16,1) + 3/2(6,4,1) + 2(6,1,1) + 1(6,5,1) + 1(10,1,1)
Spectrum 277; branches = [[2,4], (3/2,3), (1,-2)]; singlets = 10
10(1,1,1) + 2(15,1,1) + 4(4,4,1) + 16(4,1,1) + 14(1,4,1) + 2(1,1,10) + 1/2(1,16,1) + 2(6,4,1) + 1(6,5,1) + 1(10,1,1)
Spectrum 278; branches = [[2,4], (3/2,3), (1,-2)]; singlets = 26
26(1,1,1) + 1/2(15,4,1) + 8(4,4,1) + 5/2(1,4,1) + 2(1,1,10) + 6(1,5,1) + 1/2(1,16,1) + 13(6,1,1) + 1(10,1,1)
Spectrum 279; branches = [[2,4], (3/2,3), (1,-2)]; singlets = 22
22(1,1,1) + 1/2(15,4,1) + 7(4,4,1) + 4(4,1,1) + 7/2(1,4,1) + 2(1,1,10) + 6(1,5,1) + 1/2(1,16,1) + 1/2(6,4,1) + 11(6,1,1) + 1(10,1,1)
Spectrum 280; branches = [[2,4], (3/2,3), (1,-2)]; singlets = 18
18(1,1,1) + 1/2(15,4,1) + 6(4,4,1) + 8(4,1,1) + 9/2(1,4,1) + 2(1,1,10) + 6(1,5,1) + 1/2(1,16,1) + 1(6,4,1) + 9(6,1,1) + 1(10,1,1)
Spectrum 281; branches = [[2,4], (3/2,3), (1,-2)]; singlets = 14
14(1,1,1) + 1/2(15,4,1) + 5(4,4,1) + 12(4,1,1) + 11/2(1,4,1) + 2(1,1,10) + 6(1,5,1) + 1/2(1,16,1) + 3/2(6,4,1) + 7(6,1,1) + 1(10,1,1)
Spectrum 282; branches = [[2,4], (3/2,3), (1,-2)]; singlets = 10
10(1,1,1) + 1/2(15,4,1) + 4(4,4,1) + 16(4,1,1) + 13/2(1,4,1) + 2(1,1,10) + 6(1,5,1) + 1/2(1,16,1) + 2(6,4,1) + 5(6,1,1) + 1(10,1,1)
Spectrum 283; branches = [[2,4], (3/2,3), (1,-2)]; singlets = 6
6(1,1,1) + 1/2(15,4,1) + 3(4,4,1) + 20(4,1,1) + 15/2(1,4,1) + 2(1,1,10) + 6(1,5,1) + 1/2(1,16,1) + 5/2(6,4,1) + 3(6,1,1) + 1(10,1,1)
Spectrum 284; branches = [[2,4], (3/2,3), (1,-2)]; singlets = 2
2(1,1,1) + 1/2(15,4,1) + 2(4,4,1) + 24(4,1,1) + 17/2(1,4,1) + 2(1,1,10) + 6(1,5,1) + 1/2(1,16,1) + 3(6,4,1) + 1(6,1,1) + 1(10,1,1)
Spectrum 285; branches = [[2,4], (3/2,3), (1,-2)]; singlets = 14
14(1,1,1) + 1/2(15,4,1) + 6(4,4,1) + 3(4,1,1) + 1(4,5,1) + 21/2(1,4,1) + 2(1,1,10) +

$27(1,1,1) + 2(15,1,1) + 4(4,4,1) + 6(4,1,1) + 2(4,5,1) + 1(1,10,1) + 8(1,4,1) + 2(1,1,10) + 1(1,5,1) + 2(6,4,1) + 5(6,1,1) + 1(10,1,1)$
Spectrum 294; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 23

$23(1,1,1) + 2(15,1,1) + 3(4,4,1) + 10(4,1,1) + 2(4,5,1) + 1(1,10,1) + 9(1,4,1) + 2(1,1,10) + 1(1,5,1) + 5/2(6,4,1) + 3(6,1,1) + 1(10,1,1)$
Spectrum 295; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 19

$19(1,1,1) + 2(15,1,1) + 2(4,4,1) + 14(4,1,1) + 2(4,5,1) + 1(1,10,1) + 10(1,4,1) + 2(1,1,10) + 1(1,5,1) + 3(6,4,1) + 1(6,1,1) + 1(10,1,1)$
Spectrum 296; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 33

$33(1,1,1) + 2(15,1,1) + 8(4,4,1) + 1(1,10,1) + 4(1,4,1) + 2(1,1,10) + 3(1,5,1) + 8(6,1,1) + 1(6,5,1) + 1(10,1,1)$
Spectrum 297; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 29

$29(1,1,1) + 2(15,1,1) + 7(4,4,1) + 4(4,1,1) + 1(1,10,1) + 5(1,4,1) + 2(1,1,10) + 3(1,5,1) + 1/2(6,4,1) + 6(6,1,1) + 1(6,5,1) + 1(10,1,1)$
Spectrum 298; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 25

$25(1,1,1) + 2(15,1,1) + 6(4,4,1) + 8(4,1,1) + 1(1,10,1) + 6(1,4,1) + 2(1,1,10) + 3(1,5,1) + 1(6,4,1) + 4(6,1,1) + 1(6,5,1) + 1(10,1,1)$
Spectrum 299; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 21

$21(1,1,1) + 2(15,1,1) + 5(4,4,1) + 12(4,1,1) + 1(1,10,1) + 7(1,4,1) + 2(1,1,10) + 3(1,5,1) + 3/2(6,4,1) + 2(6,1,1) + 1(6,5,1) + 1(10,1,1)$
Spectrum 300; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 17

$17(1,1,1) + 2(15,1,1) + 4(4,4,1) + 16(4,1,1) + 1(1,10,1) + 8(1,4,1) + 2(1,1,10) + 3(1,5,1) + 2(6,4,1) + 1(6,5,1) + 1(10,1,1)$
Spectrum 301; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 17

$17(1,1,1) + 1/2(15,4,1) + 4(4,4,1) + 16(4,1,1) + 1(1,10,1) + 1/2(1,4,1) + 2(1,1,10) + 9(1,5,1) + 2(6,4,1) + 5(6,1,1) + 1(10,1,1)$
Spectrum 302; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 13

$13(1,1,1) + 1/2(15,4,1) + 3(4,4,1) + 20(4,1,1) + 1(1,10,1) + 3/2(1,4,1) + 2(1,1,10) + 9(1,5,1) + 5/2(6,4,1) + 3(6,1,1) + 1(10,1,1)$
Spectrum 303; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 9

$9(1,1,1) + 1/2(15,4,1) + 2(4,4,1) + 24(4,1,1) + 1(1,10,1) + 5/2(1,4,1) + 2(1,1,10) + 9(1,5,1) + 3(6,4,1) + 1(6,1,1) + 1(10,1,1)$
Spectrum 304; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 21

$21(1,1,1) + 1/2(15,4,1) + 6(4,4,1) + 3(4,1,1) + 1(4,5,1) + 1(1,10,1) + 9/2(1,4,1) + 2(1,1,10) + 5(1,5,1) + 13(6,1,1) + 1(10,1,1)$
Spectrum 305; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 17

$17(1,1,1) + 1/2(15,4,1) + 5(4,4,1) + 7(4,1,1) + 1(4,5,1) + 1(1,10,1) + 11/2(1,4,1) + 2(1,1,10) + 5(1,5,1) + 1/2(6,4,1) + 11(6,1,1) + 1(10,1,1)$
Spectrum 306; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 13

$13(1,1,1) + 1/2(15,4,1) + 4(4,4,1) + 11(4,1,1) + 1(4,5,1) + 1(1,10,1) + 13/2(1,4,1) + 2(1,1,10) + 5(1,5,1) + 1(6,4,1) + 9(6,1,1) + 1(10,1,1)$
Spectrum 307; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 9

$9(1,1,1) + 1/2(15,4,1) + 3(4,4,1) + 15(4,1,1) + 1(4,5,1) + 1(1,10,1) + 15/2(1,4,1) + 2(1,1,10) + 5(1,5,1) + 3/2(6,4,1) + 7(6,1,1) + 1(10,1,1)$
Spectrum 308; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 5

$5(1,1,1) + 1/2(15,4,1) + 2(4,4,1) + 19(4,1,1) + 1(4,5,1) + 1(1,10,1) + 17/2(1,4,1) + 2(1,1,10) + 5(1,5,1) + 2(6,4,1) + 5(6,1,1) + 1(10,1,1)$
Spectrum 309; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 1

$1(1,1,1) + 1/2(15,4,1) + 1(4,4,1) + 23(4,1,1) + 1(4,5,1) + 1(1,10,1) + 19/2(1,4,1) + 2(1,1,10) + 5(1,5,1) + 5/2(6,4,1) + 3(6,1,1) + 1(10,1,1)$
Spectrum 310; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 9

$9(1,1,1) + 1/2(15,4,1) + 4(4,4,1) + 6(4,1,1) + 2(4,5,1) + 1(1,10,1) + 25/2(1,4,1) + 2(1,1,10) + 1(1,5,1) + 13(6,1,1) + 1(10,1,1)$
Spectrum 311; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 5

$5(1,1,1) + 1/2(15,4,1) + 3(4,4,1) + 10(4,1,1) + 2(4,5,1) + 1(1,10,1) + 27/2(1,4,1) + 2(1,1,10) + 1(1,5,1) + 1/2(6,4,1) + 11(6,1,1) + 1(10,1,1)$
Spectrum 312; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 1

$1(1,1,1) + 1/2(15,4,1) + 2(4,4,1) + 14(4,1,1) + 2(4,5,1) + 1(1,10,1) + 29/2(1,4,1) + 2(1,1,10) + 1(1,5,1) + 1(6,4,1) + 9(6,1,1) + 1(10,1,1)$
Spectrum 313; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 37

$37(1,1,1) + 3(15,1,1) + 6(4,4,1) + 8(4,1,1) + 2(1,1,10) + 6(1,5,1) + 1/2(1,16,1) + 3(6,4,1)$
Spectrum 314; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 33

$33(1,1,1) + 3(15,1,1) + 6(4,4,1) + 3(4,1,1) + 1(4,5,1) + 6(1,4,1) + 2(1,1,10) + 2(1,5,1) + 1/2(1,16,1) + 2(6,4,1) + 4(6,1,1)$
Spectrum 315; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 29

$29(1,1,1) + 3(15,1,1) + 5(4,4,1) + 7(4,1,1) + 1(4,5,1) + 7(1,4,1) + 2(1,1,10) + 2(1,5,1) + 1/2(1,16,1) + 5/2(6,4,1) + 2(6,1,1)$
Spectrum 316; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 25

$25(1,1,1) + 3(15,1,1) + 4(4,4,1) + 11(4,1,1) + 1(4,5,1) + 8(1,4,1) + 2(1,1,10) + 2(1,5,1) + 1/2(1,16,1) + 3(6,4,1)$
Spectrum 317; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 27

$27(1,1,1) + 3(15,1,1) + 8(4,4,1) + 10(1,4,1) + 2(1,1,10) + 1/2(1,16,1) + 7(6,1,1) + 1(6,5,1)$
Spectrum 318; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 23

$23(1,1,1) + 3(15,1,1) + 7(4,4,1) + 4(4,1,1) + 11(1,4,1) + 2(1,1,10) + 1/2(1,16,1) + 1/2(6,4,1) + 5(6,1,1) + 1(6,5,1)$
Spectrum 319; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 19

$19(1,1,1) + 3(15,1,1) + 6(4,4,1) + 8(4,1,1) + 12(1,4,1) + 2(1,1,10) + 1/2(1,16,1) + 1(6,4,1) + 3(6,1,1) + 1(6,5,1)$
Spectrum 320; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 15

$15(1,1,1) + 3(15,1,1) + 5(4,4,1) + 12(4,1,1) + 13(1,4,1) + 2(1,1,10) + 1/2(1,16,1) + 3/2(6,4,1) + 1(6,1,1) + 1(6,5,1)$
Spectrum 321; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 27

$27(1,1,1) + 1/2(15,4,1) + 1(15,1,1) + 8(4,4,1) + 5/2(1,4,1) + 2(1,1,10) + 6(1,5,1) + 1/2(1,16,1) + 12(6,1,1)$
Spectrum 322; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 23

$23(1,1,1) + 1/2(15,4,1) + 1(15,1,1) + 7(4,4,1) + 4(4,1,1) + 7/2(1,4,1) + 2(1,1,10) + 6(1,5,1) + 1/2(1,16,1) + 1/2(6,4,1) + 10(6,1,1)$
Spectrum 323; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 19

$19(1,1,1) + 1/2(15,4,1) + 1(15,1,1) + 6(4,4,1) + 8(4,1,1) + 9/2(1,4,1) + 2(1,1,10) + 6(1,5,1) + 1/2(1,16,1) + 1(6,4,1) + 8(6,1,1)$
Spectrum 324; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 15

$15(1,1,1) + 1/2(15,4,1) + 1(15,1,1) + 5(4,4,1) + 12(4,1,1) + 11/2(1,4,1) + 2(1,1,10) + 6(1,5,1) + 1/2(1,16,1) + 3/2(6,4,1) + 6(6,1,1)$
Spectrum 325; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 11

$11(1,1,1) + 1/2(15,4,1) + 1(15,1,1) + 4(4,4,1) + 16(4,1,1) + 13/2(1,4,1) + 2(1,1,10) + 6(1,5,1) + 1/2(1,16,1) + 2(6,4,1) + 4(6,1,1)$
Spectrum 326; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 7

$7(1,1,1) + 1/2(15,4,1) + 1(15,1,1) + 3(4,4,1) + 20(4,1,1) + 15/2(1,4,1) + 2(1,1,10) + 6(1,5,1) + 1/2(1,16,1) + 5/2(6,4,1) + 2(6,1,1)$
Spectrum 327; branches = [(2,4), (3/2,3), (1,-2)]; singlets = 3

A3 x C4 x C4

Spectrum 360; branches = [(2,4), (1,0), (1,0)]; singlets = 10
10(1,1,1) + 4(4,8,1) + 4(4,1,8) + 1(1,1,27) + 1(1,27,1) + 1(20,1,1)

A3 x D5 x G2

Spectrum 361; branches = [(2,4), (1,-2), (2,4)]; singlets = 30
30(1,1,1) + 6(4,1,7) + 1(4,1,1) + 2(1,10,1) + 1(1,1,7) + 1(1,1,27) + 1(6,1,7) + 2(6,1,1) + 1(10,1,1) + 1(20',1,1)
Spectrum 362; branches = [(2,4), (1,-2), (2,4)]; singlets = 31
31(1,1,1) + 1(15,1,1) + 6(4,1,7) + 1(4,1,1) + 2(1,10,1) + 1(1,1,7) + 1(1,1,27) + 1(6,1,7) + 1(6,1,1) + 1(20',1,1)
Spectrum 363; branches = [(2,4), (1,-2), (2,4)]; singlets = 24
24(1,1,1) + 4(4,1,7) + 4(4,1,1) + 2(1,10,1) + 3(1,1,7) + 1(1,1,27) + 2(6,1,7) + 1(6,1,1) + 3(10,1,1)
Spectrum 364; branches = [(2,4), (1,-2), (2,4)]; singlets = 30
30(1,1,1) + 4(4,1,7) + 4(4,1,1) + 2(1,10,1) + 3(1,1,14) + 2(6,1,7) + 1(6,1,1) + 3(10,1,1)
Spectrum 365; branches = [(2,4), (1,-2), (2,4)]; singlets = 25
25(1,1,1) + 1(15,1,1) + 4(4,1,7) + 4(4,1,1) + 2(1,10,1) + 3(1,1,7) + 1(1,1,27) + 2(6,1,7) + 2(10,1,1)
Spectrum 366; branches = [(2,4), (1,-2), (2,4)]; singlets = 31
31(1,1,1) + 1(15,1,1) + 4(4,1,7) + 4(4,1,1) + 2(1,10,1) + 3(1,1,14) + 2(6,1,7) + 2(10,1,1)

A4 x A4 x A4

Spectrum 367; branches = [(1,0), (1,0), (1,0)]; singlets = 40
40(1,1,1) + 16(5,1,1) + 16(1,5,1) + 16(1,1,5) + 2(1,1,10) + 2(1,10,1) + 2(10,1,1)

A4 x A4 x B3

Spectrum 368; branches = [(1,0), (1,0), (1,0)]; singlets = 55
55(1,1,1) + 16(5,1,1) + 16(1,5,1) + 3(1,1,7) + 8(1,1,8) + 2(1,10,1) + 2(10,1,1)

A4 x A4 x C2

Spectrum 369; branches = [(1,0), (1,0), (3,8)]; singlets = 11
11(1,1,1) + 1(5,1,10) + 1(5,1,1) + 1(5,1,5) + 1(1,5,10) + 1(1,5,1) + 1(1,5,5) + 8(1,1,4) + 13(1,1,5) + 2(1,1,16) + 2(1,10,1) + 2(10,1,1)
Spectrum 370; branches = [(1,0), (1,0), (3,8)]; singlets = 18
18(1,1,1) + 1(5,1,10) + 1(5,1,1) + 1(5,1,5) + 1(1,5,10) + 1(1,5,1) + 1(1,5,5) + 1(1,1,10) + 2(1,1,4) + 16(1,1,5) + 3/2(1,1,16) + 2(1,10,1) + 2(10,1,1)
Spectrum 371; branches = [(1,0), (1,0), (3/2,3)]; singlets = 54
54(1,1,1) + 3(5,1,4) + 4(5,1,1) + 3(1,5,4) + 4(1,5,1) + 12(1,1,4) + 6(1,1,5) + 1/2(1,1,16) + 2(1,10,1) + 2(10,1,1)
Spectrum 372; branches = [(1,0), (1,0), (3/2,3)]; singlets = 39
39(1,1,1) + 3(5,1,4) + 4(5,1,1) + 1(1,5,4) + 7(1,5,1) + 1(1,5,5) + 22(1,1,4) + 1(1,1,5) + 1/2(1,1,16) + 2(1,10,1) + 2(10,1,1)
Spectrum 373; branches = [(1,0), (1,0), (3/2,3)]; singlets = 61
61(1,1,1) + 3(5,1,4) + 4(5,1,1) + 3(1,5,4) + 4(1,5,1) + 1(1,1,10) + 6(1,1,4) + 9(1,1,5) + 2(1,10,1) + 2(10,1,1)
Spectrum 374; branches = [(1,0), (1,0), (3/2,3)]; singlets = 46
46(1,1,1) + 3(5,1,4) + 4(5,1,1) + 1(1,5,4) + 7(1,5,1) + 1(1,5,5) + 1(1,1,10) + 16(1,1,4) + 4(1,1,5) + 2(1,10,1) + 2(10,1,1)

A4 x A4 x C3

Spectrum 375; branches = [(1,0), (1,0), (2,4)]; singlets = 13
13(1,1,1) + 2(5,1,1) + 1(5,1,14') + 2(1,5,1) + 1(1,5,14') + 3(1,1,21) + 6(1,1,6) + 2(1,1,14) + 2(1,10,1) + 2(10,1,1)
Spectrum 376; branches = [(1,0), (1,0), (2,4)]; singlets = 15
15(1,1,1) + 2(5,1,1) + 1(5,1,14') + 2(1,5,1) + 1(1,5,14') + 3(1,1,21) + 9/2(1,1,6) + 1(1,1,14') + 3/2(1,1,14) + 2(1,10,1) + 2(10,1,1)
Spectrum 377; branches = [(1,0), (1,0), (2,4)]; singlets = 17
17(1,1,1) + 2(5,1,1) + 1(5,1,14') + 2(1,5,1) + 1(1,5,14') + 3(1,1,21) + 3(1,1,6) + 2(1,1,14') + 1(1,1,14) + 2(1,10,1) + 2(10,1,1)
Spectrum 378; branches = [(1,0), (1,0), (2,4)]; singlets = 19
19(1,1,1) + 2(5,1,1) + 1(5,1,14') + 2(1,5,1) + 1(1,5,14') + 3(1,1,21) + 3/2(1,1,6) + 3(1,1,14') + 1/2(1,1,14) + 2(1,10,1) + 2(10,1,1)
Spectrum 379; branches = [(1,0), (1,0), (2,4)]; singlets = 21
21(1,1,1) + 2(5,1,1) + 1(5,1,14') + 2(1,5,1) + 1(1,5,14') + 3(1,1,21) + 4(1,1,14') + 2(1,10,1) + 2(10,1,1)

A4 x A4 x G2

Spectrum 380; branches = [(1,0), (1,0), (2,4)]; singlets = 36
36(1,1,1) + 2(5,1,7) + 2(5,1,1) + 2(1,5,7) + 2(1,5,1) + 11(1,1,7) + 1(1,1,27) + 2(1,10,1) + 2(10,1,1)
Spectrum 381; branches = [(1,0), (1,0), (2,4)]; singlets = 42
42(1,1,1) + 2(5,1,7) + 2(5,1,1) + 2(1,5,7) + 2(1,5,1) + 3(1,1,14) + 8(1,1,7) + 2(1,10,1) + 2(10,1,1)

A4 x A5 x C2

Spectrum 382; branches = [(1,0), (3/2,1), (3/2,3)]; singlets = 9
9(1,1,1) + 1(5,6,1) + 1(5,1,4) + 1(5,1,1) + 1(5,1,5) + 2(1,6,4) + 10(1,6,1) + 10(1,1,4) + 1(1,1,5) + 1/2(1,1,16) + 1(1,15,4) + 1/2(1,20,1) + 2(10,1,1)
Spectrum 383; branches = [(1,0), (3/2,1), (3/2,3)]; singlets = 16
16(1,1,1) + 1(5,6,1) + 1(5,1,4) + 1(5,1,1) + 1(5,1,5) + 2(1,6,4) + 10(1,6,1) + 1(1,1,10) + 4(1,1,4) + 4(1,1,5) + 1(1,15,4) + 1/2(1,20,1) + 2(10,1,1)
Spectrum 384; branches = [(1,0), (3/2,1), (3/2,3)]; singlets = 10
10(1,1,1) + 1(5,6,1) + 1(5,1,4) + 1(5,1,1) + 1(5,1,5) + 2(1,6,4) + 9(1,6,1) + 10(1,1,4) + 1(1,1,5) + 1/2(1,1,16) + 1(1,15,4) + 1(1,15,1) + 2(10,1,1)
Spectrum 385; branches = [(1,0), (3/2,1), (3/2,3)]; singlets = 17

17(1,1,1) + 1(5,6,1) + 1(5,1,4) + 1(5,1,1) + 1(5,1,5) + 2(1,6,4) + 9(1,6,1) + 1(1,1,10) + 4(1,1,4) + 4(1,1,5) + 1(1,15,4) + 1(1,15,1) + 2(10,1,1)

A4 x B3 x B3

Spectrum 386; branches = [(1,0), (1,0), (1,0)]; singlets = 70

70(1,1,1) + 16(5,1,1) + 3(1,7,1) + 3(1,1,7) + 8(1,1,8) + 8(1,8,1) + 2(10,1,1)

A4 x B3 x C2

Spectrum 387; branches = [(1,0), (1,0), (3/2,3)]; singlets = 57

57(1,1,1) + 3(5,1,4) + 4(5,1,1) + 3(1,7,1) + 15(1,1,4) + 6(1,1,5) + 1/2(1,1,16) + 3/2(1,8,4) + 2(1,8,1) + 2(10,1,1)

Spectrum 388; branches = [(1,0), (1,0), (3/2,3)]; singlets = 55

55(1,1,1) + 3(5,1,4) + 4(5,1,1) + 1/2(1,7,4) + 1(1,7,1) + 31/2(1,1,4) + 6(1,1,5) + 1/2(1,1,16) + 1(1,8,4) + 4(1,8,1) + 2(10,1,1)

Spectrum 389; branches = [(1,0), (1,0), (3/2,3)]; singlets = 42

42(1,1,1) + 1(5,1,4) + 7(5,1,1) + 1(5,1,5) + 3(1,7,1) + 25(1,1,4) + 1(1,1,5) + 1/2(1,1,16) + 3/2(1,8,4) + 2(1,8,1) + 2(10,1,1)

Spectrum 390; branches = [(1,0), (1,0), (3/2,3)]; singlets = 40

40(1,1,1) + 1(5,1,4) + 7(5,1,1) + 1(5,1,5) + 1/2(1,7,4) + 1(1,7,1) + 51/2(1,1,4) + 1(1,1,5) + 1/2(1,1,16) + 1(1,8,4) + 4(1,8,1) + 2(10,1,1)

Spectrum 391; branches = [(1,0), (1,0), (3/2,3)]; singlets = 64

64(1,1,1) + 3(5,1,4) + 4(5,1,1) + 3(1,7,1) + 1(1,1,10) + 9(1,1,4) + 9(1,1,5) + 3/2(1,8,4) + 2(1,8,1) + 2(10,1,1)

Spectrum 392; branches = [(1,0), (1,0), (3/2,3)]; singlets = 62

62(1,1,1) + 3(5,1,4) + 4(5,1,1) + 1/2(1,7,4) + 1(1,7,1) + 1(1,1,10) + 19/2(1,1,4) + 9(1,1,5) + 1(1,8,4) + 4(1,8,1) + 2(10,1,1)

Spectrum 393; branches = [(1,0), (1,0), (3/2,3)]; singlets = 49

49(1,1,1) + 1(5,1,4) + 7(5,1,1) + 1(5,1,5) + 3(1,7,1) + 1(1,1,10) + 19(1,1,4) + 4(1,1,5) + 3/2(1,8,4) + 2(1,8,1) + 2(10,1,1)

Spectrum 394; branches = [(1,0), (1,0), (3/2,3)]; singlets = 47

47(1,1,1) + 1(5,1,4) + 7(5,1,1) + 1(5,1,5) + 1/2(1,7,4) + 1(1,7,1) + 1(1,1,10) + 39/2(1,1,4) + 4(1,1,5) + 1(1,8,4) + 4(1,8,1) + 2(10,1,1)

A4 x B3 x C4

Spectrum 395; branches = [(1,0), (1,0), (1,0)]; singlets = 0

16(5,1,1) + 3(1,7,1) + 16(1,1,8) + 1(1,1,27) + 8(1,8,1) + 2(10,1,1)

A4 x B3 x G2

Spectrum 396; branches = [(1,0), (1,0), (2,4)]; singlets = 37

37(1,1,1) + 2(5,1,7) + 2(5,1,1) + 3(1,7,1) + 13(1,1,7) + 1(1,1,27) + 1(1,8,7) + 1(1,8,1) + 2(10,1,1)

Spectrum 397; branches = [(1,0), (1,0), (2,4)]; singlets = 43

43(1,1,1) + 2(5,1,7) + 2(5,1,1) + 3(1,7,1) + 3(1,1,14) + 10(1,1,7) + 1(1,8,7) + 1(1,8,1) + 2(10,1,1)

A4 x C2 x C4

Spectrum 398; branches = [(1,0), (3/2,3), (1,0)]; singlets = 35

35(1,1,1) + 3(5,4,1) + 4(5,1,1) + 3(1,4,8) + 3(1,4,1) + 4(1,1,8) + 1(1,1,27) + 6(1,5,1) + 1/2(1,16,1) + 2(10,1,1)

Spectrum 399; branches = [(1,0), (3/2,3), (1,0)]; singlets = 23

23(1,1,1) + 3(5,4,1) + 4(5,1,1) + 2(1,4,8) + 11(1,4,1) + 11/2(1,1,8) + 1(1,1,27) + 1/2(1,5,8) + 2(1,5,1) + 1/2(1,16,1) + 2(10,1,1)

Spectrum 400; branches = [(1,0), (3/2,3), (1,0)]; singlets = 20

20(1,1,1) + 1(5,4,1) + 7(5,1,1) + 1(5,5,1) + 3(1,4,8) + 13(1,4,1) + 4(1,1,8) + 1(1,1,27) + 1(1,5,1) + 1/2(1,16,1) + 2(10,1,1)

Spectrum 401; branches = [(1,0), (3/2,3), (1,0)]; singlets = 30

30(1,1,1) + 3(5,4,1) + 4(5,1,1) + 1(1,10,1) + 2(1,4,8) + 5(1,4,1) + 11/2(1,1,8) + 1(1,1,27) + 1/2(1,5,8) + 5(1,5,1) + 2(10,1,1)

Spectrum 402; branches = [(1,0), (3/2,3), (1,0)]; singlets = 18

18(1,1,1) + 3(5,4,1) + 4(5,1,1) + 1(1,10,1) + 1(1,4,8) + 13(1,4,1) + 7(1,1,8) + 1(1,1,27) + 1(1,5,8) + 1(1,5,1) + 2(10,1,1)

Spectrum 403; branches = [(1,0), (3/2,3), (1,0)]; singlets = 27

27(1,1,1) + 1(5,4,1) + 7(5,1,1) + 1(5,5,1) + 1(1,10,1) + 3(1,4,8) + 7(1,4,1) + 4(1,1,8) + 1(1,1,27) + 4(1,5,1) + 2(10,1,1)

Spectrum 404; branches = [(1,0), (3/2,3), (1,0)]; singlets = 15

15(1,1,1) + 1(5,4,1) + 7(5,1,1) + 1(5,5,1) + 1(1,10,1) + 2(1,4,8) + 15(1,4,1) + 11/2(1,1,8) + 1(1,1,27) + 1/2(1,5,8) + 2(10,1,1)

A4 x C2 x D5

Spectrum 405; branches = [(5/2,5), (3/2,3), (1,-2)]; singlets = 9

9(1,1,1) + 1/2(24,4,1) + 2(5,4,1) + 2(5,1,1) + 1(5,5,1) + 2(1,1,10) + 1(1,5,1) + 1/2(1,16,1) + 2(10,4,1) + 4(10,1,1) + 1(15,1,1) + 1(40,1,1)

Spectrum 406; branches = [(5/2,5), (3/2,3), (1,-2)]; singlets = 9

9(1,1,1) + 1/2(24,4,1) + 2(5,4,1) + 3(5,1,1) + 1(5,5,1) + 2(1,1,10) + 1(1,5,1) + 1/2(1,16,1) + 2(10,4,1) + 3(10,1,1) + 1(15,1,1) + 1(45,1,1)

Spectrum 407; branches = [(5/2,5), (3/2,3), (1,-2)]; singlets = 10

10(1,1,1) + 1/2(24,4,1) + 1(24,1,1) + 2(5,4,1) + 2(5,1,1) + 1(5,5,1) + 2(1,1,10) + 1(1,5,1) + 1/2(1,16,1) + 2(10,4,1) + 3(10,1,1) + 1(40,1,1)

Spectrum 408; branches = [(5/2,5), (3/2,3), (1,-2)]; singlets = 10

10(1,1,1) + 1/2(24,4,1) + 1(24,1,1) + 2(5,4,1) + 3(5,1,1) + 1(5,5,1) + 2(1,1,10) + 1(1,5,1) + 1/2(1,16,1) + 2(10,4,1) + 2(10,1,1) + 1(45,1,1)

A4 x C3 x C4

Spectrum 409; branches = [(1,0), (2,4), (1,0)]; singlets = 6

6(1,1,1) + 2(5,1,1) + 1(5,14',1) + 3(1,21,1) + 3/2(1,6,1) + 2(1,1,8) + 1(1,1,27) + 1(1,14',8) + 1/2(1,14,1) + 2(10,1,1)

Spectrum 410; branches = [(1,0), (2,4), (1,0)]; singlets = 8

8(1,1,1) + 2(5,1,1) + 1(5,14',1) + 3(1,21,1) + 2(1,1,8) + 1(1,1,27) + 1(1,14',8) + 1(1,14',1) + 2(10,1,1)

A4 x C4 x G2

Spectrum 411; branches = [(1,0), (1,0), (2,4)]; singlets = 23

23(1,1,1) + 2(5,1,7) + 2(5,1,1) + 2(1,8,7) + 2(1,8,1) + 5(1,1,7) + 1(1,1,27) + 1(1,27,1) + 2(10,1,1)

Spectrum 412; branches = [(1,0), (1,0), (2,4)]; singlets = 29

29(1,1,1) + 2(5,1,7) + 2(5,1,1) + 2(1,8,7) + 2(1,8,1) + 3(1,1,14) + 2(1,1,7) + 1(1,27,1) + 2(10,1,1)

A5 x C2 x C2

Spectrum 413; branches = [(3/2,1), (3/2,3), (3/2,3)]; singlets = 16

16(1,1,1) + 1(6,4,4) + 1(6,1,1) + 1(6,1,5) + 1(6,5,1) + 3(1,4,4) + 6(1,4,1) + 6(1,1,4) + 1/2(1,1,16) + 1/2(1,16,1) + 5/2(20,1,1)

Spectrum 414; branches = [(3/2,1), (3/2,3), (3/2,3)]; singlets = 23

23(1,1,1) + 1(6,4,4) + 1(6,1,1) + 1(6,1,5) + 1(6,5,1) + 3(1,4,4) + 6(1,4,1) + 1(1,1,10) + 3(1,1,5) + 1/2(1,16,1) + 5/2(20,1,1)

Spectrum 415; branches = [(3/2,1), (3/2,3), (3/2,3)]; singlets = 17

17(1,1,1) + 1(6,4,4) + 1(6,1,1) + 1(6,1,5) + 1(6,5,1) + 2(1,4,4) + 15/2(1,4,1) + 1/2(1,4,5) + 1(1,1,10) + 4(1,1,4) + 1(1,1,5) + 1/2(1,16,1) + 5/2(20,1,1)

Spectrum 416; branches = [(3/2,1), (3/2,3), (3/2,3)]; singlets = 30

30(1,1,1) + 1(6,4,4) + 1(6,1,1) + 1(6,1,5) + 1(6,5,1) + 1(1,10,1) + 3(1,4,4) + 1(1,1,10) + 3(1,1,5) + 3(1,5,1) + 5/2(20,1,1)

Spectrum 417; branches = [(3/2,1), (3/2,3), (3/2,3)]; singlets = 24

24(1,1,1) + 1(6,4,4) + 1(6,1,1) + 1(6,1,5) + 1(6,5,1) + 1(1,10,1) + 2(1,4,4) + 4(1,4,1) + 1(1,1,10) + 3/2(1,1,4) + 3(1,1,5) + 1/2(1,5,4) + 1(1,5,1) + 5/2(20,1,1)

Spectrum 418; branches = [(3/2,1), (3/2,3), (3/2,3)]; singlets = 18

18(1,1,1) + 1(6,4,4) + 1(6,1,1) + 1(6,1,5) + 1(6,5,1) + 1(1,10,1) + 1(1,4,4) + 11/2(1,4,1) + 1/2(1,4,5) + 1(1,1,10) + 11/2(1,1,4) + 1(1,1,5) + 1/2(1,5,4) + 1(1,5,1) + 5/2(20,1,1)

Spectrum 419; branches = [(3/2,1), (3/2,3), (3/2,3)]; singlets = 17

17(1,1,1) + 1(6,4,4) + 1(6,1,5) + 1(6,5,1) + 3(1,4,4) + 6(1,4,1) + 6(1,1,4) + 1/2(1,1,16) + 1/2(1,16,1) + 1(15,1,1) + 2(20,1,1)

Spectrum 420; branches = [(3/2,1), (3/2,3), (3/2,3)]; singlets = 24

24(1,1,1) + 1(6,4,4) + 1(6,1,5) + 1(6,5,1) + 3(1,4,4) + 6(1,4,1) + 1(1,1,10) + 3(1,1,5) + 1/2(1,16,1) + 1(15,1,1) + 2(20,1,1)

Spectrum 421; branches = [(3/2,1), (3/2,3), (3/2,3)]; singlets = 18

18(1,1,1) + 1(6,4,4) + 1(6,1,5) + 1(6,5,1) + 2(1,4,4) + 15/2(1,4,1) + 1/2(1,4,5) + 1(1,1,10) + 4(1,1,4) + 1(1,1,5) + 1/2(1,16,1) + 1(15,1,1) + 2(20,1,1)

Spectrum 422; branches = [(3/2,1), (3/2,3), (3/2,3)]; singlets = 31

31(1,1,1) + 1(6,4,4) + 1(6,1,5) + 1(6,5,1) + 1(1,10,1) + 3(1,4,4) + 1(1,1,10) + 3(1,1,5) + 3(1,5,1) + 1(15,1,1) + 2(20,1,1)

Spectrum 423; branches = [(3/2,1), (3/2,3), (3/2,3)]; singlets = 25

25(1,1,1) + 1(6,4,4) + 1(6,1,5) + 1(6,5,1) + 1(1,10,1) + 2(1,4,4) + 4(1,4,1) + 1(1,1,10) + 3/2(1,1,4) + 3(1,1,5) + 1/2(1,5,4) + 1(1,5,1) + 1(15,1,1) + 2(20,1,1)

Spectrum 424; branches = [(3/2,1), (3/2,3), (3/2,3)]; singlets = 19

19(1,1,1) + 1(6,4,4) + 1(6,1,5) + 1(6,5,1) + 1(1,10,1) + 1(1,4,4) + 11/2(1,4,1) + 1/2(1,4,5) + 1(1,1,10) + 11/2(1,1,4) + 1(1,1,5) + 1/2(1,5,4) + 1(1,5,1) + 1(15,1,1) + 2(20,1,1)

A5 x C2 x C4

Spectrum 425; branches = [(3/2,1), (3/2,3), (1,0)]; singlets = 6

6(1,1,1) + 2(6,4,1) + 1(6,1,8) + 7(6,1,1) + 1(1,10,1) + 1(1,4,8) + 1(1,4,1) + 1(1,1,8) + 1(1,1,27) + 1(1,5,8) + 1(1,5,1) + 1(15,4,1) + 1/2(20,1,1)

Spectrum 426; branches = [(3/2,1), (3/2,3), (1,0)]; singlets = 7

7(1,1,1) + 2(6,4,1) + 1(6,1,8) + 6(6,1,1) + 1(1,10,1) + 1(1,4,8) + 1(1,4,1) + 1(1,1,8) + 1(1,1,27) + 1(1,5,8) + 1(1,5,1) + 1(15,4,1) + 1(15,1,1)

B3 x B3 x B3

Spectrum 427; branches = [(1,0), (1,0), (1,0)]; singlets = 85

85(1,1,1) + 3(7,1,1) + 3(1,7,1) + 3(1,1,7) + 8(1,1,8) + 8(1,8,1) + 8(8,1,1)

B3 x B3 x C2

Spectrum 428; branches = [(1,0), (1,0), (9/2,13)]; singlets = 0

1/2(7,1,4) + 1(7,1,1) + 1/2(1,7,4) + 1(1,7,1) + 3(1,1,10) + 11(1,1,4) + 2(1,1,5) + 1(1,1,14) + 2(1,1,16) + 2(1,1,20) + 1/2(1,8,16) + 1/2(8,1,16)

Spectrum 429; branches = [(1,0), (1,0), (9/2,13)]; singlets = 4

4(1,1,1) + 1/2(7,1,4) + 1(7,1,1) + 1/2(1,7,4) + 1(1,7,1) + 3(1,1,10) + 1/2(1,1,4) + 14(1,1,5) + 1(1,1,14) + 5/2(1,1,20) + 1/2(1,8,16) + 1/2(8,1,16)

Spectrum 430; branches = [(1,0), (1,0), (9/2,13)]; singlets = 4

4(1,1,1) + 1/2(7,1,4) + 1(7,1,1) + 1/2(1,7,4) + 1(1,7,1) + 9/2(1,1,4) + 10(1,1,5) + 3(1,1,16) + 5/2(1,1,20) + 1/2(1,8,16) + 1/2(8,1,16)

Spectrum 431; branches = [(1,0), (1,0), (9/2,13)]; singlets = 5

5(1,1,1) + 1/2(7,1,4) + 1(7,1,1) + 1/2(1,7,4) + 1(1,7,1) + 9(1,1,10) + 3(1,1,4) + 4(1,1,5) + 1/2(1,1,16) + 1(1,1,35) + 1/2(1,8,16) + 1/2(8,1,16)

Spectrum 432; branches = [(1,0), (1,0), (9/2,13)]; singlets = 7

7(1,1,1) + 1/2(7,1,4) + 1(7,1,1) + 1/2(1,7,4) + 1(1,7,1) + 4(1,1,10) + 5(1,1,4) + 5(1,1,5) + 1(1,1,14) + 3/2(1,1,16) + 2(1,1,20) + 1/2(1,8,16) + 1/2(8,1,16)

Spectrum 433; branches = [(1,0), (1,0), (9/2,13)]; singlets = 7

7(1,1,1) + 1/2(7,1,4) + 1(7,1,1) + 1/2(1,7,4) + 1(1,7,1) + 1(1,1,10) + 9(1,1,4) + 1(1,1,5) + 9/2(1,1,16) + 2(1,1,20) + 1/2(1,8,16) + 1/2(8,1,16)

Spectrum 434; branches = [(1,0), (1,0), (9/2,13)]; singlets = 10

10(1,1,1) + 1/2(7,1,4) + 1(7,1,1) + 1/2(1,7,4) + 1(1,7,1) + 8(1,1,10) + 11/2(1,1,4) + 2(1,1,14) + 3/2(1,1,20) + 1/2(1,8,16) + 1/2(8,1,16)

Spectrum 435; branches = [(1,0), (1,0), (9/2,13)]; singlets = 14

14(1,1,1) + 1/2(7,1,4) + 1(7,1,1) + 1/2(1,7,4) + 1(1,7,1) + 2(1,1,10) + 3(1,1,4) + 4(1,1,5) + 4(1,1,16) + 2(1,1,20) + 1/2(1,8,16) + 1/2(8,1,16)

Spectrum 436; branches = [(1,0), (1,0), (3/2,3)]; singlets = 60

60(1,1,1) + 3(7,1,1) + 3(1,7,1) + 18(1,1,4) + 6(1,1,5) + 1/2(1,1,16) + 3/2(1,8,4) + 2(1,8,1) + 3/2(8,1,4) + 2(8,1,1)

Spectrum 437; branches = [(1,0), (1,0), (3/2,3)]; singlets = 58

58(1,1,1) + 3(7,1,1) + 1/2(1,7,4) + 1(1,7,1) + 37/2(1,1,4) + 6(1,1,5) + 1/2(1,1,16) + 1(1,8,4) + 4(1,8,1) + 3/2(8,1,4) + 2(8,1,1)

Spectrum 438; branches = [(1,0), (1,0), (3/2,3)]; singlets = 56

56(1,1,1) + 1/2(7,1,4) + 1(7,1,1) + 1/2(1,7,4) + 1(1,7,1) + 19(1,1,4) + 6(1,1,5) + 1/2(1,1,16) + 1(1,8,4) + 4(1,8,1) + 1(8,1,4) + 4(8,1,1)

Spectrum 439; branches = [(1,0), (1,0), (3/2,3)]; singlets = 67

67(1,1,1) + 3(7,1,1) + 3(1,7,1) + 1(1,1,10) + 12(1,1,4) + 9(1,1,5) + 3/2(1,8,4) + 2(1,8,1) + 3/2(8,1,4) + 2(8,1,1)

Spectrum 440; branches = [(1,0), (1,0), (3/2,3)]; singlets = 65

Spectrum 441; branches = [(1,0), (1,0), (3/2,3)]; singlets = 63

B3 x B3 x C4

B3 x B3 x G2

B3 x C2 x C4

B3 x C2 x D5

$$2(1,1,1) + 1(21,4,1) + 2(21,1,1) + 10(7,1,1) + 1(1,4,1) + 2(1,1,10) + 6(1,5,1) + 1/2(1,16,1) + 5/2(8,4,1)$$

Spectrum 470; branches = [(5/2,5), (3/2,3), (1,-2)]; singlets = 0

1(21,4,1) + 2(21,1,1) + 1/2(7,4,1) + 8(7,1,1) + 3/2(1,4,1) + 2(1,1,10) + 6(1,5,1) + 1/2(1,16,1) + 2(8,4,1) + 2(8,1,1)

Spectrum 471; branches = [(5/2,5), (3/2,3), (1,-2)]; singlets = 15

15(1,1,1) + 1/2(21,4,1) + 4(21,1,1) + 2(7,4,1) + 2(7,1,1) + 1(1,10,1) + 7/2(1,4,1) + 2(1,1,10) + 1(1,5,1) + 1(8,4,1) + 1(8,1,1) + 1(8,5,1)

Spectrum 472; branches = [(5/2,5), (3/2,3), (1,-2)]; singlets = 13

13(1,1,1) + 1/2(21,4,1) + 4(21,1,1) + 5/2(7,4,1) + 1(1,10,1) + 4(1,4,1) + 2(1,1,10) + 1(1,5,1) + 1/2(8,4,1) + 3(8,1,1) + 1(8,5,1)

Spectrum 473; branches = [(5/2,5), (3/2,3), (1,-2)]; singlets = 16

16(1,1,1) + 1/2(21,4,1) + 4(21,1,1) + 1/2(7,4,1) + 3(7,1,1) + 1(7,5,1) + 1(1,10,1) + 2(1,4,1) + 2(1,1,10) + 2(1,5,1) + 5/2(8,4,1)

Spectrum 474; branches = [(5/2,5), (3/2,3), (1,-2)]; singlets = 14

14(1,1,1) + 1/2(21,4,1) + 4(21,1,1) + 1(7,4,1) + 1(7,1,1) + 1(7,5,1) + 1(1,10,1) + 5/2(1,4,1) + 2(1,1,10) + 2(1,5,1) + 2(8,4,1) + 2(8,1,1)

B3 x C4 x G2

Spectrum 475; branches = [(1,0), (1,0), (2,4)]; singlets = 24

24(1,1,1) + 3(7,1,1) + 2(1,8,7) + 2(1,8,1) + 7(1,1,7) + 1(1,1,27) + 1(1,27,1) + 1(8,1,7) + 1(8,1,1)

Spectrum 476; branches = [(1,0), (1,0), (2,4)]; singlets = 30

30(1,1,1) + 3(7,1,1) + 2(1,8,7) + 2(1,8,1) + 3(1,1,14) + 4(1,1,7) + 1(1,27,1) + 1(8,1,7) + 1(8,1,1)

C2 x C2 x C2

Spectrum 477; branches = [(3/2,3), (3/2,3), (3/2,3)]; singlets = 34

34(1,1,1) + 3/2(4,4,4) + 3(4,4,1) + 1(4,1,4) + 2(4,1,1) + 1(1,4,4) + 2(1,4,1) + 6(1,1,5) + 1/2(1,1,16) + 1(1,5,4) + 2(1,5,1) + 1/2(1,16,1) + 1(5,1,4) + 2(5,1,1) + 1/2(16,1,1)

Spectrum 478; branches = [(3/2,3), (3/2,3), (3/2,3)]; singlets = 28

28(1,1,1) + 3/2(4,4,4) + 3(4,4,1) + 6(4,1,1) + 1(1,4,4) + 2(1,4,1) + 3/2(1,1,4) + 6(1,1,5) + 1/2(1,1,16) + 1(1,5,4) + 2(1,5,1) + 1/2(1,16,1) + 3/2(5,1,4) + 1/2(16,1,1)

Spectrum 479; branches = [(3/2,3), (3/2,3), (3/2,3)]; singlets = 22

22(1,1,1) + 3/2(4,4,4) + 3(4,4,1) + 6(4,1,1) + 6(1,4,1) + 3(1,1,4) + 6(1,1,5) + 1/2(1,1,16) + 3/2(1,5,4) + 1/2(1,16,1) + 3/2(5,1,4) + 1/2(16,1,1)

Spectrum 480; branches = [(3/2,3), (3/2,3), (3/2,3)]; singlets = 28

28(1,1,1) + 3/2(4,4,4) + 3(4,4,1) + 1(4,1,4) + 2(4,1,1) + 7/2(1,4,1) + 1/2(1,4,5) + 4(1,1,4) + 4(1,1,5) + 1/2(1,1,16) + 1(1,5,4) + 2(1,5,1) + 1/2(1,16,1) + 1(5,1,4) + 2(5,1,1) + 1/2(16,1,1)

Spectrum 481; branches = [(3/2,3), (3/2,3), (3/2,3)]; singlets = 22

22(1,1,1) + 3/2(4,4,4) + 3(4,4,1) + 6(4,1,1) + 7/2(1,4,1) + 1/2(1,4,5) + 11/2(1,1,4) + 4(1,1,5) + 1/2(1,1,16) + 1(1,5,4) + 2(1,5,1) + 1/2(1,16,1) + 3/2(5,1,4) + 1/2(16,1,1)

Spectrum 482; branches = [(3/2,3), (3/2,3), (3/2,3)]; singlets = 28

28(1,1,1) + 3/2(4,4,4) + 3(4,4,1) + 1(4,1,4) + 2(4,1,1) + 1(1,4,1) + 1(1,4,5) + 13/2(1,1,4) + 2(1,1,5) + 1/2(1,1,16) + 1/2(1,5,4) + 4(1,5,1) + 1/2(1,16,1) + 1(5,1,4) + 2(5,1,1) + 1/2(16,1,1)

Spectrum 483; branches = [(3/2,3), (3/2,3), (3/2,3)]; singlets = 22

22(1,1,1) + 3/2(4,4,4) + 3(4,4,1) + 6(4,1,1) + 1(1,4,1) + 1(1,4,5) + 8(1,1,4) + 2(1,1,5) + 1/2(1,1,16) + 1/2(1,5,4) + 4(1,5,1) + 1/2(1,16,1) + 3/2(5,1,4) + 1/2(16,1,1)

Spectrum 484; branches = [(3/2,3), (3/2,3), (3/2,3)]; singlets = 72

22(1,1,1) + 3/2(4,4,4) + 3(4,4,1) + 7/2(4,1,1) + 1/2(4,1,5) + 7/2(1,4,1) + 1/2(1,4,5) + 8(1,1,4) + 2(1,1,5) + 1/2(1,1,16) + 1(1,5,4) + 2(1,5,1) + 1/2(1,16,1) + 1(5,1,4) + 2(5,1,1) + 1/2(16,1,1)

Spectrum 485; branches = [(3/2,3), (3/2,3), (3/2,3)]; singlets = 22

22(1,1,1) + 3/2(4,4,4) + 3(4,4,1) + 7/2(4,1,1) + 1/2(4,1,5) + 1(1,4,1) + 1(1,4,5) + 21/2(1,1,4) + 1/2(1,1,16) + 1/2(1,5,4) + 4(1,5,1) + 1/2(1,16,1) + 1(5,1,4) + 2(5,1,1) + 1/2(16,1,1)

Spectrum 486; branches = [(3/2,3), (3/2,3), (3/2,3)]; singlets = 26

26(1,1,1) + 2(4,4,4) + 1(4,4,1) + 1(4,1,4) + 2(4,1,1) + 1(1,4,4) + 2(1,4,1) + 2(1,1,4) + 6(1,1,5) + 1/2(1,1,16) + 6(1,5,1) + 1/2(1,16,1) + 6(5,1,1) + 1/2(16,1,1)

Spectrum 487; branches = [(3/2,3), (3/2,3), (3/2,3)]; singlets = 20

20(1,1,1) + 2(4,4,4) + 1(4,4,1) + 6(4,1,1) + 1(1,4,4) + 2(1,4,1) + 7/2(1,1,4) + 6(1,1,5) + 1/2(1,1,16) + 6(1,5,1) + 1/2(1,16,1) + 1/2(5,1,4) + 4(5,1,1) + 1/2(16,1,1)

Spectrum 488; branches = [(3/2,3), (3/2,3), (3/2,3)]; singlets = 14

14(1,1,1) + 2(4,4,4) + 1(4,4,1) + 6(4,1,1) + 6(1,4,1) + 5(1,1,4) + 6(1,1,5) + 1/2(1,1,16) + 1/2(1,5,4) + 4(1,5,1) + 1/2(1,16,1) + 1/2(5,1,4) + 4(5,1,1) + 1/2(16,1,1)

Spectrum 489; branches = [(3/2,3), (3/2,3), (3/2,3)]; singlets = 14

14(1,1,1) + 2(4,4,4) + 1(4,4,1) + 6(4,1,1) + 7/2(1,4,1) + 1/2(1,4,5) + 15/2(1,1,4) + 4(1,1,5) + 1/2(1,1,16) + 6(1,5,1) + 1/2(1,16,1) + 1/2(5,1,4) + 4(5,1,1) + 1/2(16,1,1)

Spectrum 490; branches = [(3/2,3), (3/2,3), (3/2,3)]; singlets = 14

14(1,1,1) + 2(4,4,4) + 1(4,4,1) + 7/2(4,1,1) + 1/2(4,1,5) + 7/2(1,4,1) + 1/2(1,4,5) + 10(1,1,4) + 2(1,1,5) + 1/2(1,1,16) + 6(1,5,1) + 1/2(1,16,1) + 6(5,1,1) + 1/2(16,1,1)

Spectrum 491; branches = [(3/2,3), (3/2,3), (3/2,3)]; singlets = 37

37(1,1,1) + 1(4,4,4) + 5(4,4,1) + 1(4,1,1) + 1(4,1,5) + 1(1,4,1) + 1(1,4,5) + 1(1,1,10) + 5(1,1,4) + 1(1,1,5) + 3/2(1,5,4) + 1/2(1,16,1) + 3/2(5,1,4) + 1/2(16,1,1)

Spectrum 492; branches = [(3/2,3), (3/2,3), (3/2,3)]; singlets = 28

28(1,1,1) + 1(4,4,4) + 5(4,4,1) + 1(4,1,4) + 2(4,1,1) + 1(1,4,1) + 1(1,4,5) + 1(1,1,10) + 17/2(1,1,4) + 3/2(1,5,4) + 1/2(1,16,1) + 1(5,1,1) + 1(5,1,5) + 1/2(16,1,1)

Spectrum 493; branches = [(3/2,3), (3/2,3), (3/2,3)]; singlets = 35

35(1,1,1) + 3/2(4,4,4) + 3(4,4,1) + 1(4,1,4) + 2(4,1,1) + 1(1,4,1) + 1(1,4,5) + 1(1,1,10) + 1/2(1,1,4) + 5(1,1,5) + 1/2(1,5,4) + 4(1,5,1) + 1/2(1,16,1) + 1(5,1,4) + 2(5,1,1) + 1/2(16,1,1)

Spectrum 494; branches = [(3/2,3), (3/2,3), (3/2,3)]; singlets = 29

29(1,1,1) + 3/2(4,4,4) + 3(4,4,1) + 6(4,1,1) + 1(1,4,1) + 1(1,4,5) + 1(1,1,10) + 2(1,1,4) + 5(1,1,5) + 1/2(1,5,4) + 4(1,5,1) + 1/2(1,16,1) + 3/2(5,1,4) + 1/2(16,1,1)

Spectrum 495; branches = [(3/2,3), (3/2,3), (3/2,3)]; singlets = 29

29(1,1,1) + 3/2(4,4,4) + 3(4,4,1) + 7/2(4,1,1) + 1/2(4,1,5) + 1(1,1,10) + 2(1,1,4) + 5(1,1,5) + 1(1,5,4) + 2(1,5,1) + 1/2(1,16,1) + 1(5,1,4) + 2(5,1,1) + 1/2(16,1,1)

Spectrum 496; branches = [(3/2,3), (3/2,3), (3/2,3)]; singlets = 29

29(1,1,1) + 3/2(4,4,4) + 3(4,4,1) + 7/2(4,1,1) + 1/2(4,1,5) + 1(1,4,1) + 1(1,4,5) + 1(1,1,10) + 9/2(1,1,4) + 3(1,1,5) + 1/2(1,5,4) + 4(1,5,1) + 1/2(1,16,1) + 1(5,1,4) + 2(5,1,1) + 1/2(16,1,1)

Spectrum 497; branches = [(3/2,3), (3/2,3), (3/2,3)]; singlets = 29

29(1,1,1) + 3/2(4,4,4) + 3(4,4,1) + 1(4,1,1) + 1(4,1,5) + 1(1,1,4) + 1(1,4,5) + 1(1,1,10) + 7(1,1,4) + 1(1,1,5) + 1/2(1,5,4) + 4(1,5,1) + 1/2(1,16,1) + 1/2(5,1,4) + 4(5,1,1) + 1/2(16,1,1)

Spectrum 498; branches = [(3/2,3), (3/2,3), (3/2,3)]; singlets = 27

27(1,1,1) + 2(4,4,4) + 1(4,4,1) + 1(4,1,4) + 2(4,1,1) + 7/2(1,4,1) + 1/2(1,4,5) + 1(1,1,10) + 7(1,1,5) + 6(1,5,1) + 1/2(1,16,1) + 6(5,1,1) + 1/2(16,1,1)

Spectrum 499; branches = [(3/2,3), (3/2,3), (3/2,3)]; singlets = 21

21(1,1,1) + 2(4,4,4) + 1(4,4,1) + 6(4,1,1) + 7/2(1,4,1) + 1/2(1,4,5) + 1(1,1,10) + 3/2(1,1,4) + 7(1,1,5) + 6(1,5,1) + 1/2(1,16,1) + 1/2(5,1,4) + 4(5,1,1) + 1/2(16,1,1)

Spectrum 500; branches = [(3/2,3), (3/2,3), (3/2,3)]; singlets = 21

21(1,1,1) + 2(4,4,4) + 1(4,4,1) + 7/2(4,1,1) + 1/2(4,1,5) + 7/2(1,4,1) + 1/2(1,4,5) + 1(1,1,10) + 4(1,1,4) + 5(1,1,5) + 6(1,5,1) + 1/2(1,16,1) + 6(5,1,1) + 1/2(16,1,1)

Spectrum 501; branches = [(3/2,3), (3/2,3), (3/2,3)]; singlets = 36

36(1,1,1) + 1(4,4,4) + 4(4,4,1) + 1(4,1,4) + 1(4,1,1) + 1(4,1,5) + 1(1,4,4) + 1(1,4,1) + 1/2(1,4,5) + 11/2(1,1,4) + 1/2(1,1,16) + 3/2(1,5,4) + 1/2(1,16,1) + 1/2(5,4,1) + 1(5,1,4) + 1/2(16,1,1)

$$28(1,1,1) + 1(4,4,4) + 4(4,4,1) + 1(4,1,4) + 1(4,1,1) + 1(4,1,5) + 1(1,4,4) + 7/2(1,4,1) + 1(1,1,10) + 7(1,1,4) + 1(1,5,1) + 1(1,5,5) + 1/2(1,16,1) + 1/2(5,4,1) + 1(5,1,4) + 1/2(16,1,1)$$

[illegible]

3(1,1,1) + 3/2(4,4,4) + 1(4,4,1) + 1(4,1,4) + 5(4,1,1) + 1(4,1,5) + 4(1,4,1) + 1(1,4,5) + 1(1,1,10) + 11/2(1,1,4) + 1(1,1,5) + 1/2(1,5,4) + 4(1,5,1) + 1/2(1,16,1) + 1(5,4,1) + 2(5,1,1) + 1/2(16,1,1)
Spectrum 605; branches = {[3/2,3], (3/2,3), (3/2,3)}; singlets = 17
17(1,1,1) + 3/2(4,4,4) + 1(4,4,1) + 9(4,1,1) + 1(4,1,5) + 4(1,4,1) + 1(1,4,5) + 1(1,1,10) + 7(1,1,4) + 1(1,1,5) + 1/2(1,5,4) + 4(1,5,1) + 1/2(1,16,1) + 1(5,4,1) + 1/2(5,1,4) + 1/2(16,1,1)
Spectrum 606; branches = {[3/2,3], (3/2,3), (3/2,3)}; singlets = 29
29(1,1,1) + 3/2(4,4,4) + 1(4,4,1) + 13/2(4,1,1) + 3/2(4,1,5) + 2(1,4,4) + 1(1,4,1) + 1(1,1,10) + 3/2(1,1,4) + 3(1,1,5) + 1/2(1,5,4) + 4(1,5,1) + 1/2(1,16,1) + 1(5,4,1) + 2(5,1,1) + 1/2(16,1,1)
Spectrum 607; branches = {[3/2,3], (3/2,3), (3/2,3)}; singlets = 23
23(1,1,1) + 3/2(4,4,4) + 1(4,4,1) + 13/2(4,1,1) + 3/2(4,1,5) + 1(1,4,4) + 5(1,4,1) + 1(1,1,10) + 3(1,1,4) + 3(1,1,5) + 1(1,5,4) + 2(1,5,1) + 1/2(1,16,1) + 1(5,4,1) + 2(5,1,1) + 1/2(16,1,1)
Spectrum 608; branches = {[3/2,3], (3/2,3), (3/2,3)}; singlets = 17
17(1,1,1) + 3/2(4,4,4) + 1(4,4,1) + 13/2(4,1,1) + 3/2(4,1,5) + 9(1,4,1) + 1(1,1,10) + 9/2(1,1,4) + 3(1,1,5) + 3/2(1,5,4) + 1/2(1,16,1) + 1(5,4,1) + 2(5,1,1) + 1/2(16,1,1)
Spectrum 609; branches = {[3/2,3], (3/2,3), (3/2,3)}; singlets = 23
23(1,1,1) + 3/2(4,4,4) + 1(4,4,1) + 13/2(4,1,1) + 3/2(4,1,5) + 1(1,4,4) + 5/2(1,4,1) + 1/2(1,4,5) + 1(1,1,10) + 11/2(1,1,4) + 1(1,1,5) + 1/2(1,5,4) + 4(1,5,1) + 1/2(1,16,1) + 1(5,4,1) + 2(5,1,1) + 1/2(16,1,1)
Spectrum 610; branches = {[3/2,3], (3/2,3), (3/2,3)}; singlets = 17
17(1,1,1) + 3/2(4,4,4) + 1(4,4,1) + 13/2(4,1,1) + 3/2(4,1,5) + 13/2(1,4,1) + 1/2(1,4,5) + 1(1,1,10) + 7(1,1,4) + 1(1,1,5) + 1(1,5,4) + 2(1,5,1) + 1/2(1,16,1) + 1(5,4,1) + 2(5,1,1) + 1/2(16,1,1)
Spectrum 611; branches = {[3/2,3], (3/2,3), (3/2,3)}; singlets = 36
36(1,1,1) + 1(4,4,4) + 2(4,4,1) + 2(4,1,4) + 5/2(4,1,1) + 3/2(4,1,5) + 2(1,4,4) + 5/2(1,4,1) + 5/2(1,1,4) + 1/2(1,1,16) + 3/2(1,5,4) + 1/2(1,16,1) + 3/2(5,4,1) + 1/2(16,1,1)
Spectrum 612; branches = {[3/2,3], (3/2,3), (3/2,3)}; singlets = 4
4(1,1,1) + 3/2(4,4,4) + 21/2(4,1,1) + 3/2(4,1,5) + 21/2(1,4,1) + 21/2(1,1,4) + 1/2(1,1,16) + 3/2(1,5,4) + 1/2(1,16,1) + 3/2(5,4,1) + 1/2(16,1,1)
Spectrum 613; branches = {[3/2,3], (3/2,3), (3/2,3)}; singlets = 37
37(1,1,1) + 1(4,4,4) + 2(4,4,1) + 3(4,1,4) + 1(4,1,1) + 1(4,1,5) + 11/2(1,4,1) + 1(1,4,5) + 1(1,1,10) + 1/2(1,1,4) + 1(1,1,5) + 3/2(1,5,4) + 1/2(1,16,1) + 3/2(5,4,1) + 1/2(16,1,1)
Spectrum 614; branches = {[3/2,3], (3/2,3), (3/2,3)}; singlets = 37
37(1,1,1) + 1(4,4,4) + 2(4,4,1) + 2(4,1,4) + 5/2(4,1,1) + 3/2(4,1,5) + 1(1,4,4) + 4(1,4,1) + 1/2(1,4,5) + 1(1,1,10) + 1/2(1,1,4) + 1(1,1,5) + 3/2(1,5,4) + 1/2(1,16,1) + 3/2(5,4,1) + 1/2(16,1,1)
Spectrum 615; branches = {[3/2,3], (3/2,3), (3/2,3)}; singlets = 37
37(1,1,1) + 1(4,4,4) + 2(4,4,1) + 1(4,1,4) + 4(4,1,1) + 2(4,1,5) + 2(1,4,4) + 5/2(1,4,1) + 1(1,1,10) + 1/2(1,1,4) + 1(1,1,5) + 3/2(1,5,4) + 1/2(1,16,1) + 3/2(5,4,1) + 1/2(16,1,1)
Spectrum 616; branches = {[3/2,3], (3/2,3), (3/2,3)}; singlets = 28
28(1,1,1) + 1(4,4,4) + 2(4,4,1) + 3(4,1,4) + 1(4,1,1) + 1(4,1,5) + 1(1,4,4) + 13/2(1,4,1) + 1(1,1,10) + 4(1,1,4) + 1(1,5,1) + 1(1,5,5) + 1/2(1,16,1) + 3/2(5,4,1) + 1/2(16,1,1)
Spectrum 617; branches = {[3/2,3], (3/2,3), (3/2,3)}; singlets = 29
29(1,1,1) + 3/2(4,4,4) + 3(4,1,4) + 6(4,1,1) + 3(1,4,1) + 3/2(1,4,5) + 1(1,1,10) + 3(1,1,5) + 6(1,5,1) + 1/2(1,16,1) + 3/2(5,4,1) + 1/2(16,1,1)
Spectrum 618; branches = {[3/2,3], (3/2,3), (3/2,3)}; singlets = 29
29(1,1,1) + 3/2(4,4,4) + 2(4,1,4) + 15/2(4,1,1) + 1/2(4,1,5) + 1(1,4,4) + 3/2(1,4,1) + 1(1,4,5) + 1(1,1,10) + 3(1,1,5) + 6(1,5,1) + 1/2(1,16,1) + 3/2(5,4,1) + 1/2(16,1,1)
Spectrum 619; branches = {[3/2,3], (3/2,3), (3/2,3)}; singlets = 23
23(1,1,1) + 3/2(4,4,4) + 2(4,1,4) + 15/2(4,1,1) + 1/2(4,1,5) + 11/2(1,4,1) + 1(1,4,5) + 1(1,1,10) + 3/2(1,1,4) + 3(1,1,5) + 1/2(1,5,4) + 4(1,5,1) + 1/2(1,16,1) + 3/2(5,4,1) + 1/2(16,1,1)
Spectrum 620; branches = {[3/2,3], (3/2,3), (3/2,3)}; singlets = 23
23(1,1,1) + 3/2(4,4,4) + 2(4,1,4) + 15/2(4,1,1) + 1/2(4,1,5) + 3(1,4,1) + 3/2(1,4,5) + 1(1,1,10) + 4(1,1,4) + 1(1,1,5) + 6(1,5,1) + 1/2(1,16,1) + 3/2(5,4,1) + 1/2(16,1,1)
Spectrum 621; branches = {[3/2,3], (3/2,3), (3/2,3)}; singlets = 17
17(1,1,1) + 3/2(4,4,4) + 1(4,1,4) + 9(4,1,1) + 1(4,1,5) + 21/2(1,4,1) + 1(1,1,10) + 1/2(1,1,4) + 5(1,1,5) + 3/2(1,5,4) + 1/2(1,16,1) + 3/2(5,4,1) + 1/2(16,1,1)
Spectrum 622; branches = {[3/2,3], (3/2,3), (3/2,3)}; singlets = 29
29(1,1,1) + 3/2(4,4,4) + 1(4,1,4) + 9(4,1,1) + 1(4,1,5) + 2(1,4,4) + 1/2(1,4,5) + 1(1,1,10) + 3(1,1,5) + 6(1,5,1) + 1/2(1,16,1) + 3/2(5,4,1) + 1/2(16,1,1)
Spectrum 623; branches = {[3/2,3], (3/2,3), (3/2,3)}; singlets = 23
23(1,1,1) + 3/2(4,4,4) + 1(4,1,4) + 9(4,1,1) + 1(4,1,5) + 1(1,4,4) + 4(1,4,1) + 1/2(1,4,5) + 1(1,1,10) + 3/2(1,1,4) + 3(1,1,5) + 1/2(1,5,4) + 4(1,5,1) + 1/2(1,16,1) + 3/2(5,4,1) + 1/2(16,1,1)
Spectrum 624; branches = {[3/2,3], (3/2,3), (3/2,3)}; singlets = 17

$0(1,1,1) + 3/2(4,4,4) + 1(4,4,1) + 9(4,1,1) + 1/2(4,1,5) + 1/2(4,5,1) + 13/2(1,4,1) + 1(1,4,5) + 21/2(1,1,4) + 1/2(1,1,16) + 1/2(1,5,4) + 2(1,5,1) + 1/2(1,16,1) + 1/2(5,4,1) + 1(5,1,4) + 1/2(16,1,1)$
 Spectrum 639; branches = $\{[3/2,3], (3/2,3), (3/2,3)\}$; singlets = 43
 $43(1,1,1) + 1(4,4,4) + 3(4,4,1) + 2(4,1,4) + 1(4,1,1) + 1/2(4,1,5) + 1/2(4,5,1) + 1(1,4,4) + 3/2(1,4,5) + 1(1,1,10) + 1/2(1,1,4) + 1(1,1,5) + 1/2(1,5,4) + 2(1,5,1) + 1/2(1,16,1) + 1/2(5,4,1) + 1(5,1,4) + 1/2(16,1,1)$
 Spectrum 640; branches = $\{[3/2,3], (3/2,3), (3/2,3)\}$; singlets = 37
 $37(1,1,1) + 1(4,4,4) + 3(4,4,1) + 2(4,1,4) + 1(4,1,1) + 1/2(4,1,5) + 1/2(4,5,1) + 4(1,4,1) + 3/2(1,4,5) + 1(1,1,10) + 2(1,1,4) + 1(1,1,5) + 1(1,5,4) + 1/2(1,16,1) + 1/2(5,4,1) + 1(5,1,4) + 1/2(16,1,1)$
 Spectrum 641; branches = $\{[3/2,3], (3/2,3), (3/2,3)\}$; singlets = 37
 $37(1,1,1) + 1(4,4,4) + 3(4,4,1) + 1(4,1,4) + 5/2(4,1,1) + 1(4,1,5) + 1/2(4,5,1) + 1(1,4,4) + 5/2(1,4,1) + 1(1,4,5) + 1(1,1,10) + 2(1,1,4) + 1(1,1,5) + 1(1,5,4) + 1/2(1,16,1) + 1/2(5,4,1) + 1(5,1,4) + 1/2(16,1,1)$
 Spectrum 642; branches = $\{[3/2,3], (3/2,3), (3/2,3)\}$; singlets = 23
 $23(1,1,1) + 3/2(4,4,4) + 1(4,4,1) + 1(4,1,1) + 1(4,1,4) + 15/2(4,1,1) + 1/2(4,5,1) + 13/2(1,4,1) + 1(1,4,5) + 1(1,1,10) + 1/2(1,1,4) + 5(1,1,5) + 1/2(1,5,4) + 2(1,5,1) + 1/2(1,16,1) + 1/2(5,4,1) + 1(5,1,4) + 1/2(16,1,1)$
 Spectrum 643; branches = $\{[3/2,3], (3/2,3), (3/2,3)\}$; singlets = 29
 $29(1,1,1) + 3/2(4,4,4) + 1(4,4,1) + 2(4,1,4) + 7/2(4,1,1) + 1/2(4,5,1) + 4(1,4,1) + 3/2(1,4,5) + 1(1,1,10) + 3/2(1,1,4) + 3(1,1,5) + 4(1,5,1) + 1/2(1,16,1) + 1/2(5,4,1) + 1/2(5,1,4) + 2(5,1,1) + 1/2(16,1,1)$
 Spectrum 644; branches = $\{[3/2,3], (3/2,3), (3/2,3)\}$; singlets = 23
 $23(1,1,1) + 3/2(4,4,4) + 1(4,4,1) + 1(4,1,4) + 15/2(4,1,1) + 1/2(4,5,1) + 4(1,4,1) + 3/2(1,4,5) + 1(1,1,10) + 3(1,1,4) + 3(1,1,5) + 4(1,5,1) + 1/2(1,16,1) + 1/2(5,4,1) + 1(5,1,4) + 1/2(16,1,1)$
 Spectrum 645; branches = $\{[3/2,3], (3/2,3), (3/2,3)\}$; singlets = 23
 $23(1,1,1) + 3/2(4,4,4) + 1(4,4,1) + 1(4,1,1) + 9(4,1,1) + 1/2(4,1,5) + 1/2(4,5,1) + 1(1,4,4) + 5(1,4,1) + 1/2(1,4,5) + 1(1,1,10) + 1/2(1,1,4) + 5(1,1,5) + 1/2(1,5,4) + 2(1,5,1) + 1/2(1,16,1) + 1/2(5,4,1) + 1(5,1,4) + 1/2(16,1,1)$
 Spectrum 646; branches = $\{[3/2,3], (3/2,3), (3/2,3)\}$; singlets = 17
 $17(1,1,1) + 3/2(4,4,4) + 1(4,4,1) + 9(4,1,1) + 1/2(4,1,5) + 1/2(4,5,1) + 9(1,4,1) + 1/2(1,4,5) + 1(1,1,10) + 2(1,1,4) + 5(1,1,5) + 1(1,5,4) + 1/2(1,16,1) + 1/2(5,4,1) + 1(5,1,4) + 1/2(16,1,1)$
 Spectrum 647; branches = $\{[3/2,3], (3/2,3), (3/2,3)\}$; singlets = 35
 $35(1,1,1) + 3/2(4,4,4) + 1(4,4,1) + 2(4,1,4) + 1(4,1,1) + 1/2(4,1,5) + 1/2(4,5,1) + 1(1,4,4) + 5/2(1,4,1) + 1(1,4,5) + 1(1,1,10) + 3(1,1,5) + 4(1,5,1) + 1/2(1,16,1) + 1/2(5,4,1) + 4(5,1,1) + 1/2(16,1,1)$
 Spectrum 648; branches = $\{[3/2,3], (3/2,3), (3/2,3)\}$; singlets = 29
 $29(1,1,1) + 3/2(4,4,4) + 1(4,4,1) + 1(4,1,4) + 1(4,1,1) + 1/2(4,1,5) + 1/2(4,5,1) + 1(1,4,4) + 5/2(1,4,1) + 1(1,4,5) + 1(1,1,10) + 3/2(1,1,4) + 3(1,1,5) + 4(1,5,1) + 1/2(1,16,1) + 1/2(5,4,1) + 1/2(5,1,4) + 2(5,1,1) + 1/2(16,1,1)$
 Spectrum 649; branches = $\{[3/2,3], (3/2,3), (3/2,3)\}$; singlets = 23
 $23(1,1,1) + 3/2(4,4,4) + 1(4,4,1) + 9(4,1,1) + 1/2(4,1,5) + 1/2(4,5,1) + 1(1,4,4) + 5/2(1,4,1) + 1(1,4,5) + 1(1,1,10) + 3(1,1,4) + 3(1,1,5) + 4(1,5,1) + 1/2(1,16,1) + 1/2(5,4,1) + 1(5,1,4) + 1/2(16,1,1)$
 Spectrum 650; branches = $\{[3/2,3], (3/2,3), (3/2,3)\}$; singlets = 29
 $29(1,1,1) + 3/2(4,4,4) + 1(4,4,1) + 2(4,1,4) + 1(4,1,1) + 1/2(4,1,5) + 1/2(4,5,1) + 13/2(1,4,1) + 1(1,4,5) + 1(1,1,10) + 3/2(1,1,4) + 3(1,1,5) + 1/2(1,5,4) + 2(1,5,1) + 1/2(1,16,1) + 1/2(5,4,1) + 4(5,1,1) + 1/2(16,1,1)$
 Spectrum 651; branches = $\{[3/2,3], (3/2,3), (3/2,3)\}$; singlets = 23
 $23(1,1,1) + 3/2(4,4,4) + 1(4,4,1) + 1(4,1,4) + 1(4,1,1) + 1/2(4,1,5) + 1/2(4,5,1) + 13/2(1,4,1) + 1(1,4,5) + 1(1,1,10) + 3(1,1,4) + 3(1,1,5) + 1/2(1,5,4) + 2(1,5,1) + 1/2(1,16,1) + 1/2(5,4,1) + 1/2(5,1,4) + 2(5,1,1) + 1/2(16,1,1)$
 Spectrum 652; branches = $\{[3/2,3], (3/2,3), (3/2,3)\}$; singlets = 17
 $17(1,1,1) + 3/2(4,4,4) + 1(4,4,1) + 9(4,1,1) + 1/2(4,1,5) + 1/2(4,5,1) + 13/2(1,4,1) + 1(1,4,5) + 1(1,1,10) + 9/2(1,1,4) + 3(1,1,5) + 1/2(1,5,4) + 2(1,5,1) + 1/2(1,16,1) + 1/2(5,4,1) + 1(5,1,4) + 1/2(16,1,1)$
 Spectrum 653; branches = $\{[3/2,3], (3/2,3), (3/2,3)\}$; singlets = 29
 $29(1,1,1) + 3/2(4,4,4) + 1(4,4,1) + 2(4,1,4) + 1(4,1,1) + 1/2(4,1,5) + 1/2(4,5,1) + 4(1,4,1) + 3/2(1,4,5) + 1(1,1,10) + 4(1,1,4) + 1(1,1,5) + 4(1,5,1) + 1/2(1,16,1) + 1/2(5,4,1) + 4(5,1,1) + 1/2(16,1,1)$
 Spectrum 654; branches = $\{[3/2,3], (3/2,3), (3/2,3)\}$; singlets = 23
 $23(1,1,1) + 3/2(4,4,4) + 1(4,4,1) + 1(4,1,4) + 5(4,1,1) + 1/2(4,1,5) + 1/2(4,5,1) + 4(1,4,1) + 3/2(1,4,5) + 1(1,1,10) + 11/2(1,1,4) + 1(1,1,5) + 4(1,5,1)$

$$24(1.1.1) + 3/2(4.4.4) + 1(4.4.1) + 9(4.1.1) + 1(4.1.5) + 1(1.10.1) + 1/2(1.4.1) + 1/2(1.4.5) + 1(1.1.10) + 9/2(1.1.4) + 3(1.1.5) + 1(1.5.4) + 5(1.5.1) + 1(5.4.1) + 1/2(5.1.4) + 1/2(16.1.1)$$

$$24(1.1.1) + 3/2(4.4.4) + 1(4.4.1) + 9(4.1.1) + 1/2(4.1.5) + 1/2(4.5.1) + 1(1.10.1) + 3(1.4.1) + 1/2(1.4.5) + 1(1.1.10) + 2(1.1.4) + 5(1.1.5) + 1(1.5.4) + 3(1.5.1) + 1/2(5.4.1) + 1(5.1.4) + 1/2(16.1.1)$$

$24(1,1,1) + 3/2(4,4,4) + 1(4,4,1) + 9(4,1,1) + 1/2(4,1,5) + 1/2(4,5,1) + 1(1,10,1) + 1/2(1,4,1) + 1(1,4,5) + 1(1,1,10) + 9/2(1,1,4) + 3(1,1,5) + 1/2(1,5,4) + 5(1,5,1) + 1/2(5,4,1) + 1(5,1,1) + 1/2(16,1,1)$
Spectrum 741; branches = $\{[3/2,3], (3/2,3), (3/2,3)\}$; singlets = 30

$30(1,1,1) + 3/2(4,4,4) + 1(4,4,1) + 13/2(4,1,1) + 1(4,1,5) + 1/2(4,5,1) + 1(1,10,1) + 1(1,4,4) + 3/2(1,4,1) + 1(1,1,10) + 1/2(1,1,4) + 5(1,1,5) + 1(1,5,4) + 3(1,5,1) + 1/2(5,4,1) + 1/2(5,1,4) + 2(5,1,1) + 1/2(16,1,1)$
Spectrum 742; branches = $\{[3/2,3], (3/2,3), (3/2,3)\}$; singlets = 30

$30(1,1,1) + 3/2(4,4,4) + 1(4,4,1) + 1(4,1,4) + 5/2(4,1,1) + 1(4,1,5) + 1/2(4,5,1) + 1(1,10,1) + 11/2(1,4,1) + 1(1,1,10) + 1/2(1,1,4) + 5(1,1,5) + 3/2(1,5,4) + 1(1,5,1) + 1/2(5,4,1) + 4(5,1,1) + 1/2(16,1,1)$
Spectrum 743; branches = $\{[3/2,3], (3/2,3), (3/2,3)\}$; singlets = 24

$24(1,1,1) + 3/2(4,4,4) + 1(4,4,1) + 13/2(4,1,1) + 1(4,1,5) + 1/2(4,5,1) + 1(1,10,1) + 11/2(1,4,1) + 1(1,1,10) + 2(1,1,4) + 5(1,1,5) + 3/2(1,5,4) + 1(1,5,1) + 1/2(5,4,1) + 1/2(5,1,4) + 2(5,1,1) + 1/2(16,1,1)$
Spectrum 744; branches = $\{[3/2,3], (3/2,3), (3/2,3)\}$; singlets = 30

$30(1,1,1) + 3/2(4,4,4) + 1(4,4,1) + 1(4,1,1) + 1(4,1,4) + 5/2(4,1,1) + 1(4,1,5) + 1/2(4,5,1) + 1(1,10,1) + 3(1,4,1) + 1/2(1,4,5) + 1(1,1,10) + 3(1,1,4) + 3(1,1,5) + 1(1,5,4) + 3(1,5,1) + 1/2(5,4,1) + 4(5,1,1) + 1/2(16,1,1)$
Spectrum 745; branches = $\{[3/2,3], (3/2,3), (3/2,3)\}$; singlets = 24

$24(1,1,1) + 3/2(4,4,4) + 1(4,4,1) + 13/2(4,1,1) + 1(4,1,5) + 1/2(4,5,1) + 1(1,10,1) + 3(1,4,1) + 1/2(1,4,5) + 1(1,1,10) + 9/2(1,1,4) + 3(1,1,5) + 1(1,5,4) + 3(1,5,1) + 1/2(5,4,1) + 1/2(5,1,4) + 2(5,1,1) + 1/2(16,1,1)$
Spectrum 746; branches = $\{[3/2,3], (3/2,3), (3/2,3)\}$; singlets = 30

$30(1,1,1) + 3/2(4,4,4) + 1(4,4,1) + 1(4,1,4) + 5/2(4,1,1) + 1(4,1,5) + 1/2(4,5,1) + 1(1,10,1) + 1/2(1,4,1) + 1(1,4,5) + 1(1,1,10) + 11/2(1,1,4) + 1(1,1,5) + 1/2(1,5,4) + 5(1,5,1) + 1/2(5,4,1) + 4(5,1,1) + 1/2(16,1,1)$
Spectrum 747; branches = $\{[3/2,3], (3/2,3), (3/2,3)\}$; singlets = 24

$24(1,1,1) + 3/2(4,4,4) + 1(4,4,1) + 13/2(4,1,1) + 1(4,1,5) + 1/2(4,5,1) + 1(1,10,1) + 1/2(1,4,1) + 1(1,4,5) + 1(1,1,10) + 7(1,1,4) + 1(1,1,5) + 1/2(1,5,4) + 5(1,5,1) + 1/2(5,4,1) + 1/2(5,1,4) + 2(5,1,1) + 1/2(16,1,1)$
Spectrum 748; branches = $\{[3/2,3], (3/2,3), (3/2,3)\}$; singlets = 30

$30(1,1,1) + 3/2(4,4,4) + 1(4,4,1) + 4(4,1,1) + 3/2(4,1,5) + 1/2(4,5,1) + 1(1,10,1) + 1(1,4,4) + 3/2(1,4,1) + 1(1,1,10) + 3(1,1,4) + 3(1,1,5) + 1(1,5,4) + 3(1,5,1) + 1/2(5,4,1) + 4(5,1,1) + 1/2(16,1,1)$
Spectrum 749; branches = $\{[3/2,3], (3/2,3), (3/2,3)\}$; singlets = 24

$24(1,1,1) + 3/2(4,4,4) + 1(4,4,1) + 4(4,1,1) + 3/2(4,1,5) + 1/2(4,5,1) + 1(1,10,1) + 11/2(1,4,1) + 1(1,1,10) + 9/2(1,1,4) + 3(1,1,5) + 3/2(1,5,4) + 1(1,5,1) + 1/2(5,4,1) + 4(5,1,1) + 1/2(16,1,1)$
Spectrum 750; branches = $\{[3/2,3], (3/2,3), (3/2,3)\}$; singlets = 24

$24(1,1,1) + 3/2(4,4,4) + 1(4,4,1) + 4(4,1,1) + 3/2(4,1,5) + 1/2(4,5,1) + 1(1,10,1) + 3(1,4,1) + 1/2(1,4,5) + 1(1,1,10) + 7(1,1,4) + 1(1,1,5) + 1(1,5,4) + 3(1,5,1) + 1/2(5,4,1) + 4(5,1,1) + 1/2(16,1,1)$
Spectrum 751; branches = $\{[3/2,3], (3/2,3), (3/2,3)\}$; singlets = 38

$38(1,1,1) + 1(4,4,4) + 2(4,4,1) + 2(4,1,4) + 5/2(4,1,1) + 1(4,1,5) + 1/2(4,5,1) + 1(1,10,1) + 2(1,4,1) + 1(1,4,5) + 1(1,1,10) + 2(1,1,4) + 1(1,1,5) + 3/2(1,5,4) + 1(1,5,1) + 1(5,4,1) + 1/2(5,1,4) + 1/2(16,1,1)$
Spectrum 752; branches = $\{[3/2,3], (3/2,3), (3/2,3)\}$; singlets = 44

$44(1,1,1) + 1(4,4,4) + 2(4,4,1) + 2(4,1,4) + 3/2(4,1,5) + 1/2(4,5,1) + 1(1,10,1) + 1(1,4,4) + 1/2(1,4,1) + 1/2(1,4,5) + 1(1,1,10) + 1/2(1,1,4) + 1(1,1,5) + 3/2(1,5,4) + 1(1,5,1) + 1(5,4,1) + 2(5,1,1) + 1/2(16,1,1)$
Spectrum 753; branches = $\{[3/2,3], (3/2,3), (3/2,3)\}$; singlets = 38

$38(1,1,1) + 1(4,4,4) + 2(4,4,1) + 1(4,1,4) + 4(4,1,1) + 3/2(4,1,5) + 1/2(4,5,1) + 1(1,10,1) + 1(1,4,4) + 1/2(1,4,1) + 1/2(1,4,5) + 1(1,1,10) + 2(1,1,4) + 1(1,1,5) + 3/2(1,5,4) + 1(1,5,1) + 1(5,4,1) + 1/2(5,1,4) + 1/2(16,1,1)$
Spectrum 754; branches = $\{[3/2,3], (3/2,3), (3/2,3)\}$; singlets = 29

$29(1,1,1) + 1(4,4,4) + 2(4,4,1) + 2(4,1,4) + 5/2(4,1,1) + 1(4,1,5) + 1/2(4,5,1) + 1(1,10,1) + 1(1,4,4) + 3(1,4,1) + 1(1,1,10) + 11/2(1,1,4) + 2(1,5,1) + 1(1,5,5) + 1(5,4,1) + 1/2(5,1,4) + 1/2(16,1,1)$
Spectrum 755; branches = $\{[3/2,3], (3/2,3), (3/2,3)\}$; singlets = 23

$23(1,1,1) + 1(4,4,4) + 2(4,4,1) + 2(4,1,4) + 5/2(4,1,1) + 1(4,1,5) + 1/2(4,5,1) + 1(1,10,1) + 7(1,4,1) + 1(1,1,10) + 7(1,1,4) + 1/2(1,5,4) + 1(1,5,5) + 1(5,4,1) + 1/2(5,1,4) + 1/2(16,1,1)$
Spectrum 756; branches = $\{[3/2,3], (3/2,3), (3/2,3)\}$; singlets = 24

$24(1,1,1) + 3/2(4,4,4) + 21/2(4,1,1) + 1(4,1,5) + 1/2(4,5,1) + 1(1,10,1) + 1(1,4,4) + 3(1,4,1) + 1(1,1,10) + 1/2(1,1,4) + 5(1,1,5) + 1(1,5,4) + 3(1,5,1) + 1(5,4,1) + 1/2(5,1,4) + 1$

$$31(1\ 1\ 1) + 1(10\ 1\ 1) + 3/2(4\ 4\ 4) + 1(4\ 4\ 1) + 3(4\ 1\ 1) + 1(4\ 1\ 5) + 1(1\ 10\ 1) + 3(1\ 4\ 1) + 1(1\ 1\ 10) + 2(1\ 1\ 4) + 5(1\ 1\ 5) + 3/2(1\ 5\ 4) + 3(1\ 5\ 1) + 1(5\ 4\ 1) + 1/2(5\ 1\ 4) + 3(5\ 1\ 1)$$

$$22(\mathbf{1}, \mathbf{1}, \mathbf{1}) + 1/2(\mathbf{4}, \mathbf{4}, \mathbf{8}) + 1(\mathbf{4}, \mathbf{1}, \mathbf{8}) + 13(\mathbf{4}, \mathbf{1}, \mathbf{1}) + 1(\mathbf{4}, \mathbf{5}, \mathbf{1}) + 1(\mathbf{1}, \mathbf{10}, \mathbf{1}) + 1(\mathbf{1}, \mathbf{4}, \mathbf{8}) + 9/2(\mathbf{1}, \mathbf{4}, \mathbf{1}) + 1(\mathbf{1}, \mathbf{1}, \mathbf{27}) + 5(\mathbf{1}, \mathbf{5}, \mathbf{1}) + 3/2(\mathbf{5}, \mathbf{4}, \mathbf{1}) + 1/2(\mathbf{16}, \mathbf{1}, \mathbf{1})$$

$$14(1\bar{1}1) + 1(10\bar{1}1) + 1/2(448) + 1(418) + 12(411) + 1(1101) + 1(148) + 19/2(141) + 1(1127) + 4(151) + 1/2(541) + 2(511) + 1(551)$$

Spectrum 909; branches = [(3/2,3), (3/2,3), (1,0)]; singlets = 2

2(1,1,1) + 1(10,1,1) + 1/2(4,4,8) + 1(4,1,8) + 12(4,1,1) + 1(1,10,1) + 35/2(1,4,1) + 3/2(1,1,8) + 1(1,1,27) + 1/2(1,5,8) + 1/2(5,4,1) + 2(5,1,1) + 1(5,5,1)

C2 x C2 x C5

Spectrum 910; branches = [(3/2,3), (3/2,3), (3/2,1)]; singlets = 13

13(1,1,1) + 1/2(4,4,10) + 1/2(4,1,44) + 1/2(1,4,44) + 2(1,1,10) + 1(1,5,1) + 1/2(1,16,1) + 1(5,1,1) + 1(5,5,1) + 1/2(16,1,1)

C2 x C2 x D5

Spectrum 911; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 52

52(1,1,1) + 9(4,4,1) + 6(4,1,1) + 6(1,4,1) + 2(1,1,10) + 6(1,5,1) + 1/2(1,16,1) + 6(5,1,1) + 1/2(16,1,1)

Spectrum 912; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 46

46(1,1,1) + 8(4,4,1) + 10(4,1,1) + 15/2(1,4,1) + 2(1,1,10) + 6(1,5,1) + 1/2(1,16,1) + 1/2(5,4,1) + 4(5,1,1) + 1/2(16,1,1)

Spectrum 913; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 40

40(1,1,1) + 7(4,4,1) + 14(4,1,1) + 9(1,4,1) + 2(1,1,10) + 6(1,5,1) + 1/2(1,16,1) + 1(5,4,1) + 2(5,1,1) + 1/2(16,1,1)

Spectrum 914; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 34

34(1,1,1) + 6(4,4,1) + 18(4,1,1) + 21/2(1,4,1) + 2(1,1,10) + 6(1,5,1) + 1/2(1,16,1) + 3/2(5,4,1) + 1/2(16,1,1)

Spectrum 915; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 40

40(1,1,1) + 7(4,4,1) + 23/2(4,1,1) + 1/2(4,5,1) + 23/2(1,4,1) + 2(1,1,10) + 4(1,5,1) + 1/2(1,16,1) + 1/2(5,4,1) + 4(5,1,1) + 1/2(16,1,1)

Spectrum 916; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 34

34(1,1,1) + 6(4,4,1) + 31/2(4,1,1) + 1/2(4,5,1) + 13(1,4,1) + 2(1,1,10) + 4(1,5,1) + 1/2(1,16,1) + 1(5,4,1) + 2(5,1,1) + 1/2(16,1,1)

Spectrum 917; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 28

28(1,1,1) + 5(4,4,1) + 39/2(4,1,1) + 1/2(4,5,1) + 29/2(1,4,1) + 2(1,1,10) + 4(1,5,1) + 1/2(1,16,1) + 3/2(5,4,1) + 1/2(16,1,1)

Spectrum 918; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 28

28(1,1,1) + 5(4,4,1) + 17(4,1,1) + 1(4,5,1) + 17(1,4,1) + 2(1,1,10) + 2(1,5,1) + 1/2(1,16,1) + 1(5,4,1) + 2(5,1,1) + 1/2(16,1,1)

Spectrum 919; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 22

22(1,1,1) + 4(4,4,1) + 21(4,1,1) + 1(4,5,1) + 37/2(1,4,1) + 2(1,1,10) + 2(1,5,1) + 1/2(1,16,1) + 3/2(5,4,1) + 1/2(16,1,1)

Spectrum 920; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 16

16(1,1,1) + 3(4,4,1) + 45/2(4,1,1) + 3/2(4,5,1) + 45/2(1,4,1) + 2(1,1,10) + 1/2(1,16,1) + 3/2(5,4,1) + 1/2(16,1,1)

Spectrum 921; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 13

13(1,1,1) + 5(4,4,1) + 22(4,1,1) + 22(1,4,1) + 2(1,1,10) + 1(1,5,1) + 1/2(1,16,1) + 1(5,1,1) + 1(5,5,1) + 1/2(16,1,1)

Spectrum 922; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 59

59(1,1,1) + 9(4,4,1) + 6(4,1,1) + 1(1,10,1) + 2(1,1,10) + 9(1,5,1) + 6(5,1,1) + 1/2(16,1,1)

Spectrum 923; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 53

53(1,1,1) + 8(4,4,1) + 10(4,1,1) + 1(1,10,1) + 3/2(1,4,1) + 2(1,1,10) + 9(1,5,1) + 1/2(5,4,1) + 4(5,1,1) + 1/2(16,1,1)

Spectrum 924; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 47

47(1,1,1) + 7(4,4,1) + 14(4,1,1) + 1(1,10,1) + 1(1,10,1) + 3(1,4,1) + 9(1,5,1) + 1(5,4,1) + 2(5,1,1) + 1/2(16,1,1)

Spectrum 925; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 41

41(1,1,1) + 6(4,4,1) + 18(4,1,1) + 1(1,10,1) + 9/2(1,4,1) + 2(1,1,10) + 9(1,5,1) + 3/2(5,4,1) + 1/2(16,1,1)

Spectrum 926; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 53

53(1,1,1) + 8(4,4,1) + 15/2(4,1,1) + 1/2(4,5,1) + 1(1,10,1) + 4(1,4,1) + 2(1,1,10) + 7(1,5,1) + 6(5,1,1) + 1/2(16,1,1)

Spectrum 927; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 47

47(1,1,1) + 7(4,4,1) + 23/2(4,1,1) + 1/2(4,5,1) + 1(1,10,1) + 11/2(1,4,1) + 2(1,1,10) + 7(1,5,1) + 1/2(5,4,1) + 4(5,1,1) + 1/2(16,1,1)

Spectrum 928; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 41

41(1,1,1) + 6(4,4,1) + 31/2(4,1,1) + 1/2(4,5,1) + 1(1,10,1) + 7(1,4,1) + 2(1,1,10) + 7(1,5,1) + 1(5,4,1) + 2(5,1,1) + 1/2(16,1,1)

Spectrum 929; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 35

35(1,1,1) + 5(4,4,1) + 39/2(4,1,1) + 1/2(4,5,1) + 1(1,10,1) + 17/2(1,4,1) + 2(1,1,10) + 7(1,5,1) + 3/2(5,4,1) + 1/2(16,1,1)

Spectrum 930; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 47

47(1,1,1) + 7(4,4,1) + 9(4,1,1) + 1(4,5,1) + 1(1,10,1) + 8(1,4,1) + 2(1,1,10) + 5(1,5,1) + 6(5,1,1) + 1/2(16,1,1)

Spectrum 931; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 41

41(1,1,1) + 6(4,4,1) + 13(4,1,1) + 1(4,5,1) + 1(1,10,1) + 19/2(1,4,1) + 2(1,1,10) + 5(1,5,1) + 1/2(5,4,1) + 4(5,1,1) + 1/2(16,1,1)

Spectrum 932; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 35

35(1,1,1) + 5(4,4,1) + 17(4,1,1) + 1(4,5,1) + 1(1,10,1) + 11(1,4,1) + 2(1,1,10) + 5(1,5,1) + 1(5,4,1) + 2(5,1,1) + 1/2(16,1,1)

Spectrum 933; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 29

29(1,1,1) + 4(4,4,1) + 21(4,1,1) + 1(4,5,1) + 1(1,10,1) + 25/2(1,4,1) + 2(1,1,10) + 5(1,5,1) + 3/2(5,4,1) + 1/2(16,1,1)

Spectrum 934; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 41

41(1,1,1) + 6(4,4,1) + 21/2(4,1,1) + 3/2(4,5,1) + 1(1,10,1) + 12(1,4,1) + 2(1,1,10) + 3(1,5,1) + 6(5,1,1) + 1/2(16,1,1)

Spectrum 935; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 35

35(1,1,1) + 5(4,4,1) + 29/2(4,1,1) + 3/2(4,5,1) + 1(1,10,1) + 27/2(1,4,1) + 2(1,1,10) + 3(1,5,1) + 1/2(5,4,1) + 4(5,1,1) + 1/2(16,1,1)

Spectrum 936; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 29

29(1,1,1) + 4(4,4,1) + 37/2(4,1,1) + 3/2(4,5,1) + 1(1,10,1) + 15(1,4,1) + 2(1,1,10) + 3(1,5,1) + 1(5,4,1) + 2(5,1,1) + 1/2(16,1,1)

Spectrum 937; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 23

23(1,1,1) + 3(4,4,1) + 45/2(4,1,1) + 3/2(4,5,1) + 1(1,10,1) + 33/2(1,4,1) + 2(1,1,10) + 3(1,5,1) + 3/2(5,4,1) + 1/2(16,1,1)

Spectrum 938; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 35

35(1,1,1) + 5(4,4,1) + 12(4,1,1) + 2(4,5,1) + 1(1,10,1) + 16(1,4,1) + 2(1,1,10) + 1(1,5,1) + 6(5,1,1) + 1/2(16,1,1)

Spectrum 939; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 29

29(1,1,1) + 4(4,4,1) + 16(4,1,1) + 2(4,5,1) + 1(1,10,1) + 35/2(1,4,1) + 2(1,1,10) + 1(1,5,1) + 1/2(5,4,1) + 4(5,1,1) + 1/2(16,1,1)

Spectrum 940; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 23

23(1,1,1) + 3(4,4,1) + 20(4,1,1) + 2(4,5,1) + 1(1,10,1) + 19(1,4,1) + 2(1,1,10) + 1(1,5,1) + 1(5,4,1) + 2(5,1,1) + 1/2(16,1,1)

Spectrum 941; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 17

17(1,1,1) + 2(4,4,1) + 24(4,1,1) + 2(4,5,1) + 1(1,10,1) + 41/2(1,4,1) + 2(1,1,10) + 1(1,5,1) + 3/2(5,4,1) + 1/2(16,1,1)

Spectrum 942; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 20

20(1,1,1) + 5(4,4,1) + 22(4,1,1) + 1(1,10,1) + 16(1,4,1) + 2(1,1,10) + 4(1,5,1) + 1(5,1,1) + 1(5,5,1) + 1/2(16,1,1)

Spectrum 943; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 14

14(1,1,1) + 4(4,4,1) + 47/2(4,1,1) + 1/2(4,5,1) + 1(1,10,1) + 20(1,4,1) + 2(1,1,10) + 2(1,5,1) + 1(5,1,1) + 1(5,5,1) + 1/2(16,1,1)

Spectrum 944; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 8

8(1,1,1) + 3(4,4,1) + 25(4,1,1) + 1(4,5,1) + 1(1,10,1) + 24(1,4,1) + 2(1,1,10) + 1(5,1,1) + 1(5,5,1) + 1/2(16,1,1)

Spectrum 945; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 66

66(1,1,1) + 1(10,1,1) + 9(4,4,1) + 1(1,10,1) + 2(1,1,10) + 9(1,5,1) + 9(5,1,1)

Spectrum 946; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 60

60(1,1,1) + 1(10,1,1) + 8(4,4,1) + 4(4,1,1) + 1(1,10,1) + 3/2(1,4,1) + 2(1,1,10) + 9(1,5,1) + 1/2(5,4,1) + 7(5,1,1)

Spectrum 947; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 54

54(1,1,1) + 1(10,1,1) + 7(4,4,1) + 8(4,1,1) + 1(1,10,1) + 3(1,4,1) + 2(1,1,10) + 9(1,5,1) + 1(5,4,1) + 5(5,1,1)

Spectrum 948; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 48

48(1,1,1) + 1(10,1,1) + 6(4,4,1) + 12(4,1,1) + 1(1,10,1) + 9/2(1,4,1) + 2(1,1,10) + 9(1,5,1) + 3/2(5,4,1) + 3(5,1,1)

Spectrum 949; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 42

42(1,1,1) + 1(10,1,1) + 5(4,4,1) + 16(4,1,1) + 1(1,10,1) + 6(1,4,1) + 2(1,1,10) + 9(1,5,1) + 2(5,4,1) + 1(5,1,1)

Spectrum 950; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 54

54(1,1,1) + 1(10,1,1) + 7(4,4,1) + 11/2(4,1,1) + 1/2(4,5,1) + 1(1,10,1) + 11/2(1,4,1) + 2(1,1,10) + 7(1,5,1) + 1/2(5,4,1) + 7(5,1,1)

Spectrum 951; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 48

48(1,1,1) + 1(10,1,1) + 6(4,4,1) + 19/2(4,1,1) + 1/2(4,5,1) + 1(1,10,1) + 7(1,4,1) + 2(1,1,10) + 7(1,5,1) + 1(5,4,1) + 5(5,1,1)

Spectrum 952; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 42

42(1,1,1) + 1(10,1,1) + 5(4,4,1) + 27/2(4,1,1) + 1/2(4,5,1) + 1(1,10,1) + 17/2(1,4,1) + 2(1,1,10) + 7(1,5,1) + 3/2(5,4,1) + 3(5,1,1)

Spectrum 953; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 36

36(1,1,1) + 1(10,1,1) + 4(4,4,1) + 35/2(4,1,1) + 1/2(4,5,1) + 1(1,10,1) + 10(1,4,1) + 2(1,1,10) + 7(1,5,1) + 2(5,4,1) + 1(5,1,1)

Spectrum 954; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 42

42(1,1,1) + 1(10,1,1) + 5(4,4,1) + 11(4,1,1) + 1(4,5,1) + 1(1,10,1) + 11(1,4,1) + 2(1,1,10) + 5(1,5,1) + 1(5,4,1) + 5(5,1,1)

Spectrum 955; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 36

36(1,1,1) + 1(10,1,1) + 4(4,4,1) + 15(4,1,1) + 1(4,5,1) + 1(1,10,1) + 25/2(1,4,1) + 2(1,1,10) + 5(1,5,1) + 3/2(5,4,1) + 3(5,1,1)

Spectrum 956; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 30

30(1,1,1) + 1(10,1,1) + 3(4,4,1) + 19(4,1,1) + 1(4,5,1) + 1(1,10,1) + 14(1,4,1) + 2(1,1,10) + 5(1,5,1) + 2(5,4,1) + 1(5,1,1)

Spectrum 957; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 30

30(1,1,1) + 1(10,1,1) + 3(4,4,1) + 33/2(4,1,1) + 3/2(4,5,1) + 1(1,10,1) + 33/2(1,4,1) + 2(1,1,10) + 3(1,5,1) + 3/2(5,4,1) + 3(5,1,1)

Spectrum 958; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 24

24(1,1,1) + 1(10,1,1) + 2(4,4,1) + 41/2(4,1,1) + 3/2(4,5,1) + 1(1,10,1) + 18(1,4,1) + 2(1,1,10) + 3(1,5,1) + 2(5,4,1) + 1(5,1,1)

Spectrum 959; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 18

18(1,1,1) + 1(10,1,1) + 1(4,4,1) + 22(4,1,1) + 2(4,5,1) + 1(1,10,1) + 22(1,4,1) + 2(1,1,10) + 1(1,5,1) + 2(5,4,1) + 1(5,1,1)

Spectrum 960; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 27

27(1,1,1) + 1(10,1,1) + 5(4,4,1) + 16(4,1,1) + 1(1,10,1) + 16(1,4,1) + 2(1,1,10) + 4(1,5,1) + 4(5,1,1) + 1(5,5,1)

Spectrum 961; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 21

21(1,1,1) + 1(10,1,1) + 4(4,4,1) + 20(4,1,1) + 1(1,10,1) + 35/2(1,4,1) + 2(1,1,10) + 4(1,5,1) + 1/2(5,4,1) + 2(5,1,1) + 1(5,5,1)

Spectrum 962; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 15

15(1,1,1) + 1(10,1,1) + 3(4,4,1) + 24(4,1,1) + 19(1,4,1) + 2(1,1,10) + 4(1,5,1) + 1(5,4,1) + 1(5,5,1)

Spectrum 963; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 15

15(1,1,1) + 1(10,1,1) + 3(4,4,1) + 43/2(4,1,1) + 1/2(4,5,1) + 1(1,10,1) + 43/2(1,4,1) + 2(1,1,10) + 2(1,5,1) + 1/2(5,4,1) + 2(5,1,1) + 1(5,5,1)

Spectrum 964; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 9

9(1,1,1) + 1(10,1,1) + 2(4,4,1) + 51/2(4,1,1) + 1/2(4,5,1) + 1(1,10,1) + 23(1,4,1) + 2(1,1,10) + 2(1,5,1) + 1(5,4,1) + 1(5,5,1)

Spectrum 965; branches = [(3/2,3), (3/2,3), (1,-2)]; singlets = 3

3(1,1,1) + 1(10,1,1) + 1(4,4,1) + 27(4,1,1) + 1(4,5,1) + 1(1,10,1) + 27(1,4,1) + 2(1,1,10) + 1(5,4,1) + 1(5,5,1)

C2 x C2 x G2

Spectrum 966; branches = [(3/2,3), (3/2,3), (2,4)]; singlets = 18

18(1,1,1) + 1(4,4,7) + 2(4,1,7) + 1/2(1,4,14) + 2(1,4,1) + 1(1,1,14) + 4(1,1,7) + 6(1,5,1) + 1/2(1,16,1) + 1(5,4,1) + 2(5,1,1) + 1/2(16,1,1)

Spectrum 967; branches = [(3/2,3), (3/2,3), (2,4)]; singlets = 10

10(1,1,1) + 1(4,4,7) + 1(4,4,1) + 1/2(4,1,14) + 3(4,1,1) + 1(1,10,1) + 3/2(1,4,1) + 1(1,1,14) + 7(1,1,7) + 1(1,5,7) + 2(1,5,1) + 1/2(5,4,1) + 4(5,1,1) + 1/2(16,1,1)

Spectrum 968; branches = [(3/2,3), (3/2,3), (2,4)]; singlets = 26

26(1,1,1) + 1(4,4,7) + 2(4,1,7) + 1(1,10,1) + 3(1,4,1) + 2(1,1,7) + 1(1,1,27) + 1(1,5,7) + 2(1,5,1) + 1(5,4,1) + 2(5,1,1) + 1/2(16,1,1)

Spectrum 969; branches = [(3/2,3), (3/2,3), (2,4)]; singlets = 4

4(1,1,1) + 1(4,4,7) + 1/2(4,1,14) + 7(4,1,1) + 1(1,10,1) + 3(1,4,1) + 1(1,1,14) + 7(1,1,7) + 1(1,5,7) + 2(1,5,1) + 1(5,4,1) + 2(5,1,1) + 1/2(16,1,1)

Spectrum 970; branches = [(3/2,3), (3/2,3), (2,4)]; singlets = 10

10(1,1,1) + 1(4,4,7) + 1(4,4,1) + 1/2(4,1,14) + 1/2(4,1,1) + 1/2(4,5,1) + 1(1,10,1) + 4(1,4,1) + 1(1,1,14) + 7(1,1,7) + 1(1,5,7) + 6(5,1,1) + 1/2(16,1,1)

Spectrum 971; branches = [(3/2,3), (3/2,3), (2,4)]; singlets = 4

4(1,1,1) + 1(4,4,7) + 1/2(4,1,14) + 9/2(4,1,1) + 1/2(4,5,1) + 1(1,10,1) + 11/2(1,4,1) + 1(1,1,14) + 7(1,1,7) + 1(1,5,7) + 1/2(5,4,1) + 4(5,1,1) + 1/2(16,1,1)

Spectrum 972; branches = [(3/2,3), (3/2,3), (2,4)]; singlets = 24

24(1,1,1) + 1(10,1,1) + 1(4,4,7) + 2(4,4,1) + 1(1,10,1) + 5(1,1,7) + 1(1,1,27) + 1(1,5,7) + 2(1,5,1) + 1(5,1,7) + 2(5,1,1)

Spectrum 973; branches = [(3/2,3), (3/2,3), (2,4)]; singlets = 30

30(1,1,1) + 1(10,1,1) + 1(4,4,7) + 2(4,4,1) + 1(1,10,1) + 3(1,1,14) + 2(1,1,7) + 1(1,5,7) + 2(1,5,1) + 1(5,1,7) + 2(5,1,1)

Spectrum 974; branches = [(3/2,3), (3/2,3), (2,4)]; singlets = 18

18(1,1,1) + 1(10,1,1) + 1(4,4,7) + 1(4,4,1) + 4(4,1,1) + 1(1,10,1) + 3/2(1,4,1) + 5(1,1,7) + 1(1,1,27) + 1(1,5,7) + 2(1,5,1) + 1/2(5,4,1) + 1(5,1,7)

Spectrum 975; branches = [(3/2,3), (3/2,3), (2,4)]; singlets = 24

24(1,1,1) + 1(10,1,1) + 1(4,4,7) + 1(4,4,1) + 4(4,1,1) + 1(1,10,1) + 3/2(1,4,1) + 3(1,1,14) + 2(1,1,7) + 1(1,5,7) + 2(1,5,1) + 1/2(5,4,1) + 1(5,1,7)

Spectrum 976; branches = [(3/2,3), (3/2,3), (2,4)]; singlets = 11

11(1,1,1) + 1(10,1,1) + 1(4,4,7) + 1/2(4,1,14) + 1(4,1,1) + 1(1,10,1) + 3(1,4,1) + 1(1,1,14) + 7(1,1,7) + 1(1,5,7) + 2(1,5,1) + 1(5,4,1) + 5(5,1,1)

Spectrum 977; branches = [(3/2,3), (3/2,3), (2,4)]; singlets = 12

12(1,1,1) + 1(10,1,1) + 1(4,4,7) + 11/2(4,1,1) + 1/2(4,5,1) + 1(1,10,1) + 11/2(1,4,1) + 5(1,1,7) + 1(1,1,27) + 1(1,5,7) + 1/2(5,4,1) + 1(5,1,7)

Spectrum 978; branches = [(3/2,3), (3/2,3), (2,4)]; singlets = 18

18(1,1,1) + 1(10,1,1) + 1(4,4,7) + 11/2(4,1,1) + 1/2(4,5,1) + 1(1,10,1) + 11/2(1,4,1) + 3(1,1,14) + 2(1,1,7) + 1(1,5,7) + 1/2(5,4,1) + 1(5,1,7)

C2 x C3 x D5

Spectrum 979; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 11

11(1,1,1) + 2(4,6,1) + 2(4,1,1) + 1(4,14,1) + 1(1,21,1) + 6(1,6,1) + 2(1,1,10) + 1(1,14,1) + 1/2(1,64,1) + 1(5,6,1) + 1/2(16,1,1)

Spectrum 980; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 13

13(1,1,1) + 2(4,6,1) + 2(4,1,1) + 2(4,14,1) + 1(1,21,1) + 9/2(1,6,1) + 2(1,1,10) + 1(1,14,1) + 1/2(1,14,1) + 1/2(1,64,1) + 1(5,6,1) + 1/2(16,1,1)

Spectrum 981; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 15

15(1,1,1) + 2(4,6,1) + 2(4,1,1) + 2(4,14,1) + 1(1,21,1) + 3(1,6,1) + 2(1,1,10) + 2(1,14,1) + 1/2(1,64,1) + 1(5,6,1) + 1/2(16,1,1)

Spectrum 982; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 4

4(1,1,1) + 1(4,6,1) + 1(4,1,1) + 5/2(4,14,1) + 1(1,21,1) + 19/2(1,6,1) + 2(1,1,10) + 1/2(1,64,1) + 1/2(5,6,1) + 3(5,1,1) + 1/2(16,1,1)

Spectrum 983; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 17

17(1,1,1) + 3(4,6,1) + 3(4,1,1) + 1/2(4,14,1) + 1(4,14,1) + 3(1,21,1) + 1(1,6,1) + 2(1,1,10) + 2(1,14,1) + 1(5,6,1) + 1/2(16,1,1)

Spectrum 984; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 10

10(1,1,1) + 2(4,6,1) + 2(4,1,1) + 1(4,14,1) + 1(4,14,1) + 3(1,21,1) + 9/2(1,6,1) + 2(1,1,10) + 1(1,14,1) + 1/2(5,6,1) + 3(5,1,1) + 1/2(16,1,1)

Spectrum 985; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 1

1(1,1,1) + 1(4,6,1) + 8(4,1,1) + 1(4,14,1) + 1(4,14,1) + 3(1,21,1) + 6(1,6,1) + 2(1,1,10) + 1(1,14,1) + 1(5,6,1) + 1/2(16,1,1)

Spectrum 986; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 11

11(1,1,1) + 1/2(4,21,1) + 2(4,6,1) + 11/2(4,1,1) + 1(4,14,1) + 1(1,21,1) + 2(1,6,1) + 2(1,1,10) + 5(1,14,1) + 1(5,6,1) + 1/2(16,1,1)

Spectrum 987; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 12

12(1,1,1) + 2(4,6,1) + 2(4,1,1) + 1(4,14,1) + 1(4,14,1) + 3(1,21,1) + 3(1,6,1) + 2(1,1,10) + 1(1,14,1) + 1/2(1,14,1) + 1/2(5,6,1) + 3(5,1,1) + 1/2(16,1,1)

Spectrum 988; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 3

3(1,1,1) + 1(4,6,1) + 8(4,1,1) + 1(4,14,1) + 1(4,14,1) + 3(1,21,1) + 9/2(1,6,1) + 2(1,1,10) + 1(1,14,1) + 1/2(1,14,1) + 1(5,6,1) + 1/2(16,1,1)

Spectrum 989; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 13

13(1,1,1) + 1/2(4,21,1) + 2(4,6,1) + 11/2(4,1,1) + 1(4,14,1) + 1(1,21,1) + 1/2(1,6,1) + 2(1,1,10) + 1(1,14,1) + 9/2(1,14,1) + 1(5,6,1) + 1/2(16,1,1)

Spectrum 990; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 14

14(1,1,1) + 2(4,6,1) + 2(4,1,1) + 1(4,14,1) + 1(4,14,1) + 3(1,21,1) + 3/2(1,6,1) + 2(1,1,10) + 2(1,14,1) + 1/2(5,6,1) + 3(5,1,1) + 1/2(16,1,1)

Spectrum 991; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 5

5(1,1,1) + 1(4,6,1) + 8(4,1,1) + 1(4,14,1) + 1(4,14,1) + 3(1,21,1) + 3(1,6,1) + 2(1,1,10) + 2(1,14,1) + 1(5,6,1) + 1/2(16,1,1)

Spectrum 992; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 3

3(1,1,1) + 1(4,6,1) + 1(4,1,1) + 3/2(4,14,1) + 1(4,14,1) + 3(1,21,1) + 8(1,6,1) + 2(1,1,10) + 6(5,1,1) + 1/2(16,1,1)

Spectrum 993; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 4

4(1,1,1) + 1/2(4,21,1) + 1(4,6,1) + 9/2(4,1,1) + 3/2(4,14,1) + 1(1,21,1) + 11/2(1,6,1) + 2(1,1,10) + 4(1,14,1) + 1/2(5,6,1) + 3(5,1,1) + 1/2(16,1,1)

Spectrum 994; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 0

1/2(4,21,1) + 2(4,6,1) + 11/2(4,1,1) + 1(4,14,1) + 1(1,21,1) + 3/2(1,6,1) + 2(1,1,10) + 6(1,14,1) + 1/2(5,6,1) + 3(5,1,1) + 1/2(16,1,1)

Spectrum 995; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 6

6(1,1,1) + 1/2(4,21,1) + 1(4,6,1) + 9/2(4,1,1) + 3/2(4,14,1) + 1(1,21,1) + 4(1,6,1) + 2(1,1,10) + 1(1,14,1) + 7/2(1,14,1) + 1/2(5,6,1) + 3(5,1,1) + 1/2(16,1,1)

Spectrum 996; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 8

8(1,1,1) + 1/2(4,21,1) + 1(4,6,1) + 9/2(4,1,1) + 3/2(4,14,1) + 1(1,21,1) + 5/2(1,6,1) + 2(1,1,10) + 2(1,14,1) + 3(1,14,1) + 1/2(5,6,1) + 3(5,1,1) + 1/2(16,1,1)

Spectrum 997; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 10

10(1,1,1) + 1/2(4,21,1) + 1(4,6,1) + 9/2(4,1,1) + 3/2(4,14,1) + 1(1,21,1) + 1(1,6,1) + 2(1,1,10) + 3(1,14,1) + 5/2(1,14,1) + 1/2(5,6,1) + 3(5,1,1) + 1/2(16,1,1)

Spectrum 998; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 1

1(1,1,1) + 1/2(4,21,1) + 21/2(4,1,1) + 3/2(4,14,1) + 1(1,21,1) + 5/2(1,6,1) + 2(1,1,10) + 3(1,14,1) + 5/2(1,14,1) + 1(5,6,1) + 1/2(16,1,1)

Spectrum 999; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 17

17(1,1,1) + 7(4,1,1) + 5/2(4,14,1) + 3(1,21,1) + 1(1,6,1) + 2(1,1,10) + 2(1,14,1) + 1(5,6,1) + 1/2(16,1,1)

Spectrum 1000; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 3

3(1,1,1) + 1/2(4,21,1) + 21/2(4,1,1) + 3/2(4,14,1) + 1(1,21,1) + 1(1,6,1) + 2(1,1,10) + 4(1,14,1) + 2(1,14,1) + 1(5,6,1) + 1/2(16,1,1)

Spectrum 1001; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 1

1(1,1,1) + 1/2(4,21,1) + 7/2(4,1,1) + 2(4,14,1) + 1(1,21,1) + 6(1,6,1) + 2(1,1,10) + 2(1,14,1) + 2(1,14,1) + 6(5,1,1) + 1/2(16,1,1)

Spectrum 1002; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 3

3(1,1,1) + 1/2(4,21,1) + 7/2(4,1,1) + 2(4,14,1) + 1(1,21,1) + 9/2(1,6,1) + 2(1,1,10) + 3(1,14,1) + 3/2(1,14,1) + 6(5,1,1) + 1/2(16,1,1)

Spectrum 1003; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 19

19(1,1,1) + 3(4,14,1) + 3(1,21,1) + 3(1,6,1) + 2(1,1,10) + 1(1,14,1) + 6(5,1,1) + 1/2(16,1,1)

Spectrum 1004; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 5

5(1,1,1) + 1/2(4,21,1) + 7/2(4,1,1) + 2(4,14,1) + 1(1,21,1) + 3(1,6,1) + 2(1,1,10) + 4(1,14,1) + 1(1,14,1) + 6(5,1,1) + 1/2(16,1,1)

Spectrum 1005; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 21

21(1,1,1) + 3(4,14,1) + 3(1,21,1) + 3/2(1,6,1) + 2(1,1,10) + 1(1,14,1) + 1/2(1,14,1) + 6(5,1,1) + 1/2(16,1,1)

Spectrum 1006; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 7

7(1,1,1) + 1/2(4,21,1) + 7/2(4,1,1) + 2(4,14,1) + 1(1,21,1) + 3/2(1,6,1) + 2(1,1,10) + 5(1,14,1) + 1/2(1,14,1) + 6(5,1,1) + 1/2(16,1,1)

Spectrum 1007; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 23

23(1,1,1) + 3(4,14',1) + 3(1,21,1) + 2(1,1,10) + 2(1,14',1) + 6(5,1,1) + 1/2(16,1,1)

Spectrum 1008; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 9

9(1,1,1) + 1/2(4,21,1) + 7/2(4,1,1) + 2(4,14',1) + 1(1,21,1) + 2(1,1,10) + 6(1,14',1) + 6(5,1,1) + 1/2(16,1,1)

Spectrum 1009; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 9

9(1,1,1) + 1(10,1,1) + 1(4,6,1) + 2(4,1,1) + 2(4,14',1) + 1(1,21,1) + 15/2(1,6,1) + 2(1,1,10) + 1(1,14,1) + 1/2(1,64,1) + 3/2(5,6,1)

Spectrum 1010; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 11

11(1,1,1) + 1(10,1,1) + 1(4,6,1) + 2(4,1,1) + 2(4,14',1) + 1(1,21,1) + 6(1,6,1) + 2(1,1,10) + 1(1,14',1) + 1/2(1,14,1) + 1/2(1,64,1) + 3/2(5,6,1)

Spectrum 1011; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 13

13(1,1,1) + 1(10,1,1) + 1(4,6,1) + 2(4,1,1) + 2(4,14',1) + 1(1,21,1) + 9/2(1,6,1) + 2(1,1,10) + 2(1,14',1) + 1/2(1,64,1) + 3/2(5,6,1)

Spectrum 1012; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 2

2(1,1,1) + 1(10,1,1) + 1(4,1,1) + 5/2(4,14',1) + 1(1,21,1) + 11(1,6,1) + 2(1,1,10) + 1/2(1,64,1) + 1(5,6,1) + 3(5,1,1)

Spectrum 1013; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 15

15(1,1,1) + 1(10,1,1) + 2(4,6,1) + 3(4,1,1) + 1/2(4,14',1) + 1(4,14,1) + 3(1,21,1) + 5/2(1,6,1) + 2(1,1,10) + 2(1,14,1) + 3/2(5,6,1)

Spectrum 1014; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 17

17(1,1,1) + 1(10,1,1) + 2(4,6,1) + 3(4,1,1) + 1/2(4,14',1) + 1(4,14,1) + 3(1,21,1) + 1(1,6,1) + 2(1,1,10) + 1(1,14',1) + 3/2(1,14,1) + 3/2(5,6,1)

Spectrum 1015; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 8

8(1,1,1) + 1(10,1,1) + 1(4,6,1) + 2(4,1,1) + 1(4,14',1) + 1(4,14,1) + 3(1,21,1) + 6(1,6,1) + 2(1,1,10) + 1(1,14,1) + 1(5,6,1) + 3(5,1,1)

Spectrum 1016; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 9

9(1,1,1) + 1(10,1,1) + 1/2(4,21,1) + 1(4,6,1) + 11/2(4,1,1) + 1(4,14',1) + 1(1,21,1) + 7/2(1,6,1) + 2(1,1,10) + 5(1,14,1) + 3/2(5,6,1)

Spectrum 1017; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 5

5(1,1,1) + 1(10,1,1) + 3(4,6,1) + 4(4,1,1) + 1(4,14',1) + 3(1,21,1) + 3(1,6,1) + 2(1,1,10) + 5/2(1,14,1) + 2(5,1,1) + 1/2(5,14,1)

Spectrum 1018; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 10

10(1,1,1) + 1(10,1,1) + 1(4,6,1) + 2(4,1,1) + 1(4,14',1) + 1(4,14,1) + 3(1,21,1) + 9/2(1,6,1) + 2(1,1,10) + 1(1,14',1) + 1/2(1,14,1) + 1(5,6,1) + 3(5,1,1)

Spectrum 1019; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 1

1(1,1,1) + 1(10,1,1) + 8(4,1,1) + 1(4,14',1) + 1(4,14,1) + 3(1,21,1) + 6(1,6,1) + 2(1,1,10) + 1(1,14',1) + 1/2(1,14,1) + 3/2(5,6,1)

Spectrum 1020; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 11

11(1,1,1) + 1(10,1,1) + 1/2(4,21,1) + 1(4,6,1) + 11/2(4,1,1) + 1(4,14',1) + 1(1,21,1) + 2(1,6,1) + 2(1,1,10) + 1(1,14',1) + 9/2(1,14,1) + 3/2(5,6,1)

Spectrum 1021; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 7

7(1,1,1) + 1(10,1,1) + 3(4,6,1) + 4(4,1,1) + 1(4,14',1) + 3(1,21,1) + 3/2(1,6,1) + 2(1,1,10) + 1(1,14',1) + 2(1,14,1) + 2(5,1,1) + 1/2(5,14,1)

Spectrum 1022; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 12

12(1,1,1) + 1(10,1,1) + 1(4,6,1) + 2(4,1,1) + 1(4,14',1) + 1(4,14,1) + 3(1,21,1) + 3(1,6,1) + 2(1,1,10) + 2(1,14',1) + 1(5,6,1) + 3(5,1,1)

Spectrum 1023; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 3

3(1,1,1) + 1(10,1,1) + 8(4,1,1) + 1(4,14',1) + 1(4,14,1) + 3(1,21,1) + 9/2(1,6,1) + 2(1,1,10) + 2(1,14',1) + 3/2(5,6,1)

Spectrum 1024; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 1

1(1,1,1) + 1(10,1,1) + 1(4,1,1) + 3/2(4,14',1) + 1(4,14,1) + 3(1,21,1) + 19/2(1,6,1) + 2(1,1,10) + 1/2(5,6,1) + 6(5,1,1)

Spectrum 1025; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 13

13(1,1,1) + 1(10,1,1) + 1/2(4,21,1) + 1(4,6,1) + 11/2(4,1,1) + 1(4,14',1) + 1(1,21,1) + 1/2(1,6,1) + 2(1,1,10) + 2(1,14',1) + 4(1,14,1) + 3/2(5,6,1)

Spectrum 1026; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 2

2(1,1,1) + 1(10,1,1) + 1/2(4,21,1) + 9/2(4,1,1) + 3/2(4,14',1) + 1(1,21,1) + 7(1,6,1) + 2(1,1,10) + 4(1,14,1) + 1(5,6,1) + 3(5,1,1)

Spectrum 1027; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 9

9(1,1,1) + 1(10,1,1) + 3(4,6,1) + 4(4,1,1) + 1(4,14',1) + 3(1,21,1) + 2(1,1,10) + 2(1,14',1) + 3/2(1,14,1) + 2(5,1,1) + 1/2(5,14,1)

Spectrum 1028; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 4

4(1,1,1) + 1(10,1,1) + 1/2(4,21,1) + 9/2(4,1,1) + 3/2(4,14',1) + 1(1,21,1) + 11/2(1,6,1) + 2(1,1,10) + 1(1,14',1) + 7/2(1,14,1) + 1(5,6,1) + 3(5,1,1)

Spectrum 1029; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 6

6(1,1,1) + 1(10,1,1) + 1/2(4,21,1) + 9/2(4,1,1) + 3/2(4,14',1) + 1(1,21,1) + 4(1,6,1) + 2(1,1,10) + 2(1,14',1) + 3(1,14,1) + 1(5,6,1) + 3(5,1,1)

Spectrum 1030; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 8

8(1,1,1) + 1(10,1,1) + 1/2(4,21,1) + 9/2(4,1,1) + 3/2(4,14',1) + 1(1,21,1) + 5/2(1,6,1) + 2(1,1,10) + 3(1,14',1) + 5/2(1,14,1) + 1(5,6,1) + 3(5,1,1)

Spectrum 1031; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 24

24(1,1,1) + 1(10,1,1) + 1(4,1,1) + 5/2(4,14',1) + 3(1,21,1) + 1(1,6,1) + 2(1,1,10) + 2(1,14,1) + 1(5,6,1) + 3(5,1,1)

Spectrum 1032; branches = [(3/2,3), (2,4), (1,-2)]; singlets = 10

10(1,1,1) + 1(10,1,1) + 1/2(4,21,1) + 9/2(4,1,1) + 3/2(4,14',1) + 1(1,21,1) + 1(1,6,1) + 2(1,1,10) + 4(1,14',1) + 2(1,14,1) + 1(5,6,1) + 3(5,1,1)

C2 x C4 x C4

Spectrum 1033; branches = [(3/2,3), (1,0), (1,0)]; singlets = 4

4(1,1,1) + 3(4,8,1) + 2(4,1,8) + 2(4,1,1) + 4(1,8,1) + 11/2(1,1,8) + 1(1,1,27) + 1(1,27,1) + 1/2(5,1,8) + 2(5,1,1) + 1/2(16,1,1)

C2 x C4 x D4

Spectrum 1034; branches = [(3/2,3), (1,0), (3/2,1)]; singlets = 23

23(1,1,1) + 2(4,8,1) + 2(4,1,1) + 3/2(4,1,8_v) + 3/2(4,1,8_s) + 1/2(1,8,8_v) + 3/2(1,8,1) + 3(1,1,8_v) + 1(1,1,8_s) + 1(1,1,8_c) + 1(1,27,1) + 1/2(5,8,1) + 2(5,1,1) + 1/2(16,1,1)

Spectrum 1035; branches = [(3/2,3), (1,0), (3/2,1)]; singlets = 23

23(1,1,1) + 2(4,8,1) + 1/2(4,1,8_s) + 2(4,1,1) + 1(4,1,8_v) + 3/2(4,1,8_c) + 1/2(1,8,8_v) + 3/2(1,8,1) + 1(1,1,8_v) + 3(1,1,8_s) + 1(1,1,8_c) + 1(1,27,1) + 1/2(5,8,1) + 2(5,1,1) + 1/2(16,1,1)

Spectrum 1036; branches = [(3/2,3), (1,0), (3/2,1)]; singlets = 23

23(1,1,1) + 2(4,8,1) + 1/2(4,1,8_s) + 2(4,1,1) + 3/2(4,1,8_v) + 1(4,1,8_c) + 1/2(1,8,8_v) + 3/2(1,8,1) + 1(1,1,8_v) + 1(1,1,8_s) + 3(1,1,8_c) + 1(1,27,1) + 1/2(5,8,1) + 2(5,1,1) + 1/2(16,1,1)

Spectrum 1037; branches = [(3/2,3), (1,0), (3/2,1)]; singlets = 23

23(1,1,1) + 2(4,8,1) + 1(4,1,8_v) + 2(4,1,1) + 1/2(4,1,8_s) + 3/2(4,1,8_c) + 3/2(1,8,1) + 1/2(1,8,8_s) + 3(1,1,8_v) + 1(1,1,8_s) + 1(1,1,8_c) + 1(1,27,1) + 1/2(5,8,1) + 2(5,1,1) + 1/2(16,1,1)

Spectrum 1038; branches = [(3/2,3), (1,0), (3/2,1)]; singlets = 23

23(1,1,1) + 2(4,8,1) + 1(4,1,8_v) + 2(4,1,1) + 3/2(4,1,8_s) + 1/2(4,1,8_c) + 3/2(1,8,1) + 1/2(1,8,8_c) + 3(1,1,8_v) + 1(1,1,8_s) + 1(1,1,8_c) + 1(1,27,1) + 1/2(5,8,1) + 2(5,1,1) + 1/2(16,1,1)

[illegible]

Spectrum 1073; branches = [(3/2,3), (1,0), (3/2,1)]; singlets = 18

18(1,1,1) + 1(10,1,1) + 1(4,8,1) + 3/2(4,1,8_v) + 4(4,1,1) + 1/2(4,1,8_s) + 1(4,1,8_c) + 3(1,8,1) + 1/2(1,8,8_s) + 1(1,1,8_v) + 1(1,1,8_s) + 3(1,1,8_c) + 1(1,27,1) + 1(5,8,1) + 1(5,1,1)

Spectrum 1074; branches = [(3/2,3), (1,0), (3/2,1)]; singlets = 18

18(1,1,1) + 1(10,1,1) + 1(4,8,1) + 3/2(4,1,8_v) + 4(4,1,1) + 1(4,1,8_s) + 1/2(4,1,8_c) + 3(1,8,1) + 1/2(1,8,8_c) + 1(1,1,8_v) + 3(1,1,8_s) + 1(1,1,8_c) + 1(1,27,1) + 1(5,8,1) + 1(5,1,1)

Spectrum 1075; branches = [(3/2,3), (1,0), (3/2,1)]; singlets = 18

18(1,1,1) + 1(10,1,1) + 1(4,8,1) + 3/2(4,1,8_v) + 4(4,1,1) + 3/2(4,1,8_s) + 3(1,8,1) + 1/2(1,8,8_c) + 1(1,1,8_v) + 1(1,1,8_s) + 3(1,1,8_c) + 1(1,27,1) + 1(5,8,1) + 1(5,1,1)

C2 x D4 x D5

Spectrum 1076; branches = [(3/2,3), (3,6), (1,-2)]; singlets = 2

2(1,1,1) + 3/2(4,28,1) + 4(1,28,1) + 2(1,1,10) + 6(5,1,1) + 1/2(16,1,1)

Spectrum 1077; branches = [(3/2,3), (3,6), (1,-2)]; singlets = 9

9(1,1,1) + 1(10,1,1) + 1(4,28,1) + 1(4,8_v,1) + 2(1,28,1) + 2(1,1,10) + 1(1,56_c,1) + 1(5,1,1) + 1(5,8_c,1)

Spectrum 1078; branches = [(3/2,3), (3,6), (1,-2)]; singlets = 9

9(1,1,1) + 1(10,1,1) + 1(4,28,1) + 1(4,8_v,1) + 2(1,28,1) + 2(1,1,10) + 1(1,56_s,1) + 1(5,1,1) + 1(5,8_s,1)

Spectrum 1079; branches = [(3/2,3), (3,6), (1,-2)]; singlets = 9

9(1,1,1) + 1(10,1,1) + 1(4,28,1) + 1(4,8_v,1) + 2(1,28,1) + 2(1,1,10) + 1(1,56_v,1) + 1(5,8_v,1) + 1(5,1,1)

C2 x D5 x G2

Spectrum 1080; branches = [(3/2,3), (1,-2), (2,4)]; singlets = 38

38(1,1,1) + 6(4,1,7) + 2(1,10,1) + 7(1,1,7) + 1(1,1,27) + 6(5,1,1) + 1/2(16,1,1)

Spectrum 1081; branches = [(3/2,3), (1,-2), (2,4)]; singlets = 44

44(1,1,1) + 6(4,1,7) + 2(1,10,1) + 3(1,1,14) + 4(1,1,7) + 6(5,1,1) + 1/2(16,1,1)

Spectrum 1082; branches = [(3/2,3), (1,-2), (2,4)]; singlets = 16

16(1,1,1) + 1/2(4,1,14) + 4(4,1,7) + 7(4,1,1) + 2(1,10,1) + 1(1,1,14) + 12(1,1,7) + 6(5,1,1) + 1/2(16,1,1)

Spectrum 1083; branches = [(3/2,3), (1,-2), (2,4)]; singlets = 24

24(1,1,1) + 1(10,1,1) + 4(4,1,7) + 8(4,1,1) + 2(1,10,1) + 10(1,1,7) + 1(1,1,27) + 1(5,1,7) + 2(5,1,1)

Spectrum 1084; branches = [(3/2,3), (1,-2), (2,4)]; singlets = 30

30(1,1,1) + 1(10,1,1) + 4(4,1,7) + 8(4,1,1) + 2(1,10,1) + 3(1,1,14) + 7(1,1,7) + 1(5,1,7) + 2(5,1,1)

Spectrum 1085; branches = [(3/2,3), (1,-2), (2,4)]; singlets = 23

23(1,1,1) + 1(10,1,1) + 1/2(4,1,14) + 4(4,1,7) + 1(4,1,1) + 2(1,10,1) + 1(1,1,14) + 12(1,1,7) + 9(5,1,1)

Spectrum 1086; branches = [(3/2,3), (1,-2), (2,4)]; singlets = 2

2(1,1,1) + 1(10,1,1) + 1/2(4,1,14) + 2(4,1,7) + 15(4,1,1) + 2(1,10,1) + 1(1,1,14) + 15(1,1,7) + 1(5,1,7) + 2(5,1,1)

C3 x D5 x G2

Spectrum 1087; branches = [(2,4), (1,-2), (2,4)]; singlets = 13

13(1,1,1) + 3(21,1,1) + 3/2(6,1,7) + 2(1,10,1) + 1(1,1,7) + 1(1,1,27) + 1(14',1,7) + 1/2(14,1,7)

Spectrum 1088; branches = [(2,4), (1,-2), (2,4)]; singlets = 13

13(1,1,1) + 3(21,1,1) + 2(1,10,1) + 3(1,1,7) + 1(1,1,27) + 2(14',1,7)

Spectrum 1089; branches = [(2,4), (1,-2), (2,4)]; singlets = 19

19(1,1,1) + 3(21,1,1) + 2(1,10,1) + 3(1,1,14) + 2(14',1,7)

D5 x G2 x G2

Spectrum 1090; branches = [(1,-2), (2,4), (2,4)]; singlets = 28

28(1,1,1) + 2(10,1,1) + 4(1,7,7) + 3(1,7,1) + 3(1,1,7) + 1(1,1,27) + 1(1,27,1)

Spectrum 1091; branches = [(1,-2), (2,4), (2,4)]; singlets = 34

34(1,1,1) + 2(10,1,1) + 4(1,7,7) + 3(1,7,1) + 3(1,1,14) + 1(1,27,1)

Spectrum 1092; branches = [(1,-2), (2,4), (2,4)]; singlets = 40

40(1,1,1) + 2(10,1,1) + 3(1,14,1) + 4(1,7,7) + 3(1,1,14)

n = 4 detailed spectra

A3 x A3 x A3 x A3

Spectrum 1; branches = [(2,4), (2,4), (2,4), (2,4)]; singlets = 4

4(1,1,1,1) + 1(4,4,4,4) + 1(1,1,1,20) + 1(1,1,20,1) + 1(1,20,1,1) + 1(20,1,1,1)

A3 x A3 x A3 x D5

Spectrum 2; branches = [(2,4), (2,4), (2,4), (1,-2)]; singlets = 4

4(1,1,1,1) + 4(4,4,4,1) + 2(1,1,1,10) + 1(1,1,20,1) + 1(1,20,1,1) + 1(20,1,1,1)

A3 x A3 x A4 x A4

Spectrum 3; branches = [(2,4), (2,4), (1,0), (1,0)]; singlets = 4

4(1,1,1,1) + 1(4,4,5,1) + 1(4,4,1,5) + 6(4,4,1,1) + 2(1,1,1,10) + 2(1,1,10,1) + 1(1,20,1,1) + 1(20,1,1,1)

A3 x A4 x A4 x A4

Spectrum 4; branches = [(2,4), (1,0), (1,0), (1,0)]; singlets = 4

4(1,1,1,1) + 4(4,5,1,1) + 4(4,1,5,1) + 4(4,1,1,5) + 4(4,1,1,1) + 2(1,1,1,10) + 2(1,1,10,1) + 2(1,10,1,1) + 1(20,1,1,1)

Spectrum 5; branches = [(2,4), (1,0), (1,0), (1,0)]; singlets = 2

2(1,1,1,1) + 4(4,5,1,1) + 2(4,1,5,1) + 2(4,1,1,1) + 2(1,1,5,1) + 4(1,1,1,5) + 2(1,1,1,10) + 2(1,1,10,1) + 2(1,10,1,1) + 1(6,1,5,1) + 2(6,1,1,5) + 3(10,1,1,1)

Spectrum 6; branches = [(2,4), (1,0), (1,0), (1,0)]; singlets = 2

2(1,1,1,1) + 2(4,5,1,1) + 2(4,1,5,1) + 2(4,1,1,1) + 2(1,5,1,1) + 2(1,1,5,1) + 2(1,1,1,5) + 2(1,1,1,10) + 2(1,1,10,1) + 2(1,10,1,1) + 1(6,5,1,1) + 1(6,1,5,1) + 1(6,1,1,5) + 3(10,1,1,1)

A3 x A4 x A4 x B3

Spectrum 7; branches = [(2,4), (1,0), (1,0), (1,0)]; singlets = 3

3(1,1,1,1) + 4(4,5,1,1) + 4(4,1,5,1) + 8(4,1,1,1) + 2(4,1,1,8) + 3(1,1,1,7) + 2(1,1,10,1) + 2(1,10,1,1) + 1(20,1,1,1)

Spectrum 8; branches = [(2,4), (1,0), (1,0), (1,0)]; singlets = 7

7(1,1,1,1) + 4(4,5,1,1) + 4(4,1,5,1) + 3(4,1,1,1) + 3(1,1,1,7) + 2(1,1,1,8) + 2(1,1,10,1) + 2(1,10,1,1) + 1(6,1,1,1) + 1(6,1,1,8) + 1(10,1,1,1) + 1(20',1,1,1)

Spectrum 9; branches = [(2,4), (1,0), (1,0), (1,0)]; singlets = 8

8(1,1,1,1) + 1(15,1,1,1) + 4(4,5,1,1) + 4(4,1,5,1) + 3(4,1,1,1) + 3(1,1,1,7) + 2(1,1,1,8) + 2(1,1,10,1) + 2(1,10,1,1) + 1(6,1,1,8) + 1(20',1,1,1)

Spectrum 10; branches = [(2,4), (1,0), (1,0), (1,0)]; singlets = 5

5(1,1,1,1) + 4(4,5,1,1) + 2(4,1,5,1) + 2(4,1,1,1) + 2(1,1,5,1) + 3(1,1,1,7) + 2(1,1,1,8) + 2(1,1,10,1) + 2(1,10,1,1) + 1(6,1,5,1) + 2(6,1,1,1) + 1(6,1,1,8) + 3(10,1,1,1)

Spectrum 11; branches = [(2,4), (1,0), (1,0), (1,0)]; singlets = 1

1(1,1,1,1) + 2(4,5,1,1) + 6(4,1,1,1) + 2(4,1,1,8) + 2(1,5,1,1) + 4(1,1,5,1) + 3(1,1,1,7) + 2(1,1,10,1) + 2(1,10,1,1) + 1(6,5,1,1) + 2(6,1,5,1) + 3(10,1,1,1)

Spectrum 12; branches = [(2,4), (1,0), (1,0), (1,0)]; singlets = 6

6(1,1,1,1) + 1(15,1,1,1) + 4(4,5,1,1) + 2(4,1,5,1) + 2(4,1,1,1) + 2(1,1,5,1) + 3(1,1,1,7) + 2(1,1,1,8) + 2(1,1,10,1) + 2(1,10,1,1) + 1(6,1,5,1) + 1(6,1,1,1) + 1(6,1,1,8) + 2(10,1,1,1)

Spectrum 13; branches = [(2,4), (1,0), (1,0), (1,0)]; singlets = 7

7(1,1,1,1) + 2(15,1,1,1) + 4(4,5,1,1) + 2(4,1,5,1) + 2(4,1,1,1) + 2(1,1,5,1) + 3(1,1,1,7) + 2(1,1,1,8) + 2(1,1,10,1) + 2(1,10,1,1) + 1(6,1,5,1) + 1(6,1,1,8) + 1(10,1,1,1)

A3 x A4 x B3 x B3

Spectrum 14; branches = [(2,4), (1,0), (1,0), (1,0)]; singlets = 2

2(1,1,1,1) + 4(4,5,1,1) + 12(4,1,1,1) + 2(4,1,1,8) + 2(4,1,8,1) + 3(1,1,7,1) + 3(1,1,1,7) + 2(1,10,1,1) + 1(20,1,1,1)

Spectrum 15; branches = [(2,4), (1,0), (1,0), (1,0)]; singlets = 6

6(1,1,1,1) + 4(4,5,1,1) + 7(4,1,1,1) + 2(4,1,8,1) + 3(1,1,7,1) + 3(1,1,1,7) + 2(1,1,1,8) + 2(1,10,1,1) + 1(6,1,1,1) + 1(6,1,1,8) + 1(10,1,1,1) + 1(20',1,1,1)

Spectrum 16; branches = [(2,4), (1,0), (1,0), (1,0)]; singlets = 12

12(1,1,1,1) + 2(4,5,1,1) + 1(4,1,1,1) + 2(4,1,1,8) + 2(4,1,8,1) + 2(1,5,1,1) + 3(1,1,7,1) + 3(1,1,1,7) + 2(1,1,10,1) + 1(6,5,1,1) + 4(6,1,1,1) + 1(10,1,1,1) + 1(20',1,1,1)

Spectrum 17; branches = [(2,4), (1,0), (1,0), (1,0)]; singlets = 7

7(1,1,1,1) + 1(15,1,1,1) + 4(4,5,1,1) + 7(4,1,1,1) + 2(4,1,8,1) + 3(1,1,7,1) + 3(1,1,1,7) + 2(1,1,1,8) + 2(1,10,1,1) + 1(6,1,1,8) + 1(20',1,1,1)

Spectrum 18; branches = [(2,4), (1,0), (1,0), (1,0)]; singlets = 13

13(1,1,1,1) + 1(15,1,1,1) + 2(4,5,1,1) + 1(4,1,1,1) + 2(4,1,1,8) + 2(4,1,8,1) + 2(1,5,1,1) + 3(1,1,7,1) + 3(1,1,1,7) + 2(1,10,1,1) + 1(6,5,1,1) + 3(6,1,1,1) + 1(20',1,1,1)

Spectrum 19; branches = [(2,4), (1,0), (1,0), (1,0)]; singlets = 4

4(1,1,1,1) + 2(4,5,1,1) + 6(4,1,1,1) + 2(4,1,8,1) + 2(1,5,1,1) + 3(1,1,7,1) + 3(1,1,1,7) + 2(1,1,1,8) + 2(1,10,1,1) + 1(6,5,1,1) + 2(6,1,1,1) + 1(6,1,1,8) + 3(10,1,1,1)

Spectrum 20; branches = [(2,4), (1,0), (1,0), (1,0)]; singlets = 10

10(1,1,1,1) + 2(4,1,1,8) + 2(4,1,8,1) + 4(1,5,1,1) + 3(1,1,7,1) + 3(1,1,1,7) + 2(1,10,1,1) + 2(6,5,1,1) + 5(6,1,1,1) + 3(10,1,1,1)

Spectrum 21; branches = [(2,4), (1,0), (1,0), (1,0)]; singlets = 5

5(1,1,1,1) + 1(15,1,1,1) + 2(4,5,1,1) + 6(4,1,1,1) + 2(4,1,8,1) + 2(1,5,1,1) + 3(1,1,7,1) + 3(1,1,1,7) + 2(1,1,1,8) + 2(1,10,1,1) + 1(6,5,1,1) + 1(6,1,1,1) + 1(6,1,1,8) + 2(10,1,1,1)

Spectrum 22; branches = [(2,4), (1,0), (1,0), (1,0)]; singlets = 11

11(1,1,1,1) + 1(15,1,1,1) + 2(4,1,1,8) + 2(4,1,8,1) + 4(1,5,1,1) + 3(1,1,7,1) + 3(1,1,1,7) + 2(1,10,1,1) + 2(6,5,1,1) + 4(6,1,1,1) + 2(10,1,1,1)

Spectrum 23; branches = [(2,4), (1,0), (1,0), (1,0)]; singlets = 6

6(1,1,1,1) + 2(15,1,1,1) + 2(4,5,1,1) + 6(4,1,1,1) + 2(4,1,8,1) + 2(1,5,1,1) + 3(1,1,7,1) + 3(1,1,1,7) + 2(1,1,1,8) + 2(1,10,1,1) + 1(6,5,1,1) + 1(6,1,1,8) + 1(10,1,1,1)

Spectrum 24; branches = [(2,4), (1,0), (1,0), (1,0)]; singlets = 12

12(1,1,1,1) + 2(15,1,1,1) + 2(4,1,1,8) + 2(4,1,8,1) + 4(1,5,1,1) + 3(1,1,7,1) + 3(1,1,1,7) + 2(1,10,1,1) + 2(6,5,1,1) + 3(6,1,1,1) + 1(10,1,1,1)

Spectrum 25; branches = [(2,4), (1,0), (1,0), (1,0)]; singlets = 13

13(1,1,1,1) + 3(15,1,1,1) + 2(4,1,1,8) + 2(4,1,8,1) + 4(1,5,1,1) + 3(1,1,7,1) + 3(1,1,1,7) + 2(1,10,1,1) + 2(6,5,1,1) + 2(6,1,1,1)

A3 x B3 x B3 x B3

Spectrum 32: branches = $[(2,4), (1,0), (1,0), (1,0)]$; singlets = 16
 $16(\mathbf{1,1,1,1}) + 3(\mathbf{15,1,1,1}) + 2(\mathbf{4,1,8,1}) + 2(\mathbf{4,8,1,1}) + 3(\mathbf{1,7,1,1}) + 3(\mathbf{1,1,7,1}) + 3(\mathbf{1,1,1,7}) + 2(\mathbf{1,1,1,8}) + 4(\mathbf{6,1,1,1}) + 1(\mathbf{6,1,1,8})$

[illegible]

[illegible]

[illegible]

[illegible]

$$3(1,1,1,1) + 3/2(4,4,4,1) + 1(4,4,1,1) + 1(4,1,4,1) + 15/2(4,1,1,1) + 1/2(4,5,1,1) + 13/2(1,4,1,1) + 1(1,4,5,1) + 1(1,1,10,1) + 1/2(1,1,4,1) + 2(1,1,1,10) + 5(1,1,5,1) + 1/2(1,5,4,1) + 2(1,5,1,1) + 1/2(1,16,1,1) + 1/2(5,4,1,1) + 1(5,1,4,1) + 1/2(16,1,1,1)$$
$$9(1,1,1,1) + 3/2(4,4,4,1) + 1(4,4,1,1) + 2(4,1,4,1) + 7/2(4,1,1,1) + 1/2(4,5,1,1) + 4(1,4,1,1) + 3/2(1,4,5,1) + 1(1,1,10,1) + 3/2(1,1,4,1) + 2(1,1,1,10) + 3(1,1,5,1) + 4(1,5,1,1) + 1/2(1,16,1,1) + 1/2(5,4,1,1) + 1/2(5,1,4,1) + 2(5,1,1,1) + 1/2(16,1,1,1)$$
$$3(1,1,1,1) + 3/2(4,4,4,1) + 1(4,4,1,1) + 1(4,1,4,1) + 15/2(4,1,1,1) + 1/2(4,5,1,1) + 4(1,4,1,1) + 3/2(1,4,5,1) + 1(1,1,10,1) + 3(1,1,4,1) + 2(1,1,1,10) + 3(1,1,5,1) + 4(1,5,1,1) + 1/2(1,16,1,1) + 1/2(5,4,1,1) + 1(5,1,4,1) + 1/2(16,1,1,1)$$
$$3(1,1,1,1) + 3/2(4,4,4,1) + 1(4,4,1,1) + 9(4,1,1,1) + 1/2(4,1,5,1) + 1/2(4,5,1,1) + 1(1,4,4,1) + 5(1,4,1,1) + 1/2(1,4,5,1) + 1(1,1,10,1) + 1/2(1,1,4,1) + 2(1,1,1,10) + 5(1,1,5,1) + 1/2(1,5,4,1) + 2(1,5,1,1) + 1/2(1,16,1,1) + 1/2(5,4,1,1) -$$
$$15(1,1,1,1,1) + 3/2(4,4,4,4,1) + 1/2(4,4,1,1,1) + 2(4,1,4,1,1) + 1/2(1,1,1,1,1) + 1/2(4,1,5,1,1) + 1/2(4,5,1,1,1) + 1/2(1,4,4,1,1) + 5/2(1,4,1,1,1) + 1/2(1,4,5,1,1) + 1/2(1,1,10,1,1) + 2(1,1,1,10,1) + 3(1,1,5,1,1) + 4(1,5,1,1,1) + 1/2(1,16,1,1,1) + 1/2(5,4,1,1,1) + 4(5,1,1,1,1) + 1/2(16,1,1,1,1)$$

spectrum 223, branches = $[(S2z,S), (S2z,S), (S2z,S), (1,-2)]$, singlets = 9
 $9(1,1,1,1,1) + 3/2(4,4,4,4,1) + 1(4,4,1,1,1) + 5(4,1,1,1,1) + 1/2(4,1,5,1) + 1/2(4,5,1,1) + 1(1,4,4,1) + 5/2(1,4,1,1) + 1(1,4,5,1) + 1(1,1,1,1,0) + 3/2(1,1,4,1) + 2(1,1,1,1,0) + 3(1,1,5,1) + 4(1,5,1,1) + 1/2(1,1,6,1,1) + 1/2(5,4,1,1) + 1/2(5,1,4,1)$

Spectrum Z24, branches = [(3/2,3), (3/2,3), (3/2,3), (1,-2)], Singlets = 3
 $3(1\ 1\ 1\ 1) + 3/2(4\ 4\ 4\ 1) + 1(4\ 4\ 1\ 1) + 9(4\ 1\ 1\ 1) + 1/2(4\ 1\ 5\ 1) + 1/2(4\ 5\ 1\ 1) + 1(1\ 4\ 4\ 1) + 5/2(1\ 4\ 1\ 1) + 1(1\ 4\ 5\ 1) + 1(1\ 1\ 1\ 10) + 3(1\ 1\ 4\ 1) + 2(1\ 1\ 1\ 10) + 3(1\ 1\ 5\ 1) + 4(1\ 5\ 1\ 1) + 1/2(1\ 16\ 1\ 1) + 1/2(5\ 4\ 1\ 1) + 1(5\ 1\ 4\ 1) + 1/2(16\ 1\ 1\ 1)$

Spectrum 225: branches = [(3/2, 3), (3/2, 3), (3/2, 3), (1, -2)]; singlets = 9

Spectrum 226: branches = $\{(3/2, 3), (3/2, 3), (3/2, 3), (1, -2)\}$; singlets = 3

Spectrum $27Z$: branches = $l(3/2, 3) \quad (3/2, 3) \quad (1-2)l$: singlets = $l(2, 1, 1, 1) + 3l(2, 4, 4, 4) + 1(4, 4, 1, 1) + 1(4, 1, 4, 1) + 5(4, 1, 1, 1) + 1(2, 4, 1, 3, 1) + 1(2, 4, 3, 1, 1) + 1(3, 2, 1, 1) + 1(1, 4, 3, 1) + 1(1, 1, 1, 10, 1) + 3(1, 1, 3, 4, 1) + 2(1, 1, 1, 10, 1) + 3(1, 1, 3, 1, 1) + 1(2, 1, 5, 4, 1) + 2(1, 1, 5, 1, 1) + 1(2, 1, 10, 1, 1) + 1(2, 3, 4, 1, 1) + 1(2, 3, 1, 4, 1) + 2(3, 1, 1, 1) + 1(2, 10, 1, 1, 1)$

$$3(1,1,1,1,1) + 3(2,4,4,4,1) + 1(4,4,1,1) + 2(4,1,1,1) + 1(4,1,1,1) + 1(2,2,4,1,5,1) + 1(2,4,5,1,1) + 4(1,4,1,1) + 3(2,1,1,1,1) + 1(1,1,10,1) + 4(1,1,4,1) + 2(1,1,1,1,10) + 1(1,1,1,5,1) + 4(1,5,1,1) + 1(2,1,16,1,1) + 1(2,5,4,1,1) + 4(5,1,1,1) + 1(2,16,1,1,1)$$
$$3(1,1,1,1) + 3(2,4,4,4,1) + 1(4,4,4,1,1) + 1(4,1,4,1,1) + 5(4,1,1,1,1) + 1/2(4,1,5,1) + 1/2(4,5,1,1) + 4(1,4,1,1) + 3/2(1,4,5,1) + 1(1,1,10,1) + 11/2(1,1,4,1) + 2(1,1,1,10) + 1(1,1,5,1) + 4(1,5,1,1) + 1/2(1,16,1,1) + 1/2(5,4,1,1) + 1/2(5,1,4,1) + 2(5,1,1,1) + 1/2(16,1,1,1)$$
$$9(1.1.1.1) + 3/2(4.4.4.1) + 1/4(4.4.1.1) + 1/4(4.1.4.1) + 5/2(4.1.1.1) + 1/4(4.1.5.1) + 1/2(4.5.1.1) + 1/4(4.4.1) + 5/2(1.4.1.1) + 1/4(1.4.5.1) + 1/4(1.1.10.1) + 4(1.1.4.1) + 2(1.1.1.10) + 1/4(1.1.5.1) + 4(1.5.1.1) + 1/2(1.16.1.1) + 1/2(5.4.1.1) + 4(5.1.1.1) + 1/2(16.1.1.1)$$
$$3(1,1,1,1) + 3/2(4,4,4,1) + 1(4,4,1,1) + 13/2(4,1,1,1) + 1(4,1,5,1) + 1/2(4,5,1,1) + 1(1,4,4,1) + 5/2(1,4,1,1) + 1(1,4,5,1) + 1(1,1,10,1) + 11/2(1,1,4,1) + 2(1,1,1,10) + 1(1,1,5,1) + 4(1,5,1,1) + 1/2(1,16,1,1) + 1/2(5,4,1,1) + 1/2(5,1,4,1) + 2(5,1,1,1) + 1/2(16,1,1,1)$$
$$16(1,1,1,1) + 1(4,4,4,1) + 2(4,4,1,1) + 2(4,1,4,1) + 5/2(4,1,1,1) + 1(4,1,5,1) + 1/2(4,5,1,1) + 2(1,4,4,1) + 5/2(1,4,1,1) + 1/2(1,4,5,1) + 5/2(1,1,4,1) + 2(1,1,1,10) + 1/2(1,1,16,1) + 1(1,5,4,1) + 1/2(1,16,1,1) + 1(5,4,1,1) + 1/2(5,1,4,1) + 1/2(16,1,1,1)$$
$$17(1,1,1,1) + 1(4,4,4,1) + 2(4,4,1,1) + 3(4,1,4,1) + 1(4,1,1,1) + 1/2(4,1,5,1) + 1/2(4,5,1,1) + 11/2(1,4,1,1) + 3/2(1,4,5,1) + 1(1,1,10,1) + 1/2(1,1,4,1) + 2(1,1,1,10) + 1(1,1,5,1) + 1(1,5,4,1) + 1/2(1,16,1,1) + 1(5,4,1,1) + 1/2(5,1,4,1) + 1/2(16,1,1,1)$$
$$17(\mathbf{1},\mathbf{1},\mathbf{1},\mathbf{1}) + 1(\mathbf{4},\mathbf{4},\mathbf{4},\mathbf{1}) + 2(\mathbf{4},\mathbf{4},\mathbf{1},\mathbf{1}) + 2(\mathbf{4},\mathbf{1},\mathbf{4},\mathbf{1}) + 5/2(\mathbf{4},\mathbf{1},\mathbf{1},\mathbf{1}) + 1(\mathbf{4},\mathbf{1},\mathbf{5},\mathbf{1}) + 1/2(\mathbf{4},\mathbf{5},\mathbf{1},\mathbf{1}) + 1(\mathbf{1},\mathbf{4},\mathbf{4},\mathbf{1}) + 4(\mathbf{1},\mathbf{4},\mathbf{1},\mathbf{1}) + 1(\mathbf{1},\mathbf{4},\mathbf{5},\mathbf{1}) + 1(\mathbf{1},\mathbf{1},\mathbf{1},\mathbf{0}) + 1/2(\mathbf{1},\mathbf{1},\mathbf{4},\mathbf{1}) + 2(\mathbf{1},\mathbf{1},\mathbf{1},\mathbf{0}) + 1(\mathbf{1},\mathbf{1},\mathbf{5},\mathbf{1}) + 1(\mathbf{1},\mathbf{5},\mathbf{4},\mathbf{1}) + 1/2(\mathbf{1},\mathbf{16},\mathbf{1},\mathbf{1}) + 1(\mathbf{5},\mathbf{4},\mathbf{1},\mathbf{1}) + 1/2(\mathbf{5},\mathbf{1},\mathbf{4},\mathbf{1}) + 1/2(16$$
$$17(\mathbf{1,1,1,1}) + 1(\mathbf{4,4,4,1}) + 2(\mathbf{4,4,1,1}) + 1(\mathbf{4,1,4,1}) + 4(\mathbf{4,1,1,1}) + 3/2(\mathbf{4,1,5,1}) + 1/2(\mathbf{4,5,1,1}) + 2(\mathbf{1,4,4,1}) + 5/2(\mathbf{1,4,1,1}) + 1/2(\mathbf{1,4,5,1}) + 1(\mathbf{1,1,10,1}) + 1/2(\mathbf{1,1,4,1}) + 2(\mathbf{1,1,1,10}) + 1(\mathbf{1,1,5,1}) + 1(\mathbf{1,5,4,1}) + 1/2(\mathbf{1,16,1,1}) + 1(\mathbf{5,4,1,1}) + 1/2(\mathbf{5,1,4,1}) + 1/2(\mathbf{16,1,1,1})$$
$$17(1,1,1,1,1) + 1(4,4,4,4,1) + 2(4,4,1,1,1) + 11/2(4,1,1,1,1) + 2(4,1,5,1,1) + 1/2(4,5,1,1,1) + 3(1,4,4,4,1) + 1(1,4,1,1,1) + 1(1,1,1,10,1) + 1/2(1,1,4,4,1) + 2(1,1,1,1,10) + 1(1,1,5,1,1) + 1(1,5,4,1,1) + 1/2(1,16,1,1,1) + 1(5,4,1,1,1) + 1/2(5,1,4,1,1) + 1/2(16,1,1,1,1)$$
$$9(1.1.1.1) + 3/2(4.4.4.1) + 3(4.1.4.1) + 7/2(4.1.1.1) + 1/2(4.5.1.1) + 11/2(1.4.1.1) + 3/2(1.4.5.1) + 1(1.1.10.1) + 2(1.1.1.10) + 3(1.1.5.1) + 4(1.5.1.1) + 1/2(1.16.1.1) + 1(5.4.1.1) + 2(5.1.1.1) + 1/2(16.1.1.1)$$

Spectrum 237; branches = [(3/2,3), (3/2,3), (3/2,3), (1,-2)]; singlets = 3
 $\frac{2}{3}(1,1,1,1) + \frac{2}{3}(1,1,1,1) + \frac{2}{3}(1,1,1,1) + \frac{1}{5}(1,1,1,1) + \frac{1}{10}(1,1,1,1) + \frac{1}{10}(1,1,1,1) + \frac{1}{10}(1,1,1,1) + \frac{1}{10}(1,1,1,1) + \frac{1}{10}(1,1,1,1) + \frac{1}{10}(1,1,1,1)$

Spectrum 238; branches = [(3/2,3), (3/2,3), (3/2,3), (1,-2)]; singlets = 9

Spectrum 239; branches = [(3/2,3), (3/2,3), (3/2,3), (1,-2)]; singlets = 3

Spectrum 240; branches = [(3/2,3), (3/2,3), (3/2,3), (1,-2)]; singlets = 3

Spectrum 241; branches = [(3/2,3), (3/2,3), (3/2,3), (1,-2)]; singlets = 3

Spectrum 242; branches = [(3/2,3), (3/2,3), (3/2,3), (1,-2)]; singlets = 9

Spectrum 243: branches = $[(3/2, 3), (3/2, 3), (3/2, 3), (1, -2)]$; singlets = 3

Spectrum 244: branches = $f(3/2, 3), (3/2, 3), (1-2)!$; singlets = 3

Spectrum 3/2⁺: branches = $\{1(3/2, 3), (3/2, 3), (1, -2)\}$, singlets = 3

Spectrum 246: branches = $1(2/2, 2)$, $(2/2, 2)$, $(2/2, 2)$, $(1, -2)$; singlets = 0
 $5(1, 1, 1, 1) + 3/2(4, 4, 4, 1) + 1/4(1, 4, 1) + 1/3(4, 1, 1, 1) + 1/4(1, 5, 1) + 1/2(4, 5, 1, 1) + 1/2(4, 5, 1, 1) + 1(1, 4, 4, 1) + 4(1, 4, 1, 1) + 1(1, 4, 5, 1) + 1(1, 1, 10, 1) + 4(1, 1, 4, 1) + 2(1, 1, 1, 10) + 1(1, 1, 5, 1) + 4(1, 5, 1, 1) + 1/2(1, 16, 1, 1) + 1(5, 4, 1, 1) + 2(5, 1, 1, 1) + 1/2(16, 1, 1, 1)$

Spectrum 247: branches = $f(2/2, 2)$, $(2/2, 2)$, $(2/2, 2)$, $(1, 2)$; singlets = 2

$$3(1,1,1,1) + 3/2(4,4,4,1) + 8/3(4,1,1,1) + 3/2(4,1,5,1) + 1/2(4,5,1,1) + 2(1,4,4,1) + 5(1,4,1,1) + 1(1,1,10,1) + 3/2(1,1,4,1) + 2(1,1,1,10) + 3(1,1,5,1) + 1/2(1,5,4,1) + 2(1,5,1,1) + 1/2(1,16,1,1) + 1(5,4,1,1) + 2(5,1,1,1) + 1/2(16,1,1,1)$$
$$3(1.1.1.1) + 3/2(4.4.4.1) + 8(4.1.1.1) + 3/2(4.1.5.1) + 1/2(4.5.1.1) + 2(1.4.4.1) + 5/2(1.4.1.1) + 1/2(1.4.5.1) + 1(1.1.10.1) + 4(1.1.4.1) + 2(1.1.1.10) + 1(1.1.5.1) + 4(1.5.1.1) + 1/2(1.16.1.1) + 1(5.4.1.1) + 2(5.1.1.1) + 1/2(16.1.1.1)$$
$$9(1,1,1,1) + 1(4,4,4,1) + 5(4,4,1,1) + 1(4,1,1,1) + 1(4,1,5,1) + 1(1,10,1,1) + 1(1,1,10,1) + 10(1,1,4,1) + 2(1,1,1,10) + 1/2(1,5,4,1) + 2(1,5,1,1) + 1(1,5,5,1) + 3/2(5,1,4,1) + 1/2(16,1,1,1)$$
$$16(1,1,1,1) + 3/2(4,4,4,1) + 3(4,4,1,1) + 1(4,1,1,1) + 1(4,1,5,1) + 1(1,10,1,1) + 1(1,1,10,1) + 2(1,1,4,1) + 2(1,1,1,10) + 5(1,1,5,1) + 3/2(1,5,4,1) + 3(1,5,1,1) + 1/2(5,1,4,1) + 4(5,1,1,1) + 1/2(16,1,1,1)$$
$$8(1,1,1,1) + 2(4,4,4,1) + 1(4,4,1,1) + 7/2(4,1,1,1) + 1/2(4,1,5,1) + 1(1,10,1,1) + 1(1,1,10,1) + 3/2(1,1,4,1) + 2(1,1,1,10) + 7(1,1,5,1) + 1/2(1,5,4,1) + 7(1,5,1,1) + 6(5,1,1,1) + 1/2(16,1,1,1)$$

$$18(1,1,1,1) + 1(4,4,4,1) + 1(4,4,1,1) + 3(4,1,4,1) + 5/2(4,1,1,1) + 1(4,1,5,1) + 1/2(4,5,1,1) + 1(1,10,1,1) + 7/2(1,4,1,1) + 1(1,4,5,1) + 1(1,1,10,1) + 1/2(1,1,4,1) + 2(1,1,1,10) + 1(1,1,5,1) + 3/2(1,5,4,1) + 1(1,5,1,1) + 3/2(5,4,1,1) + 1/2(16,1,1,1)$$
$$18(1,1,1,1) + 1(4,4,4,1) + 1(4,4,1,1) + 2(4,1,4,1) + 4(4,1,1,1) + 3/2(4,1,5,1) + 1/2(4,5,1,1) + 1(1,10,1,1) + 1(1,4,4,1) + 2(1,4,1,1) + 1/2(1,4,5,1) + 1(1,1,10,1) + 1/2(1,1,4,1) + 2(1,1,1,10) + 1(1,1,5,1) + 3/2(1,5,4,1) + 1(1,5,1,1) + 3/2(5,4,1,1) + 1/2(16,1,1,1)$$
$$18(\mathbf{1,1,1,1}) + 1(\mathbf{4,4,4,1}) + 1(\mathbf{4,4,1,1}) + 1(\mathbf{4,1,4,1}) + 11/2(\mathbf{4,1,1,1}) + 2(\mathbf{4,1,5,1}) + 1/2(\mathbf{4,5,1,1}) + 1(\mathbf{1,10,1,1}) + 2(\mathbf{1,4,4,1}) + 1/2(\mathbf{1,4,1,1}) + 1(\mathbf{1,1,10,1}) + 1/2(\mathbf{1,1,4,1}) + 2(\mathbf{1,1,1,10}) + 1(\mathbf{1,1,5,1}) + 3/2(\mathbf{1,5,4,1}) + 1(\mathbf{1,5,1,1}) + 3/2(\mathbf{5,4,1,1}) + 1/2(\mathbf{16,1,1,1})$$
$$9(1,1,1,1) + 1(4,4,4,1) + 1(4,4,1,1) + 3(4,1,4,1) + 5/2(4,1,1,1) + 1(4,1,5,1) + 1/2(4,5,1,1) + 1(1,10,1,1) + 1(1,4,4,1) + 9/2(1,4,1,1) + 1(1,1,10,1) + 4(1,1,4,1) + 2(1,1,1,10) + 2(1,5,1,1) + 1(1,5,5,1) + 3/2(5,4,1,1) + 1/2(16,1,1,1)$$
$$3(1,1,1,1) + 1(4,4,4,1) + 1(4,4,1,1) + 3(4,1,4,1) + 5/2(4,1,1,1) + 1(4,1,5,1) + 1/2(4,5,1,1) + 1(1,10,1,1) + 17/2(1,4,1,1) + 1(1,1,10,1) + 11/2(1,1,4,1) + 2(1,1,1,10) + 1/2(1,5,4,1) + 1(1,5,5,1) + 3/2(5,4,1,1) + 1/2(16,1,1,1)$$
$$9(1,1,1,1) + 1(4,4,4,1) + 3(4,4,1,1) + 1(4,1,4,1) + 1(4,1,5,1) + 1(4,5,1,1) + 1(1,10,1,1) + 1(1,4,4,1) + 4(1,4,1,1) + 1(1,1,10,1) + 7(1,1,4,1) + 2(1,1,1,10) + 1(1,5,5,1) + 1(5,1,4,1) + 2(5,1,1,1) + 1/2(16,1,1,1)$$
$$3(1,1,1,1) + 1(4,4,4,1) + 3(4,4,1,1) + 4(4,1,1,1) + 1(4,1,5,1) + 1(4,5,1,1) + 1(1,10,1,1) + 1(1,4,4,1) + 4(1,4,1,1) + 1(1,1,10,1) + 17/2(1,1,4,1) + 2(1,1,1,10) + 1(1,5,5,1) + 3/2(5,1,4,1) + 1/2(16,1,1,1)$$
$$10(\mathbf{1,1,1,1}) + 3/2(\mathbf{4,4,4,1}) + 1(\mathbf{4,4,1,1}) + 4(\mathbf{4,1,1,1}) + 1(\mathbf{4,1,5,1}) + 1(\mathbf{4,5,1,1}) + 1(\mathbf{1,10,1,1}) + 1(\mathbf{1,4,4,1}) + 4(\mathbf{1,4,1,1}) + 1(\mathbf{1,1,10,1}) + 1/2(\mathbf{1,1,4,1}) + 2(\mathbf{1,1,1,10}) + 5(\mathbf{1,1,5,1}) + 1(\mathbf{1,5,4,1}) + 1(\mathbf{1,5,1,1}) + 1/2(\mathbf{5,1,4,1}) + 4(\mathbf{5,1,1,1}) + 1/2(\mathbf{16,1,1,1})$$
$$16(\mathbf{1,1,1,1}) + 3/2(\mathbf{4,4,4,1}) + 1(\mathbf{4,4,1,1}) + 1(\mathbf{4,1,4,1}) + 1(\mathbf{4,1,5,1}) + 1(\mathbf{4,5,1,1}) + 1(\mathbf{1,10,1,1}) + 1(\mathbf{1,4,4,1}) + 3/2(\mathbf{1,4,1,1}) + 1/2(\mathbf{1,4,5,1}) + 1(\mathbf{1,1,10,1}) + 3/2(\mathbf{1,1,4,1}) + 2(\mathbf{1,1,1,10}) + 3(\mathbf{1,1,5,1}) + 1/2(\mathbf{1,5,4,1}) + 3(\mathbf{1,5,1,1}) + 6(\mathbf{5,1,1,1}) + 1/2(\mathbf{16,1,1,1})$$
$$10(1,1,1,1) + 3/2(4,4,4,1) + 1(4,4,1,1) + 4(4,1,1,1) + 1(4,1,5,1) + 1(4,5,1,1) + 1(1,10,1,1) + 1(1,4,4,1) + 3/2(1,4,1,1) + 1/2(1,4,5,1) + 1(1,1,10,1) + 3(1,1,4,1) + 2(1,1,1,10) + 3(1,1,5,1) + 1/2(1,5,4,1) + 3(1,5,1,1) + 1/2(5,1,4,1) + 4(5,1,1,1) + 1/2(16,1,1,1)$$
$$10(1,1,1,1) + 3/2(4,4,4,1) + 1(4,4,1,1) + 1(4,1,4,1) + 1(4,1,5,1) + 1(4,5,1,1) + 1(1,10,1,1) + 11/2(1,4,1,1) + 1/2(1,4,5,1) + 1(1,1,10,1) + 3(1,1,4,1) + 2(1,1,1,10) + 3(1,1,5,1) + 1(1,5,4,1) + 1(1,5,1,1) + 6(5,1,1,1) + 1/2(16,1,1,1)$$
$$4(1,1,1,1) + 3/2(4,4,4,1) + 1(4,4,1,1) + 4(4,1,1,1) + 1(4,1,5,1) + 1(4,5,1,1) + 1(1,10,1,1) + 11/2(1,4,1,1) + 1/2(1,4,5,1) + 1(1,1,10,1) + 9/2(1,1,4,1) + 2(1,1,1,10) + 3(1,1,5,1) + 1(1,5,4,1) + 1(1,5,1,1) + 1/2(5,1,4,1) + 4(5,1,1,1) + 1/2(16,1,1,1)$$
$$4(1,1,1,1) + 3/2(4,4,4,1) + 1(4,4,1,1) + 4(4,1,1,1) + 1(4,1,5,1) + 1(4,5,1,1) + 1(1,10,1,1) + 3(1,4,1,1) + 1(1,4,5,1) + 1(1,1,10,1) + 7(1,1,4,1) + 2(1,1,1,10) + 1(1,1,5,1) + 1/2(1,5,4,1) + 3(1,5,1,1) + 1/2(5,1,4,1) + 4(5,1,1,1) + 1/2(16,1,1,1)$$
$$16(1,1,1,1) + 3/2(4,4,4,1) + 1(4,4,1,1) + 3/2(4,1,1,1) + 3/2(4,1,5,1) + 1(4,5,1,1) + 1(1,10,1,1) + 2(1,4,4,1) + 1(1,1,10,1) + 3/2(1,1,4,1) + 2(1,1,1,10) + 3(1,1,5,1) + 1/2(1,5,4,1) + 3(1,5,1,1) + 6(5,1,1,1) + 1/2(16,1,1,1)$$
$$10(1,1,1,1) + 3/2(4,4,4,1) + 1(4,4,1,1) + 3/2(4,1,1,1) + 3/2(4,1,5,1) + 1(4,5,1,1) + 1(1,10,1,1) + 1(1,4,4,1) + 4(1,4,1,1) + 1(1,1,10,1) + 3(1,1,4,1) + 2(1,1,1,10) + 3(1,1,5,1) + 1(1,5,4,1) + 1(1,5,1,1) + 6(5,1,1,1) + 1/2(16,1,1,1)$$
$$10(1,1,1,1,1) + 3/2(4,4,4,1) + 1(4,4,1,1) + 3/2(4,1,1,1) + 3/2(4,1,5,1) + 1(4,5,1,1) + 1(1,10,1,1) + 1(1,4,4,1) + 3/2(1,4,1,1) + 1/2(1,4,5,1) + 1(1,1,10,1) + 11/2(1,1,4,1) + 2(1,1,1,10) + 1(1,1,5,1) + 1/2(1,5,4,1) + 3(1,5,1,1) + 6(5,1,1,1) + 1/2(16,1,1,1)$$
$$4(1,1,1,1) + 3/2(4,4,4,1) + 1(4,4,1,1) + 3/2(4,1,1,1) + 3/2(4,1,5,1) + 1(4,5,1,1) + 1(1,10,1,1) + 11/2(1,4,1,1) + 1/2(1,4,5,1) + 1(1,1,10,1) + 7(1,1,4,1) + 2(1,1,1,10) + 1(1,1,5,1) + 1(1,5,4,1) + 1(1,5,1,1) + 6(5,1,1,1) + 1/2(16,1,1,1)$$
$$24(1,1,1,1) + 1(4,4,4,1) + 2(4,4,1,1) + 2(4,1,4,1) + 1(4,1,5,1) + 1(4,5,1,1) + 1(1,10,1,1) + 1(1,4,4,1) + 1/2(1,4,1,1) + 1(1,4,5,1) + 1(1,1,10,1) + 1/2(1,1,4,1) + 2(1,1,1,10) + 1(1,1,5,1) + 1(1,5,4,1) + 1(1,5,1,1) + 1/2(5,4,1,1) + 1/2(5,1,4,1) + 2(5,1,1,1) + 1/2(16,1,1,1)$$
$$18(1,1,1,1,1) + 1(4,4,4,1) + 2(4,4,1,1) + 1(4,1,4,1) + 4(4,1,1,1) + 1(4,1,5,1) + 1(4,5,1,1) + 1(1,10,1,1) + 1(1,4,4,1) + 1/2(1,4,1,1) + 1(1,4,5,1) + 1(1,1,10,1) + 2(1,1,4,1) + 2(1,1,1,1,10) + 1(1,1,5,1) + 1(1,5,4,1) + 1(1,5,1,1) + 1/2(5,4,1,1) + 1(5,1,4,1) + 1/2(16,1,1,1)$$
$$9(1.1.1.1.1) + 1(4.4.4.1.1) + 2(4.4.1.1.1) + 2(4.1.4.1.1) + 1(4.1.5.1.1) + 1(4.5.1.1.1) + 1(1.10.1.1.1) + 1(1.4.4.4.1) + 11/2(1.4.1.1.1) + 1(1.1.10.1.1) + 11/2(1.1.4.1.1) + 2(1.1.1.1.10) + 1(1.5.5.1.1) + 1/2(5.4.1.1.1) + 1/2(5.1.4.1.1) + 2(5.1.1.1.1) + 1/2(16.1.1.1.1)$$
$$3(1,1,1,1,1) + 1(4,4,4,1,1) + 2(4,4,1,1,1) + 1(4,1,4,1,1) + 4(4,1,1,1,1) + 1(4,1,5,1,1) + 1(4,5,1,1,1) + 1(1,1,10,1,1) + 1(1,4,4,1,1) + 11/2(1,4,1,1,1) + 1(1,1,10,1,1) + 7(1,1,4,1,1) + 2(1,1,1,1,10) + 1(1,5,5,1,1) + 1/2(5,4,1,1,1) + 1(5,1,4,1,1) + 1/2(16,1,1,1,1)$$
$$4(1.1.1.1.1) + 3/2(4.4.4.1) + 8(4.1.1.1) + 1(4.1.5.1) + 1(4.5.1.1) + 1(1.10.1.1) + 1(1.4.4.1) + 11/2(1.4.1.1) + 1(1.1.10.1) + 1/2(1.1.4.1) + 2(1.1.1.10) + 5(1.1.5.1) + 1(1.5.4.1) + 1(1.5.1.1) + 1/2(5.4.1.1) + 1/2(5.1.4.1) + 2(5.1.1.1) + 1/2(16.1.1.1)$$
$$4(1.1.1.1) + 3/2(4.4.4.1) + 8(4.1.1.1) + (4.1.5.1) + 1(4.5.1.1) + 1(1.10.1.1) + 1(1.4.4.1) + 3(1.4.1.1) + 1/2(1.4.5.1) + 1(1.1.10.1) + 3(1.1.4.1) + 2(1.1.1.10) + 3(1.1.5.1) + 1/2(1.5.4.1) + 3(1.5.1.1) + 1/2(5.4.1.1) + 1/2(5.1.4.1) + 2(5.1.1.1) + 1/2(16.1.1.1)$$
$$10(1.1.1.1.1) + 3/2(4.4.4.1.1) + 1/12(4.4.1.1.1) + 3/2(4.1.5.1.1) + 1(4.5.1.1.1) + 1(1.10.1.1.1) + 3/2(1.4.4.1.1) + 3/2(1.4.1.1.1) + 1(1.1.10.1.1) + 3/2(1.1.4.1.1) + 2(1.1.1.1.10) + 3(1.1.5.1.1) + 1/2(1.5.4.1.1) + 3(1.5.1.1.1) + 1/2(5.4.1.1.1) + 4(5.1.1.1.1) + 1/2(16.1.1.1.1)$$
$$4(1.1.1.1.1) + 3/2(4.4.4.1) + 11/2(4.1.1.1) + 3/2(4.1.5.1) + 1(4.5.1.1) + 1(1.10.1.1) + 1(1.4.4.1) + 11/2(1.4.1.1) + 1(1.1.10.1) + 3(1.1.4.1) + 2(1.1.1.10) + 3(1.1.5.1) + 1(1.5.4.1) + 1(1.5.1.1) + 1/2(5.4.1.1) + 4(5.1.1.1) + 1/2(16.1.1.1)$$
$$4(1.1.1.1.1) + 3(2.4.4.4.1) + 11/2(4.1.1.1.1) + 3/2(4.1.5.1) + 1(4.5.1.1) + 1(1.1.0.1.1) + 1(1.4.4.1) + 3(1.4.1.1) + 1/2(1.4.5.1) + 1(1.1.1.0.1) + 11/2(1.1.4.1) + 2(1.1.1.1.0) + 1(1.1.5.1) + 1/2(1.5.4.1) + 3(1.5.1.1) + 1/2(5.4.1.1) + 4(5.1.1.1) + 1/2(16.1.1.1)$$
$$18(1,1,1,1,1) + 1(4,4,4,1) + 1(4,4,1,1) + 1(4,1,4,1) + 11/2(4,1,1,1) + 3/2(4,1,5,1) + 1(4,5,1,1) + 1(1,10,1,1) + 2(1,4,4,1) + 1/2(1,4,1,1) + 1/2(1,4,5,1) + 1(1,1,2,0,1) + 1/2(1,1,4,1) + 2(1,1,1,1,0) + 1(1,1,5,1) + 1(1,5,4,1) + 1(1,5,1,1) + 1(5,4,1,1) + 1/2(5,1,4,1) + 1/2(16,1,1,1)$$
$$16(1.1.1.1) + 3/2(4.4.4.1) + 3(4.1.1.1) + 3/2(4.1.5.1) + 3/2(4.5.1.1) + 1(1.10.1.1) + 3(1.4.4.1) + 1(1.1.10.1) + 2(1.1.1.10) + 3(1.1.5.1) + 3(1.5.1.1) + 6(5.1.1.1) + 1/2(16.1.1.1)$$
$$10(1,1,1,1,1) + 3/2(4,4,4,1) + 3(4,1,1,1,1) + 3/2(4,1,5,1) + 3/2(4,5,1,1) + 1(1,1,1,1,1) + 2(1,4,4,1) + 4(1,4,1,1) + 1(1,1,1,1,1) + 3/2(1,1,4,1) + 2(1,1,1,1,1) + 3(1,1,5,1) + 1/2(1,5,4,1) + 1(1,5,1,1) + 6(5,1,1,1) + 1/2(16,1,1,1)$$
$$\text{spectrum } 349, \text{ branches } - [(3/2, 3), (3/2, 3), (1, -2)], \text{ singlets } = 4$$

$$4(1.1.1.1) + 3/2(4.4.4.1) + 3(4.1.1.1) + 3/2(4.1.5.1) + 3/2(4.5.1.1) + 1(1.10.1.1) + 1(1.4.4.1) + 11/2(1.4.1.1) + 1/2(1.4.5.1) + 1(1.1.10.1) + 11/2(1.1.4.1) + 2(1.1.1.10) + 1(1.5.1.1) + 1/2(1.5.4.1) + 1(1.5.1.1) + 6(5.1.1.1) + 1/2(16.1.1.1)$$
$$9(11111) + 1(10111) + 2(4441) + 3/2(4111) + 1/2(4151) + 1(11011) + 3/2(1411) + 1(11101) + 3/2(1141) + 2(11110) + 7(1151) + 1/2(1541) + 7(1511) + 1/2(5411) + 7(5111)$$
$$19(1,1,1,1) + 1(10,1,1,1) + 1(4,4,4,1) + 3(4,4,1,1) + 1/2(4,1,1,1) + 3/2(4,1,5,1) + 1(1,10,1,1) + 1/2(4,1,4,1) + 1/2(4,4,5,1) + 1(1,1,10,1) + 5(1,1,4,1) + 2(1,1,1,10) + 1(1,1,5,1) + 2(1,5,4,1) + 1(1,5,1,1) + 1(5,4,1,1) + 1(5,1,4,1) + 1(5,1,1,1)$$
$$10(1\mathbf{1}\mathbf{1}\mathbf{1}\mathbf{1}) + 1(10\mathbf{1}\mathbf{1}\mathbf{1}\mathbf{1}) + 1(4\mathbf{4}\mathbf{4}\mathbf{1}\mathbf{1}) + 3(4\mathbf{4}\mathbf{1}\mathbf{1}\mathbf{1}) + 1(4\mathbf{1}\mathbf{4}\mathbf{1}\mathbf{1}) + 4(4\mathbf{1}\mathbf{1}\mathbf{1}\mathbf{1}) + 1(1\mathbf{1}\mathbf{0}\mathbf{1}\mathbf{1}) + 1/2(1\mathbf{1}\mathbf{4}\mathbf{1}\mathbf{1}) + 1/2(1\mathbf{4}\mathbf{5}\mathbf{1}\mathbf{1}) + 1(1\mathbf{1}\mathbf{1}\mathbf{5}\mathbf{1}) + 6(1\mathbf{1}\mathbf{4}\mathbf{1}\mathbf{1}) + 2(1\mathbf{1}\mathbf{1}\mathbf{1}\mathbf{0}) + 2(1\mathbf{1}\mathbf{5}\mathbf{1}\mathbf{1}) + 2(1\mathbf{5}\mathbf{4}\mathbf{1}\mathbf{1}) + 1(1\mathbf{5}\mathbf{1}\mathbf{1}\mathbf{1}) + 1(5\mathbf{4}\mathbf{1}\mathbf{1}\mathbf{1}) + 1(5\mathbf{1}\mathbf{5}\mathbf{1}\mathbf{1})$$
$$4(11111) + 1(10111) + 1(4441) + 3(4411) + 11/2(4111) + 1/2(4151) + 1(11011) + 1/2(1411) + 1/2(1451) + 1(11101) + 10(1141) + 2(11110) + 2(1541) + 1(1511) + 1(5411) + 1(5151)$$

$$17(1,1,1,1) + 1(10,1,1,1) + 3/2(4,4,4,1) + 1(4,4,1,1) + 1(4,1,4,1) + 3/2(4,1,1,1) + 1/2(4,1,5,1) + 1(1,10,1,1) + 1/2(1,4,1,1) + 1/2(1,4,5,1) + 1(1,1,10,1) + 1/2(1,1,4,1) + 2(1,1,1,10) + 5(1,1,5,1) + 1(1,5,4,1) + 5(1,5,1,1) + 1(5,4,1,1) + 1/2(5,1,4,1) + 3(5,1,1,1)$$
$$11(1,1,1,1) + 1(10,1,1,1) + 3/2(4,4,4,1) + 1(4,4,1,1) + 11/2(4,1,1,1) + 1/2(4,1,5,1) + 1(1,10,1,1) + 1/2(1,4,1,1) + 1/2(1,4,5,1) + 1(1,1,10,1) + 2(1,1,4,1) + 2(1,1,1,10) + 5(1,1,5,1) + 1(1,5,4,1) + 5(1,5,1,1) + 1(5,4,1,1) + 1(5,1,4,1) + 1(5,1,1,1)$$
$$11(\mathbf{1,1,1,1}) + 1(\mathbf{10,1,1,1}) + 3/2(\mathbf{4,4,4,1}) + 1(\mathbf{4,4,1,1}) + 3(\mathbf{4,1,1,1}) + 1(\mathbf{4,1,5,1}) + 1(\mathbf{1,10,1,1}) + 3(\mathbf{1,4,1,1}) + 1(\mathbf{1,1,10,1}) + 2(\mathbf{1,1,4,1}) + 2(\mathbf{1,1,1,10}) + 5(\mathbf{1,1,5,1}) + 3/2(\mathbf{1,5,4,1}) + 3(\mathbf{1,5,1,1}) + 1(\mathbf{5,4,1,1}) + 1/2(\mathbf{5,1,4,1}) + 3(\mathbf{5,1,1,1})$$
$$11(\mathbf{1,1,1,1}) + 1(\mathbf{10,1,1,1}) + 3/2(\mathbf{4,4,4,1}) + 1(\mathbf{4,4,1,1}) + 3(\mathbf{4,1,1,1}) + 1(\mathbf{4,1,5,1}) + 1(\mathbf{1,10,1,1}) + 1/2(\mathbf{1,4,1,1}) + 1/2(\mathbf{1,4,5,1}) + 1(\mathbf{1,1,10,1}) + 9/2(\mathbf{1,1,4,1}) + 2(\mathbf{1,1,1,10}) + 3(\mathbf{1,1,5,1}) + 1(\mathbf{1,5,4,1}) + 5(\mathbf{1,5,1,1}) + 1(\mathbf{5,4,1,1}) + 1/2(\mathbf{5,1,4,1}) + 3(\mathbf{5,1,1,1})$$
$$11(1,1,1,1) + 1(10,1,1,1) + 3/2(4,4,4,1) + 1(4,4,1,1) + 1/2(4,1,1,1) + 3/2(4,1,5,1) + 1(1,10,1,1) + 3(1,4,1,1) + 1(1,1,10,1) + 9/2(1,1,4,1) + 2(1,1,1,10) + 3(1,1,5,1) + 3/2(1,5,4,1) + 3(1,5,1,1) + 1(5,4,1,1) + 5(5,1,1,1)$$
$$11(1,1,1,1) + 1(10,1,1,1) + 3/2(4,4,4,1) + 1(4,4,1,1) + 1/2(4,1,1,1) + 3/2(4,1,5,1) + 1(1,10,1,1) + 1/2(1,4,1,1) + 1/2(1,4,5,1) + 1(1,1,10,1) + 7(1,1,4,1) + 2(1,1,1,10) + 1(1,1,5,1) + 1(1,5,4,1) + 5(1,5,1,1) + 1(5,4,1,1) + 5(5,1,1,1)$$
$$19(1.1.1.1) + 1(10.1.1.1) + 1(4.4.4.1) + 2(4.4.1.1) + 1(4.1.4.1) + 1/2(4.1.1.1) + 3/2(4.1.5.1) + 1(1.10.1.1) + 2(1.4.1.1) + 1/2(1.4.5.1) + 1(1.1.10.1) + 7/2(1.1.4.1) + 2(1.1.1.10) + 1(1.1.5.1) + 2(1.5.4.1) + 1(1.5.1.1) + 3/2(5.4.1.1) + 1/2(5.1.4.1) + 1(5.1.1.1)$$
$$19(\mathbf{1,1,1,1}) + 1(\mathbf{10,1,1,1}) + 1(\mathbf{4,4,4,1}) + 2(\mathbf{4,4,1,1}) + 2(\mathbf{4,1,1,1}) + 2(\mathbf{4,1,5,1}) + 1(\mathbf{1,10,1,1}) + 1(\mathbf{1,4,4,1}) + 1/2(\mathbf{1,4,1,1}) + 1(\mathbf{1,1,10,1}) + 7/2(\mathbf{1,1,4,1}) + 2(\mathbf{1,1,1,10}) + 1(\mathbf{1,1,5,1}) + 2(\mathbf{1,5,4,1}) + 1(\mathbf{1,5,1,1}) + 3/2(\mathbf{5,4,1,1}) + 1/2(\mathbf{5,1,4,1}) + 1(\mathbf{5,1,1,1})$$
$$11(1,1,1,1) + 1(10,1,1,1) + 3/2(4,4,4,1) + 1(4,1,4,1) + 1/2(4,1,1,1) + 1/2(4,1,5,1) + 1(1,10,1,1) + 2(1,4,1,1) + 1/2(1,4,5,1) + 1(1,1,10,1) + 1/2(1,1,4,1) + 2(1,1,1,10) + 5(1,1,5,1) + 1(1,5,4,1) + 5(1,5,1,1) + 3/2(5,4,1,1) + 1/2(5,1,4,1) + 1(5,1,1,1)$$
$$5(1,1,1,1) + 1(10,1,1,1) + 3/2(4,4,4,1) + 7(4,1,1,1) + 1(4,1,5,1) + 1(1,10,1,1) + 9/2(1,4,1,1) + 1(1,1,10,1) + 2(1,1,4,1) + 2(1,1,1,10) + 5(1,1,5,1) + 3/2(1,5,4,1) + 3(1,5,1,1) + 3/2(5,4,1,1) + 1/2(5,1,4,1) + 1/5(1,1,1,1)$$
$$5(1,1,1,1) + 1(10,1,1,1) + 3/2(4,4,4,1) + 7(4,1,1,1) + 1(4,1,5,1) + 1(1,10,1,1) + 2(1,4,1,1) + 1/2(1,4,5,1) + 1(1,1,10,1) + 9/2(1,1,4,1) + 2(1,1,1,10) + 3(1,1,5,1) + 1(1,5,4,1) + 5(1,5,1,1) + 3/2(5,4,1,1) + 1/2(5,1,4,1) + 1(5,1,1,1)$$
$$5(1,1,1,1) + 1(10,1,1,1) + 3/2(4,4,4,1) + 9/2(4,1,1,1) + 3/2(4,1,5,1) + 1(1,10,1,1) + 9/2(1,4,1,1) + 1(1,1,10,1) + 9/2(1,1,4,1) + 2(1,1,1,10) + 3(1,5,1,1) + 3/2(1,5,4,1) + 3(1,5,1,1) + 3/2(5,4,1,1) + 3(5,1,1,1)$$
$$19(1,1,1,1,1) + 1(10,1,1,1) + 1(4,4,4,1) + 1(4,4,1,1) + 1(4,1,4,1) + 2(4,1,1,1) + 2(4,1,5,1) + 1(1,10,1,1) + 1(1,4,4,1) + 2(1,4,1,1) + 1(1,1,10,1) + 2(1,1,4,1) + 2(1,1,1,10) + 1(1,1,5,1) + 2(1,5,4,1) + 1(1,5,1,1) + 2(5,4,1,1) + 1(5,1,1,1)$$
$$19(1,1,1,1,1) + 1(1,0,1,1,1) + 1(4,4,4,1) + 3(4,4,1,1) + 1/2(4,1,1,1) + 1(4,1,5,1) + 1/2(4,5,1,1) + 1(1,1,0,1,1) + 1/2(1,4,1,1) + 1(1,4,5,1) + 1(1,1,1,1,0) + 5(1,1,4,1) + 2(1,1,1,1,1,0) + 1(1,1,5,1) + 3/2(1,5,4,1) + 1(1,5,1,1) + 1/2(5,4,1,1) + 3/2(5,1,4,1) + 1(5,1,1,1)$$

Spectrum 368: branches = $[(3/2, 3), (3/2, 3), (3/2, 3), (1, -2)]$; singlets = 10

$$1/(1 \ 1 \ 1 \ 1) + 1/(10 \ 1 \ 1 \ 1) + 1/(4 \ 4 \ 4 \ 1) + 3/(4 \ 4 \ 1 \ 1) + 1/(4 \ 1 \ 4 \ 1) + 3/(2 \ 4 \ 1 \ 1) + 1/2(4 \ 5 \ 1 \ 1) + 1/(1 \ 10 \ 1 \ 1) + 1/2(1 \ 4 \ 1 \ 1) + 2(1 \ 1 \ 1 \ 10) + 3/2(1 \ 5 \ 4 \ 1) + 1(1 \ 5 \ 1 \ 1) + 1/2(5 \ 4 \ 1 \ 1) + 2(5 \ 1 \ 1 \ 1) + 1(5 \ 1 \ 5 \ 1)$$

Spectrum 369; branches = $\{[(3/2, 3), (3/2, 3), (1, -2)]$; singlets = 4
 $4(1, 1, 1, 1) + 1/10(1, 1, 1, 1) + 1/4(4, 4, 1) + 3/4(4, 1, 1) + 1/12(4, 1, 1, 1) + 1/2(4, 5, 1, 1) + 1/1(1, 0, 1, 1) + 1/2(1, 4, 1, 1) + 1/1(4, 5, 1) + 1/1(1, 1, 0, 1) + 10(1, 1, 4, 1) + 2(1, 1, 1, 10) + 3/2(1, 5, 4, 1) + 1/1(5, 1, 1) + 1/2(5, 4, 1, 1) + 1/2(5, 1, 4, 1) + 1/5(5, 1, 5, 1)$

Spectrum 370: branches = $\{f(3/2, 3), (3/2, 3), (3/2, 3), (1, -2)\}$; signalets = 11

Spectrum 371: branches = $\{(3/2, 3), (3/2, 3), (3/2, 3), (1, -2)\}$; *sinolets* = 11

$$11(1,1,1,1) + 1(10,1,1,1) + 3/2(4,4,4,1) + 1(4,4,1,1) + 3(4,1,1,1) + 1/2(4,1,5,1) + 1/2(4,5,1,1) + 1(1,10,1,1) + 1/2(1,4,1,1) + 1(1,4,5,1) + 1(1,1,10,1) + 9/2(1,1,4,1) + 2(1,1,1,10) + 3(1,1,5,1) + 1/2(1,5,4,1) + 5(1,5,1,1) + 1/2(5,4,1,1) + 1(5,1,4,1) + 3(5,1,1,1)$$
$$11(1,1,1,1) + 1(10,1,1,1) + 3/2(4,4,4,1) + 1(4,4,1,1) + 1/2(4,1,1,1) + 1(4,1,5,1) + 1/2(4,5,1,1) + 1(1,10,1,1) + 1/2(1,4,1,1) + 1(1,4,5,1) + 1(1,1,10,1) + 7(1,1,4,1) + 2(1,1,1,10) + 1(1,1,5,1) + 1/2(1,5,4,1) + 5(1,5,1,1) + 1/2(5,4,1,1) + 1/2(5,1,4,1) + 5(5,1,1,1)$$

Spectrum 373; branches = [(3/2,3), (3/2,3), (3/2,3), (1,-2)]; singlets = 19

$$19(1,1,1,1,1) + 1(10,1,1,1,1) + 1(4,4,4,4,1) + 2(4,4,1,1,1) + 1(4,1,4,1,1) + 1/2(4,1,1,1,1) + 1(4,1,5,1) + 1/2(4,5,1,1) + 1(1,10,1,1) + 2(1,4,1,1) + 1(1,4,5,1) + 1(1,1,10,1) + 7/2(1,1,4,1) + 2(1,1,1,1,10) + 1(1,1,5,1) + 3/2(1,5,4,1) + 1(1,5,1,1) + 1(5,4,1,1) + 1(5,1,4,1) + 1(5,1,1,1)$$
$$19(1,1,1,1) + 1(10,1,1,1) + 1(4,4,4,1) + 2(4,1,1,1) + 2(4,1,1,1) + 3/2(4,1,5,1) + 1/2(4,5,1,1) + 1(1,10,1,1) + 1(1,4,4,1) + 1/2(1,4,1,1) + 1/2(1,4,5,1) + 1(1,1,10,1) + 7/2(1,1,4,1) + 2(1,1,1,10) + 1(1,1,5,1) + 3/2(1,5,4,1) + 1(1,5,1,1) + 1(5,4,1,1) + 1(5,1,4,1) + 1(5,1,1,1)$$

Spectrum $3/5$, branches = $[(3/2, 3), (3/2, 3), (3/2, 3), (1, -2)]$, singlets = 10
 $10(1\ 1\ 1\ 1\ 1) + 1(10\ 1\ 1\ 1\ 1) + 1(4\ 4\ 4\ 1\ 1) + 2(4\ 4\ 1\ 1\ 1) + 1(4\ 1\ 4\ 1\ 1) + 1/2(12\ 1\ 1\ 1\ 1) + 1/2(4\ 5\ 1\ 1\ 1) + 1(1\ 10\ 1\ 1\ 1) + 1(1\ 4\ 4\ 1\ 1) + 1/2(1\ 4\ 1\ 1\ 1) + 1/2(1\ 4\ 5\ 1\ 1) + 1(1\ 1\ 10\ 1\ 1) + 9/2(1\ 1\ 4\ 1\ 1) + 2(1\ 1\ 5\ 1\ 1) + 3/2(1\ 5\ 4\ 1\ 1) + 1(1\ 5\ 1\ 1\ 1) + 1/5(4\ 1\ 1\ 1\ 1) + 1/5(1\ 5\ 1\ 1\ 1)$

$$10(1,1,1,1) + 1(10,1,1,1) + 1(4,4,4,1) + 2(4,4,1,1) + 1(4,1,4,1) + 11/2(4,1,1,1) + 1/2(4,5,1,1) + 1(1,10,1,1) + 1(1,4,4,1) + 1/2(1,4,1,1) + 1/2(1,4,5,1) + 1(1,1,10,1) + 9/2(1,1,4,1) + 2(1,1,1,10) + 2(1,1,5,1) + 3/2(1,5,4,1) + 1(1,5,$$

Spectrum 376; branches = [(3/2,3), (3/2,3), (3/2,3), (1,-2)]; singlets = 4

$$\text{Spectrum } 377: \text{branch} = \frac{5}{2}(3, 2), \frac{3}{2}(2, 1), \frac{1}{2}(1, 2); \text{singlet} = 4$$

Spectrum 377; branches = $[(3/2,3), (3/2,3), (1,-2)]$; singlets = 4
 $4(1,1,1,1) + 1(1/10,1,1) + 1(4,4,4,4) + 2(4,4,1,1) + 7(4,1,1,1) + 1/2(4,1,5,1) + 1/2(4,5,1,1) + 1(1,10,1,1) + 1(1,4,4,1) + 1/2(1,4,1,1) + 1/2(1,4,5,1) + 1(1,1,10,1) + 17/2(1,1,4,1) + 2(1,1,1,10) + 2/2(1,5,4,1) + 1(1,5,1,1) + 1/5(1,1,1,1) + 1/5(1,5,1,1)$

$$4(1,1,1,1,1) + 1(10,1,1,1,1) + 1(4,4,4,1,1) + 2(4,4,1,1,1) + 7(4,1,1,1,1) + 1/2(4,1,5,1) + 1/2(4,5,1,1) + 1(1,10,1,1) + 1(1,4,4,1) + 1/2(1,4,1,1) + 1/2(1,4,5,1) + 1(1,1,10,1) + 1/2(1,1,4,1) + 2(1,1,1,10) + 3/2(1,5,4,1) + 1(1,5,1,1) + 1(5,4,1,1) + 1(5,1,5,1)$$

Spectrum 378; branches = [(3/2,3), (3/2,3), (3/2,3), (1,-2)]; singlets = 10

$$10(1,1,1,1,1) + 1(10,1,1,1,1) + 1(4,4,4,1,1) + 2(4,4,1,1,1) + 1(4,1,4,1,1) + 1/2(4,1,1,1,1) + 1(4,1,5,1) + 1/2(4,5,1,1) + 1(1,10,1,1) + 1(1,4,4,1) + 3(1,4,1,1) + 1(1,1,10,1) + 7(1,1,4,1) + 2(1,1,1,10) + 2(1,5,1,1) + 1(1,5,5,1) + 1(5,4,1,1) + 1(5,1,4,1) + 1$$

Spectrum 379; branches = [(3/2,3), (3/2,3), (3/2,3), (1,-2)]; singlets = 4

$$4(1.1.1.1.1) + 1(10.1.1.1.1) + 1(4.4.4.4.1) + 2(4.4.1.1.1) + 1(4.1.4.1.1) + 1/2(4.1.1.1.1) + 1(4.1.5.1) + 1/2(4.5.1.1) + 1(1.10.1.1.1) + 7(1.4.1.1) + 1(1.1.10.1) + 17/2(1.1.4.1) + 2(1.1.1.1.10) + 1/2(1.5.4.1) + 1(1.5.5.1) + 1(5.4.1.1) + 1(5.1.4.1) + 1(5.1.1.1)$$

Spectrum 380; branches = [(3/2,3), (3/2,3), (3/2,3), (1,-2)]; singlets = 5

$$5(1.1.1.1.1) + 1(10.1.1.1) + 3/2(4.4.4.1) + 7(4.1.1.1) + 1/2(4.1.5.1) + 1/2(4.5.1.1) + 1(1.10.1.1) + 9/2(1.4.1.1) + 1/2(1.4.5.1) + 1(1.1,10.1) + 2(1.1.4.1) + 2(1.1,1.10) + 5(1.1.5.1) + 1(1.5.4.1) + 3(1.5.1.1) + 1(5.4.1.1) + 1(5.1.4.1) + 1(5.1.1.1)$$

Spectrum 381; branches = [(3/2,3), (3/2,3), (3/2,3), (1,-2)]; singlets = 5

Spectrum 381: branches = $[(3/2, 3), (3/2, 3), (3/2, 3), (1, -2)]$; singlets = 5
 $5(1, 1, 1, 1) + 1(1/10, 1, 1, 1) + 3(2/4, 4, 4, 1) + 9(2/4, 1, 1, 1) + 1(4, 1, 5, 1) + 1/2(4, 5, 1, 1) + 1(1/2, 1, 1, 1) + 9/2(1, 4, 1, 1) + 1/2(1, 4, 5, 1) + 1(1, 1, 10, 1) + 9/2(1, 1, 4, 1) + 2(1, 1, 1, 10) + 3(1, 1, 5, 1) + 1(1, 5, 4, 1) + 3(1, 5, 1, 1) + 1/5(4, 1, 1) + 1/2(5, 1, 4, 1) + 3(5, 1, 1, 1)$

Spectrum $382: \text{branches} = [(3/2,3), (3/2,3), (3/2,3), (1,-2)]$; $\text{sinlets} = 19$

$$19(1,1,1,1,1) + 1(10,1,1,1) + 1(4,4,4,1) + 1(4,4,1,1) + 1(4,1,4,1) + 2(4,1,1,1) + 3/2(4,1,5,1) + 1/2(4,5,1,1) + 1(1,10,1,1) + 1(1,4,4,1) + 2(1,4,1,1) + 1/2(1,4,5,1) + 1(1,1,10,1) + 2(1,1,4,1) + 2(1,1,1,10) + 1(1,1,5,1) + 3/2(1,5,4,1) + 1(1,5,1,1) + 3/2(5,4,1,1) + 1/2(5,1,4,1) +$$

1(5,1,1,1)

Spectrum 3B2: branches = $[(2/2, 2), (2/2, 2), (2/2, 2), (1, 2)]$; singlets = 10

Spectrum 383, branches = [(3/2,3), (3/2,3), (3/2,3), (1,-2)], Singlets = 19
 19(1.1.1.1)+1(1.10.1.1)+1(1.4.4.1)+1(1.4.4.1)+1(1.4.1.4)+2(4.1.1.1)+1(4.1.5.1)+1(4.5.1.1)+1(1.10.1.1)+1(1.4.4.1)+2(1.4.1.1)+1(1.4.5.1)+1(1.1.10.1)+2(1.1.4.1)+2(1.1.1.10)+1(1.1.5.1)+1(1.5.4.1)+1(1.5.1.1)+1(5.4.1.1)+1(5.1.4.1)+1(5.1.1.1)

Spectrum $384: 10 \text{ branches} = [(3/2, 3), (3/2, 3), (1, -2)]$; *singlets* = 4

$$4(1,1,1,1) + 1(10,1,1,1) + 1(4,4,4,1) + 1(4,4,1,1) + 1(4,1,4,1) + 7(4,1,1,1) + 1(4,5,1,1) + 1(1,10,1,1) + 1(1,4,4,1) + 2(1,4,1,1) + 1(1,4,5,1) + 1(1,1,10,1) + 7(1,1,4,1) + 2(1,1,1,10) + 1(1,5,4,1) + 1(1,5,1,1) + 1(5,4,1,1) + 1(5,1,5,1)$$

Spectrum 385; branches = [(3/2,3), (3/2,3), (1,-2), (2,4)]; singlets = 6

$$6(1,1,1,1) + 1(4,4,1,7) + 2(4,1,1,7) + 1(1,10,1,1) + 3(1,4,1,1) + 2(1,1,10,1) + 2(1,1,1,7) + 1(1,1,1,27) + 1(1,5,1,7) + 2(1,5,1,1) + 1(5,4,1,1) + 2(5,1,1,1) + 1/2(16,1,1,1)$$

Spectrum 333, substance 1: $(\frac{1}{2}, \frac{1}{2})$, $(\frac{1}{2}, \frac{1}{2})$, $(\frac{1}{2}, \frac{1}{2})$, $(\frac{1}{2}, \frac{1}{2})$, singlets

$$4(\mathbf{1,1,1,1}) + 1(\mathbf{10,1,1,1}) + 1(\mathbf{4,4,1,7}) + 2(\mathbf{4,4,1,1}) + 1(\mathbf{1,10,1,1}) + 2(\mathbf{1,1,10,1}) + 5(\mathbf{1,1,1,7}) + 1(\mathbf{1,1,1,27}) + 1(\mathbf{1,5,1,7}) + 2(\mathbf{1,5,1,1}) + 1(\mathbf{5,1,1,7}) + 2(\mathbf{5,1,1,1})$$

Spectrum 387; branches = [(3/2,3), (3/2,3), (1,-2), (2,4)]; singlets = 10

$$10(\mathbf{1,1,1,1}) + 1(\mathbf{10,1,1,1}) + 1(\mathbf{4,4,1,7}) + 2(\mathbf{4,4,1,1}) + 1(\mathbf{1,10,1,1}) + 2(\mathbf{1,1,10,1}) + 3(\mathbf{1,1,1,14}) + 2(\mathbf{1,1,1,7}) + 1(\mathbf{1,5,1,7}) + 2(\mathbf{1,5,1,1}) + 1(\mathbf{5,1,1,7}) + 2(\mathbf{5,1,1,1})$$

Spectrum 388; branches = [(3/2,3), (3/2,3), (1,-2), (2,4)]; singlets = 4

$$4(\mathbf{1,1,1,1}) + 1(\mathbf{10,1,1,1}) + 1(\mathbf{4,4,1,7}) + 1(\mathbf{4,4,1,1}) + 4(\mathbf{4,1,1,1}) + 1(\mathbf{1,10,1,1}) + 3/2(\mathbf{1,4,1,1}) + 2(\mathbf{1,1,10,1}) + 3(\mathbf{1,1,1,14}) + 2(\mathbf{1,1,1,7}) + 1(\mathbf{1,5,1,7}) + 2(\mathbf{1,5,1,1}) + 1/2(\mathbf{5,4,1,1}) + 1(\mathbf{5,1,1,7})$$

$n \geq 5$ Scan Result

The five-factor catalogue is empty, and the shadow argument excludes all higher-factor spectra

Quantity	Value
$n = 5$ spectra	0
Gauge-quintuple sectors	0
Branch classes	0
Detailed entries	0

Consequence

No anomaly-free $n = 5$ spectra were found. Consequently, no spectra exist for any $n \geq 5$.

Indeed, every admissible model with $n > 5$ would yield a nonempty five-factor shadow after deleting the additional factors and replacing their vector multiplets by drones. This contradicts the empty complete $n = 5$ catalogue.

Anomaly-vector convention. $\eta = ((0,1),(1,0))$, $a = (-2,-2)$, $b_R = (3/2,-5)$.

Input verification. The $n = 5$ catalogue reports zero spectra, zero gauge-quintuple sectors, zero branch classes, and zero detailed entries.